Clinical Study Protocol

Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

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Regulatory Agency Identifier Number(s)

IND: 151,117

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This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

Redacted for Public Disclosure

Protocol Number: D967JC00001

Amendment Number: 04

Study Treatments: Trastuzumab deruxtecan (T-DXd; AZD4552; DS-8201a), durvalumab, pertuzumab, paclitaxel, tucatinib.

The non-proprietary name of T-DXd is trastuzumab deruxtecan except in the United States where it is fam-trastuzumab deruxtecan-nxki.

Study Phase: Phase 1b/Phase 2

Short Title: Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan in Combination with other Anti-cancer Agents in Patients with Metastatic HER2-positive Breast Cancer

Acronym: DESTINY-Breast07

Study Physician Name and Contact Information will be provided separately

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PROTOCOL AMENDMENT SUMMARY OF CHANGES TABLE

DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Amendment 4

Overall Rationale for the Amendment:

The reason for CSP amendment 4 is to align with updated post-trial access reporting requirements for adverse events. Changes are provided in the table below.

Section #	**Description of Change**	**Brief Rationale**
NON-SUBSTANTIAL CHANGES		
Core CSP Section 6.7 Section 8 Section 8.3.12 Section 8.3.13	Updated wording to include collection of non-serious AEs and special situations after final DCO.	To clarify requirements for adverse event and special situation reporting after final DCO.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Protocol Title: A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07).

Short Title: Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan in Combination with other Anti-cancer Agents in Patients with Metastatic HER2-positive Breast Cancer.

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

EU CT: 2023-505309-18-00

Rationale:

Breast cancer is the second most common cancer in the world, the most frequent women's cancer, and the fifth most common cause of death from cancer (GLOBOCAN 2018). HER2-positive breast cancer, defined by amplification of the *HER2* gene and/or overexpression at the protein level, comprises approximately 20% of all breast cancers. There are multiple treatment options for HER2-positive breast cancer patients in the adjuvant and neoadjuvant settings; however, once resistance develops against HER2-targeting therapy, there are limited effective treatment options. Therefore, novel treatment options with better efficacy are needed for patients with HER2-positive advanced/MBC.

Trastuzumab deruxtecan (T-DXd; also known as DS-8201a), a HER2-targeting antibody-drug conjugate, has shown impressive monotherapy efficacy in HER2-positive breast cancer, with an ORR of 60.9% and mPFS of 16.4 months in late-line MBC (Modi et al 2020), leading to initial approval by the US FDA in 2019. Given the efficacy of T-DXd monotherapy in late line metastatic HER2-positive breast cancer, it is of interest to explore combinations with T-DXd that can lead to enhanced efficacy. In this Phase 1b/2 study, combinations of T-DXd with other agents that have the potential to increase anti-tumour activity will be explored based on the MOA of each compound and preclinical data for each combination.

Objectives and Endpoints

Some objectives and endpoints are module-specific, as indicated in Table 1.

Table 1 Objectives and Endpoints

Objectives	Module/Part	Endpoints/variables
Primary		
• To assess the safety and tolerability, and determine RP2D for the T-DXd combinations	Modules 1 to 5/Part 1	• AEs, SAEs, DLTs, laboratory findings
• To assess the safety and tolerability of T-DXd monotherapy and T-DXd combinations	All modules/Part 2	• AEs, SAEs, laboratory findings
Secondary		
• To assess anti-tumour activity of T-DXd combinations and T-DXd monotherapy	All modules/Part 1 and 2	• ORR is defined as the proportion of patients who have a CR or PR, as determined by the Investigator at local site per RECIST 1.1.
	All modules/Part 1 and 2	• PFS is defined as time from the date of randomisation until the date of progression as assessed by the Investigator at local site per RECIST 1.1, or death due to any cause.
	All modules/Part 2	• PFS2 is defined as time from the date of randomisation until the date of progression on next line treatment (the earliest of the progression event subsequent to first subsequent anticancer therapy) or death; second progression will be defined according to local standard clinical practice.
	All modules/Part 2	• DoR is defined as time from the date of first documented response until the date of documented progression or death in the absence of disease progression.
	All Modules/Part 2	• OS is defined as time from the date of randomisation until the date of death due to any cause.
• To assess the PK of T-DXd, total anti-HER2 antibody and MAAA-1181a for T-DXd monotherapy and in combination	Modules 1 to 5/Part 1 All modules/Part 2	• Serum concentration of T-DXd, total anti-HER2 antibody and MAAA-1181a
• To assess the PK of combination treatments	Modules 1 to 5/Part 1 and Modules 1 to 6/Part 2	• Serum concentration of durvalumab, pertuzumab, plasma concentration of paclitaxel, tucatinib
• To investigate the immunogenicity of T-DXd	Modules 1 to 5/Part 1 All modules/Part 2	• Presence of ADAs for T-DXd

Table 1 Objectives and Endpoints

Objectives	Module/Part	Endpoints/variables
• To investigate the immunogenicity of durvalumab	Module 1 and 4/Parts 1 and 2	• Presence of ADAs for durvalumab
• To investigate the immunogenicity of pertuzumab	Module 2/Parts 1 and 2	• Presence of ADAs for pertuzumab

AEs = adverse events; ADA = anti-drug antibody; CR = complete response; DLT = dose-limiting toxicity; DoR = duration of response; HER2 = Human epidermal growth factor 2; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; PFS2 = progression-free survival after next-line treatment; PK = pharmacokinetics; PR = partial response; RECIST 1.1 = Response Evaluation Criteria In Solid Tumours, version 1.1; RP2D = recommended phase 2 dose; SAEs = serious adverse event; T-DXd = trastuzumab deruxtecan.

For exploratory objectives and endpoints, see Section 3.

Overall Design

The study is modular in design allowing assessment of the safety, tolerability and anti-tumour activity of T-DXd in combination with other anti-cancer agents. Patients will be enrolled globally at approximately 110 sites in approximately 10 countries. New combination treatment arms may be added during the course of the study as additional modules, via a protocol amendment, as novel agents of interest become available for testing in this HER2-positive advanced/MBC population. Up to a total of 10 treatment combinations may be evaluated in this study.

Combination-treatment Modules 1 to 5 will have 2 parts: a dose-finding phase (Part 1), and a dose expansion phase (Part 2); the Part 2 dose-expansion phase will use the RP2D determined in Part 1. There is no Part 1 for the combination-treatment Module 6. Patients in Module 6 (T-DXd + tucatinib in patients with active brain metastases) will receive study treatment at the RP2D determined in Part 1 of Module 5 (T-DXd + tucatinib in patients with no/stable brain metastases).

Part 1 of each module will enrol patients with locally assessed HER2-positive advanced/MBC in second-line or later and patients will be Sponsor-assigned. The dose-escalation/de-escalation decision rules in the dose-finding phase will be based on an mTPI-2 algorithm (Guo et al 2017), and an SRC will review DLTs observed from all evaluable patients who have received the same dose of treatment. Combinations will move from Part 1 to Part 2 once the RP2D has been determined. Alternatively, Part 1 of Modules 1 and 4 may not need to be performed in this study if an RP2D is identified from BEGONIA and DESTINY-Breast08, respectively.

In Part 2 of Modules 0 to 5, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/metastatic disease (ie, 1L) will be randomly assigned to a treatment module.

Random assignment of patients will be stratified by PD-L1 expression (positive vs negative), hormone receptor status (positive vs negative), and disease status (recurrent vs *de novo*) in Modules 0 to 5.

In Modules 6 and 7, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer in the first- or second-line treatment setting (1L or 2L) will be randomly assigned to a treatment module.

An interim safety analysis will be performed separately in Part 2 of Modules 0 to 6 after the first CCI evaluable patients (for Modules 0 to 4) or the first CCI evaluable patients (for Modules 5 to 6) all have had the opportunity for 2 months of treatment or have discontinued or withdrawn from treatment.

Additional interim analyses may be further performed separately in Part 2 for select modules to support development purposes. A final analysis will be performed when the last patient in overall has been followed for at least 12 months or all patients have discontinued or withdrawn.

Disclosure Statement: This is a Phase 1b/2, parallel, open-label, modular, multicentre, international study assessing the safety and tolerability of T-DXd combinations in patients with HER2-positive advanced/MBC. There is no blinding in any of the modules.

Patient Population: The target population of interest in this study is patients with HER2-positive (as per ASCO/CAP 2018 guidelines) advanced/MBC and includes cohorts with no/stable brain metastases, as well as modules with active brain metastases.

Number of Patients:

A minimum of CCI and a maximum of CCI evaluable patients can be enrolled in each dose level in Part 1. In Part 2 Modules 0 to 4 inclusive, 50 or 75 patients will be enrolled including a minimum of CCI patients with evaluable paired biopsies in each module. In Part 2 Module 5, 35 patients will be enrolled including a minimum of CCI patients with evaluable paired biopsies. In Part 2 Modules 6 and 7, thirty-five patients will be enrolled and there is no requirement for a minimum number of evaluable paired biopsies to be obtained.

Refer to Section 9.2 for further details on sample size determination and Section 8.6 for details on obtaining the minimum number of CCI samples.

Intervention Groups and Duration:

A modular structure was chosen for this study to allow for flexibility to open and close different T-DXd combinations at a different pace, depending on whether the RP2D is available and ready to expand. This protocol refers to the following modules and study

treatments. For specific information on each of the study treatments, please refer to the relevant module.

- Module 0 (core CSP) – T-DXd monotherapy (Part 2 only)
- Module 1 (Appendix O) – T-DXd + durvalumab
- Module 2 (Appendix P) – T-DXd + pertuzumab
- Module 3 (Appendix Q) – T-DXd + paclitaxel
- Module 4 (Appendix R) -T-DXd + durvalumab + paclitaxel
- Module 5 (Appendix S) – T-DXd + tucatinib*
- Module 6 (Appendix T) – T-DXd + tucatinib in patients with active brain metastases (Part 2 only)*
- Module 7 (Appendix U) – T-DXd monotherapy in patients with active brain metastases (Part 2 only)

* Modules 5 and 6 will only be conducted at sites where tucatinib is commercially available or central supply of tucatinib is operationally feasible.

The core CSP contains common features applicable to all modules and provides all the information relevant to the T-DXd monotherapy arm in patients with no/stable brain metastases. Therefore, Module 0 is provided in the core CSP. Other module-specific information for Modules 1 to 7 is provided in the respective module documents.

In Part 1, patients will be Sponsor-assigned using an IRT to one of the modules as indicated in Table 2. In Part 2, patients will be randomly assigned to a module that is open for enrolment at the time. Patients in Modules 1 to 5 will receive study treatment at the RP2D as determined in Part 1. There is no Part 1 for the combination-treatment Module 6. Patients in Module 6 (T-DXd + tucatinib in patients with active brain metastases) will receive study treatment at the RP2D determined in Part 1 of Module 5 (T-DXd + tucatinib in patients with no/stable brain metastases).

There is no dose-finding Part 1 in the monotherapy Modules 0 and 7 because the dose of T-DXd is established in HER2-positive breast cancer. Patients in Module 0 and Module 7 will receive T-DXd monotherapy at 5.4 mg/kg.

Patients should continue study treatment until a study treatment discontinuation criterion is met.

The combination of T-DXd (5.4 mg/kg Q3W) and durvalumab (1120 mg Q3W) is being studied in TNBC patients in the BEGONIA clinical trial (NCT03742102). Safety data from that study will be considered such that Module 1 Part 1 may not need to be performed in this

study if an RP2D is identified in that study. For more details regarding Module 1 Part 1, refer to Section 4.1.2 in the core CSP and to Module 1.

The combination of T-DXd (5.4 mg/kg Q3W), durvalumab (1120 mg Q3W), and paclitaxel (60 mg/m^2 QW) is being studied in HER2-low patients in the DESTINY-Breast08 clinical trial (NCT04556773). Safety data from that study will be considered such that Module 4 Part 1 may not need to be performed in this study if an RP2D is identified in that study. For more details regarding Module 4 Part 1, refer to Section 4.1.2 in the core CSP and to Module 4.

Table 2 Modules 1 to 5 Part 1, Dose Finding

	mTPI-2 escalation/de-escalation			
	Dose Level (1)	**Dose Level (-1)**	**Dose Level (2)**	**Dose Level (-2)**
Module 1 [a]	T-DXd: 5.4 mg/kg IV infusion Q3W Durvalumab: CCI mg IV QW	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QW	NA	NA
Module 2	T-DXd: 5.4 mg/kg IV infusion Q3W Pertuzumab: 1x loading dose CCI mg, CCI mg IV infusion Q3W	T-DXd: CCI mg/kg IV infusion Q3W Pertuzumab: 1x loading dose CCI mg, CCI mg IV infusion Q3W	NA	NA
Module 3	T-DXd: CCI mg/kg IV infusion Q3W Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Paclitaxel: CCI mg/m^2 IV QW	NA
Module 4 [b]	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QW Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QW Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QW Paclitaxel: CCI mg/m^2 IV QW	NA
Module 5	T-DXd: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid	T-DXd: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid	NA	T-DXd: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid

[a] Module 1 Part 1 may not need to be performed in this study if an RP2D is identified from BEGONIA.
[b] Starting dose level will be dependent on Module 3 Part 1. Alternatively, Module 4 Part 1 may not need to be performed in this study if an RP2D is identified from DESTINY-Breast08.

bid = twice daily; IV = intravenous; mTPI-2 = modified toxicity probability interval-2; NA = not applicable; PO = orally; QW = every week; Q3W = every 3 weeks; RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan.

Follow-up of patients post discontinuation of study treatment:

After study treatment discontinuation, all patients will undergo an end of treatment visit (within 7 days of discontinuation) and will be followed up for safety assessments 40 days (+ 7 days) (or as otherwise specified in the respective modules) (ie, until the safety follow-up visit).

Patients who have discontinued study treatment in the absence of RECIST 1.1-defined by Investigator assessment radiological progression will be followed up with tumour assessments according to the SoA in the respective modules until RECIST 1.1-defined PD or death regardless of whether or not the patient started a subsequent anticancer therapy, unless they have withdrawn consent to study related assessments.

In addition, all patients will be followed up for survival status after treatment discontinuation every 3 months (± 14 days) from the safety follow-up visit until death, withdrawal of consent, or the end of the study, as per the SoA in the respective modules.

In addition, patients in Part 2 of all modules may be followed up for PFS2 after treatment discontinuation every 3 months (± 14 days) from the date of the safety follow-up visit until death, withdrawal of consent, or the end of the study, as per the SoA in the respective modules.

See Section 6.7 for a description of assessments following study DCOs.

Data Monitoring Committee: A study-specific SRC (Section 9.6) will review the initial safety profile from each combination treatment in Part 1 of each module after 3 to 6 DLT evaluable patients have completed the first 21-day cycle or had a DLT during the first 21-day cycle. The SRC will continue to review all dose-finding data until an RP2D for the respective module is determined. The SRC will also review interim safety data in Part 2 of Modules 0 to 6.

Statistical methods

The study is exploratory in nature, and there is no formal hypothesis testing involved. Safety analyses of Parts 1 and 2 for each module will be performed using the SAF; immunogenicity analyses will also be performed using the SAF. Efficacy analyses will be performed using FAS or EAS. Analysis of PK parameters will be based on the PK Analysis Set.

There are no primary efficacy endpoints. Secondary efficacy endpoints include ORR, PFS, PFS2, DoR, and OS.

Confirmed ORR is defined as the percentage of patients with a confirmed Investigator assessed response of CR or PR, based on RECIST 1.1. The ORR will be reported using the point estimate and the 2-sided 80% CIs of ORR will be calculated by the Clopper-Pearson

method. The differences between the combination treatment arms against the T-DXd monotherapy will be reported using the point estimates as well as their 2-sided 80% CIs by the Miettinen-Nurminen method (Miettinen and Nurminen 1985). The odds ratio for the comparisons regarding ORR and their corresponding 80% CIs will be calculated using logistic regression models adjusted for all stratification factors in Modules 0 to 5. Treatment assignment will be the only covariate of the logistic models of Modules 6 and 7 as no stratification factors are defined.

Unconfirmed ORR is defined as the percentage of patients with an unconfirmed Investigator assessed response of CR or PR and will be analysed using the same methodology as confirmed ORR.

Kaplan-Meier estimates will be provided for PFS, PFS2, DoR and OS, with median and the 2-sided 80% CIs for the median using Brookmeyer and Crowley method for each treatment arm. Exploratory comparisons between the T-DXd combinations and the T-DXd monotherapy will be provided using stratified log-rank tests. The HR and its CIs will be estimated from a stratified Cox Proportional Hazards model (ties = Efron) with the CIs calculated using a profile likelihood approach.

An interim safety analysis will be performed separately in Part 2 of Modules 0 to 6 after the first [redacted] evaluable patients (for Modules 0 to 4) or the first [redacted] evaluable patients (for Modules 5 and 6) all have had the opportunity for 2 months of treatment or have discontinued or withdrawn from treatment. All available data from these evaluable patients up to the interim analysis will be used. Safety will be evaluated for each combination treatment and the assessment will entail review of AEs, SAEs and treatment-related AEs leading to discontinuations, dose interruptions, dose delays, or dose reductions. Enrolment in each module will continue during the conduct of the interim safety analysis up to a total of [redacted] patients unless the stopping criterion for Modules 1 to 6 described below is met. Enrolment beyond [redacted] patients in any given module may recommence when the interim analysis is completed and the SRC has determined that enrolment may continue.

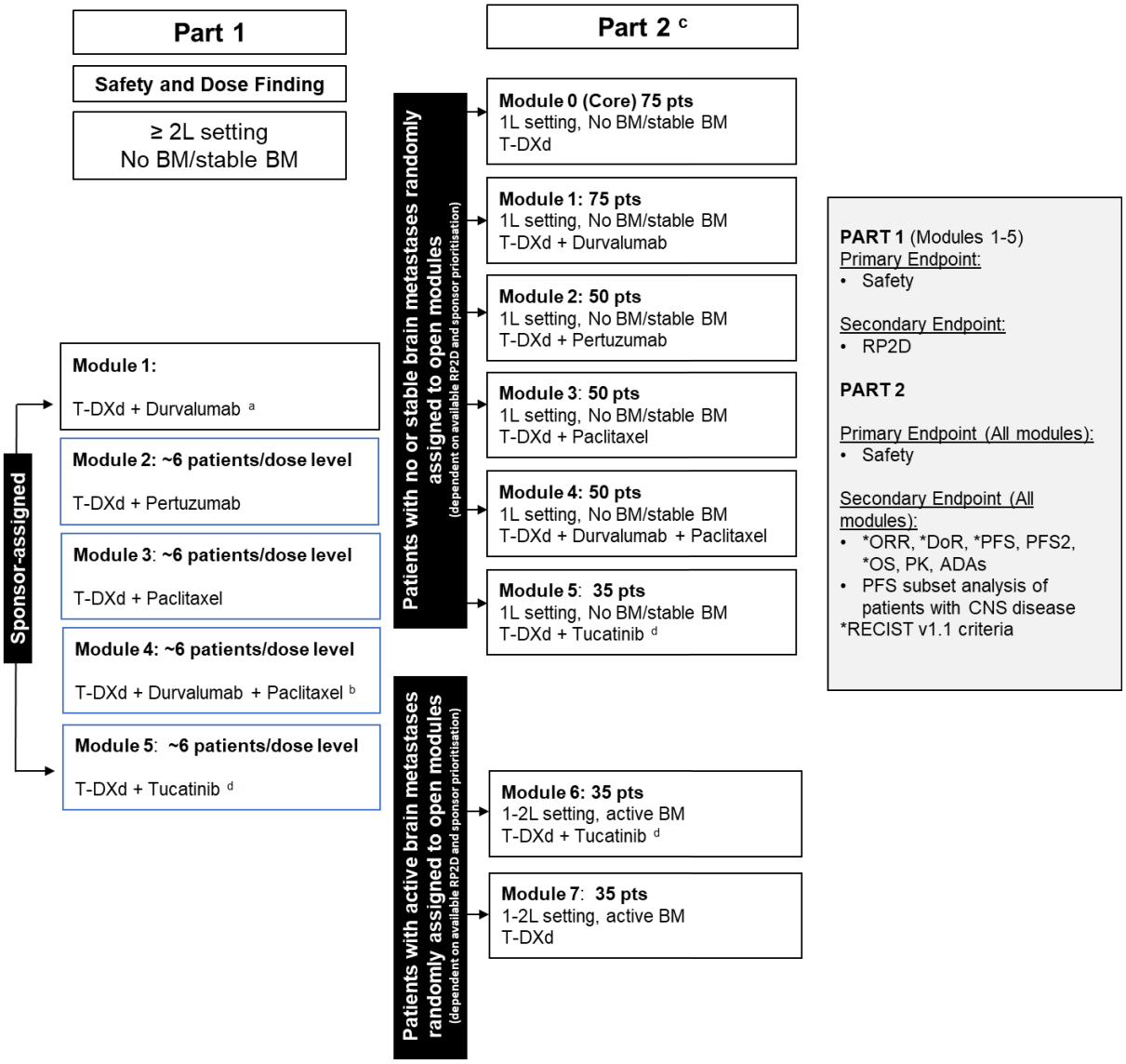
At the time of interim safety analysis, Modules 1 to 6 will be assessed for stopping based on the frequency of treatment-related AEs leading to discontinuation of all study treatments within a combination. Enrolment into Modules 1 to 6 will be paused if a pre-specified stopping criterion is met (Section 9.5). If the stopping threshold is met, enrolment in the module will be paused and an ad hoc SRC will be convened. All safety results will be presented to the SRC and a decision if the module needs to be permanently stopped from further enrolment will be made based on a comprehensive review by the SRC of safety data and the totality of the benefit-risk profile of the treatment regimen. No stopping rules will be applied to Modules 0 or 7, which are evaluating T-DXd monotherapy, for which a substantial body of safety data is already available (see T-DXd IB).

Additional interim analyses may be performed separately in Part 2 for select modules to support development purposes. A final analysis will be performed when the last patient in overall has been followed for at least 12 months or all patients have discontinued or withdrawn.

Safety data will be summarised descriptively unless otherwise specified.

1.2 Schema

Figure 1 Study Design



[a] Module 1 Part 1 may not be conducted in this study if the RP2D for the T-DXd + durvalumab combination is determined in the BEGONIA study (NCT03742102), a 1st line study in metastatic TNBC patients with HER2-low expression.

[b] Module 4 Part 1 is ungated by RP2D determination from Module 3. Alternatively, Module 4 Part 1 may not need to be performed in this study if an RP2D is identified from DESTINY-Breast08.

[c] For modules other than Module 0, Module 6 and Module 7, Part 2 of a module will be initiated after RP2D is determined in Part 1 of that module. In Module 6, the RP2D for the T-DXd + tucatinib combination will be determined in Part 1 of Module 5. Patients in Module 0 and Module 7 will receive T-DXd monotherapy at 5.4 mg/kg.

[d] Modules 5 and 6 will only be conducted at sites where tucatinib is commercially available or central supply of tucatinib is operationally feasible.

Note for Part 2: A total of [redacted] or [redacted] biopsies are required, depending on module. If the number of paired biopsies is less than the number required when targeted enrolment is achieved, additional patients may be enrolled beyond the target enrolment with mandatory biopsies until the required number of paired biopsies per module are met.

ADA = anti-drug antibody; BM = brain metastases; CNS = central nervous system; DoR = duration of response; HER2 = human epidermal growth factor receptor 2; 1L = first line; 1-2L = first or second line; 2-L = second line; ORR = objective response rate; OS = overall survival; PFS = progression free survival; PFS2 = progression-free survival after next-line treatment; PK = pharmacokinetics; RECIST = Response Evaluation Criteria in Solid Tumours; RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan; TNBC = triple-negative breast cancer.

1.3 Schedule of Activities

The table for screening assessments for the study (Table 3) is applicable to all modules. For procedures to be conducted in the treatment period and follow-up period for individual treatment modules refer to Table 4 and Table 5 in this core document for Module 0, and to the SoA tables in the relevant module documents for Modules 1 to 7. Details for procedures common to all modules are described in Section 8 of this core CSP. For details of treatment specific procedures for Modules 1 to 7 refer to Section 8 of the respective module documents.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

Unless all eligibility criteria have been met, patients must not be assigned to treatment and must not be treated. When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

All patients are required to sign the main ICF. After signing the main ICF, patients will begin the screening/baseline procedures. Screening will take place for up to 28 days from the date of enrolment. At the end of screening/baseline procedures, patients who pass the eligibility criteria will be assigned to treatment. Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures” (Section 5.4).

An optional pre-screening ICF may be signed by patients to permit tumour tissue sample collection for central PD-L1 testing and/or central retrospective HER2 status testing prior to the 28-day screening window. At the time of signing the pre-screening ICF, investigators should ensure that there is a reasonable possibility that the patient would be a candidate for this study based on available information (eg, medical history and availability of adequate tumour sample). When a pre-screening ICF is signed, it is recommended that the main ICF and other study procedures not be started until the sample is ready for submission to the central laboratory for PD-L1 and/or central retrospective HER2 testing. The tumor sample

collected during pre-screening may be submitted to fulfill the remaining mandatory sample requirements for biomarker analysis at study entry, as noted in core CSP Section 8.6.1, after the main ICF is signed.

1.3.1 Enrolment/Screening Period – All Modules

Table 3 Schedule of Activities for Screening/Baseline – All Modules

Procedures	Screening/Baseline	CSP Cross-reference
	D -28 to -1	
Informed consent: main study	X	5.1, 5.2
Informed consent: genetic sample and analysis (optional) [a]	X	8.7
Tumour assessments		
Tumour assessment including chest, abdomen, and pelvis by RECIST 1.1; Submit baseline images to iCRO for collect and hold (Part 2 only)	X	8.1.1, 8.1.2
Brain contrast MRI [b]	X	8.1.1
Whole body bone (scintigraphy) scan	X	8.1.1
Study procedures and assessments		
Inclusion and exclusion criteria [c]	X	5.1, 5.2
Surgical history	X	5.1, 5.2
Prior cancer therapies	X	5.1, 5.2
Medical history [d]	X	5.1, 5.2
Demography	X	5.1
High-resolution CT of the chest [e]	X	8.2.5.2
Pulmonary function test [f]	X	8.2.5.2
SpO_2	X	8.2.5.2
Full physical examination	X	8.2.1
ECOG PS	X	8.2.5.3
ECHO/MUGA for LVEF evaluation	X	8.2.5.1
12-lead ECG (triplicate)	X	8.2.3
Vital signs	X	8.2.2
Height	X	8.2.1
Weight	X	8.2.1
AE/SAE [g]	X	8.3
Concomitant medications	X	6.5
Laboratory assessments		
Clinical safety laboratory assessments (clinical chemistry, haematology, urinalysis) [h]	X	8.2.4, Table 11
Coagulation	X	8.2.4, Table 11
Serum pregnancy test (WOCBP only) [i]	X	8.2.4
Hepatitis B / Hepatitis C serology	X	8.2.4

Table 3 Schedule of Activities for Screening/Baseline – All Modules

Procedures	Screening/Baseline	CSP Cross-reference
	D -28 to -1	
HIV screening (as per local regulations)	CCI	8.2.4
Ophthalmologic assessments		8.2.5.4
Biomarker assessments		
Tissue sample from the metastatic setting: either a newly acquired biopsy [j] (where clinically feasible) or the most recent archival tissue (biomarker testing will include, but is not limited to, PD-L1 and exploratory biomarkers including CCI		8.6.1, 8.6.2
Archival tissue sample of primary tumour (where available)		8.6.1
Blood sample for CCI		8.6.1
Blood sample for CCI		8.6.1
Blood sample for CCI		8.6.1
Blood sample for CCI		8.6.1
Blood sample for CCI		8.6.1
Blood sample for CCI		8.6.1
Blood sample to CCI		8.6.1

[a] The blood sample for genetic research will be obtained at CCI (refer to the on-treatment SoA in the respective modules). If, for any reason, the sample is not drawn at CCI, it may be taken at CCI .

[b] Brain imaging by IV contrast-enhanced MRI will be required for all patients at screening/baseline. For Modules 0-5, patients with untreated brain metastases identified either on the baseline brain imaging obtained during the screening period, or identified prior to signing the ICF, are not eligible for the study (refer to exclusion criteria #10; Section 5.2). Those patients who are enrolled with baseline stable brain metastases will receive regularly scheduled follow-up brain scans while on study. For Modules 6 and 7, untreated patients with brain metastases are eligible, if the criteria #18 is met (Module 6, Section 5.1).

[c] For enrolment into the study, adequate organ and bone marrow function should be confirmed within 14 days before treatment assignment: (see inclusion criterion #10; Section 5.1)

[d] To include history, type and frequency of tobacco use, e-cigarette use, vaping (including dates).

[e] A non-contrast HRCT scan of the chest is preferred if feasible, otherwise a non-contrast CT is acceptable. The HRCT/CT chest scan will be performed for all patients, in addition to standard CT/MRI scans of chest/abdomen/pelvis for tumour assessments.

[f] Pulmonary function test as a minimum should include basic spirometry. (Minimum requirement of: FVC [L], FVC % predicted, FEV1 [L], FEV1 % predicted, FEV1/FVC%. Optional components to include PEF, FEV6, TLC, and DLCO). DLCO will be performed for all patients at screening/baseline.

[g] For patients who sign the pre-screening ICF, only SAEs directly related to tissue screening procedure (ie, if a patient undergoes a tumour biopsy) will be reported during the pre-screening period.

[h] Clinical chemistry assessments at screening include troponin (local assessment), amylase and lipase, and TSH assessments (refer to Table 11).

[i] For WOCBP, a negative result from a serum pregnancy test must be available at the screening visit. A pregnancy test (urine or serum test per institutional guideline) 72 hours before study entry is required. Within 72 hours before study entry, if a positive urine pregnancy test result is confirmed using a serum test in a female patient of child-bearing potential, then the patient should not be treatment-assigned.

[j] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to Section 8.6.1 and 8.6.2). Tumour sample collection does not need to be repeated if the patient went through pre-screening and a tumour sample was collected at pre-screening.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in Section 8.
AE = adverse event; C = cycle; CSP = clinical study protocol; CT = computed tomography; CCI CCI ; D = day; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; FFPE = formalin-fixed and paraffin-embedded; HER2 = human epidermal growth factor receptor 2; HIV = human immunodeficiency virus; HOSPAD = hospital admission; iCRO = imaging Contract Research Organisation; LVEF = left ventricular ejection fraction; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; CCI ; PD-L1 = programmed death-ligand 1; RECIST = Response Evaluation Criteria in Solid Tumours; SAE = serious adverse event; SpO_2 = pulse oximetry; TSH = thyroid stimulating hormone; WOCBP = women of child-bearing potential.

1.3.2 Treatment Period - Module 0 only

For the SoA for the treatment period of Module 0 Part 2, refer to Table 4. For the SoA for the treatment period for the other modules in this study, refer to Section 1.3 in the respective module documents.

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
Study procedures and assessments												
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.2
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2
Full physical examination	X [b]				X [b]		X [b]		X [b]		X	8.2.1
Weight	X [b]				X [b]		X [b]		X [b]		X	8.2.1
ECOG PS	X [b]				X [b]		X [b]		X [b]		X	8.2.5.3
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3
Laboratory assessments [c]												
Pregnancy test [d]	X				X		X		X		X	8.2.4
Clinical safety laboratory	X [b]		X	X	X [b]		X [b]		X [b]		X	8.2.4, Table 11

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
assessments (clinical chemistry, haematology)												
Urinalysis	As clinically indicated											Table 11
Coagulation	As clinically indicated											Table 11
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3
ECHO/MUGA (LVEF) [f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1
Troponin [g] (local assessment)	As clinically indicated										X	8.2.5.1
Ophthalmologic assessments [p]	As clinically indicated										X	8.2.5.4
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)												
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycles 4, 6, and 8)			8.5.1
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
Blood sample for T-DXd ADA assessment in serum [j]	X				X				X (C4, then every 4 cycles)		X	8.5.2
Study treatments administered												
T-DXd	5.4 mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1
Tumour assessments												
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.										X	7.1.2, 8.1.1
Whole body bone scan	As clinically indicated											8.1.1

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
Biomarker assessments												
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [k] (if not collected at screening)	X [l]											8.6.1, 8.6.2
Paired on-treatment biopsy					X [l]							8.6.1, 8.6.2
Tumour biopsy on disease progression											X (At progression only)	8.6.2
Blood sample for CCI	CCI [redacted]											8.6.1
Blood sample for CCI												8.6.1

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
CCI	CCI											
Blood sample for CCI												8.6.1
Blood sample for CCI												8.6.1
Blood sample for CCI												8.6.1
Blood sample for CCI [m]												8.6.1
Blood sample to CCI												8.6.1
Blood sample for CCI												8.7

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
CCI [redacted]												
Other assessments and procedures												
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [o]	X											8.1.5.1
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3
CANTAB	X								X (to be performed at Month 3, Month 6 and Month 12 for patients with stable brain metastases at baseline).		X	8.1.6.1
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2

Table 4 Schedule of Activities During the Treatment Period - Module 0 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days			
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons							
					BI	EI	BI	EI	BI	EI		
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening laboratory assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.
[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, EOT and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8.
[i] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[j] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[k] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to Section 8.6.1 and 8.6.2)
[l] Refer to Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[m] At time of suspected CCI [redacted], an additional sample should be taken.
[n] The sample for genetic research will be obtained at CCI [redacted]. If, for any reason, the sample is not drawn at CCI, it may be taken at CCI [redacted] CCI [redacted]. Only CCI [redacted] should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.

[o] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use in case needed, in accordance with device training.

[p] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB = Cambridge Neurophysiological Test Automated Battery; CNS = central nervous system; COVID-19 = coronavirus 2019-nCoV; CSP = clinical study protocol; CT = computed tomography; CCI D = day; ECG = electrocardiogram; DNA = deoxyribonucleic acid; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; E/D = Early Study Treatment/Discontinuation; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumour; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI ; T-DXd = trastuzumab deruxtecan.

1.3.3 Follow-up Period – Module 0 only

For the SoA for the follow-up period of Module 0 Part 2, refer to Table 5. For the SoA for the follow-up period for the other modules in this study, refer to Section 1.3 in the respective module documents.

Table 5 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 0

Procedures	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP
Vital signs	X		8.2.2
Weight	X		8.2.1
Full physical examination	X		8.2.1
SpO_2	X		8.2.5.2
ECOG PS	X		8.2.5.3
12-lead ECG	X		8.2.3
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4, Table 11
Urinalysis	As clinically indicated		Table 11
Coagulation tests	As clinically indicated		Table 11
Pregnancy test	X	As clinically indicated	8.2.4
Blood sample for T-DXd ADA analysis in serum	X		8.5.2
Concomitant medications	X		6.5
AEs/SAEs	X		8.3
PFS2 assessment	X	X	7.1.2.1, 8.1.3
Health care resource use (HOSPAD)	X		8.8

Table 5 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 0

Procedures	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: • To be completed daily up to and including the first survival visit.		8.1.5.2
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit • To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3
MDASI-BT		X (month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.2
NANO		X (month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.3
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5
Survival status	X	X	7.1.2.1, 8.1.4

Table 5 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 0

Procedures	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+ 7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumour; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

Trastuzumab deruxtecan (T-DXd; also known as DS-8201a) is a HER2-targeting ADC in development as a candidate therapy for breast cancer and other HER2-expressing tumours. The non-proprietary name for T-DXd is trastuzumab deruxtecan except in the United States where it is fam-trastuzumab deruxtecan-nxki.

2.1 Study Rationale

Before the discovery of HER2-directed therapies, an increased level of HER2 was a poor prognostic factor in breast cancer, leading to high recurrence rates and increased mortality (Chia et al 2008). In the past 20 years, significant progress has been made in the treatment of the disease, however HER2-positive MBC is still incurable and better HER2-targeting therapies are needed to move towards longer survival and cure.

T-DXd has shown impressive monotherapy efficacy in HER2-positive breast cancer, with an ORR of 60% and mPFS of 16.4 months in late-line MBC (Modi et al 2020), leading to its initial approval by the US FDA in 2019. Ongoing pivotal T-DXd monotherapy studies include DESTINY-Breast02 (NCT03523585) and DESTINY-Breast03 (NCT03529110), comparing T-DXd to physician's choice of chemotherapy and T-DM1, respectively.

There is an unmet need for targeted therapies in treatment of HER2-positive breast cancer to move towards longer patient survival and disease cure. Given the efficacy of T-DXd monotherapy in late-line metastatic HER2-positive breast cancer, it is of interest to explore combinations with T-DXd that can lead to enhanced efficacy. In this Phase 1b/2 study, combinations of T-DXd with other agents that have the potential to increase anti-tumour activity will be explored based on the MOA of each compound and preclinical data for the combination of each combination.

In addition to establishing tolerability, this study will also explore preliminary efficacy and establish biomarker data critical for optimising patient selection for T-DXd combinations. This will provide critical information when moving from late-line to first-line MBC treatment and further to treatment in early breast cancer, where treatments are typically given in multi-drug regimens to achieve maximal efficacy and potential for cure. This study may support the future development of T-DXd combinations in different breast cancer disease settings.

2.2 Background

2.2.1 Background on the Disease to be Treated

Breast cancer is the second most common cancer in the world, the most frequent women's cancer, with an estimated 2 million new cases in 2018 globally (11.6% of all new cancers), and the fifth most common cause of death from cancer, with an estimated 626,000 deaths in

2018 (GLOBOCAN 2018). According to 2019 estimates, over 268,000 women in the US will be diagnosed with breast cancer and 42,000 will die from the disease (Siegel et al 2019). Despite advances in diagnosis and treatment, around 6% of women diagnosed with breast cancer in the US present with *de novo* metastatic disease and up to 30% of women with early stage non---MBC will develop metastatic disease (O'Shaughnessy 2005). MBC remains incurable, with an estimated 5-year OS of 25% (Cardoso et al 2018).

HER2-positive breast cancer, defined by amplification of the *HER2* gene (ISH+, see Appendix F) and/or overexpression at the protein level (IHC 3+, or IHC 2+/ISH+), comprises approximately 20% of all breast cancers (Arteaga et al 2011). Herceptin (trastuzumab), the first approved HER2-targeting mAb, which is indicated for both the advanced (Slamon 2001) and early disease (Piccart-Gebhart et al 2005) settings, changed the natural history of the disease, significantly improving clinical outcomes. In the metastatic setting, the addition of trastuzumab to standard chemotherapy improved mPFS from 4.6 to 7.4 months, and OS from 20.3 months to 25.1 months (Slamon 2001).

Since trastuzumab, other HER2-targeted agents have been developed such as lapatinib, a TKI, and pertuzumab, an anti-HER2 humanised mAb. Lapatinib provided an mPFS of 8.4 months when combined with capecitabine in late-line locally advanced or metastatic disease (Geyer et al 2006). Pertuzumab demonstrated an mPFS of 18.5 months and mOS of 56.5 months in the first-line metastatic setting in combination with trastuzumab and chemotherapy (Swain et al 2015), as well as a pCR rate of 46% in the neoadjuvant setting (Gianni et al 2012); this established the combination as SOC in both indications. T-DM1 is an ADC that is currently approved in the second-line metastatic setting, based on mPFS of 9.6 months vs 6.4 months with lapatinib plus capecitabine (Verma et al 2012), as well as in the post-neoadjuvant setting in patients with residual disease after neoadjuvant therapy, based on improved invasive disease -free survival at 3 years of 88% in the T-DM1treated arm vs 77% in the trastuzumab- -treated arm (von Minckwitz et al 2019). Another TKI, neratinib, is approved for extended adjuvant therapy (Martin et al 2017).

Taken together, approvals of HER2-targeted therapies have vastly improved outcome in HER2-positive breast cancer, however new agents are needed to extend survival in the metastatic setting and increase cure rates in the early disease setting.

2.2.2 T-DXd

As of 28 March 2019, AstraZeneca and Daiichi Sankyo Company, Limited (Daiichi Sankyo) entered into a joint global development and collaboration agreement for T-DXd, a HER2-targeting ADC that is being developed as a therapeutic candidate for HER2-expressing tumours.

T-DXd consists of an anti-HER2 antibody, MAAL9001, covalently linked to ~8 molecules of MAAA-1162a (GGFG tetrapeptide cleavable linker and a topoisomerase I inhibitor [MAAA-1181a]). The antibody MAAL9001 has the same amino acid sequence as HERCEPTIN® (trastuzumab), and thus T-DXd is similarly targeted to HER2expressing tumours. The drug MAAA-1181a, a derivative of exatecan, is released after internalisation and leads to apoptosis of the target tumour cells via the inhibition of topoisomerase I.

Due to incorporation of a novel linker, T-DXd achieves a higher DAR of approximately 8 with homogeneous conjugation of MAAA-1181a, compared with other currently approved ADCs, which have a DAR of 3 to 4 (Ogitani et al 2016). In addition, the cleavable linker in T-DXd is stable in plasma, conferring a favourable safety profile as observed in nonclinical toxicology rat and monkey studies.

T-DXd exhibits HER2-specific anti-tumour activity via a MOA that combines the mAb- specificity with the broad cytotoxicity of the released drug. After binding to HER2 and internalisation, T-DXd is cleaved by lysosomal enzymes preferentially expressed in tumour cells and releases the drug MAAA-1181a in the cytoplasm. MAAA-1181a is an -exatecan- derivative with greater potency than SN-38, the active metabolite of irinotecan (Ogitani et al 2016). T-DXd is expected to exhibit anti-tumour activity through MAAA-1181a-induced apoptosis and, potentially, the ADCC activity of MAAL-9001, which leads to the inhibition of Akt phosphorylation.

There are completed and ongoing clinical trials with T-DXd, either alone or in combination, across multiple HER2-expressing tumour types including breast cancer, gastric cancer, NSCLC, and colorectal cancer (please refer to the T-DXd IB for a list of completed and ongoing trials and most recent patient exposure data).

For details on the schematic structures, molecular formula, finished product of T-DXd and the overall safety and clinical efficacy data from different studies that involved T-DXd, refer to the most recent IB.

2.2.3 Combination Treatments

Refer to the respective modules for background information on each of the respective combination treatments used in this study.

2.3 Benefit/Risk Assessment

The potential risks of T-DXd and of the classes of HER2-targeting agents and topoisomerase I inhibitors are provided in Sections 2.3.1, 2.3.2, and 2.3.3. Section 2.3.4 provides an assessment of the potential benefits that are associated with T-DXd, while Section 2.3.5 provides a high-level assessment of the benefit-risk of the T-DXd combinations proposed in this study, including combinations with immunotherapies, additional

HER2-targeting agents, and with chemotherapy. More detailed information about the known and expected benefits and potential risks of T-DXd may be found in the T-DXd IB, while more detailed information regarding benefits and risks for the specific combination treatments proposed in this study, including combination with durvalumab (Modules 1 and 4), pertuzumab (Module 2), paclitaxel (Modules 3 and 4), and tucatinib (Modules 5 and 6), are provided in the respective modules. The study will be performed in accordance with ethical principles that have their origin in the Declaration of Helsinki and are consistent with ICH GCP, and applicable regulatory requirements.

2.3.1 Risk Assessment

2.3.1.1 Potential risks of T-DXd

Based on cumulative review of safety data from nonclinical, clinical, epidemiologic information, scientific literature and taking into account biological plausibility, ILD/pneumonitis and neutropenia, including febrile neutropenia, are considered important identified risks associated with administration of T-DXd.

Based on the available pre-clinical data, review of the cumulative literature, and reported toxicities for the same class of agents, the important potential risks for T-DXd are LVEF decrease (re-labelled as ‘Left ventricular dysfunction’ as the undesirable clinical outcome of LVEF reductions, in accordance with the Revision 2 of the EMA guidelines on Good Pharmacovigilance Practice EMA GVP guidelines, Module V). This re-labelling of the risk does not affect the nature or monitoring methods of the LVEF decrease as a potential risk associated with T-DXd and embryo-foetal toxicity.

In the T-DXd clinical development program, specific inclusion/exclusion criteria and monitoring/management guidelines are currently in place to mitigate the important identified risks of ILD/pneumonitis and neutropenia, including febrile neutropenia, and important potential risks of LVEF decrease and embryo foetal toxicity.

ILD/pneumonitis and LVEF decrease are considered as AESIs and are closely monitored in the T-DXd clinical development program.

Keratitis is considered a potential risk for T-DXd. Other identified risks for T-DXd are infusion-related reactions, haematological toxicity (thrombocytopenia, anaemia, leukopenia, and lymphopenia), pulmonary/respiratory AEs (cough, dyspnoea, upper respiratory tract infection, and epistaxis), gastrointestinal AEs (abdominal pain, constipation, diarrhoea, dyspepsia, nausea, stomatitis, and vomiting), hepatic adverse events (hepatic function abnormality, alanine aminotransferase [ALT] increased, aspartate aminotransferase [AST] increased, alkaline phosphatase increased, and blood bilirubin increased), skin adverse events (alopecia, rash, and pruritis), pneumonia, dry eye, dehydration, decreased appetite, hypokalemia, dizziness, fatigue, peripheral oedema, pyrexia, and headache.

T-DXd has not been studied in subjects with moderate/severe hepatic impairment. Any patient could be at risk for ILD, but a higher incidence of Grade 1 and 2 ILD/pneumonitis has been observed in patients with moderate renal impairment at baseline. Patients with moderate renal impairment should be monitored carefully. ILD education is to be reemphasized in patients with moderate renal impairment. Physicians should consider more frequent patient contact in between visits (at least once in between cycles) in order to reinforce ILD education and the importance of prompt reporting of symptoms, until the Investigator is confident the patient is familiar with the requirements.

Trasutuzumab deruxtecan is an emetogenic anticancer agent, which includes delayed nausea and/or vomiting. Nausea and vomiting are identified risks and occur very commonly with trastuzumab deruxtecan.

These identified and potential risks are generally manageable through dose modification and routine clinical practice.

T-DXd has demonstrated a generally acceptable safety profile in late-line patients. In the DESTINY-Breast01 study, patients had received a median of 6 prior lines of treatments. The most common AEs of Grade 3 or higher that occurred in more than 5% of the patients were decreased neutrophil count (20.7%), anaemia (8.7%), nausea (7.6%), decreased white blood cell count (6.5%), decreased lymphocyte count (6.5%), and fatigue (6.0%). Twenty-eight patients (15.2%) discontinued treatment because of an AE (Modi et al 2020). In DESTINY-Breast01, out of [redacted] patients with [redacted] of therapy [redacted] patients ([redacted]) discontinued study treatment due to AE (unpublished data). The risks associated with T-DXd have been communicated in the IB and appropriate risk minimisation measures are in place, and exploration of the safety of T-DXd in a larger set of earlier-line patients is warranted.

2.3.2 Potential risks of HER2-targeted agents

Several agents that target HER2 and prevent its activation or heterodimerisation have been developed and marketed for the treatment of HER2-positive cancers. These include the monoclonal antibodies trastuzumab (Herceptin®) and pertuzumab (Perjeta®), the ADC T-DM1 (Kadcyla®), and HER1- and 2-associated TKI, lapatinib (Tykerb®), and neratinib (Nerlynx®). The safety profile of these HER2-targeted agents has been well described. The main safety risks identified in patients receiving HER2-targeted products are described below; these could potentially be expected to occur in patients receiving T-DXd.

Cardiotoxicity: Patients treated with trastuzumab are at increased risk for developing CHF (NYHA Class II to IV) or asymptomatic cardiac dysfunction, including LVEF decrease. Cardiac dysfunction, mainly asymptomatic LVEF decrease, has also been observed with pertuzumab in combination with trastuzumab. Similarly, cardiac dysfunction has been

observed in patients receiving T-DM1, at a lower incidence than in patients receiving trastuzumab. Most cases have been asymptomatic decreases in LVEF. Cardiac dysfunction with lapatinib has occurred mainly in patients receiving the combination of trastuzumab and lapatinib and has consisted of predominantly asymptomatic LVEF decrease.

Pulmonary toxicity: Cases of pulmonary toxicity, including ILD/pneumonitis, have been observed in patients receiving trastuzumab, T-DM1, and lapatinib. Occasionally, these cases have been severe in nature and have resulted in fatal outcomes. Risk factors associated with ILD/pneumonitis include prior or concomitant therapy with other antineoplastic therapies known to be associated with it such as taxanes, gemcitabine, vinorelbine, and radiation therapy.

Hypersensitivity/infusion-related reactions: The administration of therapeutic proteins is associated with a risk of hypersensitivity and/or infusion reactions. Hypersensitivity/infusion-related reactions have been reported with trastuzumab, pertuzumab and T-DM1. These can range from mild reactions to severe anaphylactic shock with fatal outcome, as has been the case for trastuzumab.

Hepatic toxicity: Cases of hepatic toxicity have occurred with T-DM1, lapatinib, and trastuzumab. In patients receiving T-DM1, hepatic toxicity has manifested mainly as transient asymptomatic liver transaminase elevations, although serious cases of drug-induced liver failure and nodular regenerative hyperplasia have also been reported. Lapatinib has also been associated with serious cases of drug-induced liver injury.

Haematological toxicity: Haematological toxicity has been observed with all HER2-targeted therapies. Neutropenia, febrile neutropenia, leukopenia, and anaemia have occurred commonly with trastuzumab, pertuzumab, and T-DM1. Thrombocytopenia, including Grade 3 and 4, is a common occurrence in T-DM1-treated patients. Although rare, serious haemorrhagic events have been reported in the setting of thrombocytopenia. Lower rates of thrombocytopenia have also occurred with trastuzumab and pertuzumab when used, in combination with chemotherapy.

Refer to Appendix L for T-DXd TMGs.

2.3.3 Potential Risks of Topoisomerase I inhibitors

MAAA-1181a is a derivative of exatecan (DX-8951f), a topoisomerase I inhibitor. Other products of the same class include irinotecan and topotecan. Exatecan is a campothecin derivative, which has previously been developed by the former Daiichi Pharmaceuticals Co., Ltd. as an anticancer therapy.

The main risks associated with the use of topoisomerase I inhibitors include haematological and gastrointestinal toxicities. Haematological toxicities, manifesting as neutropenia, febrile

neutropenia, anaemia, thrombocytopenia and pancytopenia are commonly observed. An increased risk of infections, including neutropenic colitis and neutropenic sepsis has been reported with these agents.

Acute and delayed onset diarrhoea, which can be severe and lead to dehydration, have been associated with topoisomerase I inhibitors. Other significant risks include ILD/pneumonitis, liver impairment, immune system disorders and alopecia. Acute cholinergic syndrome, manifesting as diarrhoea and other cholinergic symptoms has been reported with irinotecan.

The safety profile of exatecan is broadly similar to the safety profile of other topoisomerase I inhibitors, with haematological toxicities and gastrointestinal toxicities being the most significant groups of events.

2.3.4 Benefit Assessment

2.3.4.1 Potential benefits of T-DXd

Based on preliminary clinical observations in a Phase 1 study (Study DS8201-A-J101), T-DXd demonstrated anti-tumour activity in HER2-expressing cancers, including breast cancer and gastric cancer, and a generally acceptable safety profile in these populations. As of the DCO date of 01 Feb 2019, confirmed ORR by independent central review of 52.5% (95% CI: 43.1, 61.8) was shown among the 118 patients with HER2-positive breast cancer. In the recently published study DS8201-A-U201 (DESTINY-Breast01) a response to T-DXd therapy (5.4 mg/kg) was reported in 112 patients (60.9%; 95% CI, 13.8, 16.9) with HER2-positive MBC who had received previous treatment with T-DM1 and had median 6 lines of prior therapy. The median response duration was 14.8 months, and the median duration of PFS was 16.4 months. Estimated OS was 93.9% (95% CI: 89.3, 96.6) at 6 months and 86.2% (95% CI: 79.8, 90.7) at 12 months; the median OS was not reached at the time of the data cut-off (Modi et al 2020).

2.3.5 General Benefit/Risk Assessment of T-DXd Combinations

Combination with Immunotherapy: Based on data from recent clinical trials, combining HER2 ADCs with immunotherapy such as checkpoint inhibitors (eg, anti-PD-L1, anti-PD-1 antibodies, etc.) appear to be well tolerated. In KATE2, 53% of patients experienced ≥ Grade 3 AEs on TDM1 + atezolizumab vs 45% for T-DM1 + placebo and the most common of which were thrombocytopenia (13% vs 4%), AST increased (9% vs 3%), and anaemia (8% vs 0%), respectively. The use of immunotherapy comes with the risk of imAEs. On KATE2, the most common imAE was immune-related rash in 32% of patients receiving T-DM1 + atezolizumab vs 15% in combination with placebo followed by immune-related- hypothyroidism in 16% vs 7%, respectively (Emens et al 2019b). Similarly, in PANACEA, 29% of patients receiving pembrolizumab + trastuzumab experienced a ≥ Grade 3 treatment related AE (Loi et al 2019). Overall, both studies demonstrated that combination with immunotherapy has an acceptable safety profile while offering efficacy

benefits for patients and warrants further study with additional HER2-targeted agents such as T-DXd.

Combination with other HER2-targeting agents: Combinations of multiple HER2-targeted agents have shown promise in the clinic. Trastuzumab and pertuzumab, which target HER2 signaling in slightly different ways, have shown clinical benefit when used in combination with chemotherapy and have become SOC in the first-line setting for HER2-positive MBC (Baselga et al 2012). T-DM1, which is SOC in the second-line setting, showed improved tolerability with or without pertuzumab in the MARIANNE trial with 48.6%, 47.1%, and 55.8% of patients experiencing a ≥ Grade 3 AE in the T-DM1 + pertuzumab, T-DM1 alone, and trastuzumab + taxane arms, respectively (Perez et al 2019). This was also observed in the neoadjuvant setting with the KRISTINE trial where patients in the T-DM1 + pertuzumab arm experienced fewer ≥ Grade 3 AEs than patients receiving docetaxel, carboplatin, and trastuzumab plus pertuzumab (Hurvitz et al 2019). ADCs and TKIs have also been combined previously. A combination of T-DM1, lapatinib and nab-paclitaxel has been tested in the neoadjuvant breast cancer setting and showed similar AEs compared with the combination of trastuzumab, pertuzumab and paclitaxel (Patel et al 2019). These trials suggest that HER2-targeted agents can safely be combined and further evaluation of these and newer HER2-targeted agents with T-DXd is needed.

Combination with chemotherapy: Lastly, combination of HER2-targeted agents with chemotherapy is already SOC in the neoadjuvant (Gianni et al 2012), adjuvant (von Minckwitz et al 2017), and first-line metastatic setting (Baselga et al 2012) for HER2-positive breast cancer, demonstrating its effectiveness and utility in the clinic. In a study evaluating T-DM1 + paclitaxel with or without pertuzumab, patients experienced similar AEs in both arms, with dry eye, alopecia, epistaxis, diarrhoea and rash occurring less frequently in the T-DM1 + paclitaxel arm vs T-DM1 + paclitaxel + pertuzumab arm (Krop et al 2016). These results suggest HER2-targeted ADCs in combination with chemotherapy are well tolerated and support further evaluation in HER2-positive MBC.

Overall, HER2-targeted agents in combination with immunotherapy, other HER2-targeted agents (eg, pertuzumab), and chemotherapy have an acceptable safety profile and are associated with clinical benefit in the clinic. Additional studies are needed to further demonstrate the utility of these combinations for patients.

2.3.6 Overall Benefit/Risk Conclusion

There is an unmet medical need for better therapies in treatment of first and second-line HER2-positive MBC to move towards longer survival. There are multiple treatment options for HER2-positive breast cancer patients in the adjuvant and neoadjuvant settings, however once resistance develops against HER2-targeting therapy, there are limited effective treatment options. Re-challenge with previously administered therapies can be attempted, but their

efficacy may be compromised, usually depending on the disease-free interval between the last adjuvant therapy and metastatic disease (Metzger-Filho et al 2016). For patients who present with *de novo* Stage IV disease, responses to SOC therapy may be better than in patients with recurrent disease, but disease progression eventually occurs. New combination regimens may provide prolongation of survival and improvement of quality of life.

The important identified risks associated with administration of T-DXd are ILD/pneumonitis and neutropenia, including febrile neutropenia; LVEF decrease, and embryo-foetal toxicity are considered important potential risks. To specifically mitigate the incidence of pulmonary toxicities, strict inclusion/exclusion criteria have been included in this CSP, prohibiting most patients with pre-existing pulmonary co-morbidities from entering the study. In addition, baseline pulmonary function tests will be performed for all patients. For haematological toxicities, the use of growth factors is allowed at the discretion of the Investigator. Patients will be monitored closely throughout the study and clinical and laboratory assessments are performed before every cycle. TMGs are added to assist with the management of the most commonly seen AEs (Appendix L).

The emergence of the coronavirus 2019-nCoV (COVID-19) presents a potential safety risk for patients, therefore, several risk-mitigation factors have been implemented in this study. Details regarding instructions related to COVID-19 and a more detailed description of benefit/risk considerations relevant to COVID-19 are provided in Appendix M.

T-DXd has the potential to provide meaningful clinical benefit. Considering the measures to minimise risks to patients, the benefit-risk assessment supports the proposed study. This study aims to explore the efficacy and safety of T-DXd combinations. Combination-specific benefit-risk assessments are detailed in the equivalent sections in the respective modules.

3 OBJECTIVES AND ENDPOINTS

Table 6 Objectives and Endpoints

Objectives	Module/Part	Endpoints/Variables
Primary		
• To assess the safety and tolerability, and determine RP2D for the T-DXd combinations	Modules 1 to 5/Part 1	• AEs, SAEs, DLTs, laboratory findings
• To assess the safety and tolerability of T-DXd monotherapy and T-DXd combinations	All modules/Part 2	• AEs, SAEs, laboratory findings
Secondary		
• To assess anti-tumour activity of T-DXd combinations and T-DXd monotherapy	All modules/Part 1 and 2	• ORR is defined as the proportion of patients who have a CR or PR, as determined by the Investigator at local site per RECIST 1.1.
	All modules/Part 1 and 2	• PFS is defined as time from the date of randomisation until the date of progression as assessed by the Investigator at local site per RECIST 1.1, or death due to any cause.
	All modules/Part 2	• PFS2 is defined as time from the date of randomisation until the date of progression on next line treatment (the earliest of the progression event subsequent to first subsequent anticancer therapy) or death; second progression will be defined according to local standard clinical practice.
	All modules/Part 2	• DoR is defined as time from the date of first documented response until the date of documented progression or death in the absence of disease progression.
	All modules/Part 2	• OS is defined as time from the date of randomisation until the date of death due to any cause.
• To assess the PK of T-DXd, total anti-HER2 antibody and MAAA-1181a for T-DXd monotherapy and in combinations	Modules 1 to 5/Part 1 All modules/Part 2	• Serum concentration of T-DXd, total anti-HER2 antibody and MAAA-1181a
• To assess the PK of combination treatments	Modules 1 to 5/Part 1 and Modules 1 to 6/Part 2	• Serum concentration of durvalumab, pertuzumab, plasma concentration of paclitaxel, tucatinib
• To investigate the immunogenicity of T-DXd	Modules 1 to 5/Part 1 All modules/Part 2	• Presence of ADAs for T-DXd

Table 6 Objectives and Endpoints

Objectives	Module/Part	Endpoints/Variables
• To investigate the immunogenicity of durvalumab	Module 1 and 4/Parts 1 and 2	• Presence of ADAs for durvalumab
• To investigate the immunogenicity of pertuzumab	Module 2/Parts 1 and 2	• Presence of ADAs for pertuzumab
Exploratory		
• To assess the efficacy of T-DXd monotherapy and T-DXd combinations in terms of prevention of developing CCI CCI in patients with no CCI at baseline	CCI	BMFS: Time from date of randomisation until the date of diagnosis of CCI CCI according to investigator assessment using RECIST 1.1 or death by any cause.
• To collect blood and tissue samples (including evaluable CCI for each treatment combination) for defining biological responses to T-DXd and/or combination therapy by CCI and CCI with CCI and/or CCI CCI and CCI and CCI and CCI • Additional blood samples will be collected on development of ILD/pneumonitis symptoms or suspected ILD/pneumonitis development and/or diagnosis for additional CCI biomarker analysis.	CCI CCI	CCI
• To explore and optimise CCI	CCI CCI	
• To explore and optimise CCI	CCI CCI	• ILD/pneumonitis symptom and impact eDiary (CCI and CCI • At-home pulse oximetry • Qualitative patient interviews
• To assess the efficacy of T-DXd combined with tucatinib at CCI CCI	CCI	• CCI as determined per RECIST 1.1 • CCI as determined per RECIST 1.1 • CCI : the time from the date of randomisation until the date of documented disease progression in CCI according to investigator assessment using RECIST 1.1 or death by any cause

Table 6 Objectives and Endpoints

Objectives	Module/Part	Endpoints/Variables
• To assess the efficacy of T-DXd in combinations at CCI	CCI	• Time from the date of randomisation to second progression or death among patients who have CCI at baseline, then received local therapy for isolated CCI and continued study treatment • Time from the date of the first isolated CCI to second progression or death among patients who have CCI at baseline, then received local therapy for isolated CCI and continued study treatment
• To assess the efficacy of T-DXd combined with CCI in patients with CCI	CCI	• CCI per RANO-BM • CCI per RANO-BM • Time to progression in brain (CCI) per RANO-BM
• To assess the efficacy of T-DXd monotherapy and T-DXd in combination with other anti-cancer agents on CCI	CCI	• Descriptive analyses on CANTAB, MDASI-BT, and NANO selected scales
• To assess the efficacy of T-DXd combined with CCI in patients with CCI	CCI	• Time to next CCI local therapy is defined as time from randomization to the start date of the next CCI local therapy • Time to CCI is defined as time from randomisation to the start date of CCI

AEs = adverse events; ADA = anti-drug antibody; BMFS = brain metastases free-survival; CANTAB = Cambridge Neuropsychological Test Automated Battery; CCI; CR = complete response; CCI; DLT = dose-limiting toxicity; DoR = duration of response; CCI; IHC = immunohistochemistry; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumour; NANO = Neurologic Assessment in Neuro-Oncology; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; PFS2 = progression-free survival after next-line treatment; PK = pharmacokinetics; PR = partial response; RANO-BM = Response Assessment in NeuroOncology – Brain Metastases; RECIST 1.1 = Response Evaluation Criteria In Solid Tumours, version 1.1; RP2D = recommended phase 2 dose; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SAEs = serious adverse event; T-DXd = trastuzumab deruxtecan; CCI.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC.

4.1.1 Modular Protocol Structure

The details for the modules in this study are listed below. Modules 0 and 7 are T-DXd monotherapy modules. The study also comprises 5 T-DXd treatment combinations in 6 modules, Modules 1 to 6. Modules 1 to 6 may be initiated at different times per Sponsor discretion.

- Module 0 (core CSP) – T-DXd monotherapy (Part 2 only)
- Module 1 (Appendix O) – T-DXd + durvalumab
- Module 2 (Appendix P) – T-DXd + pertuzumab
- Module 3 (Appendix Q) – T-DXd + paclitaxel
- Module 4 (Appendix R) -T-DXd + durvalumab + paclitaxel
- Module 5 (Appendix S) – T-DXd + tucatinib*
- Module 6 (Appendix T) – T-DXd + tucatinib in patients with active brain metastases (Part 2 only)*
- Module 7 (Appendix U) – T-DXd monotherapy in patients with active brain metastases (Part 2 only)

* Modules 5 and 6 will only be conducted at sites where tucatinib is commercially available or central supply of tucatinib is operationally feasible.

This study consists of 2 parts: a dose-finding phase (Part 1), and a dose-expansion phase (Part 2). There is no dose-finding Part 1 in the monotherapy modules (Modules 0 and 7) because the dose of T-DXd is established in HER2-positive breast cancer. The Part 2 dose-expansion phases will use the RP2D of the treatment combination that was determined in Part 1 of each Module (Figure 1). There is no Part 1 in Module 6 (T-DXd + tucatinib in patients with active brain metastases). Patients in Module 6 will receive study treatment at the RP2D determined in Part 1 of Module 5 (T-DXd + tucatinib in patients with no/stable brain metastases).

The core CSP provides:

- All information applicable to all modules.
- The screening SoA table applicable to all modules (see Section 1.3.1)
- The on-treatment and follow-up SoA tables for Module 0, (see Sections 1.3.2).

T-DXd combination treatment-specific information for Modules 1 to 6, including the on-treatment and follow-up SoA tables, is provided in the respective module documents.

Patients will be enrolled globally at approximately 110 sites in approximately 10 countries. New combination treatments may be added during the course of the study as additional modules, via a protocol amendment, as novel agents of interest become available for testing in this HER2-positive advanced/MBC population. Up to a total of 10 treatment combinations may be evaluated in this study.

4.1.2 Part 1 Study Design Overview – Dose Finding

Part 1 study design details applicable to Modules 1 to 5 are provided below. For further module-specific details, refer to the respective modules. In Part 1, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled to determine the RP2D.

In the dose-finding phase, dose-escalation/de-escalation decision rules will be based on an mTPI-2 algorithm (Guo et al 2017) with 30% target toxicity rate and ± 5% equivalence radius (Table 7). An SRC (see Section 9.6) will review DLTs observed from all evaluable patients who have received the same dose of treatment. Following the rules described in Table 7, the SRC will decide whether the next [redacted] patients should be dosed at an escalated, a de-escalated, or at the same dose level. The SRC will continue to review all dose-finding data until an RP2D is determined. Escalation/de-escalation decisions regarding a dose level require data collected from a minimum of [redacted] evaluable patients treated at the dose level of interest unless unacceptable toxicity is encountered in the first [redacted] patients prior to enrolment of the [redacted] For a dose level that has fewer than [redacted] patients who are DLT-evaluable, patients will be replaced. A maximum of [redacted] evaluable patients can be enrolled at each dose level according to Table 7. At the highest protocol-specified dose level, enrolment may be stopped if no more than one DLT is observed in the first [redacted] evaluable patients.

Upon completion of the dose-escalation/de-escalation process, final DLT rates at each dose level will be estimated by isotonic regression (Ji et al 2010). Given the DLT estimates for each dose level, the RP2D will be selected from all tested dose levels that have not been previously declared to be unsafe with a DU (current dose is unacceptably toxic) decision according to the mTPI-2 decision table (Table 7). With this constraint, the RP2D will be determined as the dose level with the DLT estimate closest to the target toxicity level of 30%. At the discretion

of the Sponsor, dose-escalation/de-escalation may be stopped before an RP2D is reached. In this case, the recommended dose for the expansion arms may be chosen based on all available data, including but not limited to, assessment of PK, safety, or efficacy data.

Treatment combinations will move from Part 1 to Part 2 once the RP2D has been determined.

Table 7 Modified Toxicity Probability-2 (mTPI-2) Interval Decision Rules

CCI

[a] The columns indicating the actions based on data from CCI patients are included to reflect the completeness of the mTPI−2 design. However, in this study, a minimum of CC patients will be enrolled at each dose level, unless unacceptable toxicity is seen.
[b] The original mTPI algorithm was S for these cases and is modified to either CCI according to whether the observed DLT rate is < or ≥ the target toxicity of 30%, respectively.
[c] The original mTPI algorithm was S for this case and is modified CCI as the observed DLT rate is 50%.
D = de-escalate to the next lower dose level; DLT = dose-limiting toxicity; DU = current dose is unacceptably toxic. De-escalate to a lower dose and the current dose will never be used again in the trial; E = escalate to the next higher dose level; mTPI = modified toxicity probability interval; S = stay at the current dose level.

4.1.2.1 DLT Definitions

Dose-Limiting Toxicity

A DLT is defined as any TEAE not attributable to disease or disease-related processes, that occurs during the DLT evaluation period (Day 1 to 21 in Cycle 1 of Part 1 of each module) and is Grade 3 or greater according to NCI-CTCAE v5.0, with the exceptions as defined below:

For haematological toxicities, a DLT is defined as follows:

- Grade 4 neutrophil count decreased lasting > 7 days
- Grade 4 neutrophil count decreased with haematological growth factors prescribed
- Febrile neutropenia
- Grade 4 anaemia

- Grade 4 platelet count decreased
- Grade 3 platelet count decreased lasting > 7 days
- Grade 3 platelet count decreased with clinically significant haemorrhage
- Grade 4 lymphocyte count decreased lasting ≥ 14 days

Note: Administration of haemopoietic growth factors within the DLT assessment period is at Investigator discretion according to local practice and as required for patient safety. If it is deemed necessary to use prophylactic and/or therapeutic G-CSF during the DLT period to support the patient it will be allowed but it will constitute either DLT or non-evaluable per the dissertation of SRC.

For hepatic organ toxicities, a DLT is defined as follows:

- Grade 4 AST or ALT increased
- AST or ALT ≥ 3 × ULN, if accompanied by Grade > 2 blood bilirubin increased
- In patients without liver metastases, AST or ALT > 5 × ULN lasting > 3 days
- In patients with liver metastases, AST or ALT > 5 × ULN lasting > 3 days, if the baseline level was ≤ 3 × ULN
- In patients with liver metastases, AST or ALT > 8 × ULN lasting > 3 days, if the baseline level was > 3 × ULN

For non-haematological, non-hepatic toxicities, a DLT is defined as follows:

- Symptomatic CHF
- LVEF decline to < 40%, or > 20% drop from baseline
- Other Grade ≥ 3 non-haematological, non-hepatic toxicities

The following TEAEs are NOT considered DLTs:

- Grade 3 fatigue lasting < 7 days
- Grade 3 nausea, vomiting, diarrhoea, or anorexia that has resolved to Grade ≤ 2 within 3 days
- Isolated laboratory findings not associated with signs or symptoms, including but not limited to, Grade 3/4 ALP increased, hyperuricemia, serum amylase increased, lipase increased, and Grade 3 hyponatremia lasting < 72 hours developed from normal or Grade 1 at baseline
- Grade 3 lymphocyte count decreased

- AEs that are clearly not related to study treatment (eg, car accidents) per Investigator's assessment are not considered DLTs.

Once the Investigators or other site personnel indicate a **DLT** in the EDC system, an automated email alert is sent to the designated AstraZeneca representative.

DLT-Evaluable Patient

A DLT-evaluable patient is one who:

- Has received one full prescribed dose of T-DXd, and ≥ 65% of the prescribed dose of other study treatment for Modules 1, 2, 3, and 4 or ≥ 75% for Module 5, and has completed the first 21-day cycle, or
- Has received T-DXd and other study treatment (any amount) and had a DLT during the first 21-day cycle.

If a patient experiences a DLT and has only received T-DXd and not any other study treatment in the combination modules, this patient will not be considered DLT evaluable.

4.1.3 Part 2 Study Design Overview – Dose Expansion

Part 2 study design details applicable to all modules are provided below. For further module-specific details, refer to the respective modules.

In Part 2 of Modules 0 to 5, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/metastatic disease (ie, 1L) will be randomly assigned to a treatment module to determine preliminary evidence of anti-tumour activity, and the potential of the studied combination to become a therapeutic option for this patient population in the 1L setting. These modules will enrol patients with no or stable brain metastases.

In Modules 6 and 7, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer in the first- or second-line treatment setting (1L or 2L) will be randomly assigned to a treatment module to determine preliminary evidence of anti-tumour activity. These modules will enrol patients with active brain metastases.

Part 2 of each module will begin only after determination of the RP2D in Part 1, except for the monotherapy modules (Modules 0 and 7). Patients in Module 6 will be enrolled after the RP2D has been determined in Module 5. Approximately 35, 50, or 75 patients will be enrolled in Part 2 of each module (as noted in Figure 1). If the number of paired biopsies is less than the number required when targeted enrolment is achieved, additional patients may be enrolled beyond the target enrolment with mandatory biopsies until the required number of paired biopsies per module are met (for details, please see Section 8.6.1 and Section 8.6.2). Patients

will be randomly assigned to a module that is open for enrolment in Part 2 at the time. A randomisation method with dynamically changing randomisation ratios will be employed to account for fluctuation in the number of treatment arms open for enrolment over the course of the study. Randomisation will be stratified by PD-L1 expression (positive vs negative), hormone receptor status (positive vs negative), and disease status (recurrent vs *de novo*) in Modules 0 to 5. An interim safety analysis will be carried out for Modules 0 to 6 and stopping rules are in effect for Modules 1 to 6 (please refer to Section 9.5 for more details about these interim safety analyses).

4.1.3.1 Stratification factors

Patients in Part 2 of Modules 0 to 5 will be stratified by PD-L1 expression (PD-L1-positive vs PD L1-negative [PD-L1-positive is defined by ≥ 1% IHC]), hormone receptor status (positive vs negative), and disease status (recurrent vs *de novo*). Patients with active brain metastases will be randomised to Modules 6 and 7 without stratification due to limited sample size.

Recurrent disease status is defined as:

- Previously treated in adjuvant/neoadjuvant setting with chemotherapy, HER2-targeted agents or endocrine therapy
- Includes previously treated HER2-negative patients who now have HER2-positive disease in the metastatic setting.

De novo disease is defined as: no prior systemic therapy in adjuvant/neoadjuvant setting with chemotherapy, HER2-targeted agents, or endocrine therapy.

4.2 Scientific Rationale for Study Design

Rationale for this study is provided in Section 2.1. The rationale for study design, evaluation of T-DXd in the chosen patient population, and for the endpoints of this study are described below. Refer to the respective module documents for details on rationale for use of the respective treatment combinations.

4.2.1 Rationale for Study Design

This study consists of T-DXd monotherapy cohorts (Module 0 and Module 7) and 5 treatment combinations in 6 modules (Modules 1 to 6), which represent T-DXd with different combination partners or patient populations. Modules 6 and 7 will enrol patients with active brain metastases. A modular structure was chosen for this study to allow for flexibility to open and close different T-DXd combinations at a different pace, depending on whether the RP2D is available and ready to expand. In addition, some modules may be prioritised.

The core CSP contains common features applicable to all modules and provides all of the information relevant to the T-DXd monotherapy arm. Therefore Module 0 is provided in the

core CSP. Other module-specific information for Modules 1 to 7 is provided in the respective module documents.

Patients in Part 1 of Modules 1 to 5 will be Sponsor-assigned to treatment. All modules will be randomly assigned in Part 2 as randomisation is used to control the potential bias in the efficacy assessment of T-DXd monotherapy and T-DXd combinations. The larger target size (N = ~75) for Modules 0 and 1 will provide a more accurate estimate for the efficacy and safety, and will allow the Sponsor to better explore the prevalence and the predictive values of factors such as disease status and PD-L1 expression in this patient population. The efficacy, safety and biomarker results from all modules will support pivotal combination trials in metastatic and early breast cancer.

While T-DXd and immune checkpoint inhibitor combinations are already being tested, there are no clinical data for the other classes (chemotherapy and HER2-targeted agents) of compounds in combination with T-DXd. Depending on the safety profile, future studies may be pursued for each of these combinations in the metastatic or early-stage settings.

4.2.2 Rationale for T-DXd in HER2-positive advanced/MBC

Based on the results of the DS8201-A-U201 (ORR of 60.9%, mPFS of 16.4 months) study in late-line patients, it is hypothesised that T-DXd can provide greater benefit than the current SOC (docetaxel, trastuzumab and pertuzumab; mPFS 18.5 months) to patients with HER2-positive advanced/MBC in the first-line setting, either as a monotherapy or combined with other anti-cancer agents (Baselga et al 2012, Modi et al 2020). Many drugs have demonstrated improved efficacy moving from late-line to earlier. For example, T-DM1 showed mPFS of 6.2 months in third-line patients (Krop et al 2014), mPFS of 9.6 months in the second-line (Verma et al 2012) and mPFS of 14.1 months in first-line patients (Perez et al 2017). Therefore, it is reasonable to expect that the mPFS of 16.4 months in patients with median of 6 lines of prior therapy will translate into significant extension of mPFS using T-DXd with or without combination therapy in the first-line metastatic setting.

Although there are multiple treatment options for HER2-positive breast cancer patients in the adjuvant and neoadjuvant settings, however once resistance develops against HER2-targeting therapy, there are limited effective treatment options. Therefore, novel treatment options with better efficacy are needed for patients with HER2-positive advanced/MBC.

4.2.3 Rationale for T-DXd Combinations

The purpose of this Phase 1b/2 study is to characterise the safety and efficacy profile of novel T-DXd based combinations and provide data to inform future pivotal studies in the first-line metastatic and early-stage settings in breast cancer. As the current SOC in these settings is combination therapy, it is reasonable to test T-DXd-based combinations with immune

checkpoint inhibitors, as well as HER2-targeted therapies and chemotherapies that are already approved in HER2-postive breast cancer.

Studies evaluating PD-L1 expression in breast cancer have focused primarily on TNBC and evaluation in HER2-positive breast cancer has been limited. A recent study found that 17.5% of the HER2-positive breast cancer patients evaluated were PD-L1 positive and those patients responded well to neoadjuvant chemotherapy with trastuzumab (Kurozumi et al 2019). This suggests that exploration of immunotherapy with checkpoint inhibitors (eg, anti-PD-L1, anti-PD-1 antibodies, etc.) in HER2-positive MBC may hold clinical benefit. In KATE2, a randomised Phase 2 study evaluating SOC T-DM1 in combination with atezolizumab, an anti-PD-L1 antibody, vs T-DM1 + placebo in HER2-positive advanced or MBC patients, showed promising preliminary results. An mPFS of 8.2 months and 6.8 months (HR 0.82, p=0.332) for T-DM1 + atezolizumab and T-DM1 + placebo, respectively, was demonstrated in the overall study population but mOS had not been reached. PFS, however, was numerically greater with the combination in an exploratory analysis of a subgroup of patients who had PD-L1 expression in immune cells (median PFS: 8.5 vs 4.1 months; stratified HR, 0.60; 95% CI: 0.32, 1.11) (Emens et al 2019a). Additionally, immune checkpoint inhibitors were evaluated in PANACEA, a Phase 1b/2 trial evaluating pembrolizumab, an anti-PD-1 antibody, in combination with trastuzumab in a population refractory to trastuzumab. OS benefit was seen in patients with PD-L1-positive tumours (Loi et al 2019). Thus, these results warrant further evaluation to determine the benefit of combining immunotherapy with other HER2-targeted agents such as T-DXd, which has shown strong efficacy results in late-line HER2-positive MBC, following progression with T-DM1 and trastuzumab.

Combining multiple HER2-targeted agents for the treatment of HER2-positive MBC has also been evaluated and shown promise in the clinic. A combination of trastuzumab, taxane and pertuzumab is approved for the first-line treatment of HER2-positive MBC based on the results from the CLEOPATRA trial (Swain et al 2015). An evaluation of ADCs such as TDM1 in combination with pertuzumab was conducted in MARIANNE, a randomised Phase 3 study studying T-DM1 with or without pertuzumab vs trastuzumab + taxanes- in previously untreated HER2-positive MBC. In this study, T-DM1 and T-DM1 with pertuzumab- showed noninferior efficacy compared to trastuzumab plus taxane with an mPFS- of 14.1 months and 15.2 months compared to 13.7 months, respectively (Perez et al 2017). However, ORR was higher in trastuzumab + taxane (67.9%) than T-DM1 (59.7%) and TDM1 + pertuzumab- (64.2%) and did not translate to significant differences in OS (Perez et al 2019). T-DM1 + pertuzumab was also evaluated in the neoadjuvant setting vs SOC (trastuzumab, pertuzumab, and chemotherapy) in KRISTINE, a randomised Phase 3 study, and was well tolerated; however, improvement in pCR rate of patients was not observed (Hurvitz et al 2018). CCI [redacted] (data on file).

Similarly, combinations of ADCs and TKIs are also of interest. The TEAL study showed that T-DM1, lapatinib and nab-paclitaxel can be safely combined (Patel et al 2019). Preclinical data indicate that neratinib may increase the internalisation of the cytotoxic payload in HER2-amplified or mutant lung cancers (Li et al 2020). The most interesting clinical data on TKIs comes from tucatinib. In the HER2CLIMB study, the median PFS in patients receiving tucatinib, trastuzumab, and capecitabine was 7.8 months (95% CI: 7.5, 9.6) compared to 5.6 months (95% CI: 4.2, 7.1) for patients enrolled on the control arm (placebo, trastuzumab, and capecitabine [HR 0.54; 95% CI: 0.42, 0.71; p<0.00001]). Significant anti-tumour activity was seen in patients with brain metastases, in particular (Murthy et al 2020). Tucatinib has also been combined with T-DM1. In a Phase 1b study the combination showed acceptable toxicity and preliminary anti-tumour activity in heavily pre-treated patients with HER2-positive disease (Borges et al 2018). Based on the body of available data, an exploration of T-DXd in combination with other HER2-targeted agents in patients with HER2-positive MBC is warranted.

Taxanes are the cornerstone of the treatment regimens used in of HER2-positive breast cancer in the neoadjuvant (Gianni et al 2012), adjuvant (von Minckwitz et al 2017) and MBC (Baselga et al 2012) settings. Taxanes are commonly combined with all classes of HER2-targeted therapy, including antibodies, such as trastuzumab and pertuzumab as well as TKIs, such as lapatinib (Di Leo et al 2008). In addition, paclitaxel has been safely combined with T-DM1 (Krop et al 2016). Taxanes have also been previously combined with other topoisomerase I inhibitors, such as irinotecan, due to complementary MOAs (Stathopoulos et al 2005) and the combination showed a high response rate and acceptable safety profile in lung cancer. CCI
(data on file). Together these data provide a rationale for combining T-DXd and paclitaxel from both a safety and efficacy perspective.

4.2.4 Rationale for Study Endpoints

The primary objective of the study is to evaluate safety and tolerability of T-DXd in combination with other anti-cancer agents in patients with advanced or metastatic HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer.

The safety and tolerability of T-DXd-based combination regimens or T-DXd monotherapy will be assessed by the standard safety endpoints (Table 6). Careful consideration has been given to the mitigation of risks related to the mode of action, the nature of the target, and potential overlapping AEs, which will be closely monitored during the study.

Anti-tumour activity is a secondary objective of the study, and will be assessed by evaluating standard efficacy endpoints, such as ORR, PFS, PFS2, DoR and OS (Table 6).

The efficacy of T-DXd ± combinations in terms of prevention of developing brain metastasis in participants with no brain metastases at baseline will be assessed by BMFS as an exploratory endpoint. In addition, the efficacy of T-DXd monotherapy and T-DXd combined with tucatinib in participants with active CCI at baseline will be assessed in Modules 6 and 7, summarised by CCI, time to next CCI local therapy and time to CCI as exploratory endpoints.

ILD/pneumonitis-specific clinical outcome assessments will be included for exploratory purposes. ILD/pneumonitis is a known AE of T-DXd that occurs in approximately 13% of patients. Little is known about the baseline respiratory functioning of the population receiving T-DXd. Historically, ILD/pneumonitis is considered an idiopathic TEAE, whose MOA is poorly defined. Therefore, within this trial a multi-modality approach will be applied, including a digital patient diary for respiratory symptom collection, at-home pulse oximetry, site-based PFTs, and qualitative patient interviews of consenting diagnosed ILD/pneumonitis cases. The aim of the clinical outcome assessments is to better characterise ILD/pneumonitis and the progression of these cases.

The study will also investigate variations in PK and exploratory biomarker profiles, and their relationship to drug effect. These biomarkers may be derived from DNA, RNA, proteins and/or metabolites. There are many potential benefits of this research, including the possibility of identifying patients most likely to benefit from treatment, explaining outliers or non-responders, and potential adverse reactions related to drug exposure. This research may result in an understanding of the impact of variation between individuals, and how this knowledge can be utilised to bring better T-DXd drug combinations into the clinic.

4.3 Justification for Dose

The justification for T-DXd dose, which is used in all modules of this study, is provided below. Refer to the respective modules for dose justification for the combination study treatments.

4.3.1 T-DXd Dose Rationale

The target dose of T-DXd in Part 1 of each combination treatment module, and the dose for Module 0 and Module 7, in this study is 5.4 mg/kg. A T-DXd dose of 5.4 mg/kg has been granted accelerated approval in the United States for treatment of HER2-positive MBC and is the dose for which additional national and regional marketing authorisations are currently being filed. During dose-finding if the initial dose is not tolerated in combination with the proposed doses of the combination partners, a dose de-escalation will be undertaken (refer to Section 4.1.2). Due to the better benefit-risk profile seen in breast cancer at the 5.4 mg/kg dose compared to higher doses, T-DXd doses above 5.4 mg/kg will not be explored in this study.

For information on dose modifications for T-DXd, see Section 6.6.

4.4 End of Study Definition

A patient is considered to have completed the study when he/she has completed his/her last scheduled visit or last scheduled procedure shown in the SoA in the respective modules.

See Section 6.7 for details on patient management following the final DCO as well as following study completion.

Patients may be withdrawn from the study if the study itself is stopped. The study or any individual module may be stopped if, in the judgment of AstraZeneca, study patients are placed at undue risk because of clinically significant findings. The study may be terminated at individual centres if the study procedures are not being performed according to ICH GCP, or if recruitment is slow.

For the purpose of Clinical Trial Transparency (CTT) the definition of the end of the study differs under FDA and EU regulatory requirements:

- **European Union requirements** define study completion as the last visit of the last subject for any protocol related activity.
- **Food and Drug Administration requirements** defines two completion dates:
    - Primary Completion Date – the date that the final participant is examined or receives an intervention for the purposes of final collection of data for the primary outcome measure, whether the clinical study concluded according to the pre-specified protocol or was terminated. In the case of clinical studies with more than one primary outcome measure with different completion dates, this term refers to the date on which data collection is completed for all of the primary outcomes.
    - Study Completion Date – the date the final participant is examined or receives an intervention for purposes of final collection of data for the primary and secondary outcome measures and AEs (for example, last participant's last visit), whether the clinical study concludes according to the pre-specified protocol or is terminated.

AstraZeneca will continue to supply study treatment in the continued access phase of this study while, in the opinion of the Investigator, the participant is benefiting. Patients who remain on study treatment after the final DCO will be managed according to standard clinical practice by an Investigator. All SAE data is to be collected and reported to AstraZeneca via the paper form until the end of the follow-up period.

In the event that product development reaches a point where alternative product supply options become available, then these alternative product supply options will be discussed by AstraZeneca with the Investigator. AstraZeneca will work with the Investigator to transition the participant(s) to alternative supply, where possible.

In the event that a roll-over or safety extension study is available at the time of the final DCO and database closure, participant(s) currently receiving treatment with study treatment may then be transitioned to such a study, and the current study would reach its end. The roll-over or extension study would ensure treatment continuation with visit assessments per its protocol, as applicable. Any participant who would be eligible to move to such a study would be given a new informed consent, as applicable.

5 STUDY POPULATION

The target population of interest in this study is patients with pathologically documented HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC and includes cohorts with no/stable brain metastases, as well as modules with active brain metastases.

Each patient should meet all of the inclusion criteria and none of the exclusion criteria for this study in order to be assigned to a study treatment module. Under no circumstances can there be exceptions to this rule. Patients who do not meet entry requirements are screen failures (see Section 5.4). For procedures for withdrawal of incorrectly enrolled patients see Section 6.3.2.

In this protocol, enrolled patients are defined as those who sign informed consent, including the pre-screening ICF. Patients in Part 1 of Modules 1 to 5 will be Sponsor-assigned to treatment. Randomised patients in Part 2 are defined as those who undergo randomisation and receive a randomisation number (see Section 6.3). For procedures for withdrawal or incorrectly randomised patients, see Section 6.3.2.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

Eligibility criteria for all modules are listed below with a few exceptions where noted. Any module-specific eligibility criteria are listed in the respective modules. Where there are differences, the specific module takes precedence over the core.

5.1 Inclusion Criteria

Informed Consent

1. Capable of giving signed informed consent as described in Appendix A which includes compliance with the requirements and restrictions listed in the ICF and in this CSP.
2. Provision of signed and dated written Optional Genetic Research Information informed consent prior to collection of samples for optional genetic research that supports the AstraZeneca Genomic Initiative (Appendix D).

Age

3. Patient must be ≥ 18 years of age at the time of signing the informed consent.

Type of Patient and Disease Characteristics

4. Male or female patients who have pathologically documented breast cancer that:
   a. Is advanced/unresectable (patients that can be treated with curative intent are not eligible) or metastatic.
   b. Is HER2-positive (IHC 3+ or IHC 2+/ISH+) based on local assessment according to ASCO/CAP guidelines (Wolff et al 2018). The local HER2 result must be from a tumour sample obtained in the metastatic setting.
   c. Is documented as hormone receptor-positive (either oestrogen receptor and/or progesterone receptor positive [oestrogen receptor or progesterone receptor ≥ 1%]) or negative as per ASCO/CAP guidelines (Allison et al 2020) in the metastatic setting.
5. Patients must provide an adequate tissue sample: either FFPE tissue blocks (preferred) or at least 20 freshly-cut unstained serial tumour slides. An adequate sample must be provided from the metastatic setting; either a newly acquired biopsy or the most recent archival tumour tissue. For patients with relapsed disease an archival sample from the primary tumour (where available) must be submitted in addition to any more recent archival tissue provided. Refer to the Pathology Laboratory Manual for further details.
6. ECOG PS of 0 or 1.
7. **Part 1 of each module**:
   a. Radiological or objective evidence of disease progression on or after the last systemic therapy prior to starting study treatment.
   b. At least 1 prior treatment line in metastatic setting required. A ‘treatment line’ may consist of endocrine therapy, chemotherapy and/or anti-HER2 therapy, or a combination of any of these. Prior taxane, trastuzumab and pertuzumab exposure is allowed.
8. **Part 2 of Modules 0 to 5** (refer to Module 6 and Module 7 for module-specific eligibility criteria): No prior lines of therapy for advanced/MBC allowed. (Additional requirement: at least 12 months since last adjuvant HER2-targeted therapy, or chemotherapy required). Prior taxane, trastuzumab and pertuzumab exposure is allowed in the adjuvant or neoadjuvant setting.
9. LVEF ≥ 50% within 28 days before treatment assignment.
10. Adequate organ and bone marrow function within 14 days before treatment assignment as listed in Table 8. All parameters must meet the inclusion criteria on the same day and must be the most recent results available.

Table 8 Laboratory Values for Adequate Organ and Bone Marrow Function

Item	Laboratory value
Haemoglobin	≥ 9 g/dL (patients requiring ongoing transfusions or growth factor support to maintain haemoglobin ≥ 9 g/dL are not eligible. Red blood cell transfusion is not allowed within 1 week prior to screening assessment)

Table 8 Laboratory Values for Adequate Organ and Bone Marrow Function

Item	Laboratory value
Absolute neutrophil count	≥ 1,500/mm^3. (G-CSF administration is not allowed within 1 week prior to screening assessment)
Platelet count	≥ 100,000/mm^3 (platelet transfusion is not allowed within 1 week prior to screening assessment)
Total bilirubin	≤ 1.5 × ULN if no liver metastases or < 3 x ULN in the presence of documented Gilbert's Syndrome (unconjugated hyperbilirubinemia) or liver metastases at baseline
ALT and AST	≤ 2.5 × ULN, or < 5 × ULN in patients with liver metastasis
Serum albumin	≥ 2.5 g/dL
Creatinine clearance	≥ 30 mL/min, as calculated using the Cockcroft and Gault equation: $CrCl\ (mL/min) = \frac{[140 - age\ (years)] \times weight\ (kg)}{72 \times serum\ creatinine\ (mg/dL)}$ {× 0.85 for females}
INR/PT and either PTT or aPTT	≤ 1.5 × ULN

ALT = alanine aminotransferase; aPTT = activated partial thromboplastin time; AST = aspartate aminotransferase; G-CSF = granulocyte colony stimulating factor; INR = international normalised ratio; PT = prothrombin time; PTT = partial thromboplastin time; ULN = upper limit of normal.

11. Minimum life expectancy of 12 weeks at screening.
12. **Modules 0 to 5** (refer to Module 6 and Module 7 for module-specific eligibility criteria): At least one lesion, not previously irradiated, that qualifies as a RECIST 1.1 TL at baseline (see Appendix G).
13. Has adequate treatment washout period before randomisation, defined in Table 9.

Table 9 Adequate Treatment Washout Periods

Treatment	Washout period
Major surgery	≥ 4 weeks
Radiation therapy [a]	Including palliative stereotactic radiation therapy to chest: ≥ 4 weeks (palliative stereotactic radiation therapy to other anatomic areas: ≥ 2 weeks)
Targeted agents and small molecules	≥ 2 weeks or 5 half-lives, whichever is longer
Chemotherapy, immunotherapy, hormonal therapy (including hormone replacement therapy)	≥ 3 weeks; ≥ 6 weeks for nitrosoureas or mitomycin C
Antibody-based anti-cancer therapy	≥ 4 weeks, with the exception of RANKL inhibitors (eg, denosumab for the treatment of complications resulting from bone metastases)
Chloroquine/Hydroxychloroquine	> 14 days

Table 9 Adequate Treatment Washout Periods

Treatment	Washout period

[a] Module specific criteria applicable to patients with stable brain metastases in Modules 0 to 5 and all patients in Module 6 and 7. Refer to respective inclusion criteria in Section 5.1.
RANKL = Receptor activator of nuclear factor-kB (RANK), its ligand.

Reproduction

14. Evidence of post-menopausal status or negative serum pregnancy test for females of child-bearing potential who are sexually active with a non-sterilised male partner.
    (a) For women of child-bearing potential, a negative result for serum pregnancy test (test must have a sensitivity of at least 25 mIU/mL) must be available at the screening visit and urine beta-human chorionic gonadotropin pregnancy test prior to each administration of study treatment.

    (b) Women will be considered post-menopausal if they have been amenorrhoeic for 12 months without an alternative medical cause. The following age-specific requirements apply:
        (i) Women < 50 years of age will be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of exogenous hormonal treatments and if they have luteinizing hormone and follicle-stimulating hormone levels in the postmenopausal range for the institution or underwent surgical sterilization (bilateral oophorectomy, bilateral salpingectomy or complete hysterectomy).
        (ii) Women ≥ 50 years of age will be considered post-menopausal if they have been amenorrheic for 12 months or more following cessation of all exogenous hormonal treatments, had radiation-induced menopause with last menses > 1 year ago, had chemotherapy-induced menopause with last menses > 1 year ago, or underwent surgical sterilization (bilateral oophorectomy, bilateral salpingectomy or complete hysterectomy).

    (c) Women who are surgically sterile (ie, bilateral salpingectomy, bilateral oophorectomy, or complete hysterectomy) are eligible.

15. Females of childbearing potential (defined as those who are not surgically sterile or post-menopausal) who are sexually active with a non-sterilized male partner must use at least one highly effective method of contraception (Table 26) from the time of screening and must agree to continue using such precautions for 7 months after the last dose of study treatment; cessation of birth control after this point should be discussed with a responsible physician. Not all methods of contraception are highly effective. Female patients must refrain from egg cell donation and breastfeeding while on study and for 7 months after the last dose of study treatment. Not engaging in sexual activity for the duration of the study and drug washout period is an acceptable practice, provided it is the preferred and usual

lifestyle choice of the participant; however, periodic or occasional abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception.

16. **Modules 0, 1 and Modules 5 to 7** (refer to Modules 2, 3, and 4 for module-specific eligibility criteria): Non-sterilised male patients who are sexually active with a female partner of childbearing potential must use a condom with spermicide from screening to 4 months after the final dose of study treatment. Not engaging in sexual activity for the duration of the study and drug washout period is an acceptable practice, provided it is the preferred and usual lifestyle choice of the participant; however, periodic or occasional abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception (Appendix H 2). It is strongly recommended for the female partners of a male patient to also use at least one highly effective method of contraception throughout this period, as described in Table 26. In addition, male patients should refrain from fathering a child or freezing or donating sperm during the study and for 4 months after the last dose of study treatment in Modules 0, 1 and 5 to 7. The contraception duration for males after the last dose of study treatment in Modules 2, 3 and 4 is longer. (See Section 5.1 of Modules 2, 3, and 4 for module-specific eligibility criteria and also Appendix H 2). Preservation of sperm should be considered prior to enrollment in this study.
17. Female patients must not donate, or retrieve for their own use, ova from the time of screening and throughout the study treatment period, and for at least 7 months after the final study treatment administration. Preservation of ova may be considered prior to enrolment in this study.

CNS Inclusion

18. **Modules 0 to 5** (refer to Modules 6 and 7 for module specific eligibility criteria): Participant must have either:
    (a) No evidence of brain metastases.
    (b) Previously treated stable brain metastases.
        (i) Progressing brain metastases are not allowed in Module 0 to 5 (refer to Modules 6 and 7 for module-specific eligibility criteria).
    (c) Untreated brain metastases are not allowed in Module 0 to 5 (refer to Module 6 and Module 7 for module-specific eligibility criteria).

19. Participants with active or stable brain metastases must be neurologically stable:
    (a) Be receiving the equivalent of dexamethasone $\leq$ 2 mg/day
    (b) If receiving an anticonvulsant regimen, the regimen must have been stable for $\geq$ 14 days

20. **Modules 0 to 5**: Participants with stable brain metastases (prior CNS therapy and CNS disease is stable) or newly identified lesions found on contrast brain MRI performed must be:

(a) ≥ 14 days since SRS or gamma knife
(b) ≥ 21 days since WBRT

5.2 Exclusion Criteria

The below exclusion criteria are applicable to all modules. Patients are excluded from the study if any of the following criteria apply:

Medical Conditions

1. Uncontrolled intercurrent illness, including but not limited to, ongoing or active infection, uncontrolled or significant cardiovascular disease, serious chronic gastrointestinal conditions associated with diarrhoea, or psychiatric illness/social situations that would limit compliance with study requirements, substantially increase risk of incurring AEs or compromise the ability of the patient to give written informed consent.
2. Uncontrolled or significant cardiovascular disease including any of the following:
   (a) Patients with a medical history of myocardial infarction within 6 months before enrolment or symptomatic CHF (NYHA Class II to IV)
   (b) Uncontrolled hypertension
   (c) Uncontrolled and/or clinically important cardiac arrhythmias
   (d) Corrected QT interval by Fridericia's method (QTcF) prolongation to > 470 ms (females) or > 450 ms (male) based on average of screening triplicate 12-lead ECG
   (e) Patients with troponin levels above ULN at screening (as defined by the manufacturer), and without any myocardial infarction related symptoms, should have a cardiologic consultation before enrolment to rule out myocardial infarction
   (f) History of autoimmune myocarditis
3. Has a history of (non-infectious) ILD/pneumonitis that required steroids, has current ILD/pneumonitis, or where suspected ILD/pneumonitis cannot be ruled out by imaging at screening.
4. Lung criteria:
   (a) Lung-specific intercurrent clinically significant illnesses including, but not limited to, any underlying pulmonary disorder (eg, pulmonary emboli within 3 months of the study enrolment, severe asthma, severe COPD, restrictive lung disease, pleural effusion etc.)
   (b) Any autoimmune, connective tissue or inflammatory disorders (eg, Rheumatoid arthritis, Sjogren's, sarcoidosis etc.) where there is documented, or a suspicion of pulmonary involvement at the time of screening. Full details of the disorder should be recorded in the eCRF for patients who are included in the study.
   (c) Prior pneumonectomy.

5. Has known HIV infection, or active hepatitis B or C infection, such as those with serologic evidence of viral infection within 28 days of Cycle 1 Day 1. Subjects with past or resolved hepatitis B virus (HBV) infection who are anti-HBc positive (+) are eligible only if they are HBsAg negative (-). Patients positive for HCV antibody are eligible only if polymerase chain reaction is negative for HCV RNA. Subjects should be tested for HIV prior to enrolment if required by local regulations or IRB/EC.
6. History of another primary malignancy except for:
   (a) Malignancy treated with curative intent and with no known active disease ≥ 5 years before the first dose of study treatment
   (b) Adequately treated non-melanoma skin cancer or lentigo maligna without evidence of disease
   (c) Adequately treated carcinoma in situ without evidence of disease
7. Active or prior documented autoimmune or inflammatory disorders (including inflammatory bowel disease [eg, colitis or Crohn's disease], diverticulitis [with the exception of diverticulosis], systemic lupus erythematosus, sarcoidosis syndrome, Wegener syndrome [granulomatosis with polyangiitis], Graves' disease, and rheumatoid arthritis, hypophysitis, uveitis, etc). The following are exceptions to this criterion:
   (a) Patients with vitiligo or alopecia
   (b) Patients with hypothyroidism (eg, following Hashimoto syndrome) stable on hormone replacement
   (c) Patients with any chronic skin condition that does not require systemic therapy
   (d) Patients without active disease in the last 5 years, who may also be included but only after consultation with the Study Physician
   (e) Patients with celiac disease controlled by diet alone, who may also be included but only after consultation with the Study Physician
8. Uncontrolled infection requiring IV antibiotics, antivirals, or antifungals
9. Patients with spinal cord compression or a history of leptomeningeal carcinomatosis.

CNS Exclusion

10. **Modules 0 to 5** (refer to Modules 6 and 7 for module-specific eligibility criteria): Has untreated brain metastasis identified either on the baseline brain imaging obtained during screening period or identified prior to signing the ICF for Modules 0 to 5.

Prior/Concomitant Therapy

11. Patients who have received prior immune checkpoint inhibitors.
12. Receipt of live attenuated vaccine within 30 days prior to the first dose of study treatment (mRNA and replication deficient adenoviral vaccines are not considered attenuated live vaccines). **Note**: Patients, if enrolled, should not receive live vaccine whilst receiving study treatment and up to 30 days after the last dose of study treatment.

13. Unresolved toxicities from previous anticancer therapy, defined as toxicities (other than alopecia) not yet resolved to Grade ≤ 1 or baseline. Patients with chronic Grade 2 toxicities may be eligible per the discretion of the Investigator after consultation with the Study Physician or designee (eg, Grade 2 chemotherapy-induced neuropathy). Patients with irreversible toxicity not reasonably expected to be exacerbated by treatment may be included only after consultation with the Study Physician.
14. Any concurrent chemotherapy, study treatment, or biologic for cancer treatment.
15. Prior treatment with the following:
    (a) an ADC that comprises of an exatecan derivative that is a topoisomerase I inhibitor.
    (b) tucatinib.

Prior/Concurrent Clinical Study Experience

16. Participation in another clinical study with a study intervention administered in the last 30 days prior to randomisation or concurrent enrolment in another clinical study, unless it is observational (non-interventional) or during the follow-up period of an interventional study.
17. Has a history of severe hypersensitivity reactions to any of the study treatments, inactive ingredients in the drug product or to monoclonal antibodies.

Other Exclusions

18. Involvement in the planning and/or conduct of the study (applies to both AstraZeneca staff and/or staff at the study site).
19. Judgment by the Investigator that the patients should not participate in the study if the patient is unlikely to comply with study procedures, restrictions and requirements.
20. Ineligible for any of the combination partner therapies.
21. History of allogeneic organ transplantation.
22. Patients with prior use of immunosuppressive medication within 14 days prior to first study dose, except for intranasal and inhaled corticosteroids or systemic corticosteroids at doses less than 10 mg/day of prednisone or equivalent.
23. Female patients who are pregnant or breastfeeding.
24. Unable to swallow pills or has significant gastrointestinal disease which would preclude the adequate oral absorption of medications.
25. Has used a strong CYP3A4 or CYP2C8 inhibitor within 5 half-lives of the inhibitor prior to first dose of study treatment or have used a strong CYP3A4 or CYP2C8 inducer within 5 days prior to first dose of study treatment (see Appendix J 3).
26. Unable for any reason to undergo contrast MRI of the brain.

5.3 Lifestyle Considerations

The following restrictions apply while the patient is receiving study treatment and for the specified times before and after:

1. Patients must follow the contraception requirements outlined in Appendix H.
2. Patients should not donate blood or blood components while participating in this study and up to and including the safety follow-up visit (as defined in each respective module).
3. Restrictions relating to concomitant therapies are described in Appendix J 2.
4. Preservation of sperm/ova should be considered prior to enrolment in this study.

5.4 Screen Failures

Screen failures are defined as patients who consent to participate in the clinical study but are not subsequently assigned to treatment. A minimal set of screen failure information is required to ensure transparent reporting of screen failure patients to meet the Consolidated Standards of Reporting Trials publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria, and any SAEs.

Patients may be rescreened a single time. Rescreened patients should be assigned the same patient number (ie, E-code) as for the initial screening. However, rescreening data should be documented so that its effect on study results, if any, can be assessed. All assessments must be repeated for rescreening unless they are within 28 days of treatment assignment.

These patients should have the reason for study withdrawal recorded in eCRF as “eligibility criteria not fulfilled” (ie, patient does not meet the required inclusion/exclusion criteria). This reason for study withdrawal is only valid for screen failures (ie, patients who are not assigned to treatment).

6 STUDY INTERVENTION

Study intervention (referred to as study treatment in this CSP) is defined as any investigational intervention(s), marketed product(s), placebo, or medical device(s) intended to be administered to a study patient according to the CSP. Study treatment in this study refers to T-DXd monotherapy or in combination.

6.1 Study Intervention(s) Administered

AstraZeneca will supply T-DXd. Refer to the respective modules for supply information for the combination treatments. Details for T-DXd, which will be administered in all modules in this study, are described in this section. Refer to the respective module documents for details of the study treatments and dosing schedules applicable to each module.

T-DXd dose modifications are described in Section 6.6.

For information on the study treatment to be administered in this study, please refer to Table 10 for T-DXd, and to the respective module documents for combination treatments.

Table 10 Study Treatment – All Modules

Intervention Name	T-DXd
Type	ADC
Dosage Presentation	Vial
Unit Dose Strength(s)	Powder for concentrate for solution for infusion CCI mg/vial
Dosage Level(s)	CCI or 5.4 mg/kg [a]
Route of Administration	IV
Use	Experimental
Sourcing	Provided centrally by the Sponsor
Packaging and Labelling	Study treatment will be provided in CCI mg vials in carton. Each vial and carton will be labelled as required per country requirements [b]

[a] Module 0 (T-DXd monotherapy) will use a dose of 5.4 mg/kg.
[b] Label text for T-DXd (DS-8201a) will show "DS-8201a" depending on the agreed product name used in the respective approved study master label document. All naming conventions for these compounds are correct during this transitional period.
ADC = antibody-drug conjugate; IV = intravenous; T-DXd = trastuzumab deruxtecan

Duration of Treatment

Patients in each module will receive study treatment as stipulated in the respective SoA for each module until any discontinuation criteria are met (see Section 7.1).

Crossover within the study will not be permitted.

6.2 Preparation/Handling/Storage/Accountability of Interventions

1. The Investigator or designee (eg, pharmacist) must confirm appropriate temperature conditions have been maintained during transit for all study treatment received and any discrepancies are reported and resolved before use of the study treatment.
2. Only patients enrolled in the study may receive study treatment and only authorised site staff may supply or administer study treatment. All study treatment must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labelled storage conditions with access limited to the Investigator and authorised site staff.

3. The Investigator, institution, or the head of the medical institution (where applicable) is responsible for study treatment accountability, reconciliation, and record maintenance (ie, receipt, reconciliation, and final disposition records).
4. Further guidance and information for the final disposition of unused study treatments are provided in the Pharmacy Manual.

6.2.1 T-DXd Preparation and Administration

T-DXd will be supplied by AstraZeneca as a CCI mg lyophilised powder for concentrate for solution for infusion. Following reconstitution with sterile water for injection the reconstituted solution contains 20 mg/mL T-DXd in 25 mM histidine/histidine HCl, 90 mg/mL sucrose, 0.03% (w/v) polysorbate 80; it has a pH of 5.5 and density of 1.04 g/mL. The post-reconstitution label-claim volume is 5 mL.

The reconstituted product is a clear CCI CCI

Preparation of T-DXd

The dose of T-DXd for administration must be prepared by the Investigator's or site's designated study intervention manager using aseptic technique in compliance with local regulations and site requirements. Total time from needle puncture of the T-DXd vial to the start of administration must not exceed 24 hours. Of this time, not more than 4 hours may be at room temperature, with the remaining time at 2°C to 8°C (36°F to 46°F), otherwise a new dose must be prepared from new vials.

T-DXd should be handled in accordance with practices required for hazardous drugs (e.g. chemotherapy). Reconstituted T-DXd should be diluted in an infusion bag containing 100 mL of 5% dextrose solution. Incompatibilities have been observed with saline for injection and **must not** be used for dose preparation.

Following preparation and during administration, the prepared IV bag must be covered by light protection cover.

Refer to the Pharmacy Manual for detailed information on dose preparation and handling procedures for T-DXd.

Administration of T-DXd

It is recommended that participants receive prophylactic anti-emetic agents prior to infusion of T-DXd and on subsequent days. Antiemetics such as 5-HT3 antagonists or NK1 receptor antagonists and/or steroids (eg, dexamethasone) should be considered and administered in accordance with the prescribing information or institutional guidelines.

T-DXd will be administered using an IV bag containing 5% (w/v) dextrose injection infusion solution and delivered through an IV administration set with a 0.2- or 0.22-µm filter. The standard infusion time for T-DXd is approximately 90 minutes ± 10 minutes for the first infusion. If the first infusion is well tolerated and the patient does not experience an infusion-related reaction, then the minimum infusion time for subsequent cycles is 30 minutes. However, if there are interruptions during infusion, the total allowed time should not exceed 3 hours at room temperature.

Patients in Module 0 and Module 7 will receive a dose of 5.4 mg/kg T-DXd Q3W, and patients in the combination treatment modules will receive a dose of either CCI or CCI mg/kg T-DXd Q3W (refer to the respective modules documents for more details). The number of treatment cycles with T-DXd is not fixed. Upon commencing study treatment, patients will continue receiving T-DXd until RECIST 1.1 disease progression, withdrawal of consent or any of the discontinuation criteria are met.

The patient's weight at screening (baseline) will be used to calculate the initial dose. If, during the course of treatment, the patient's weight changes by ≥ ± 10%, the patient's dose will be recalculated based on the patient's updated weight.

T-DXd should be administered before other study treatments in this study. Other study treatments should not be co-administered with T-DXd through the same infusion line.

Monitoring of T-DXd administration

Patients will be monitored during and after infusion of T-DXd. Vital signs will be measured according to the SoA in each of the module documents.

Management of study treatment-related toxicities are described in Section 8.3.16.1. As with any biologic product, allergic reactions to dose administration are possible. Therefore, appropriate drugs and medical equipment to treat acute anaphylactic reactions must be immediately available, and study personnel must be trained to recognise and treat anaphylaxis.

6.2.2 Storage of T-DXd

The Investigator, or an appropriate delegate, will ensure that all study treatment is stored in a secured area, at appropriate temperatures and as specified on the label, and in accordance with applicable regulatory requirements. A calibrated temperature monitoring device will be used to record the temperature conditions in the drug storage facility. A temperature log will be used to record the temperature of the storage area. Temperature excursions outside the permissible range listed in the clinical supply packaging are to be reported to the Study Monitor upon detection. Storage conditions stated in the respective IBs may be superseded by the label storage instructions.

T-DXd vials are to be stored at 2°C to 8°C (36°F to 46 F) and must not be frozen. T-DXd must be kept in original packaging until use to prevent prolonged light exposure.

6.3 Measures to Minimise Bias: Randomisation and Blinding

This study is open label and blinding is not applicable.

6.3.1 Patient Enrolment and Randomisation

In Part 1 (not applicable for Module 0, Module 6, or Module 7), all patients will be Sponsor-assigned to study treatment using an IRT that will assign them depending on open treatment modules. Before the study is initiated, directions for the IRT will be provided to each site.

If a patient withdraws from the study, then his/her enrolment/randomisation E-code cannot be reused. Patients withdrawing during the DLT evaluation period will be replaced for the purpose of DLT assessment; withdrawn patients outside the DLT evaluation period will not be replaced.

Investigators should keep a record (ie, the patient screening log) of patients who enter pre-screening and/or screening.

Between Days -28 to -1, the Investigator or suitably trained delegate will do the following:

- Obtain signed ICF before any study-specific procedures are performed (see Section 8). If laboratory or imaging procedures were performed for alternate reasons prior to signing consent, these can be used for screening purposes with the consent of the patient. However, all screening laboratory and imaging results must have been obtained within 28 days of treatment assignment. For patients with a single TL, if screening biopsy is collected prior to screening imaging for baseline tumour assessment, allow approximately 2 weeks for healing before imaging scans are acquired. An optional pre-screening ICF may be signed by patients to permit tumour tissue sample collection for central PD-L1 testing and/or central retrospective HER2 status testing prior to the 28-day screening window. At the time of signing the pre-screening ICF, investigators should ensure that there is a reasonable possibility that the patient would be a candidate for this study based on available information (eg, medical history and availability of adequate tumour sample). When a pre-screening ICF is signed, it is recommended that the main ICF and other study procedures not be started until the sample is ready for submission to the central laboratory for PD-L1 testing. The tumor sample collected during pre-screening may be submitted to fulfill the remaining mandatory sample requirements for biomarker analysis at study entry, as noted in core CSP Section 8.6.1, after the main ICF is signed.
- Obtain a unique 7-digit enrolment number (E-code) at pre-screening or screening, through the IRT in the following format: PPD

PPD). This number is the patient's unique identifier and is used to identify the patient on the eCRFs.

- Determine patient's eligibility (see Sections 5.1 and 5.2).
- Obtain signed ICF for genetic research study (optional).

Procedures after a patient is confirmed to be eligible are described in Sections 6.3.1.1 and 6.3.1.2.

If the patient is ineligible and not assigned to any module, the IRT should be contacted to terminate the patient in the system.

6.3.1.1 Enrolment Procedures Specific to Modules with a Dose-finding Phase (Part 1)

This section is not applicable to Module 0, Module 6, or Module 7.

At baseline, once the patient is confirmed to be eligible, the Investigator or suitably trained delegate will do the following:

- Obtain a unique treatment assignment number via the IRT.
- The system will assign the eligible patient to one of the open treatment modules.
- Each treatment in Part 1 of each of the modules will involve dose-escalation/de-escalation. The number of patients enrolled in each dose level will depend on how many patients are evaluable for DLTs and how many patients have DLTs.

Patients will begin treatment on Day 1. Treatment should start no more than 3 days after treatment assignment. Patients must not be allocated to treatment, or treatment initiated, unless all eligibility criteria have been met.

6.3.1.2 Enrolment Procedures Specific to Part 2 in Each Module

This section is applicable to all modules.

At baseline, once the patient is confirmed to be eligible, the Investigator or suitably trained delegate will do the following:

- Obtain a unique treatment assignment number via the IRT.
- The system will subsequently randomly assign the eligible patient to one of the open modules.

6.3.2 Procedures for Handling Incorrectly Enrolled Patients

When a patient does not meet eligibility criteria but is assigned to treatment in error, or incorrectly started on treatment, the Investigator should inform the AstraZeneca Study

Physician immediately, and a discussion should occur between the AstraZeneca Study Physician and the Investigator regarding whether to continue or discontinue the patient from treatment. The AstraZeneca Study Physician must ensure all decisions are appropriately documented, and that the potential benefit-to-risk profile remains positive for the patient.

6.3.3 Methods for Assigning Patients to Treatment Groups (Randomisation)

This is an open-label study. The actual treatment given to patients in Part 1 and Part 2 will be determined by the assignment scheme in the IRT.

Treatment assignment codes will be assigned as patients become eligible for treatment assignment. When study treatment is provided centrally by the Sponsor, the IRT will provide the kit identification number to be allocated to the patient at each treatment visit.

Allocation of patients in Part 2 of all modules will be determined by the randomisation scheme in the IRT. The randomisation scheme will be produced by a computer software program that incorporates a standard procedure for generating randomisation numbers. One randomisation list will be produced for each of the randomisation strata. A blocked randomisation will be generated, and all centres will use the same list in order to minimise any imbalance in the number of patients assigned to each treatment arm. If a patient withdraws from the study, then his/her randomisation E-code cannot be reused.

Randomisation of patients to Modules 0 to 5 will be stratified. As patients become eligible for randomisation, randomisation codes will be assigned within each stratum per stratification by:

- PD-L1 expression (PD-L1-positive [defined by ≥ 1% IHC] vs PD-L1-negative)
- Hormone receptor status (positive vs negative)
- Disease status (recurrent vs *de novo*, defined in Section 4.1.3.1). Both subgroups are capped at 60% in either direction, such that potential imbalance in terms of these stratification factors is avoided.

Randomisation to Modules 6 and 7 will not be stratified due to limited sample size. When study treatment is provided centrally by the Sponsor, the IRT will provide the kit identification number to be allocated to the patient at each treatment visits.

6.4 Study Intervention Compliance

When the individual dose for a patient is prepared from a bulk supply, the preparation of the dose will be confirmed by a second member of the study site staff.

Patients will receive study treatment directly from the Investigator or designee, under medical supervision. The date, and time if applicable, of dose administered in the clinic will be recorded in the source documents and recorded in the eCRF. The dose of study treatment and

study patient identification will be confirmed at the time of dosing by a member of the study site staff other than the person administering the study treatment.

The Investigator or pharmacist must retain records of all study treatments administered at the site. The Study Monitor will check these records to confirm compliance with the protocol administration schedule. Deviation(s) from the prescribed dosage regimen should be recorded in the eCRF. The reason should also be documented.

The Investigational Product Storage Manager is responsible for managing the study treatment from receipt by the study site until the destruction or return of all unused study treatments.

6.5 Concomitant Therapy

Any concomitant treatment, procedure, or other medication considered necessary by the Investigator for the patient's safety and wellbeing, or vaccine (including over-the-counter or prescription medicines, vitamins, and/or herbal supplements) or other specific categories of interest that the patient is receiving from the time of screening or receives during the study including the 40 days (+ 7 days) follow-up period (or as otherwise specified in the respective modules) following the last dose of study treatment must be recorded in the eCRF along with:

- Reason for use.
- Dates of administration including start and end dates.
- Dosage information including dose and frequency.

The AstraZeneca Study Physician should be contacted if there are any questions regarding concomitant therapies.

If any concomitant therapy is administered due to new or unresolved AE, it should be recorded.

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after patients have completed 8 or more cycles of T-DXd for patients with HR-positive tumours, as per local practice.

Patients must be instructed not to take any medications, including over-the-counter products, without first consulting with the Investigator. Concomitant use of dietary supplements, medications not prescribed by the Investigator, and alternative/complementary treatments is discouraged, but not prohibited.

Guidance regarding potential interactions with concomitant medications is provided in Appendix J 1. T-DXd-specific restricted, prohibited, and permitted concomitant medications/therapies are described in more detail in Appendix J 2; refer to the respective

module documents for details on concomitant medications specific to the treatment combinations.

6.5.1 Rescue Medication

Refer to this section in the respective module documents for rescue medication associated with specific study treatments.

6.6 Dose Modification

Each module, except Module 0, Module 6, and Module 7, will have a dose-finding (Part 1) phase. The dose-escalation/de-escalation decision rules in Part 1 will be based on an mTPI-2 algorithm and an SRC will review DLTs observed from all evaluable patients at each dose level. Patients should be dosed at an escalated, a de-escalated, or at the same dose level following the rules described in Table 7. Refer to Section 4.1.2, and to the same section in the respective module documents for more details.

In Part 1 of each module, any dose reductions due to DLTs will be evaluated in new cohorts of patients; patients may not switch from one cohort to another. A patient who develops a toxicity during the DLT-evaluation period would follow the appropriate dose level modifications and TMGs as described in Sections 6.6.1.1 and 8.3.16.1 respectively.

6.6.1 Individual Patient Dose Modifications – All Modules (Part 1 and Part 2)

6.6.1.1 T-DXd

The combination of T-DXd with durvalumab and/or paclitaxel, pertuzumab, and tucatinib, CCI [redacted]. Furthermore, CCI [redacted]. The Investigator should use clinical judgement when attributing a toxicity to one or more of the agents. Guidance regarding T-DXd dose reductions and delays is included below. Also see dosing modification and TMGs in Appendix L. Dose delays may be entered into the eCRF as unscheduled visits.

The following dose modifications apply to T-DXd:

- CCI [redacted]

CCI

Dose modification for suspected ILD/pneumonitis cases

- CCI
-

See Section 8.2.5.5 for tests required if ILD/pneumonitis is suspected.

CCI

Refer to the TMGs (Appendix L) for details on dose modifications for different toxicities associated with T-DXd.

Dose modification criteria for suspected or confirmed COVID-19

Please see Appendix M for dose modification and management plan for patients with confirmed or suspected COVID-19 who are being treated with T-DXd.

6.6.2 Other Study Treatments

For details on dose modifications for the other study treatments in this study, refer to the equivalent sections in the respective modules.

6.7 Intervention after the End of the Study

As described in Section 4.4, the study will remain open until all patients have discontinued study treatment and completed their last expected visit/contact.

After the final DCO for this study, AstraZeneca will continue to supply open-label drug to patients receiving T-DXd, and centrally-supplied combination treatments as per the respective modules, up to the time that they discontinue the treatment for whatever reason (for definitions of discontinuation criteria see Section 7.1).

Participants should be followed according to the institution's standard of care assessments and as deemed appropriate by the investigators. It is recommended that investigators continue to observe participants at the frequency used prior to the final DCO. No further data collection is required, except for reporting of non-serious AEs, SAEs, and special situations (i.e. pregnancy, overdoses, medication error, drug abuse and misuse), if applicable, until the end of post-trial access, or until consent has been withdrawn.

The Clinical Study database will be closed to any new data and a paper report form will be used for SAE reporting. SAE data should continue to be collected until the end of the safety follow-up period which is 40+7 days after the discontinuation of all study interventions. Pregnancies will be reported until 7 months after the last dose and overdoses will be reported until the last dose.

Protocol dose modification and stopping criteria are to be followed while the patient is receiving study drug. A change in the dose or schedule of study drug should occur only for safety reasons, based on the investigator’s judgment, and should generally follow the approach for dose reduction and discontinuation as described in this protocol.

Provisions will be in place for patients still enrolled at the end of the study to continue to receive study treatment if, in the opinion of the Investigator, they are continuing to receive benefit from treatment.

In the event that product development reaches a point where alternative product supply options become available, then these alternative product supply options will be discussed by AstraZeneca with the Investigator. AstraZeneca will work with the Investigator to transition the participant(s) to alternative supply, where possible.

In the event that a roll-over or safety extension study is available at the time of the final DCO and database closure, patients currently receiving study treatment may be transitioned to such a study, and the current study would reach its end. The roll-over or safety extension study would ensure treatment continuation with visits and assessments per its protocol, as applicable. Any patient who would be proposed to move to such a study would be asked to sign a new ICF, as applicable.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

7.1 Discontinuation of Study Intervention

It may be necessary for a patient to permanently discontinue (definitive discontinuation) study treatment. If study treatment is permanently discontinued, the patient will remain in the study to be evaluated for follow-up assessments as per the SoA in the respective modules. The Investigator should instruct the patient to contact the site before or at the time if study treatment is stopped. A patient that decides to discontinue study treatment will always be asked about the reason(s) and the presence of any AEs. The reason for discontinuation should be documented in the source document and the appropriate section of the eCRF.

Patients who have permanently discontinued from further receipt of study treatment will need to be discontinued from the IRT. All study treatment should be returned by the patient at their next on-site study visit or unscheduled visit.

Patients must be discontinued from study treatment in the following situations:

- RECIST 1.1-defined radiological progression outside CNS (refer to Section 8.1.1 and Appendix G).
- Those patients who receive local therapy (radiation) for a primary CNS-progression, providing there is no other systemic progression may be allowed to continue study assigned treatment until a second progression (brain or body) is observed. Study treatment will be discontinued at the time of second progression (see Section 7.1.3).
- Investigator determination that the patient is no longer benefiting from study treatment.
- An AE that, in the opinion of the Investigator or AstraZeneca, contraindicates further dosing.
- Any AE that meets criteria for discontinuation defined in the dose modification guidelines for management of study treatment-related toxicities. Patients experiencing any of the following AEs will not be permitted to restart study treatment (see also Section 6.6):
  - ILD or non-infectious pneumonitis Grade 2 and higher
  - Any Grade 4 haematological toxicity with significant clinical symptoms that do not resolve with treatment within 4 weeks. Resuming the study treatment may be possible if the toxicity resolves, and in consultation with the Study Physician.
- Patient decision. The patient is at any time free to discontinue treatment, without prejudice to further treatment. A patient who discontinues treatment is normally expected to continue to participate in the study (eg, for safety and survival follow-up) unless they specifically withdraw their consent to all further participation in any study procedures and assessments (see Section 7.2).

- Severe non-compliance with the CSP as judged by the Investigator or AstraZeneca.
- Pregnancy or intent to become pregnant.
- Initiation of subsequent anticancer therapy, including another investigational agent
- Symptomatic deterioration (global deterioration of health status) without objective evidence of PD according to RECIST 1.1
- Investigator decision
- Death
- Patients who are incorrectly initiated on study treatment. A new patient will be enrolled to achieve the planned number of evaluable patients in each cohort. The AstraZeneca Study Physician must ensure all decisions are appropriately documented. The Investigator should document in the medical record as appropriate.
- Study terminated by Sponsor
- Patient lost to follow-up

All patients who are discontinued from study treatment should complete CSP-specified procedures for discontinuation of study treatment (see Section 7.1.1) and follow-up procedures (see Section 1.3 in the respective modules). Discontinued patients will be followed for survival, either through direct contacts or by collecting public records (eg, death certificates) as allowed by local laws.

The EOT visit should be performed as soon as the patient is permanently discontinued from study treatment (see SoA in the respective modules). The reason for discontinuation should be documented in the source document and the appropriate section of the eCRF. Note that discontinuation from study treatment is NOT the same thing as a withdrawal from the study.

See the SoA in each module for data to be collected at the time of study intervention discontinuation (ie, the EOT visit) and follow-up and for any further evaluations that need to be completed. The reason for discontinuation should be documented in the source document and the appropriate section of the eCRF.

Crossover within the study is not permitted.

If a patient discontinues treatment with the combination treatment(s) due to toxicity before disease progression they may continue with T-DXd as monotherapy as long as the AEs have resolved within the study allowed time period, they are continuing to show clinical benefit as judged by the Investigator, and in the absence of discontinuation criteria. If T-DXd is permanently discontinued due to toxicity, before disease progression, the combination partner treatment(s) may be continued at the discretion of the Investigator, if the AEs are not related to the partner drug.

7.1.1 Procedures for Discontinuation of Study Treatment

Discontinuation of study treatment does not impact the patient's participation in the study. The patient should continue attending subsequent study visits, and data collection should continue according to the CSP. If the patient does not agree to continue in-person study visits, a modified follow-up must be arranged to ensure the collection of endpoints and safety information. This follow-up could be a telephone contact with the patient, a contact with a relative or treating physician, or information from medical records. The approach taken should be recorded in the medical records. A patient that agrees to modified follow-up is not considered to have withdrawn consent or to have withdrawn from the study.

Patients who are permanently discontinued from further receipt of study treatment, regardless of the reason, will be identified as having permanently discontinued treatment. Patients who are permanently discontinued will enter follow-up (for details, see Section 1.3 in the respective modules). Patients who have permanently discontinued from further receipt of study treatment need to be recorded in the IRT.

Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ±1 week for the first 48 weeks from treatment assignment, and then Q9W ±1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression. (see Section 1.3 in the respective modules).

If a patient is discontinued for RECIST 1.1-defined progression, additional scans are to be completed per standard practice post-progression.

7.1.2 Follow-up of Patients Post Discontinuation of Study Treatment

All patients who discontinue the study treatment will be followed up for safety assessments after their last dose of study treatment for the time period specified in the respective modules. The follow-up period for T-DXd is 40 days (+ 7 days) after last dose of T-DXd. Additional assessments to be performed at the time of the safety follow-up are detailed in the SoA in the respective modules. For ILD/pneumonitis, safety follow-up will continue until the resolution of ILD/pneumonitis.

Patients who have discontinued study treatment prior to objective RECIST 1.1-defined radiological progression, regardless of whether or not they have commenced subsequent anticancer therapy, will be followed up with tumour assessments as indicated in the SoA in the respective module documents until RECIST 1.1-defined PD or death regardless of whether or not the patient started a subsequent anticancer therapy, unless they have withdrawn all consent to study related assessments. A repeat contrast MRI of the brain is required in all

patients following treatment discontinuation (unless a contrast brain MRI was performed within 30 days of treatment discontinuation or prior documentation of CNS progression while on study).

7.1.2.1 Follow-up of Patients to PFS2

For Part 2 of all modules, following objective progression that is confirmed by Investigator assessment, patients will have their subsequent progression status recorded at the 40-day (+ 7 days) follow-up visit (or as otherwise specified in the respective modules), and every 3 months (± 14 days) thereafter per local standard clinical practice to assess PFS2. Assessments will be performed according to the local practice, and formal RECIST 1.1 measurements will not be collected for assessment of PFS2. See Section 8.1.3 for additional information.

Those patients who experience isolated progression in the brain with no systemic progression and who undergo local therapy will be allowed to continue study assigned treatment until a second progression (brain or body) is observed. These patients will be followed beyond the second progression to assess PFS2, which is defined by the time of progression on next line of therapy (see Section 7.1.3).

7.1.2.2 Follow-up for Survival

Patients will be followed up for survival status as indicated in the SoA in the respective modules until death, withdrawal of consent, or the end of the study. Survival information may be obtained via telephone contact with the patient or the patient's family, or by contact with the patient's current physician or by checking publicly registries, as available and in accordance with local regulations. Additional assessments to be performed at the time of survival follow-up are detailed in the SoA in each module.

7.1.3 Continuation on Study Treatment After CNS-Only Progression

If a patient is found to have isolated progression in the CCI and does not have progression of disease CCI, the patient may be eligible to continue on study treatments after completion of CCI of the CCI CCI to allow for clinical benefit. Treatment with T-DXd must be temporarily stopped during local treatment. This is applicable to all patients with CCI CCI Continuation of study treatment requires discussion with and documented approval from the AstraZeneca Study Physician or delegate and may continue until CCI CCI. The patient will remain on the same treatment arm assigned initially, and may continue on study provided the following criteria are met and the patient continues to receive clinical benefit:

- The patient is not experiencing any worsening of cancer-related symptoms. Patients who are clinically deteriorating and unlikely to receive further benefit from continued treatment should discontinue study treatment.
- The patient is tolerating study drug.
- Review and concurrence by the AstraZeneca Study Physician or delegate.
- Patient has no evidence of unequivocal systemic progression.
- Patient has CCI .

Study treatment may be held up to CCI weeks to allow CCI holds must be discussed and approved by the AstraZeneca Study Physician or delegate. All study treatments are to be held CCI week prior to planned CCI . Plans for CCI and CCI study treatments before and after local therapy will require discussion with, and documented approval from, the AstraZeneca Study Physician or delegate. Washout periods following radiation therapy are outlined in Table 9.

Every effort should be made to avoid radiation or surgery to target lesions in the CCI unless clinically necessary in the opinion of the investigator. The rationale for this is that target lesions, CCI , cannot be adequately assessed for subsequent response to systemic therapy. Because of this, if a patient continues on assigned study therapy after CCI , special consideration must be given for evaluation of the treated target lesion and the impact on the overall RECIST 1.1. assessment.

Following CCI , the Investigators will continue to follow the patients and report CCI or systemic progression. Patients who are eligible and undergo CCI directed therapy while on study should maintain their regular assessment schedule, including monitoring of all AEs. All scans should continue as scheduled for brain, chest, abdomen and pelvis (see SoA Section 1.3).

7.2 Patient Withdrawal from the Study

A patient may withdraw from the study at any time at his/her own request or may be withdrawn at any time at the discretion of the Investigator for safety, behavioural, compliance, or administrative reasons. This is expected to be uncommon.

A patient who considers withdrawing from the study must be informed by the Investigator about modified follow-up options to ensure the collection of endpoints and safety information including new AEs and follow-up on any ongoing AEs and concomitant medications (eg, telephone contact at 40 days (+ 7 days) after study treatment is discontinued, a contact with a relative or treating physician, or information from medical records).

At the time of withdrawal from the study, if possible, an Early Study Treatment Discontinuation visit should be conducted, as shown in the SoAs in the respective modules. See the SoAs for data to be collected at the time of study withdrawal and follow-up and for any further evaluations that need to be completed.

The patient will discontinue the study treatment and be withdrawn from the study at that time.

If the patient withdraws consent for disclosure of future information, the Sponsor may retain and continue to use any data collected before such a withdrawal of consent.

If a patient withdraws from the study, it should be confirmed if he/she still agrees for existing samples to be used in line with the original consent. If he/she requests withdrawal of consent for use of samples, destruction of any samples taken and not tested should be carried in line with what was stated in the informed consent and local regulation. The Investigator must document the decision on use of existing samples in the site study records and inform the Global Study Team.

7.3 Lost to Follow-up

A patient will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and no contact has been established by the time the study is completed (see Section 4.4), such that there is insufficient information to determine the patient's status at that time.

Patients who decline to continue participation in the study, including telephone contact, should be documented as "withdrawal of consent" rather than "lost to follow-up". Investigators should document attempts to re-establish contact with missing patients throughout the study period. If contact with a missing patient is re-established, the patient should not be considered lost to follow-up and evaluations should resume according to the protocol.

The following actions must be taken if a patient fails to return to the clinic for a required study visit:

- The site must attempt to contact the patient and reschedule the missed visit as soon as possible and counsel the patient on the importance of maintaining the assigned visit schedule and ascertain whether or not the patient wishes to and/or should continue in the study.
- Before a patient is deemed lost to follow-up, the Investigator or designee must make every effort to regain contact with the patient (where possible, 3 telephone calls and, if necessary, a certified letter to the patient's last known mailing address or local equivalent methods). These contact attempts should be documented in the patient's medical record.

- Efforts to reach the patient should continue until the end of the study. Should the patient continue to be unreachable, he/she will be considered to have withdrawn from the study.
- Site personnel, or an independent third party, will attempt to collect the vital status of the patient during survival follow-up within legal and ethical boundaries for all patients assigned to treatment, including those who did not get study treatment. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented, and the patient will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

Discontinuation of specific sites or of the study as a whole are handled as part of Appendix A.

8 STUDY ASSESSMENTS AND PROCEDURES

An electronic database capture system will be used for data collection and query handling. The Investigator will ensure that data are recorded on the eCRFs as specified in the CSP and in accordance with the instructions provided, ensure the accuracy, completeness, and timeliness of the data recorded and of the provision of answers to data queries according to the Clinical Study Agreement, and will sign the completed eCRFs. The RAVE Web Based Data Capture system will be used for data collection and query handling. A copy of the completed eCRFs will be archived at the study site.

Study procedures and their timing are summarised in the SoA as follows:

- For screening assessments, which are applicable to all modules, see Table 3 in this core CSP.
- For on-treatment and follow-up SoA tables for Module 0, see Table 4 and Table 5, respectively, in this core CSP.
- For on-treatment and follow-up SoA tables for the combination treatment (Modules 1-6) and monotherapy treatment (Module 7), refer to the respective module documents.

The study assessments described in the following subsections are applicable to all modules in this study. Refer to the respective modules for study assessments that are module specific.

Note: A cycle of study treatment will be defined as 21 days of dosing.

Written informed consent and any locally required privacy act document authorisation must be obtained prior to performing any protocol-specific procedures, including screening/baseline evaluations. Data collection following study analysis until the end of the study is described below.

- Protocol waivers or exemptions are not allowed.

- Immediate safety concerns should be discussed with the Sponsor immediately upon occurrence or awareness to determine if the patient should continue or discontinue study treatment.
- Adherence to the study design requirements, including those specified in the SoA in the respective modules, is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential patients meet all eligibility criteria. The Investigator will maintain a screening log to record details of all patients screened and to confirm eligibility or record reasons for screening failure, as applicable.
- Procedures conducted as part of the patient's routine clinical management (eg, blood count) and obtained before signing of the ICF may be utilised for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the time frame defined in the SoA in the respective modules.
- Data management will be performed by AstraZeneca or a delegate according to the Data Management Plan.
- AEs and medical/surgical history will be classified according to the terminology of the latest version of MedDRA. Medications will be classified according to the World Health Organisation Drug Classification and coding will be performed by AstraZeneca or a delegate. The data collected through third party sources will be obtained and reconciled against study data.
- Data queries will be raised for inconsistent, impossible, or missing data. All entries to the study database will be available in an audit trail. The data will be validated as defined in the Data Management Plan. Quality control procedures will be applied to each stage of data handling to ensure that all data are reliable and have been processed correctly. The Data Management Plan will also clarify the roles and responsibilities of the various functions and personnel involved in the data management process.
- When all data have been coded, validated, signed, and locked, a clean file will be declared.
- Monthly SAE reconciliation reports are produced and reconciled with the Patient Safety database and/or the investigational site.

Data Collection Following Study Analysis until the End of the Study

After the final DCO and database closure only non-serious AEs, SAEs, and special situations (i.e. pregnancy, overdoses, medication error, drug abuse and misuse) will be reported for the purposes of this study, if applicable (see Section 8.3.12).

8.1 Efficacy Assessments

Core efficacy assessments applicable to all modules are described in the subsections below. Refer to this section in the respective modules for details of module-specific efficacy assessments.

Refer to Section 1.3 in this core document for the screening SoA table, which is applicable to all modules. Refer to Section 1.3 in the respective modules for the on-treatment and follow-up SoA tables, which provide the assessment schedule for each module.

8.1.1 Imaging Tumour Assessments

Efficacy assessments of ORR, DoR, PFS, PFS2, and OS will be evaluated based on Investigator RECIST 1.1 tumour assessments.

Tumour assessments use images from CT (preferred) or MRI, with IV contrast, of the chest, abdomen, and pelvis, and MRI with IV contrast for brain, collected during screening/baseline and at regular (follow-up) intervals during study treatment. Any other areas of disease involvement should be additionally imaged at screening based on known metastasis sites or by the signs and symptoms of individual patients. The imaging modality used for baseline tumour assessment, CT/MRI for chest, abdomen and pelvis, and IV contrast-enhanced MRI for brain (gadolinium), should be kept the same consistently at each subsequent follow-up assessment throughout the study if possible. It is important to follow the tumour assessment schedule as closely as possible (refer to the SoA tables in the respective modules) relative to randomisation.

A bone scan (scintigraphy) will be performed for all patients at baseline and subsequent scans will be performed as clinically indicated to evaluate possible new bone lesions.

Regularly scheduled follow-up brain scans are mandatory for all patients who are enrolled with baseline stable or active brain metastases from the date of treatment assignment until disease progression as per SoA (Section 1.3) Patients without brain metastases do not need additional brain scans for subsequent tumour assessments, unless clinically indicated. New lesions may be identified also by 18F-Fluorodeoxyglucose-PET/CT scans, X-ray, bone (scintigraphy) scans. Digital copies of all scans should be maintained at the site as source documents. Digital images are sent to the Imaging CRO by electronic transfer.

Screening/baseline imaging should be performed no more than 28 days before start of study treatment and ideally should be performed as close as possible to and prior to the start of study treatment. Treatment continues until RECIST 1.1-defined radiological progression (refer to Appendix G), and scanning/tumour assessments continue throughout treatment until RECIST 1.1-defined radiological progression by Investigator assessment. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be

performed. Following investigator determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment, if it is feasible. This additional scan may be sent for BICR review. Additional scans are to be completed per standard practice post-progression. Patients with CNS-only progression (without systemic progression) may continue on study treatment after initial CNS-only progression as described in Section 7.1.3.

The RECIST 1.1 assessments of baseline images identify TLs (defined as measurable) and NTLs. On-study images are evaluated for TLs and NTLs chosen at baseline, and for NLs when they appear. This allows determination of follow-up TL response, NTL lesion response, the presence of unequivocal NLs, and overall time point responses (CR, PR, SD, PD, or NE).

8.1.2 Central Reading of Scans

Images, including unscheduled visit scans, will be collected on an ongoing basis and sent to an AstraZeneca-appointed iCRO for storage. Digital copies of all original scans should be stored at the Investigator site as source documents. Electronic image transfer from the sites to the iCRO is strongly encouraged. A blinded independent central review (BICR) of images will be performed at the discretion of AstraZeneca. Guidelines for image acquisition, de-identification, storage of digital copies at the investigative site (as source documents), and transfer to the iCRO will be provided in a separate document.

In addition, RANO-BM will also be assessed in Modules 6 and 7 (See Appendix N).

The management of patients will be based in part upon the results of the tumour assessments conducted by the Investigator.

8.1.3 Time to Second Progression or Death

For Part 2 of all modules, following objective progression that is confirmed by Investigator assessment, patients will have their subsequent progression status recorded at the 40-day (+7 days) follow-up visit (or as otherwise specified in the respective modules), and every 3 months (± 14 days) thereafter per local standard clinical practice to assess PFS2. A patient's PFS2 status is defined according to the local practice and may involve any of: objective radiological progression (preferred), symptomatic progression, or death. Scans will be performed according to the local practice and formal RECIST 1.1 measurements will not be collected for assessment of PFS2. The second progression event must have occurred during or after treatment with a subsequent treatment after the progression event used for the primary variable PFS or death. The date of PFS2 assessment and investigator opinion of progression status (progressed or non-progressed) at each assessment will be recorded in the source documents and the eCRF.

Those patients who experience isolated progression in the brain, with no other systemic progression, and undergo local therapy will be allowed to continue study assigned treatment until a second progression (brain or body) is observed. These patients will be followed beyond the second progression to assess PFS2, which is defined by the time of progression on next line therapy (see Section 7.1.3).

8.1.4 Overall Survival

Assessments for survival will be conducted every 3 months (±14 days) from the date of the 40-day (+ 7 days) follow-up visit (or as otherwise specified in the respective modules), until death, withdrawal of consent, or study closure, whichever occurs first. Survival information may be obtained via telephone contact with the patient, patient's family, by contact with the patient's current physician, or local death registries as described in Section 7.1.2.2. The details of first and subsequent therapies for cancer, after discontinuation of treatment, will also be collected.

In addition, patients on treatment or in survival follow-up will be contacted following the DCO for the final analysis and all subsequent survival analyses to provide complete survival data. These contacts should generally occur within 7 days of the DCO.

8.1.5 ILD/pneumonitis-related Clinical Outcome Assessments

Clinical Outcome Assessments will be used for exploratory purposes to better characterise ILD/pneumonitis and the progression of ILD/pneumonitis cases. A multi-modal digital approach will be employed involving completion of ePRO questionnaires and at home pulse oximetry. The ePRO questionnaires will include a daily symptom diary (3 MDASI items: cough, shortness of breath and chest tightness/heaviness), and for those who have a suspected diagnosis of ILD/pneumonitis, the SGRQ-I.

In addition, a subset of patients diagnosed with ILD/pneumonitis will participate in interviews to further characterise ILD/pneumonitis and its progression.

ePRO questionnaires, at-home pulse oximetry and patient interviews will be completed in patients' native or preferred language using the respective linguistic version(s) of the questionnaires when available, according to the SoA in Section 1.3 of the respective modules.

More details on the ePRO questionnaires, at-home pulse oximetry and patient interviews are provided below. The ePRO questionnaires are provided in Appendix I.

8.1.5.1 Administration of ePRO Questionnaires

ePRO questionnaires will be self-administered by patients using their own electronic device via a mobile app. In cases where a patient does not have access to a compatible device to allow the use of the app, an electronic handheld device will be provided to the patient. ePRO

questionnaires will be completed at the timepoints indicated in the SoA in Section 1.3 of the respective modules. Patients will complete the ePRO questionnaires at home at approximately the same time each day. If the allowable window to complete the ePRO questionnaires overlaps with a study site visit, the assessment may also be completed at the site visit.

The time to complete the MDASI symptom diary is approximately 2 minutes, and the time to complete the SGRQ-I is approximately 10 minutes.

The below instructions should be followed when collecting PRO data via electronic device:

- The research nurse or appointed site staff must explain to patients the importance of completing the questionnaires and inform them that these questions are being asked to find out, directly from them, how they feel. The research nurse or appointed site staff should also stress that the information is not routinely shared with study staff. Therefore, if patients have any medical problems, they should discuss them with the doctor or research nurse separately from the PRO assessment.
- The research nurse or appointed site staff must train the patient on how to download and use the app, and in cases where a device is provided, the use of the ePRO device, using the materials and training provided by the ePRO vendor. Guidance also should be provided on who to call if there are problems with the app or device when completing the questionnaires at home.
- PRO symptom diary reporting will be initiated at the Cycle 1, Day 1 visit before dosing, as specified in the SoA in the respective modules. It is important to have a device set up in advance of each patient's first treatment visit, ideally at least the day before, to ensure that a charged and functioning device is available in case the patient needs to use a provisioned device.
- All questionnaires should be completed using an electronic device (patient's own device or a device provided by the site).
- ePRO questionnaires completed at Cycle, 1 Day 1 or at the time of a visit must be completed before treatment administration and ideally before any discussions of health status to avoid biasing the patient's responses to the questions. As feasible, site staff should also ensure ePRO questionnaires are completed prior to other study procedures, such as collection of laboratory samples, to further minimise bias.
- For ePRO questionnaires completed at a site visit, the patient should complete them in a quiet and private location and with time to complete them at their own speed.
- The research nurse or appointed site staff must remind patients that there are no right or wrong answers and must not interpret or clarify the meaning of items to avoid bias.
- The patient should not receive help from relatives, friends, or clinic staff to choose answers on the ePRO questionnaires.

- If the patient is unable to read the questionnaire (eg, is blind, illiterate or not fluent in the available language), or if the linguistic version is not available for patient native or preferred language that patient should be exempted from completing ePRO questionnaires but may still participate in the study. If the patient cannot complete the ePRO questionnaires due to reasons other than being blind, illiterate or not fluent in the available language, or the linguistic version of a questionnaire is not available the AstraZeneca study team must be contacted to determine if they can be exempted.
- Site staff must administer questionnaires available in the language that the patient speaks and understands. Questions should not be read in an available language and translated into another language for the patient.
- Patients will receive reminders to ensure compliance with the assessment schedules.
- If baseline (Cycle 1 Day 1) questionnaires are not completed, the reason for not completing the assessment should be documented in the eCRF.
- The research nurse or appointed site staff must monitor compliance, since minimising missing data is a key aspect of study success. Compliance must be checked regularly and at each study visit to identify problems early. If the site receives a notification regarding the patient's compliance, a check-in call from the study site to ask the patient if he or she has any difficulties is highly recommended. A solution to enhance/resolve compliance should be discussed with the patient. Discussions and compliance review should be reflected in source documents.

8.1.5.2 MDASI Symptom Diary (cough, shortness of breath, chest heaviness/tightness)

The MDASI is a validated, multi-item, cancer-specific ePRO questionnaire capturing symptom severity and interference (Cleeland et al 2000). The MDASI symptom library allows for specific symptom items to be selected to create a tailored instrument to address a specific research need. Three items from the MDASI symptom library will be used to capture the core ILD/pneumonitis symptoms of cough, shortness of breath, and chest heaviness/tightness. Each item is rated on an 11-point numeric rating scale, with higher scores indicating greater symptom severity.

These assessments correspond to the study exploratory objective and will not be shared with the site during the course of the trial. These data will not be used for any clinical care or treatment decisions during the trial, and the analyses results may be presented in a separate report (other than the CSR).

8.1.5.3 At-home Pulse Oximetry

An electronic, wireless pulse oximetry device will be provided to each patient at the Cycle 1 Day 1 visit. The research nurse or appointed site staff will train patients on the use of the device. The patient will complete a resting and walking assessment at the Cycle 1 Day 1 visit

before dosing. At-home pulse oximetry testing will be performed by the patient once daily while resting and once weekly while walking, in accordance with the SoA in Section 1.3 of the respective modules.

For the walking assessment, patients will need to walk in their homes for 3 minutes. Patients will walk at their own pace on level ground and with any walking aid they would normally use. Patients should wear the pulse oximeter and start their application for a 3-minute timer. At completion of time or in case of early discontinuation, patients should immediately take a reading before the pulse rate drops. The walk should be discontinued if at any time the patient experiences severe shortness of breath, anterior chest pain, or if the patient feels that he or she is unable to continue walking. The research nurse or appointed site staff will check the patient's adherence to the use of the pulse oximetry regularly and at each site visit.

These assessments correspond to the study exploratory objective and will not be shared with the site during the course of the trial. These data will not be used for any clinical care or treatment decisions during the trial, and the analyses results may be presented in a separate report (other than the CSR).

8.1.5.4 SGRQ-I

The original SGRQ is a commonly used, validated instrument capturing HRQoL for patients with chronic respiratory disease. The SGRQ-I is an IPF-specific version of the instrument developed and validated for use among patients with IPF, a type of ILD (Yorke et al 2010). The SGRQ-I will be used to assess the HRQoL among patients who have suspected or have been diagnosed with ILD/pneumonitis. It includes 34 of the original SGRQ items determined to be most reliable for assessing the HRQoL of patients with IPF. The instrument yields 3 domain scores (symptoms, activity and impact) as well as a total score, with scores ranging from 0-100. Higher scores indicate greater impairment in HRQoL.

These assessments correspond to the study exploratory objective and will not be shared with the site during the course of the trial. These data will not be used for any clinical care or treatment decisions during the trial, and the analyses results may be presented in a separate report (other than the CSR).

8.1.5.5 ILD/pneumonitis Qualitative Patient Interviews

All fluent English or Spanish-speaking patients from selected countries who have been diagnosed with ILD/pneumonitis will complete a one-on-one interview to characterise their ILD/pneumonitis experience, including symptoms and associated impacts. Data saturation, which is the point at which interviews no longer yield new, relevant information, is the standard by which adequate sample size is determined in qualitative research. Saturation is typically, but not always, reached by 20 interviews. If data saturation is reached prior to all ILD/pneumonitis patients from selected countries being interviewed, further interviews may be discontinued.

Interviews will be conducted by a third-party vendor contracted by AstraZeneca via telephone using a semi-structured interview guide. Interviews will be conducted during a site visit as close to the ILD/pneumonitis diagnosis as is feasible and after the patient has been informed of the diagnosis, and no later than 3 months post ILD/pneumonitis diagnosis. If conducting the interview during a site visit is not possible, contact information will be provided to the vendor, with prior patient consent, to schedule and conduct the interview outside of the site visit. The interview will last approximately 60 minutes and will be audio-recorded with the patient's prior consent.

During the interview, patients will be asked a series of open-ended questions on their experience with ILD/pneumonitis, including symptoms and their impact on patients' lives, followed by probes as necessary to elicit more detailed information. Rating scales may also be used. Patients may also be asked to review ILD/pneumonitis-specific questions for relevance and appropriateness for capturing ILD/pneumonitis symptoms and impacts.

Interview transcripts will be developed for qualitative analysis. AstraZeneca will have access to de-identified transcripts but not to the interview recordings. For interviews conducted in Spanish, transcripts will be created and translated into English. Transcripts will be de-identified and coded using a qualitative analysis software tool (Atlas.ti). All qualitative data will be stored on the vendor's secure server in a password-protected file for 10 years before being deleted. Due to the exploratory and qualitative nature of the data and analysis, the results will be presented in a separate report (not the CSR) and the data (transcripts), will not be entered into the study database. No identifiable data will be reported.

At the beginning and end of each interview, patients will be advised to report any AE to the Investigator or designee. AEs reported to Investigators will be handled following standard AstraZeneca AE reporting procedures.

8.1.6 Additional Clinical Outcomes Assessments

8.1.6.1 CANTAB

This assessment is for patients who have stable and active brain metastases at baseline. CANTAB is a computerised gamified test battery including well-established and *de novo* (with normative value) tests and utilizes a touch screen technology and press pad to assess neurocognitive function, including attention, visual memory, spatial memory, executive function and reaction time. It takes less than 8 minutes to complete.

8.1.6.2 MDASI-BT

This assessment is for patients who have stable and active brain metastases at baseline. The MDASI-BT is a patient-completed 22-item questionnaire assessing the severity of multiple brain tumour-related symptoms and the impact of these symptoms on daily functioning in patients with symptoms caused by brain tumour including brain metastases associated with

breast cancer (Armstrong et al 2009) and its treatment. Along with the 13 core constitutional symptom items (eg, fatigue, pain) and 6 interference items including impact on enjoyment of life and general activities, the MDASI-BT also assesses 9 symptoms specific to brain tumours (eg, weakness on one side of the body, seizures, problems with vision). Items are rated on a 11-point numerical rating scale within the last 24 hours prior to evaluation. The instrument yields domain scores (interference score) and also symptom-specific scores (Miettinen and Nurminen 1985). It takes less than 5 minutes to complete.

These study assessments will only be completed by participants if a linguistically validated version is available in their language for the country in which they live.

8.1.6.3 NANO

This assessment is for patients who have stable and active brain metastases at baseline. The NANO scale is a quantifiable evaluation of 9 relevant domains of neurologic function likely to be impacted by supratentorial, infratentorial, and brainstem lesions based on direct observation and testing by a clinician conducted during routine office visits. The score defines overall response criteria. It takes less than 5 minutes to complete by physicians (Nayak et al 2017).

8.2 Safety Assessments

Planned timepoints for all safety assessments at screening are provided in the SoA in Section 1.3 in this document and are applicable to all modules. Refer to Section 1.3.2 and Section 1.3.3 in this document for Module 0 and Section 1.3 in the respective modules for the on-treatment and follow-up SoA tables.

8.2.1 Physical Examinations

A full physical examination will be performed and include assessments of the following; general appearance, respiratory, cardiovascular, abdomen, skin, head and neck (including ears, eyes, nose and throat), lymph nodes, thyroid, musculoskeletal (including spine and extremities), urogenital, dermatological, gastrointestinal, endocrine, haematologic/lymphatic, and neurological systems. Targeted physical examinations are to be used by the Investigator on the basis of clinical observations and symptomatology.

Physical examination, and height and weight assessments will be performed at timepoints as specified in the SoA; Investigators should pay special attention to clinical signs related to previous serious illnesses, new or worsening abnormalities may qualify as AEs, see Section 8.3.5 for details.

8.2.2 Vital Signs

Vital signs will be performed at timepoints as specified in the SoA. Temperature, pulse rate, respiratory rate, and blood pressure will be assessed.

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the patient in a quiet setting without distractions (eg, television, cell phones).

Vital signs will be evaluated and recorded in eCRF. Standard infusion time for T-DXd is approximately 90 minutes for first infusion and 30 minutes thereafter if the first infusion was well tolerated; however, if there are interruptions during infusion, the total allowed infusion time should not exceed 3 hours at room temperature. If the infusion takes longer than 90 minutes, then blood pressure and pulse measurements should follow the principles as described below or be taken more frequently if clinically indicated. An hour observation period is recommended after the first infusion of T-DXd and the combination partner treatments that are administered intravenously.

First Infusion of Study Treatment

On the first day of study treatment administration, patients' vital signs will be monitored and collected/recorded in the eCRF before and after infusion of each study treatment as follows:

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 30 minutes ± 10 minutes after end of the T-DXd infusion (during the 1-hour observation period after the infusion).

Subsequent Infusions of Study Treatment

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 7 minutes ± 5 minutes after end of the T-DXd infusion (during the 15-minute observation period after the infusion).

Vital signs during initial and subsequent T-DXd and combination partner infusions should be monitored, and any clinically significant findings at other times to those described above should be reported via an unscheduled visit form. Any changes in vital signs should be recorded as an AE (if applicable). Situations in which vital signs results should be reported as AEs are described in core CSP Section 8.3.5. Refer to this section in the respective modules for details of module-specific vital sign assessments.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.2.3 Electrocardiograms

Single 12-lead ECGs will be performed at timepoints as specified in the SoA after the patient has been resting semi-supine for at least 5 minutes and recorded while the patient remains in that position using an ECG machine that measures parameters including PR, QRS, QT, and

QTcF intervals. At screening, ECGs will be obtained in triplicate (all 3 within a 15-minute time period, approximately 5 minutes apart). Whether or not measurement is performed, date performed, results, and findings for each parameter is to be recorded in the eCRF.

All ECGs should be assessed by the Investigator as to whether they are clinically significantly abnormal. Any clinically significant abnormalities detected require triplicate ECG results.

Situations in which ECG results should be reported as AEs are described in Section 8.3.5.

8.2.4 Clinical Safety Laboratory Assessments

Blood samples for determination of clinical chemistry, haematology, coagulation, and urinalysis will be taken at the visits indicated in the SoA.

Additional safety samples may be collected if clinically indicated at the discretion of the Investigator. The date, time of collection and results (values, units and reference ranges) will be recorded on the appropriate eCRF. Laboratory assessments for off-schedule study treatment dosing may be entered in the eCRF as unscheduled visits.

The clinical chemistry, haematology and urinalysis will be performed at a local laboratory at or near to the Investigator site. Sample tubes and sample sizes may vary depending on laboratory method used and routine practice at the site.

Other safety laboratory tests include assessment for pregnancy (serum at screening and urine at other timepoints), hepatitis B and C serology and HIV antibody test. Pregnancy tests may be performed at the site using a licensed test (urine or serum pregnancy test) and should be performed within 72 hours before treatment assignment. If a positive urine pregnancy test result is confirmed using a serum test in a female patient of child-bearing potential, then the patient should not be treatment assigned. Abnormal clinically significant laboratory results should be repeated as soon as possible (preferably within 24 to 48 hours).

The laboratory variables in Table 11 will be measured:

Table 11 Laboratory Safety Variables

Clinical Chemistry (Serum or Plasma)	
Albumin	Lactate dehydrogenase
ALP [b]	Magnesium [a]
ALT [b]	Potassium
AST [b]	Protein, total
Bicarbonate (if available) [a]	Sodium
Bilirubin, total [b]	Blood urea nitrogen or urea, depending on local practice

Table 11 Laboratory Safety Variables

Clinical Chemistry (Serum or Plasma)	
Calcium	Troponin
Chloride [a]	TSH [c]
Creatinine [a, e]	T_3 free [c, d]
Glucose (random)	T_4 free [c, d]
Gamma glutamyltransferase	Amylase and lipase [c]
Haematology	
Haemoglobin	Platelet count
White blood cell count	Differential white blood cell count [f]
Haematocrit	Red blood cell count
Urinalysis [g]	
Colour and appearance	pH and specific gravity
Blood	Microscopy assessment (if indicated)
Bilirubin	Glucose
Ketones	Protein
Coagulation	
Coagulation variables (aPTT, PTT, and PT/INR) [h]	

[a] Bicarbonate (where available), chloride, creatinine clearance, gamma glutamyltransferase, magnesium, testing are to be performed at baseline, on Day 1 (unless all screening laboratory clinical chemistry assessments are performed within 3 days prior to Day 1), and if clinically indicated.
[b] Tests for ALT, AST, alkaline phosphatase, and total bilirubin must be conducted and assessed concurrently. If total bilirubin is ≥ 2 × ULN (and no evidence of Gilbert's syndrome), then fractionate into direct and indirect bilirubin.
[c] After screening, TSH, T_3, T_4, amylase and lipase will be measured in Modules 1 and 4 only.
[d] Free T_3 and free T_4 will only be measured if TSH is abnormal or if there is clinical suspicion of an AE related to the endocrine system.
[e] Creatinine clearance will be calculated by the site using Cockcroft-Gault (using actual body weight).
[f] Differential white blood cell count: neutrophils, lymphocytes, monocytes, eosinophils, basophils will be recorded.
[g] Microscopy is preferred to investigate white blood cells, with use of high-power field for red and white blood cells; dipstick can be used as well.
[h] Coagulation parameters are to be assessed at baseline and as clinically indicated.
AE = adverse event; ALP = alkaline phosphatase; ALT = alanine aminotransferase; aPTT = activated partial thromboplastin time; AST = aspartate aminotransferase; INR = international normalised ratio; PT = prothrombin time; PTT = partial thromboplastin time; T_3 = triiodothyronine; T_4 = thyroxine; TSH = thyroid stimulating hormone; ULN = upper limit of normal.

The Investigator should assess the available results with regard to clinically relevant abnormalities in documentation. Any clinically significant abnormal laboratory values should be repeated as clinically indicated and recorded on the eCRF. Situations in which laboratory safety results should be reported as AEs are described in Section 8.3.5.

All patients with Grade 3 or 4 laboratory values at the time of completion or discontinuation from study treatment must be followed up and have further tests performed until the laboratory values have returned to Grade 1 or 2, unless these values are not likely to improve because of the underlying disease.

Note: In case a patient shows an AST or ALT ≥ 3 × ULN together with TBL ≥ 2 × ULN please refer to Appendix E, for further instructions. “Actions required in cases of increases in liver biochemistry and evaluation of HL”, for further instructions.”

8.2.5 Other Safety Assessments

8.2.5.1 Echocardiogram/Multi-gated Acquisition Scan

An ECHO or MUGA scan to assess left ventricular function will be performed as detailed in the screening SoA in Section 1.3 and the SoA in the respective modules.

The modality of the cardiac function assessments must be consistent for a given patient (ie, if ECHO is used for the screening assessment for a given patient, then ECHO should also be used for subsequent scans for that patient). The patients should also be examined using the same machine and operator whenever possible, and quantitative measurements should be taken (ie, accurate to 1% and not estimated to 5%).

If a patient has had an ECHO or MUGA performed within 4 weeks prior to treatment discontinuation, the discontinuation visit ECHO/MUGA scan is not required unless clinically indicated. If a patient has any clinically significant decrease in LVEF (greater than 10 percentage points to below 50%), there should be follow-up within 4 weeks until resolution.

Blood samples for troponin (preferably high-sensitivity troponin-T) will be collected as indicated in the SoA for each respective module and if at any time a participant reports signs or symptoms suggesting CHF, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed. At baseline, if troponin levels are above the ULN and below the level of myocardial infarction defined by the manufacturer (CTCAE Grade 1), no repeat testing is required. Only Grade 3 troponin needs to be repeated.

Situations in which ECHO or MUGA results should be reported as AEs are described in Section 8.3.5.

8.2.5.2 Pulmonary Assessments

Pulse oximetry (SpO_2) should be obtained at the visits as shown in the screening SoA in Section 1.3 and the SoA in the respective modules. SpO_2 should be evaluated by the investigator or the delegated physician before infusion and at the end of infusion, following the same guidance described for vital signs in Section 8.2.2.

Pulmonary function tests should include basic spirometry at a minimum with optional additional components as mentioned in Table 12.

Table 12 Spirometry Components

Required spirometry components	Optional spirometry components
FVC (L)	PEF
FVC% predicted	FEV6
FEV1 (L)	TLC
FEV1% predicted	RV
FEV1/FVC%	---
DLCO	---

DLCO = diffusion capacity of the lungs for carbon monoxide; FEV = forced expiratory volume; FEV1 = FEV-1 second; FEV6 = FEV-6 seconds; FVC = forced vital capacity; PEF = peak expiratory flow; L = litres; RV = residual volume; TLC = total lung capacity

DLCO will be performed for all patients at screening/baseline. In event of suspected ILD/pneumonitis, refer to Section 8.2.5.5 for additional pulmonary assessments.

HRCT of the chest will be performed at screening, and if ILD/pneumonitis is suspected. Chest CT and/or chest HRCT scans will be reviewed separately for safety for the presence of ILD/pneumonitis prior to administration of the next scheduled dose of T-DXd. A non-contrast HRCT scan of the chest is preferred if feasible, otherwise a non-contrast CT is acceptable. The HRCT/CT chest scan will be performed for all patients, in addition to standard CT/MRI scans of chest/abdomen/pelvis for tumour assessments. If both a non-contrast chest HRCT/CT scan for assessment of ILD/pneumonitis and a diagnostic IV contrast enhanced chest CT scan for tumour response assessment (as part of chest abdomen pelvis imaging) are to be acquired in the same imaging session, HRCT/CT should be performed first.

8.2.5.3 ECOG Performance Status

ECOG PS will be assessed as detailed in the screening SoA in Section 1.3 and the SoA in the respective modules based on the following:

1. Fully active; able to carry out all usual activities without restrictions.
2. Restricted in strenuous activity, but ambulatory and able to carry out light work or work of a sedentary nature (eg, light housework or office work).
3. Ambulatory and capable of self-care, but unable to carry out any work activities; up and about more than 50% of waking hours.
4. Capable of only limited self-care; confined to bed or chair more than 50% of waking hours.
5. Completely disabled; unable to carry out any self-care and totally confined to bed or chair.

Any significant change from baseline or screening must be reported as an AE (Section 8.3.5).

8.2.5.4 Ophthalmologic Assessments

Ophthalmologic assessments will occur as detailed in the screening SoA in Section 1.3 and the SoA in the respective modules, and will include visual acuity testing, slit lamp examination and fundoscopy.

8.2.5.5 ILD/Pneumonitis Investigation

If new or worsening pulmonary symptoms (eg, dyspnoea, cough or fever) or radiological abnormality suggestive of ILD/pneumonitis is observed, study treatment should be interrupted, and a full investigation is required as described in the T-DXd TMGs (Appendix L). Evaluations should include:

- Signs and symptoms (cough, shortness of breath, and pyrexia, etc.)
- Detailed past medical history, including concomitant medications
- Physical examination, including auscultation of lung field.
- Arterial blood gases if clinically indicated
- PFTs (Section 8.2.5.2) including FVC and CO diffusing capacity / DLCO, and pulse oximetry (SpO_2)
- Bronchoscopy and bronchoalveolar lavage as clinically indicated and feasible
- HRCT
- Pulmonologist consultation (infectious disease consultation as clinically indicated)
- Clinical laboratory tests:
  - COVID-19 test
  - Arterial blood gases if clinically indicated
  - Blood culture, blood cell count (CBC), differential white blood cell count (WBC), C-reactive protein (CRP)
  - One blood sample collection for PK as soon as ILD/pneumonitis is suspected, if feasible
- If feasible, blood samples for exploratory biomarker analysis will be collected from patients, as described in the SoA, as soon as ILD/pneumonitis is suspected.
- Other tests could be considered, as needed.

The results of the full diagnostic workup (including HRCT, blood and sputum culture, haematological parameters, etc.) will be captured in the eCRF. It is strongly recommended to perform a full diagnostic workup to exclude alternative causes such as lymphangitic carcinomatosis, infection, allergy, cardiogenic oedema, or pulmonary haemorrhage. In the

presence of confirmatory HRCT scans where other causes of respiratory symptoms have been excluded, a diagnosis of ILD/pneumonitis should be considered and the TMGs should be followed. Troponin measurements will be done to rule out cardiac aetiology.

The following assessments should be performed, if feasible, to enhance the investigation and diagnosis of potential cases of pneumonitis. The results of the assessment will be collected.

- Other items
  - When ILD/pneumonitis is suspected during study treatment, the following markers should be measured where possible:
    - ILD/pneumonitis markers (KL-6, SPD) and βDglucan
    - Tumour markers: particular tumour markers that are related to disease progression
    - Additional clinical chemistry: C-reactive protein, lactate dehydrogenase

8.3 Adverse Events and Serious Adverse Events

The Principal Investigator(s) is responsible for ensuring that all staff involved in the study are familiar with the content of this section.

The definitions of an AE or SAE can be found in Appendix B.

AEs will be reported by the patient (or, when appropriate, by a caregiver, surrogate, or the patient's legally authorised representative).

The Investigator and any designees are responsible for detecting, documenting, recording, and reporting events that meet the definition of an AE.

Care will be taken not to introduce bias when detecting AEs and/or SAEs. Open-ended and non-leading verbal questioning of the patient is the preferred method to inquire about AE occurrences.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

For patients who sign the pre-screening ICF, only SAEs directly related to tissue screening procedure (ie, if a patient undergoes a tumour biopsy) will be reported during pre-screening period.

All AEs and SAEs (other than ILD/pneumonitis) will be collected from the time of signing the main ICF throughout the treatment period and including the safety follow-up (which is 40 [+7] days after the discontinuation of all study treatments [or as otherwise specified in the respective modules] or until initiation of another therapy). For ILD/pneumonitis, safety follow-up will be continued until resolution of ILD/pneumonitis. If an event that starts after

the defined safety follow-up period noted above is considered to be due to late-onset toxicity to study treatment, then it should be reported as an AE or SAE, as applicable. Collection and reporting of AEs and SAEs after the final DCO is described in Section 8.3.12.

If the Investigator becomes aware of an SAE with a suspected causal relationship to the study treatment that occurs after the end of the clinical study in a patient treated by him or her, the Investigator shall, without undue delay, report the SAE to the Sponsor.

The following types of events should be reported by the Investigator in eCRF AE and AE-related page(s) in the clinical study database within 24 hours of becoming aware for the purposes of reporting in the global safety database:

- SAEs.
- All potential ILD/pneumonitis cases; including both serious and non-serious potential ILD/pneumonitis cases (potential ILD/pneumonitis is defined in the current Site Manual List of MedDRA Preferred Terms).
- Hepatic events (both serious and non-serious) which meet the PHL criteria defined as an elevated (ALT or AST) ≥ 3 × ULN and an elevated TBL ≥ 2 × ULN that may occur either at different timepoints or simultaneously during the study. A targeted questionnaire is built within the eCRF to collect relevant additional information for these potential cases.
- Overdose.
- Medication error.
- Pregnancy.

8.3.2 Follow-up of AEs and SAEs

Any AEs that are unresolved at the patient's last AE assessment in the study are followed up by the Investigator for as long as medically indicated, but without further recording in the eCRF. AstraZeneca retains the right to request additional information for any patient with ongoing AE(s)/SAE(s) at the end of the study, if judged necessary.

AE variables

The following variables will be collected for each AE:

- AE (verbatim).
- The date when the AE started and stopped.
- Changes in CTCAE grade (report only the maximum CTCAE grade for a calendar day).
- Whether the AE is serious or not (Appendix B).
- Investigator causality rating against the study treatment(s) (yes or no).

- Action taken with regard to study treatment(s).
- Administration of treatment for the AE.
- Outcome.

In addition, the following variables will be collected for SAEs:

- Date AE met criteria for SAE.
- Date Investigator became aware of SAE.
- Seriousness criteria.
- Date of hospitalisation.
- Date of discharge.
- Probable cause of death.
- Date of death.
- Autopsy performed.
- Causality assessment in relation to study procedure(s).
- Causality assessment to other medication.
- Description of the SAE.

The grading scales found in the NCI CTCAE v5.0 will be utilised for all events with an assigned CTCAE grading. For those events without assigned CTCAE grades, the recommendation in the CTCAE criteria that converts mild, moderate, and severe events into CTCAE grades should be used. A copy of the CTCAE can be downloaded from the Cancer Therapy Evaluation Program website (http://ctep.cancer.gov).

8.3.3 Causality Collection

The Investigator should assess causal relationship between study treatment and each AE, and answer “yes” or “no” to the question “Do you consider that there is a reasonable possibility that the event may have been caused by the investigational product?”

For SAEs, causal relationship should also be assessed for other medication and study procedures. Note that for SAEs that could be associated with any study procedure the causal relationship is implied as “yes”.

For a guide to the interpretation of the causality question refer to Appendix B.

8.3.4 Adverse Events Based on Signs and Symptoms

All AEs spontaneously reported by the patient or reported in response to the open question from the study site staff: “Have you had any health problems since the previous visit/you were

last asked?", or revealed by observation will be collected and recorded in the eCRF. When collecting AEs, the recording of diagnoses is preferred (when possible) to recording a list of signs and symptoms. However, if a diagnosis is known and there are other signs or symptoms that are not generally part of the diagnosis, the diagnosis and each sign or symptom will be recorded separately.

8.3.5 Adverse Events Based on Examinations and Tests

The results from the CSP-mandated laboratory tests, vital signs, physical examinations, ECGs, and ECHO/MUGA scans will be summarised in the CSR.

Deterioration as compared with baseline in protocol-mandated laboratory values, vital signs, and ECGs should therefore only be reported as AEs if they fulfil any of the SAE criteria, are the reason for discontinuation of treatment with the study treatment or are considered to be clinically relevant as judged by the Investigator (which may include but not limited to consideration as to whether treatment or non-planned visits were required or other action was taken with the study treatment, eg, dose adjustment or study treatment interruption).

If deterioration in a laboratory value/vital sign is associated with clinical signs and symptoms, the sign or symptom will be reported as an AE and the associated laboratory result/vital sign will be considered as additional information. Wherever possible the reporting Investigator uses the clinical, rather than the laboratory term (eg, anaemia vs low haemoglobin value). For any diagnosis of the significant clinical outcome of 'Left ventricular dysfunction', a valid or qualifying reduction of LVEF from baseline (as measured by MUGA or ECHO) should be confirmed and included in the AE report. In addition, any other clinically relevant deteriorations in non-mandated parameters should be reported as AE(s), with associated laboratory results.

Deterioration of a laboratory value, which is unequivocally due to PD, should not be reported as an AE/SAE.

Any new or aggravated clinically relevant abnormal medical finding at a physical examination as compared with the baseline assessment will be reported as an AE unless unequivocally related to the disease under study.

8.3.6 Hy's Law

Cases where a patient shows elevations in liver biochemistry may require further evaluation and occurrences of AST or ALT $\geq 3 \times$ULN together with TBL $\geq 2\times$ULN may need to be reported as SAEs. Please refer to Appendix E for further instruction on cases of increases in liver biochemistry and evaluation of HL.

8.3.7 Disease Progression

Disease progression can be considered as a worsening of a patient's condition attributable to the disease for which the study treatment is being studied. It may be an increase in the severity of the disease under study and/or increases in the symptoms of the disease. The development of new lesions, or progression of existing metastasis to the primary cancer under study should be considered as PD and not an AE. Disease progression/worsening of the patient's breast cancer will not be recorded as an AE on the Adverse Event eCRF. Death due to disease progression should be recorded on the Death eCRF as such.

8.3.8 Disease Under Study

Diseases under study commonly occur in studies of chronic diseases with a variable pattern, eg, asthma, COPD, rhinitis, neuropsychiatric conditions such as depression, seizure disorders or multiple sclerosis, cardiac disorders such as angina or heart failure. Symptoms of disease under study are those which might be expected to occur as a direct result of breast cancer. Events that are unequivocally due to disease under study should not be reported as an AE during the study unless they meet SAE criteria or lead to discontinuation of the study treatment.

8.3.9 New Cancers

The development of a new cancer should be regarded as an AE and will generally meet at least one of the serious criteria. New primary cancers are those that are not the primary reason for the administration of study treatment and are identified after the patient's inclusion in this study. They do not include metastases of the original cancer.

8.3.10 Deaths

All deaths that occur during the study treatment period, or within the protocol-defined follow-up period after the administration of the last dose of study treatment, must be reported as follows:

- Death clearly resulting from PD should be documented in the eCRF in the Statement of Death page. It should not be reported as an SAE.
- Where death is not due (or not clearly due) to PD under study, the AE causing the death must be reported as an SAE within 24 hours. It should also be documented in the Statement of Death page in the eCRF. The report should contain a comment regarding the co-involvement of PD, if appropriate, and should assign the main and contributory causes of death.
- Deaths with an unknown cause should always be reported as an SAE and documented in the Statement of Death page in the eCRF, but every effort should be made to determine a cause of death. A post-mortem may be helpful in the assessment of the cause of death,

and if performed, a copy of the post-mortem results should be forwarded to AstraZeneca Patient Safety or its representative within the usual time frames.

Deaths occurring after the protocol-defined follow-up period after the administration of the last dose of study treatment should be documented in the Statement of Death page. If the death occurred as a result of an event that started after the defined follow-up period and the event is considered to be due to a late-onset toxicity to study treatment, then it should also be reported as an SAE.

8.3.11 Adverse Events of Special Interest

AESIs are events of scientific and medical interest specific to the further understanding of the study treatment safety profile and require close monitoring and rapid communication by the Investigators to AstraZeneca. An AESI can be serious or non-serious. All AESIs will be recorded in the eCRF. Serious AESIs will be recorded and reported as per Section 8.3.13. The rapid reporting of AESIs allows ongoing surveillance of these events in order to characterise and understand them in association with the use of a study treatment.

AESIs will be recorded on the eCRF using a recognised medical term or diagnosis that accurately reflects the event. AEs will be assessed by the Investigator for severity, relationship to the study treatment, possible aetiologies, and whether the event meets criteria for an SAE and therefore requires immediate notification to AstraZeneca. If an AESI evolves into a condition that meets the regulatory definition of "serious" it will be reported on the SAE Report Form.

Based on the available preclinical and clinical data, review of the cumulative literature, reported toxicities for the same class of agents and biological plausibility, ILD/pneumonitis and LVEF decrease are considered to be AESIs. Additional relevant information regarding the AESIs ILD/pneumonitis and LVEF decrease, for the T-DXd clinical program regardless of seriousness is to be collected through the targeted questionnaires within the clinical study database.

ILD/Pneumonitis

ILD/pneumonitis is considered an important identified risk based on a comprehensive cumulative review of potential ILD/pneumonitis cases reviewed by the independent ILD Adjudication Committee (see Section 9.6.2), the available safety data from the clinical development program, available data from recent epidemiology/literature, biological plausibility, and safety information from drugs of similar class. Refer to the current T-DXd IB for a summary of preliminary clinical study data.

HRCT and pulmonary function will be measured at baseline. If the AE is suspected to be ILD/pneumonitis during the study, study treatment should be interrupted pending further evaluations. Evaluations for ILD/pneumonitis are outlined in Section 8.2.5.5.

Refer to Appendix K for guidelines on management of drug-induced ILD/pneumonitis.

LVEF decrease

LVEF decrease in association with T-DXd is considered to be an important potential risk based on the available preclinical data, literature and available safety information for drugs of similar class. Refer to the current T-DXd IB for a summary of preliminary clinical trial data.

- LVEF will be measured by either ECHO/MUGA scan to evaluate left ventricular function. All ECHOs/MUGAs will be evaluated by the Investigator or delegated physician for monitoring cardiac function.
- Blood samples will be collected for troponin (preferably high-sensitivity troponin-T) at screening, EOT, and if at any time a participant reports signs or symptoms suggesting CHF, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
- ECGs will be performed at screening, prior to administration of study treatment for every 4th cycle (once) and if an abnormality is detected. Triplicate ECGs will be performed at screening. Subsequent ECGs will be performed in triplicate only if abnormalities are noted. Twelve-lead ECGs will be performed, and standard ECG parameters will be measured, including RR, PR, QT intervals, and QRS duration. All ECGs must be evaluated by Investigator or delegated physician for the presence of abnormalities. Whether or not measurement is performed, date performed, results, and findings for each parameter is to be recorded in the eCRF.

Additional relevant information regarding the AESIs ILD/pneumonitis and LVEF decrease for the T-DXd clinical program, regardless of seriousness, is to be collected within the clinical study database.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

For patients continuing to receive study treatment after the final DCO, AEs and SAEs will be collected, if applicable, and reported until the end of the Post-trial Access period or until consent has been withdrawn. In addition, it is recommended that Investigators monitor the patient's safety laboratory results periodically during treatment with study treatment in order to manage AEs, consistent with the dose modification guidelines for management of study treatment-related toxicities (see Section 6.6). All data after the final DCO and database closure will be recorded in the patient notes but, with the exception of non-serious AEs, SAEs, and

special situations (i.e. pregnancy, overdoses, medication error, drug abuse and misuse), will not otherwise be reported for the purposes of this study.

All SAEs that occur in patients still receiving study treatment (or within the 40 days [+ 7 days] following the last dose of study treatment [or as otherwise specified in this section for specific treatments in the respective modules]) after the final DCO must be reported as detailed in Section 8.3.13.

8.3.13 Reporting of Serious Adverse Events

All SAEs have to be reported, whether or not considered causally related to the study treatment, or to the study procedure(s). All SAEs will be recorded in the eCRF during study conduct. After study completion, when EDC is closed and SAE needs to be reported, paper SAE form should be used.

If any SAE occurs in the course of the study, then Investigators or other site personnel inform the appropriate AstraZeneca representatives within one day ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

The designated AstraZeneca representative works with the Investigator to ensure that all the necessary information is provided to the AstraZeneca Patient Safety data entry site **within 1 calendar day** of initial receipt for fatal and life-threatening events **and within 5 calendar days** of initial receipt for all other SAEs.

For fatal or life-threatening AEs where important or relevant information is missing, active follow-up is undertaken immediately. Investigators or other site personnel inform AstraZeneca representatives of any follow-up information on a previously reported SAE within 1 calendar day ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

Once the Investigators or other site personnel indicate an AE is serious in the EDC system, an automated email alert is sent to the designated AstraZeneca representative.

If the EDC system is not available, then the Investigator or other study site staff reports an SAE to the appropriate AstraZeneca representative by telephone followed by completion of a paper SAE form.

The AstraZeneca representative will advise the Investigator/study site staff on how to proceed.

For further guidance on the definition of an SAE, see Appendix B.

The reference documents for definition of expectedness/listedness for the study treatments used in this study are:

- The current IB for T-DXd

- The current IB for durvalumab
- The label of the current effective version of the SmPC for paclitaxel, Hospira UK Limited.
- The label of the current effective version of the SmPC for Perjeta, Roche Products Limited.
- The label of the current effective version of the SmPC for tucatinib, Seagen, Inc.

8.3.14 Pregnancy

All pregnancies and outcomes of pregnancy with conception dates following the first date of study treatment, including pregnancy in the partner of male patients, should be reported to AstraZeneca.

8.3.14.1 Maternal Exposure

If a patient becomes pregnant during the course of the study, study treatment should be discontinued immediately.

Pregnancy itself is not regarded as an AE unless there is a suspicion that the study treatment under study may have interfered with the effectiveness of a contraceptive medication. Congenital abnormalities/birth defects and spontaneous miscarriages should be reported and handled as SAEs. Elective abortions without complications should not be handled as AEs. The outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth or congenital abnormality) occurring from the date of the first dose of study treatment until 7 months after the last dose of study treatment should be followed up and documented even if the patient was discontinued from the study.

If any pregnancy occurs in the course of the study, then the Investigator or other site personnel informs the appropriate AstraZeneca representatives within 1 day, ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

The designated AstraZeneca representative works with the Investigator to ensure that all relevant information is provided to the AstraZeneca Patient Safety data entry site within 1 or 5 calendar days for SAEs (see Section 8.3.13) and within 30 days for all other pregnancies.

The same timelines apply when outcome information is available.

8.3.14.2 Paternal Exposure

Non-sterilised male patients who intend to be sexually active with a female partner of child-bearing potential should refrain from fathering a child or freezing or donating or banking sperm for the duration of the study (from the time of screening) and for 4 months after the last dose of study treatment in Modules 0 to 1 and Modules 5 to 7 or 6 months after the last dose of study treatment in Modules 2 to 4.

Pregnancy of the patient's partner is not considered to be an AE. However, the outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth, or congenital abnormality) occurring from the date of the first dose of study treatment until 4 months (Modules 0 to 1 and Modules 5 to 7) or 6 months (Modules 2 to 4) after the last dose of study treatment should be followed up and documented in the medical record and provided to the AstraZeneca Patient Safety data entry site. Consent from the partner must be obtained before the information is collected and reported to AstraZeneca.

Where a report of pregnancy is received, prior to obtaining information about the pregnancy, the Investigator must obtain the consent of the patient's partner. The local study team should adopt the Master Pregnant Partner Form in line with local procedures/requirements and submit it to the relevant Regulatory Authority/IRBs/IECs prior to use.

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

8.3.15.1 Timelines

If an event of medication error, drug abuse or drug misuse occurs during the study, then the investigator or other site personnel informs the appropriate AstraZeneca representatives within **1 calendar day**, ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

The designated AstraZeneca representative works with the Investigator to ensure that all relevant information is completed within **1** (Initial Fatal/Life-threatening or follow-up Fatal/Life-threatening) or **5** (other serious initial and follow-up) **calendar days** if there is an SAE associated with the event of medication error, drug abuse, or drug misuse (see Section 8.3.13) and **within 30 days** for all other events.

8.3.15.2 Medication Error

For the purposes of this clinical study a medication error is an **unintended** failure or mistake in the treatment process for a study intervention that either causes harm to the participant or has the potential to cause harm to the participant.

The full definition and examples of a Medication Error can be found in Appendix B.

8.3.15.3 Drug Abuse

Drug abuse is the persistent or sporadic **intentional**, non-therapeutic excessive use of study intervention for perceived reward or desired non-therapeutic effect.

The full definition and examples of drug abuse can be found in Appendix B.

8.3.15.4 Drug Misuse

Drug misuse is the **intentional** and inappropriate use (by a study participant) of study intervention for medicinal purposes outside of the authorised product information, or for

unauthorised study intervention(s), outside the intended use as specified in the protocol and includes deliberate administration of the product by the wrong route.

The full definition and examples of Drug Misuse can be found in Appendix B.

8.3.16 Management of Study Treatment-related Toxicities

Refer to Appendix L and the following general guidance for management of toxicities:

- Treat each of the toxicities with maximum supportive care. For withholding study treatment, refer to Appendix L.
- If the symptoms promptly resolve with supportive care, consideration should be given to continuing the same dose of the assigned study treatment along with appropriate continuing supportive care. If medically appropriate, dose modifications are permitted. See Section 6.6.
- All dose modifications should be documented with clear reasoning and documentation of the approach taken.
- Interruption and/or discontinuation of T-DXd due to toxicities will result in the interruption and/or discontinuation of combination study treatments until resolved. Refer to Appendix L.

All toxicities will be graded according to NCI CTCAE v5.0.

Refer to this section in the respective modules for treatment-specific toxicity management information.

8.3.16.1 Specific Toxicity Management and Dose Modification Information – T-DXd

All dose modifications (interruption, reduction and/or discontinuation) should be based on the worst preceding toxicity (NCI CTCAE v5.0). Specific criteria for interruption, re-initiation, dose reduction and/or discontinuation of T-DXd are listed in Appendix L, which is applicable only to TEAEs that are assessed as related to use of T-DXd by the Investigator(s).

For non-drug related TEAEs, follow standard clinical practice. Appropriate clinical experts should be consulted as deemed necessary.

All confirmed or suspected COVID-19 infection events must be recorded in the eCRF. Please refer to Appendix M for additional information on dose modification for suspected or confirmed COVID-19 infection for patients treated with T-DXd.

ILD/Pneumonitis Management Guidance

Please refer to the ILD/pneumonitis management summary flow chart in Appendix K for the management of drug-induced ILD/pneumonitis. All potential ILD/pneumonitis cases should

be reported within 24 hours; including both serious and non-serious potential ILD/pneumonitis cases (potential ILD/pneumonitis is defined in the current Site Manual List of MedDRA Preferred Terms).

ILD/pneumonitis should be ruled out if a patient develops an acute onset of new or worsening pulmonary or other related signs/symptoms such as dyspnoea, cough or fever. If the AE is confirmed to have an aetiology other than ILD/pneumonitis, follow the management guidance outlined in the dose modification section of the CSP (Section 6.6).

If the AE is suspected to be ILD/pneumonitis, treatment with study treatment should be interrupted pending further evaluations. Evaluations should include those outlined in Section 8.2.5.5. As soon as ILD/pneumonitis is suspected, corticosteroid treatment should be started promptly as per clinical treatment guidelines.

If the AE is confirmed to be ILD/pneumonitis, follow the management guidance outlined in Appendix K and Appendix L. All events of ILD/pneumonitis regardless of severity or seriousness will be followed until resolution including after drug discontinuation.

To ensure adequate and relevant evaluation, systematic additional data collection will be conducted for all cases that will be brought for evaluation. This additional data collection will cover a more in-depth relevant medical history (eg, smoking, radiation, COPD, and other chronic lung conditions), diagnostic evaluation, treatment and outcome of the event. This data collection will be triggered for AEs reported using selected MedDRA Preferred Terms.

LVEF Management Guidance:

Left ventricular function will be measured by either ECHO or MUGA scan. All ECHOs/MUGAs will be evaluated by the Investigator or delegated physician for monitoring cardiac function.

- Troponin-T will be measured at screening and at EOT, and as needed based on patient reported cardiac signs or symptoms suggesting CHF, myocardial infarction, or other causes of cardiac myocyte necrosis. If 12-lead ECG is abnormal, follow institutional guidelines.
- ECGs will be performed and standard ECG parameters will be measured, including RR, PR, QT intervals, and QRS duration. ECG will be performed in triplicate if 12-lead ECG is abnormal. All ECGs must be evaluated by Investigator or delegated physician for the presence of abnormalities prior to the injection of study treatment at every cycle. Whether or not measurement is performed, date performed, results, and findings for each parameter is to be recorded in the eCRF.

Table 13 Dose Reduction Levels of T-DXd

Dose Level 1	Dose Level -1	Dose Level -2
5.4 mg/kg	4.4 mg/kg	3.2 mg/kg

Note: if 4.4mg/kg is RP2D, then the only dose reduction level is 3.2 mg/kg.
RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan.

Once the dose of T-DXd has been reduced because of toxicity, all subsequent cycles should be administered at that lower dose level unless further dose reduction is required. More than 2 dose reductions are not allowed, and the study treatment will be discontinued if further toxicity that meets the requirement for dose reduction occurs.

Dose Interruption and Modification/TMGs

Every effort should be made to limit study drug delay, however, in circumstances of adverse event management or medical intervention, the study drug can be held up to 18 weeks (126 days) from the last T-DXd dose. If a patient is assessed as requiring a dose delay of longer than 18 weeks, a discussion with the Study Physician is required to continue administration.
During this time, scheduled CT/MRI scans should continue as per protocol, and patients should fulfil all of the following criteria:

- Study drug may be resumed with confirmation of continued benefit per RECIST 1.1. Scans should be performed at the frequency defined per protocol while the drug is being held.
- At minimum, 1 restaging scan must be done within 6 weeks prior to restarting the study drug.
- IP(s) is/are restarted within the guidance of the TMGs for T-DXd and any combination agents, if appropriate.
- No prohibited concomitant medications have been administered since the last dose of T-DXd.

Treatment cycles for a patient for whom T-DXd dosing is temporarily withheld for any reason may have future cycles scheduled based on the date of the last T-DXd dose.

In addition, Investigators may consider dose reductions or discontinuations of T-DXd according to the patient's condition and after discussion with the Study Physician or designee. Refer to the TMGs in Appendix L.

8.4 Overdose

Overdose, defined as the accidental or intentional administration of any dose of a product that is considered both excessive and medically important. An "excessive and medically important" overdose includes any overdose in which either an SAE, a non-serious AE, or no

AE occurs and is considered by the Investigator as clinically relevant, ie, poses an actual or potential risk to the patient.

Overdose is always serious. By definition, an overdose is medically important, which meets the seriousness criterion of important medical event. An overdose can occur with or without an AE. AEs can either be serious or non-serious. Details of the overdose including T-DXd dosage, clinical course, associated AEs, and outcome must be captured in the narrative form of the CRF within EDC. Use of T-DXd of 20% more than the intended w/v dose is considered to be an overdose. Refer to this section in the respective modules for details on overdose of the other study treatments. There is currently no specific treatment in the event of overdose of the study treatments used in the study, and possible symptoms of overdose are not established. The Investigator will use clinical judgement to treat any overdose. Investigators should be advised that any patient who receives a higher dose than that intended should be monitored closely, managed with appropriate supportive care, and followed up expectantly.

- An overdose with associated AEs is recorded as the AE diagnosis/symptoms on the relevant AE modules in the eCRF and on the Overdose eCRF module.
- An overdose without associated symptoms is only reported on the Overdose eCRF module.

If an overdose with T-DXd or other study treatments, with or without associated AEs/SAEs, occurs during the course of the study, then the Investigator or other site personnel must inform the appropriate AstraZeneca representative immediately, or **no later than 24 hours** of when he or she becomes aware of it, using the designated Safety e-mail address or by fax.

The designated AstraZeneca representative works with the Investigator to ensure that all relevant information is provided to the AstraZeneca Patient Safety data entry site **within 1 or 5 calendar days** for overdoses associated with an SAE (see Section 8.3.13.) and **within 30 days** for all other overdoses.

8.5 Human Biological Samples

Instructions for the collection, handling, storage and shipping of biological samples will be provided in the study-specific Laboratory Manual. Samples should be stored in a secure storage space with adequate measures to protect confidentiality.

Samples will be stored for a maximum of 15 years from the end of the study (as defined in the CSP) in line with consent and local requirements, after which they will be destroyed/repatriated.

- Samples will be disposed of after the Bioanalytical Report finalisation or 6 months after issuance of the draft Bioanalytical Report (whichever is earlier), unless consented for future analyses.

  - Samples may be disposed of or anonymised by pooling. Additional analyses may be conducted on the anonymised, pooled samples to further evaluate and validate the analytical method. Any results from such analyses may be reported separately from the CSR.
- Remaining ADA sample aliquots will be retained at AstraZeneca or its designee for a maximum of 15 years post-CSR publication. Additional use includes but is not limited to further characterisation of any ADAs, confirmation and/or requalification of the assay as well as additional assay development work. The results from future analysis will not be reported in the CSR.

For further details on Handling of Human Biological Samples, see Appendix C.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of serum concentrations of T-DXd, total HER2-antibody, and MAAA1181a in all modules, and concentrations of combination treatments in serum (durvalumab, pertuzumab) or plasma (paclitaxel, tucatinib and its metabolite) in Modules 1 to 6, at timepoints as specified in the SoA of the respective treatment modules.

Samples may be collected at additional timepoints during the study if warranted and agreed upon between the Investigator and the Sponsor, eg, for safety reasons. The timing of sampling may be altered during the course of the study based on newly available data (eg, to obtain data closer to the time of peak or trough matrix concentrations) to ensure appropriate monitoring.

Samples collected for analyses of serum concentrations of T-DXd or combination treatments may also be used to evaluate safety or efficacy aspects related to concerns arising during or after the study.

Blood samples for PK will be analysed by a designated third party on behalf of AstraZeneca. Samples will be collected, labelled, stored, and shipped as detailed in the Laboratory Manual.

8.5.1.1 Determination of Drug Concentration

Samples for determination of drug concentration in serum will be assayed by bioanalytical test sites operated by or on behalf of AstraZeneca, using an appropriately validated bioanalytical method. Full details of the analytical method used will be described in a separate Bioanalytical Report.

Incurred sample reproducibility analysis, if any, will be performed alongside the bioanalysis of the test samples. The results from the evaluation, if performed, will be reported in a separate Bioanalytical Report.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd (all modules), durvalumab (Module 1 and Module 4) and pertuzumab (Module 2) in serum will be assayed by bioanalytical test sites operated by or on behalf of AstraZeneca, using an appropriately validated bioanalytical method. Full details of the methods used will be described in a separate report. For information on combination treatment specific ADA assessments, refer to the respective module documents. If required, the baseline samples may also be used for additional method validation activities (e.g. cut points).

Samples for ADA may also be further tested for characterisation of the ADA response and/or other exploratory safety biomarkers.

Details on sample processing, handling, shipment, and storage will be provided in the Laboratory Manual.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

8.6.1 Collection of Mandatory Samples for Biomarker Analysis

Patient consent to the study includes participation in the mandatory biomarker assessment components of the study.

Samples for biomarker assessment are required and will be collected from all patients in this study as specified in the screening SoA in Section 1.3 and the SoA in the respective modules. For all tumour specimens, FFPE tissue blocks are preferred. If it is not possible to provide a tissue block, at least 20 freshly-cut unstained serial tumour slides are to be provided. The following mandatory samples will be collected from all patients, including screen failures where possible:

- **Parts 1 and 2:**
  - **Tumour samples for study entry:**
    - Patients must provide an adequate tissue sample from the metastatic setting: either FFPE tissue blocks (preferred) or at least 20 freshly-cut unstained serial tumour slides. An adequate sample must be provided from either a newly acquired biopsy or the most recent archival tumour tissue in the metastatic setting.
    - For patients with relapsed disease an archival sample from the primary tumour (where available) must be submitted in addition to any more recent archival tissue provided.

- **On-treatment paired biopsies:**
  - Paired biopsies in Part 1 are optional (see Section 8.6.2)
  - In Part 2, a minimum of [redacted] evaluable paired tumour biopsies are required in Modules 0 to 4, and a minimum of [redacted] evaluable paired tumour biopsies are required in Module 5. In Modules 1 to 5 where there is an equivalent Part 1 module, any paired biopsy samples taken in Part 1 will be counted towards the total requirement for the module. If the number of paired biopsies is less than the number required when targeted enrolment is achieved, additional patients may be enrolled beyond the target enrolment with mandatory biopsies until the required number of paired biopsies per module are met.
  - On treatment biopsies will be taken for those patients with accessible lesions that have provided optional informed consent (see Section 8.6.2) and provided either an optional newly acquired pre-treatment tumour sample or an archival sample obtained in the metastatic setting. Paired pre-treatment and on treatment tumour samples must be obtained from the same lesion. Accessible lesions are defined as tumour lesions that are biopsiable and amenable to repeat biopsy. On treatment biopsies will be collected within a 7-day window prior to and including Cycle 2 Day 1.

Tumour tissue samples listed above could be used for IHC and proteomics (including, but not limited to HER2, related family members and topoisomerase I [TOPO-1], and PD-L1 expression), sequencing and gene expression analysis.

PD-L1 testing (positive vs negative, cut-off at ≥ 1% IHC, based on tumour and/or immune cells) will be conducted by one of AstraZeneca's preferred vendors. Slides for PD-L1 testing must be < 6 months old.

For Module 6 and Module 7, retrospective central HER2 testing may be performed using on-market IHC and ISH assays. Both assays will be used according to the manufacturer's instructions for use.

Blood sample collection

- Blood sample for the isolation of plasma to enable analysis and interpretation of [redacted] to characterise changes in [redacted] between baseline and on-treatment as a [redacted].

- The buffy coat layer obtained during the plasma isolation process of the baseline sample may be retained and analysed to assess and interpret [redacted], where local regulations allow.
  - [redacted] samples will be analysed for [redacted] and may be used to develop and validate future *in vitro* diagnostic tests to identify patients most likely to respond to the treatment. The sample is requested [redacted].
  - [redacted] samples taken during treatment and a final sample taken at disease progression will be used for additional exploratory research which may include but is not limited to interrogation of [redacted] as well as the [redacted] and potential [redacted].
- Blood sample for the assessment of [redacted] such as, but not limited to, [redacted], to assess a range of [redacted] that may [redacted]. Plasma may also be used for the detection/quantification of [redacted]
- Blood sample for [redacted] analysis to understand [redacted] and to assess [redacted] that may [redacted]
- Blood sample for [redacted] repertoire measurement is currently planned to include, but is not limited to, clonality and diversity.
- Blood sample for [redacted] analysis to assess a range of [redacted] biomarkers that may [redacted]. These biomarkers may include but are not limited to [redacted] repertoire analysis and analysis of [redacted] biomarkers associated with immunomodulatory effects.
- Blood sample for [redacted].
- Blood samples for [redacted] will be taken to obtain [redacted]

Comparisons will be made between baseline and other measures to determine if biomarkers (or combination of markers) are prognostic or predictive of outcomes associated with treatment.

For further details on Handling of Human Biological Samples, including storage, re-use and destruction, refer to Appendix C and the Laboratory Manual.

8.6.2 Collection of Optional Biomarker Samples

Collection of optional samples for biomarker assessment is also part of this study as specified in the screening SoA (Section 1.3), and the on-treatment SoA in the same section in each module and is subject to agreement to optional consent. The following optional samples will be collected from consenting patients:

Tumour biopsy on study entry:

- When a patient enters the trial with a biopsiable tumour, a newly acquired on-study tumour biopsy sample from metastasis or relapsed locally advanced disease should be obtained in patients that have signed the additional optional consent. This on-study sample should be taken prior to dosing with study treatment. Tumour lesions used for newly acquired biopsies at baseline should not be the same lesions used as RECIST 1.1 TLs, unless there are no other lesions suitable for biopsy and, in this instance, only core needle (not excisional/incisional) biopsy is allowed.

Paired biopsies:

- **Parts 1 and 2:**
  - An on-treatment tumour sample will be taken where the patient has provided informed consent and has a pre-treatment tumour sample newly acquired or from an archival sample obtained in the metastatic setting. The paired on-treatment biopsy can be collected CCI [redacted]. Paired pre-treatment and on-treatment tumour samples must be obtained from the same lesion.
- **Part 2:**
  - Paired tumour biopsies considered evaluable will be obtained from a minimum of CCI [redacted] patients in Modules 0 to 4, and CCI [redacted] patients from Module 5. If the number of paired biopsies is less than the number required when targeted enrolment is achieved, additional patients may be enrolled beyond the target enrolment with mandatory biopsies until the required number of paired biopsies per module are met.

Tumour biopsy on disease progression:

- A further on-study tumour biopsy sample should be taken at documented RECIST progression in patients that have provided the additional optional consent. Biopsies at progression may be particularly valuable when there is a marked phenotypic change in a particular lesion, and Investigators are encouraged to contact AstraZeneca in these cases.

The on-study provision of tumour tissue is encouraged only if clinically appropriate and not considered detrimental to patient care.

For further details on Handling of Human Biological Samples, including storage, re-use and destruction, refer to Appendix C and the Laboratory Manual.

8.6.3 Other Study-Related Biomarker Assessments

Already collected samples may be analysed on different biomarkers thought to play a role in efficacy and safety outcomes including, but not limited to, CCI , or CCI CCI (protein, CCI DNA) and/or specific candidate genes/genome-wide analysis for DNA, RNA to evaluate their association with observed clinical responses to study treatments as well as analysing whether the presence of antibodies to SARS-CoV-2 will provide any noted differences with the safety profile of patients treated with T-DXd or other anti-cancer agents. The presence of viruses such as, but not limited to, the SARS-CoV-2 virus may also be investigated.

Additional exploratory analyses may be undertaken on patients' samples to identify other biomarkers of sensitivity and resistance to study treatments and our understanding of cancer.

For further details on Handling of Human Biological Samples, including storage, re-use and destruction, refer to Appendix C and the Laboratory Manual.

8.7 Optional Genomics Initiative Sample

Collection of optional samples for Genomics Initiative research is also part of this study as specified in the SoA and is subject to agreement in the ICF addendum. Patients who do not wish to participate in the genetic research may still participate in the study. Blood sample for DNA isolation will be collected as stipulated in the SoA in the respective modules from patients who have consented to participate in the genetic analysis component of the study. Participation is optional.

In the event of DNA extraction failure or blood sample damage, a replacement genetic blood sample may be requested from the patient. Signed informed consent will be required to obtain a replacement sample unless it was included in the original consent.

See Appendix D for information regarding the storage and destruction of Genomics Initiative genetic sample. Details on processes for collection and shipment and destruction of these samples can be found either in the appendices or in the Laboratory Manual.

8.8 Medical Resource Utilisation and Health Economics

Medical resource utilisation and health economics data, associated with medical encounters, will be collected in the eCRF by the Investigator and study-site personnel for all patients throughout the study. CSP-mandated procedures, tests, and encounters are excluded.

The assessment of health care resource use will increase the understanding regarding the relationship between treatment and tumour-related cancer symptoms on resource use, such as the need for palliative procedures to address obstruction or bleeding. This will be captured and analysed to inform submissions to payers.

The impact of treatment and disease on healthcare resource use will be captured in this study on an event-driven basis. The HOSPAD module in eCRF will be used to collect information on key health care resource use that a patient receives that is not part of the study. The data may be used to support health technology assessments and payer related submissions.

At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred. Where any visits have occurred, the site should complete the HOSPAD. This review should be done at every scheduled clinic visit up to and including the post study treatment discontinuation follow-up visit. If a patient discontinues study treatment for reasons other than RECIST 1.1 progression, the HOSPAD form should continue to be administered until progression has been confirmed. Study-mandated visits should not be included in the HOSPAD.

The data collected may be used to conduct exploratory economic analyses and will include:

- Number and duration of medical care encounters, including surgeries, and other selected procedures (inpatient and outpatient)
- Duration of hospitalisation (total days or length of stay, including duration by wards [eg, intensive care unit])
- Number and type of diagnostic and therapeutic tests and procedures
- Outpatient medical encounters and interventions (including physician or emergency room visits, tests and procedures, and medications).

9 STATISTICAL CONSIDERATIONS

9.1 Statistical Hypotheses

The study is exploratory in nature, and there is no formal hypothesis testing involved.

9.2 Sample Size Determination

For all modules in this study, the sample sizes of Part 1 and Part 2 are not based on a formal statistical hypothesis.

For Part 1 in each module, safety is the primary endpoint. A minimum of [redacted] patients will be enrolled to a dose level and the maximum number of patients to be enrolled in any given dose level will be capped at [redacted] patients. Dose-escalation/de-escalation will be determined by using a mTPI-2 algorithm (Table 7) with a target DLT rate of 30% and an equivalence interval of (25%, 35%). A dose level will be considered unsafe, with no additional patients enrolled at that dose level, if it has an estimated 95% or more probability of exceeding the target DLT rate of 30% (ie, P [DLT > 30%|data] > 95%). A minimum of [redacted] patients will be enrolled to a dose level and evaluated through the DLT evaluation period before a dose-escalation/dose--expansion/de-escalation decision can be made (unless unacceptable toxicity is encountered prior to the enrolment of [redacted] patients). At the highest protocol-specified dose level, enrolment may be stopped if no more than one (including one) DLT is observed in the first [redacted] evaluable patients.

For Part 2 of Module 0 and Module 1, approximately 40 to 75 patients will be randomised. There is a 53% to 98% probability of observing at least one AE from 75 subjects if the true incidence rate is 1% to 5%.

Assuming 10% of the randomised patients will not provide evaluable efficacy data and the observed ORR of 80%, the 2-sided 80% CI for ORR is (72.4% to 86.2%). If the true underlying ORR is 85%, there is 96% chance to observe a lower 80% CI bound above 70%. Table 14 provides the 80% CIs for some observed ORRs.

Table 14 Confidence Intervals of ORR

ORR	70%	75%	80%	85%	90%
Exact Binomial 80% CI	(61.8%, 77.3%)	(67.0%, 81.8%)	(72.4%, 86.2%)	(77.9%, 90.4%)	(83.6%, 94.4%)

ORR = objective response rate; CI = confidence interval.

For Part 2 of Module 2, Module 3 and Module 4, approximately 40 to 50 patients will be randomised. There is a 39% to 92% probability of observing at least one AE from 50 subjects if the true incidence rate is 1% to 5%.

Assuming 10% of the randomised patients will not provide evaluable efficacy data and the observed ORR of 80%, the planned sample size of 50 patients, will provide a 2-sided 80% CI of (70.3%, 87.6%) for ORR. If the true underlying ORR is 85%, the lower bound of the 80% CI of ORR estimated from 50 patients will be greater than 70% with 87% probability. Table 15 provides the 80% CIs for some observed ORRs.

Table 15 Confidence Intervals of ORR

ORR	70%	75%	80%	85%	90%
Exact Binomial 80% CI	(59.6%, 79.0%)	(64.9%, 83.4%)	(70.3%, 87.6%)	(75.9%, 91.6%)	(81.7%, 95.3%)

ORR = objective response rate; CI = confidence interval.

For Part 2 of Modules 5, 6 and 7, approximately 35 patients will be randomised. There is a 30% to 83% probability of observing at least one AE from 35 subjects if the true incidence rate is 1% to 5%.

Assuming 10% of the randomised patients will not provide evaluable efficacy data and the observed ORR of 80%, the planned sample size of 35 patients, will provide a 2-sided 80% CI of (68.0%, 88.9%) for ORR. If the true underlying ORR is 85%, the lower bound of the 80% CI of ORR estimated from 35 patients will be greater than 70% with 65% probability. Table 16 provides the 80% CIs for some observed ORRs.

Table 16 Confidence Intervals of ORR

ORR	70%	75%	80%	85%	90%
Exact Binomial 80% CI	(57.3%, 80.7%)	(62.6%, 84.9%)	(68.0%, 88.9%)	(73.6%, 92.7%)	(79.5%, 96.1%)

ORR = objective response rate; CI = confidence interval.

Additional patients may be enrolled to ensure up to [redacted] evaluable paired biopsies are obtained in each module (refer to Section 8.6 for details).

Stratification Factors

In Part 2 of Modules 0 to 5, patients will be randomised by stratification factors PD-L1 status (positive vs negative), hormone receptor status (positive vs negative) and disease status (recurrent vs *de novo*). For further details on stratification refer to Section 6.3.3.

9.3 Populations for Analyses

The populations upon which the analyses are based are defined in Table 17.

Table 17 Populations for Analysis

Population/Analysis set	Description	
	Part 1	Part 2
FAS [a]	All patients who are assigned to treatment.	All patients who are randomised to treatment. Patients who are randomised but do not subsequently go on to receive study treatment are included in the analysis in the treatment group to which they are randomised.

Table 17 Populations for Analysis

Population/Analysis set	Description	
	Part 1	Part 2
EAS [a]	All patients who are assigned to treatment and received at least one dose of study treatment and had evaluable pre- and post- treatment efficacy data.	All patients who are randomised to treatment and received at least one dose of study treatment and had evaluable pre- and post-treatment efficacy data.
SAF	All patients who received at least 1 dose of study treatment (at least 1 study treatment [T-DXd, pertuzumab, paclitaxel, durvalumab, or tucatinib]).	
PK Analysis Set	All patients who received at least one dose of T-DXd therapy per protocol and had at least one post-dose evaluable PK data point [b] for T-DXd or combination treatments.	

[a] Additional patients enrolled to accommodate the collection of mandatory biopsies per treatment combination will be included in the FAS or EAS.
[b] Including unscheduled PK assessments related to ILD/pneumonitis and intake of chloroquine or hydroxychloroquine.
EAS = efficacy analysis set; FAS = full analysis set; ILD = interstitial lung disease; PK = pharmacokinetic; SAF = safety analysis set; T-DXd = trastuzumab deruxtecan.

9.4 Statistical Analyses

Analyses will be performed by AstraZeneca or its representative. The SAP will be prepared prior to First Patient In and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints including primary and key secondary endpoints. Any deviations from this plan will be reported in the CSR.

An interim safety analysis will be performed separately in Part 2 of Modules 0 to 6 (see Section 9.4.4.5). Additional interim analyses may be further performed separately in Part 2 for select modules to support development purposes. A final analysis will be performed when the last patient in overall has been followed for at least 12 months or all patients have discontinued or withdrawn. Additional patients may be enrolled for paired biopsies. Data from these additional patients will be included in any subsequent data cut, including the data cut for the CSR regardless of their follow-up time. The CSR will be updated for each module separately when data from the module matures.

9.4.1 General Considerations

Safety analyses of Part 1 (not applicable for Module 0, Module 6 or Module 7) and Part 2 for each module will be performed using the SAF, and efficacy analyses will be performed on the

FAS or EAS. Analysis of PK parameters will be based on the PK Analysis Set. See Table 18 for a summary of the of outcome variables and analysis populations.

Table 18 Summary of outcome variables and analysis populations

Outcome variable	Populations
Demography data	FAS
Efficacy data	
PFS, PFS2, and OS	FAS
ORR [a], DoR [a]	EAS
Safety data	
Exposure, AEs, laboratory measurements, vital signs, and physical examinations	SAF
PK data	PK Analysis Set
Immunogenicity data	SAF

[a] Patients who are evaluable for the analysis of DoR are those who responded in the ORR analysis.
AEs = adverse events; CNS-DoR = central nervous system duration of response; DoR = duration of response; EAS =efficacy analysis set; FAS = full analysis set; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; PFS2 = progression-free survival after next-line treatment; PK = pharmacokinetic; SAF = safety analysis set.

Depending on the extent of any impact, summaries of data relating to patients diagnosed with COVID-19, and impact of COVID-19 on study conduct (in particular missed visits, delayed or discontinued study treatment, and other protocol deviations) may be generated. More detail will be provided in the SAP.

9.4.2 Efficacy

9.4.2.1 Primary Endpoint(s)

There are no primary efficacy endpoints.

9.4.2.2 Secondary Endpoint(s)

9.4.2.2.1 Objective Response Rate (Parts 1 and 2)

Confirmed ORR is defined as the percentage of patients with a confirmed Investigator-assessed response of CR or PR, based on RECIST 1.1. The ORR will be reported for Part 1 and 2 of each module using the point estimate and the 2-sided 80% CIs of ORR will be calculated by the Clopper-Pearson method. For Part 2 of each module, the differences between the combination treatments against the T-DXd monotherapy will be reported using the point estimates as well as their 2-sided 80% CIs by the Miettinen-Nurminen method (Miettinen and Nurminen 1985). The odds ratio for the comparisons regarding ORR and their corresponding 80% CIs will be calculated using logistic regression models adjusted for all stratification factors in Modules 0 to 5. Treatment assignment will be the only covariate of the logistic models in Modules 6 and 7 as no stratifications are considered.

In Part 2 of each module, unconfirmed ORR is defined as the percentage of patients with an unconfirmed Investigator-assessed response of CR or PR and will be analysed using the same methodology as confirmed ORR.

9.4.2.2.2 Progression-Free Survival (Parts 1 and 2)

PFS is defined as the time from the date of randomisation until the date of objective PD or death. Detailed censoring rules for PFS will be specified in the SAP.

Kaplan-Meier estimates will be provided for PFS with median and the 2-sided 80% CIs for the median using Brookmeyer and Crowley method for each treatment arm. Exploratory comparisons between the T-DXd combinations and T-DXd monotherapy will be provided using stratified log-rank tests as an exploratory analysis. The HR and its CI will be estimated from a stratified Cox Proportional Hazards model (ties = Efron) with the CI calculated using a profile likelihood approach.

9.4.2.2.3 Progression-Free Survival after Next-line Treatment (Part 2 Only)

In Part 2 of each module, PFS2 is defined as time from randomisation until the date of progression on next-line treatment (the earliest of the progression event subsequent to first subsequent anticancer therapy) or death; second progression will be defined according to local standard clinical practice. Detailed censoring rules for PFS2 will be specified in the SAP.

Kaplan-Meier estimates will be provided for PFS2, with median and the 2-sided 80% CIs for the median using Brookmeyer and Crowley method for each treatment. Exploratory comparisons between T-DXd + durvalumab (Module 1) and T-DXd monotherapy (Module 0), and T-DXd + pertuzumab (Module 2) and T-DXd monotherapy (Module 0), will be provided using stratified log-rank tests. The HR and its CI will be estimated from a stratified Cox Proportional Hazards model (ties = Efron) with the CI calculated using a profile likelihood approach.

9.4.2.2.4 Duration of Response (Part 2 Only)

In Part 2 of each module, duration of response (per RECIST 1.1 using Investigator assessment) is defined as the time from the date of the first documentation of CR or PR to the date of the first documentation of PD or death in the absence of disease progression. DoR will be measured for responding patients (CR or PR) only. Detailed censoring rules for duration of response will be specified in the SAP.

DoR will be analysed using the same methodology specified for PFS (see Section 9.4.2.2.2).

9.4.2.2.5 Overall Survival (Part 2 Only)

In Part 2, OS is defined as the time from the date of randomisation until death due to any cause. Any patient not known to have died at the time of analysis will be censored based on the last recorded date on which the patient was known to be alive.

Note: Survival calls will be made following the date of DCO for the analysis (these contacts should generally occur within 7 days of the DCO). If participants are confirmed to be alive or if the death date is after the DCO date, then these participants will be censored at the date of DCO.

OS will be analysed using the same methodology specified for PFS (see Section 9.4.2.2.2).

9.4.2.2.6 Subgroup Analysis

Subgroup analysis will be conducted for all modules for ORR, DoR, PFS, and OS. Analyses will be performed by the following factors:

- PD-L1-positive vs PD-L1-negative (PD-L1-positive is defined by ≥ 1% IHC) for analyses regarding Modules 0 and 1 only.
- Hormone receptor status (positive vs negative).
- Disease status (recurrent vs de novo, as defined in Section 4.1.3.1).
- Brain metastases status at baseline (present vs absent).

All the subgroup analyses except those regarding brain metastases status will be presented by study modules. The subgroup analysis for patients with and without brain metastases at baseline will be presented for each treatment. Therefore, patients from Modules 0 and 7 receiving T-DXd monotherapy and the patients from Modules 5 and 6 receiving T-DXd + tucatinib will be combined for the analysis respectively.

Other baseline variables may also be assessed if there is clinical justification, or an imbalance is observed between the treatment groups. The purpose of the subgroup analyses is to assess the consistency of treatment effect across expected prognostic and/or predictive factors.

The subgroup analyses for the stratification factors will be based on the values entered into the IRT; all other factors will be based on values recorded on the eCRF, or from the third-party vendor data.

The results of subgroup analyses will be summarised and presented using forest plots, along with the results of the overall analyses. More details about the subgroup analyses will be provided in the SAP.

9.4.3 Safety

The primary endpoint in both parts of the study is safety. Full details of AE analyses will be provided in the SAP.

Safety data from Part 1 (not applicable for Module 0) and Part 2 from all modules will be analysed separately. Safety data from Part 1 will be analysed as soon as they become available.

- AEs, including SAEs, AEs leading to discontinuation, AEs leading to dose reductions, and AESIs, will be listed individually by patient and coded using MedDRA Preferred Terms from the most recent version of MedDRA that will have been released for execution at AstraZeneca.
- AEs will be presented for each treatment group by SOC and/or Preferred Term, covering number and percentage of patients reporting at least one event and number of events where appropriate.
- AEs occurring prior to start of study treatment, TEAEs, and post-treatment AEs will be presented separately.
- An overview of AEs will be presented for each treatment group the number and percentage of patients with any AE, AEs with outcome of death, SAEs, and AEs leading to discontinuation of study treatment, as well as AEs leading to study treatment dose interruptions, AEs leading to study treatment dose reduction and AEs leading to withdrawal from study as well as the number of individual occurrences in those categories.
- Separate AE tables will be provided taking into consideration the relationship to study treatment as assessed by the Investigator, CTCAE grade, seriousness, death, and events leading to discontinuation of study treatment as well as other action taken related to study treatment, AESIs, other significant AEs, and timing of events.
- An additional table will present the number and percentage of patients with the most common AEs. Most common AEs will be defined in the SAP.
- In accordance with the requirements of the FDA, a separate table will present non-serious AEs occurring in more than 5% of patients in any treatment group.
- Key patient information will be presented for patients with AEs with outcome of death, serious AEs, and AEs leading to discontinuation of study treatment.
- An AE listing for the SAF will cover details for each individual AE.
- The following events are considered treatment-emergent:
  - AEs with an onset date on or after first dose of study treatment
  - Worsening of pre-existing events on or after first dose of study treatment until 47 days after last dose of the study treatment

Vital Signs

Vital sign parameters will be presented for each treatment group. Summary statistics for continuous variables will cover number of patients, mean, standard deviation, minimum, median, and maximum. Frequency tables and shift tables will cover number and percentage of patients in the respective category.

For each scheduled post-baseline visit, descriptive statistics for all vital sign parameters will be presented for observed values and the change from baseline.

Details of vital sign analyses will be provided in the SAP.

Laboratory Parameters

Laboratory parameters will be presented for each treatment group. Summary statistics for continuous variables will cover number of patients, mean, standard deviation, minimum, median, and maximum. Frequency tables and shift tables will cover number and percentage of patients in the respective category.

For each scheduled post-baseline visit, descriptive statistics for all clinical chemistry and haematology parameters will be presented for observed values and change from baseline.

Elevation in liver parameters for assessment of Hy's Law will be summarised and reported appropriately if potential cases have been identified during the course of the study.

A frequency table for urinalysis will present the number of patients reporting at least one treatment-emergent increase in baseline category. A shift table for urinalysis will present the baseline assessment against the maximum on-treatment category.

Supportive laboratory listings will cover observed values and changes from baseline for each individual patient as well as abnormalities.

Other safety data will be assessed in terms of, physical examination, pulmonary function tests, RECIST 1.1 assessments and ECHO/MUGA.

Details of laboratory analyses will be provided in the SAP.

9.4.4 Other Analyses

9.4.4.1 Biomarkers

Biomarker status will be assessed for patients in each module according to pre-specified criteria that may be detailed in the SAP. The relationship of HER2 expression, PD-L1 expression and, if applicable, of exploratory biomarkers to clinical outcomes including, but not restricted to, PFS, ORR, and OS may be presented. Biomarker exploratory analyses may be described in a separate analysis plan and the results of this biomarker assessment will be reported either in the CSR itself or as an addendum, or separately in a scientific report or publication. The results of this biomarker assessment may be pooled with biomarker data from other studies with the study treatment to generate hypotheses to be tested in future research.

9.4.4.2 Pharmacokinetic Data

PK concentration data for T-DXd, total anti-HER2 antibody, and MAAA-1181a (all modules) and combination treatments (Modules 1 to 6) will be listed for each sampling time for each patient and each dosing day, and a summary will be provided for all evaluable patients in the PK analysis set. Descriptive statistics may be calculated.

9.4.4.3 Immunogenicity Data

Immunogenicity results will be listed by patient, and a summary will be provided by the number and percentage of patients who develop detectable ADA to T-DXd (all modules), durvalumab (Modules 1 and 4), and pertuzumab (Module 2). The immunogenicity titre and neutralising ADA data will be listed for samples confirmed positive for the presence of anti-T-DXd, anti-durvalumab, and anti-pertuzumab antibodies. The effect of immunogenicity as well as the effect of its neutralising properties on PK, pharmacodynamics, efficacy, and safety will be evaluated, if the data allow.

9.4.4.4 Clinical Outcome Assessments

ePRO questionnaires and at-home pulse oximetry are included for exploratory purposes only to further characterise ILD/pneumonitis. The analyses may be described in a separate analysis plan and presented in a separate report.

MDASI-BT scores will be derived per scoring provided by the developer. NANO Scale will be based on clinician-reported assessment of the participants' neurologic functioning. Derived data will be received from the CANTAB computerised tests per user agreement. Descriptive analyses for MDASI-BT, NANO and CANTAB will be detailed in the SAP and results interpreted according to developers' guidance.

9.4.4.5 BMFS (Modules 0 to 5 Part 2 Only)

The efficacy of T-DXd ± combinations in terms of prevention of developing brain metastasis in patients with no brain metastases at baseline will be assessed by BMFS. Brain metastases -free survival is defined as the time from the date of randomisation until the date of diagnosis of brain metastasis according to Investigator assessment using RECIST 1.1 or death by any cause. Kaplan-Meier estimates will be constructed. The BMFS event time will be censored if neither brain metastases nor death are recorded before the EOT (end of treatment) assessment. The median BMFS time, incidence of brain metastases (24-month BMFS) will be reported using point estimates and their 2-sided 80% CIs. In addition, a sensitivity analysis regarding BMFS will be performed using cumulative incidence curves, treating non-brain progression and death as competing events.

9.4.4.6 CNS Efficacy Endpoints (Modules 6 and 7 Part 2 Only)

CCI [redacted]

CCI [redacted]

CCI

CCI

9.4.4.7 Time to Second Progression beyond CNS Progression (Only for Patients with Brain Metastases at Baseline)

CCI

9.5 Interim Analyses

An interim safety analysis will be performed separately in Part 2 of Modules 0 to 6 after the first CCI evaluable patients (for Modules 0 to 4) or the first CCI evaluable patients (for Modules 5 and 6) all have had the opportunity for 2 months of treatment or have discontinued or withdrawn from treatment. All available data from these evaluable patients up to the interim analysis will be used. Safety will be evaluated for each combination treatment and the assessment will entail review of AEs, SAEs and treatment-related AEs leading to discontinuations, dose interruptions, dose delays, or dose reductions. Enrolment in each module will continue during the conduct of the interim safety analysis up to a total of CCI patients unless the stopping criterion for Modules 1 to 6 described below is met. Enrolment beyond CCI patients in any given module may recommence when the interim analysis is completed and the SRC has determined that enrolment may continue.

Additional interim analyses may be further performed separately in Part 2 for select modules to support development purposes.

9.5.1 Stopping Rules

A module will be assessed on the following stopping criterion at the time of the interim analysis (ie, after the first 10 patients in Part 2 of Modules 5 and 6, and 20 patients in Modules 1, 2, 3, and 4 have had at least 2 months of follow-up from the date of first dose or

have discontinued or withdrawn from treatment). No stopping rules will be applied to Modules 0 or 7, which are evaluating T-DXd monotherapy, for which a substantial body of safety data is already available (see T-DXd IB). Enrolment in a module will be paused if there is more than 95% probability that the discontinuation rate, for any study treatment, due to treatment-related AE exceeds 20% (ie, P[true rate > 20%|data] > 95%). Assuming a Beta(1,1) prior distribution for the discontinuation rate, this equates to observing a discontinuation rate due to treatment-related AE of at least 5/10 (50%) with 10 patients and at least 7/20 (35%) with 20 patients.

If the stopping threshold is met, enrolment in the module will be paused and an ad hoc SRC will be convened. All safety results will be presented to the SRC and a decision if the module needs to be permanently stopped from further enrolment will be made based on a comprehensive review by the SRC of safety data and the totality of the benefit-risk profile of the treatment regimen.

Patients already dosed may be allowed to continue treatment based on a case-by-case assessment at the discretion of the Investigator, and if the Investigator deems that a patient may continue to benefit from treatment.

9.6 Data Monitoring Committee

The safety of all AstraZeneca clinical studies is closely monitored on an ongoing basis by AstraZeneca representatives in consultation with Patient Safety. Issues identified will be addressed; for instance, this could involve amendments to the CSP and letters to Investigators. Instead of a Data Monitoring Committee, this study will have the SRC described in more detail in Section 9.6.1 below.

9.6.1 Safety Review Committee

For those modules with a Part 1, an SRC will review the initial safety profile from each combination treatment after 3 to 6 DLT-evaluable patients have completed the first 21-day cycle or had a DLT during the first 21-day cycle. Additional details on the use of the SRC during the Part 1 dose-escalation/de-escalation period are provided in Section 4.1.2. For Part 2 of Modules 0 to 4, the SRC will also be requested to review interim safety data from the first 20 evaluable patients. The study team will assess data from Part 2 of Modules 5 and 6 against the pre-specified early stopping rules described in Section 9.5. An ad hoc SRC will be convened for these modules if the stopping rules are met. A decision of an early stop due to safety concern may be suggested by the SRC based on their review of all available safety data. The SRC Charter for this study will define the exact membership, and who should be present for decisions to be made by the SRC.

The SRC will include but is not limited to the following:

- The AstraZeneca Study Physician or delegate.
- The Principal Investigator or delegate from sites that have enrolled at least 1 patient.
- Independent SRC chair who is not participating in this study.

In addition, other physicians may be invited, including the Global Safety Physician or delegate. The Clinical Pharmacology Scientist, Study Statistician, Patient Safety Scientist, and other delegates may also be invited as appropriate. The SRC Charter for this study will define the exact membership and who should be present for decisions to be made.

Further internal or external experts may be consulted by the SRC, as necessary. The Global Safety Physician or delegate should be present at the SRC if there are safety issues for discussion.

9.6.2 ILD Adjudication Committee

An ILD Adjudication Committee and Charter will be established to review all cases of potential ILD/pneumonitis. To ensure adequate and relevant evaluation, additional data may be collected to fully characterise medical history (eg, smoking, radiation and pulmonary history), diagnostic evaluation, treatment, and outcome of the event.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Appendix A Regulatory, Ethical, and Study Oversight Considerations

A 1 Regulatory and Ethical Considerations

- This study will be conducted in accordance with the protocol and with the following:
  - Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organisations of Medical Sciences International Ethical Guidelines
  - Applicable ICH GCP Guidelines
  - Applicable laws and regulations
- The protocol, protocol amendments, ICF, IB, and other relevant documents (eg, advertisements) must be submitted to an IRB/IEC by the Investigator and reviewed and approved by the IRB/IEC before the study is initiated.
- Any amendments to the protocol will require IRB/IEC and applicable Regulatory Authority approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study patients.
- AstraZeneca will be responsible for obtaining the required authorisations to conduct the study from the concerned Regulatory Authority. This responsibility may be delegated to a Contract Research Organisation but the accountability remains with AstraZeneca.
- The Investigator will be responsible for providing oversight of the conduct of the study at the site and adherence to requirements of 21 CFR 312.120, ICH guidelines, the IRB/IEC, European Regulation 536/2014 for clinical studies (if applicable), and all other applicable local regulations.

Regulatory Reporting Requirements for Serious Adverse Events

- Prompt notification by the Investigator to the Sponsor of a SAE is essential so that legal obligations and ethical responsibilities towards the safety of patients and the safety of a study treatment under clinical investigation are met.
- The Sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study treatment under clinical investigation. The Sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRB/IEC, and Investigators.
- Investigator safety reports must be prepared for suspected unexpected serious adverse reactions according to local regulatory requirements and Sponsor policy and forwarded to Investigators as necessary.

- An Investigator who receives an Investigator safety report describing a SAE or other specific safety information (eg, summary or listing of SAEs) from the Sponsor will review and then file it along with the IB or other documents and will notify the IRB/IEC, if appropriate according to local requirements.

Regulatory Reporting Requirements for Serious Breaches

- Prompt notification by the investigator to AstraZeneca of any (potential) serious breach of the protocol or regulations is essential so that legal and ethical obligations are met.
  - A 'serious breach' means a breach likely to affect to a significant degree the safety and rights of a participant or the reliability and robustness of the data generated in the clinical study.
- If any (potential) serious breach occurs in the course of the study, investigators or other site personnel will inform the appropriate AstraZeneca representatives immediately after they become aware of it.
- In certain regions/countries, AstraZeneca has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about such breaches.
  - AstraZeneca will comply with country-specific regulatory requirements relating to serious breach reporting to the regulatory authority, IRB/IEC, and investigators. If EU Clinical Trials Regulation 536/2014 applies, AstraZeneca is required to enter details of serious breaches into the EMA CTIS. It is important to note that redacted versions of serious breach reports will be available to the public via CTIS.
- The investigator should have a process in place to ensure that:
  - The site staff or service providers delegated by the investigator/institution are able to identify the occurrence of a (potential) serious breach.
  - A (potential) serious breach is promptly reported to AstraZeneca or delegated party, through the contacts (e-mail address or telephone number) provided by AstraZeneca.

A 2 Financial Disclosure

Investigators and sub-Investigators will provide the Sponsor with sufficient, accurate financial information as requested to allow the Sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

A 3 Informed Consent Process

- An optional pre-screening ICF may be signed by patients to permit tumour tissue sample collection for central PD-L1 testing and/or central retrospective CCI status testing prior to the 28-day screening window.
- The Investigator or his/her representative will explain the nature of the study to the patient or his/her legally authorised representative and answer all questions regarding the study.
- Patients must be informed that their participation is voluntary, and they are free to refuse to participate and may withdraw their consent at any time and for any reason during the study. Patients or their legally authorised representative will be required to sign a statement of informed consent that meets the requirements of 21 Code of Federal Regulations 50, local regulations, ICH guidelines, Health Insurance Portability and Accountability Act requirements, where applicable, and the IRB/IEC or study centre.
- The medical record must include a statement that written informed consent was obtained before the patient was enrolled in the study and the date the written consent was obtained. The authorised person obtaining the informed consent must also sign the ICF.
- Patients must be re-consented to the most current version of the ICF(s) during their participation in the study.
- A copy of the ICF(s) must be provided to the patient or the patient's legally authorised representative.
- If a patient declines to participate in a voluntary exploratory genetic research component of the study, there will be no penalty or loss of benefit to the patient and he/she will not be excluded from other aspects of the study.
- If a patient's partner becomes pregnant during or within 4 months after the last dose of study treatment, the partner is asked to sign the "Adult Study Informed Consent Form for Pregnant Partners of Study Patients" and provide information about the pregnancy accordingly. Also refer to Section 8.3.14.

Patients who are rescreened are required to sign a new ICF.

The ICF will contain a separate section that addresses and documents the collection and use of any mandatory and/or optional human biological samples. The Investigator or authorised designee will explain to each patient the objectives of the analysis to be done on the samples and any potential future use. Patients will be told that they are free to refuse to participate in any optional samples or the future use and may withdraw their consent at any time and for any reason during the retention period.

A 4 Data Protection

- Patients will be assigned a unique identifier by AstraZeneca. Any patient records or datasets that are transferred to AstraZeneca will contain the identifier only; patient names or any information which would make the patient identifiable will not be transferred.
- The patient must be informed that their personal study-related data will be used by AstraZeneca in accordance with local data protection law. The level of disclosure and use of their data must also be explained to the patient in the informed consent.
- The patient must be informed that their medical records may be examined by Clinical Quality Assurance auditors or other authorised personnel appointed by AstraZeneca, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.
- The patient must be informed that data will be collected only for the business needs. We will only collect and use the minimum amount of personal data to support our business activities and will not make personal data available to anyone (including internal staff) who is not authorised or does not have a business need to know the information.
- The patient must be informed that in some cases their data may be pseudonymised. The GDPR defines pseudonymisation as the processing of personal data in such a way that the personal data can no longer be attributed to a specific individual without the use of additional information, provided that such additional information is kept separately and protected by technical and organisational measures to ensure that the personal data are not attributed to an identified or identifiable natural person.

Unless previously specified, the biomarker data will have unknown clinical significance and AstraZeneca will not provide biomarker assessment results to patients, their family members, any insurance company, any employer, a clinical study Investigator, a general physician, or any other third party, unless required to do so by law; however, AstraZeneca may share data and biosamples with research partners.

The patient’s samples will not be used for any purpose other than those described in the CSP.

Personal Data Breaches

A ‘personal data breach’ means a breach of security leading to the accidental or unlawful destruction, loss, alteration, unauthorised disclosure of, or access to, personal data transmitted, stored or otherwise processed.

- In compliance with applicable laws, the Data Controller[1] for the processing activity where the personal data breach occurred (AstraZeneca or respectively the site), will notify the data protection authorities without undue delay within the legal terms provided for such notification and within the prescribed form and content.
- Whilst AstraZeneca has processes in place to deal with personal data breaches it is important that investigators that work with AstraZeneca have controls in place to protect patient data privacy.

The Investigator should have a process in place to ensure that:

- allow site staff or service providers delegated by the investigator/institution to identify the occurrence of a (potential) personal data breaches.
- Any (potential) personal data breach is promptly reported to AstraZeneca or delegated party, through the contacts (e-mail address or telephone number) provided by AstraZeneca.

AstraZeneca and the site must demonstrate that they:

- have taken all necessary steps to avoid personal data breaches and
- have undertaken measures to prevent such breaches from occurring in the first place and to mitigate the impact of occurred data breaches (eg, applying encryption, maintaining and keeping systems and IT security measures up-to-date, regular reviews and testing, regular training of employees, and developed security policies and standards).
- where possible, have developed an internal data breach reporting and investigation process and internal protocols with guidance on how to respond swiftly and diligently to the occurrence of a personal data breach.
- where it has not been possible to develop an internal data breach reporting and investigation process, the site follows AstraZeneca's instructions.

Notification of personal Data Breach to participants:

- notification to participants is done by the site for the data breaches that occurred within the processing activities for which the site is the Data Controller and for data breaches occurred within the processing activities of AstraZeneca as the Data Controller, the notification is done in collaboration with the site and is performed by the site and/or Principal Investigator, acting on behalf of AstraZeneca, so that AstraZeneca has no access to the identifying personal information of the participants. The site and/or Principal Investigator shall conduct the notification by contacting the participants using the information that they gave for communication purposes in clinical research.

[1] The **data controller** determines the **purposes** for which and the **means** by which personal data is processed, as defined by the European Commission

- If a personal data breach occurs in a processor's systems, engaged by AstraZeneca, the processor under contractual obligations with AstraZeneca promptly and in due course after discovering the breach notifies AstraZeneca and provides full cooperation with the investigation. In these cases, to the extent AstraZeneca is the Data Controller for the processing activity where the breach occurred, it will be responsible for the notification to data protection authorities and, if applicable, to participants. If the personal data breach needs to be notified to the participants, the notification to participants is done in collaboration with the site and is performed by the site and/or Principal Investigator, acting on behalf of the Sponsor, so that AstraZeneca has no access to the identifying personal information of the participants.
- If a personal data breach involving an AstraZeneca's representative device (i.e Study Monitor laptop), AstraZeneca representative will provide AstraZeneca with all of the information needed for notification of the breach, without disclosing data that allows AstraZeneca directly or indirectly to identify the participants. The notification will be done by AstraZeneca solely with the information provided by the Study Monitor and in no event with access to information that could entail a risk of re-identification of the participants. If the data breach must be notified to the data subjects, the notification will be done directly by the Study Monitor in collaboration with the site and/or Principal Investigator, acting on behalf of the Sponsor, so that AstraZeneca has no access to the identifying personal information of the participants. The contract between AstraZeneca and the Study Monitor shall expressly specify these conditions.
- The contract between the site and AstraZeneca for performing the clinical research includes the provisions and rules regarding who is responsible for coordinating and directing the actions in relation to the breaches and performing the mandatory notifications to authorities and participants, where applicable.

A 5 Committees Structure

The safety of all AstraZeneca clinical studies is closely monitored on an on-going basis by AstraZeneca representatives in consultation with Patient Safety. Issues identified will be addressed; for instance, this could involve amendments to the CSP and letters to Investigators.

A 6 Dissemination of Clinical Study Data

Any results both technical and lay summaries for this trial, will be submitted to EU CTIS within a year from global End of Trial Date in all participating countries, due to scientific reasons, as otherwise statistical analysis is not relevant.

A description of this clinical study will be available on http://astrazenecagrouptrials.pharmacm.com and http://www.clinicaltrials.gov as will the summary of the main study results when they are available. The clinical study and/or

summary of main study results may also be available on other websites according to the regulations of the countries in which the main study is conducted.

A 7 Data Quality Assurance

- All patient data relating to the study will be recorded on the CRF unless transmitted to AstraZeneca or designee electronically (eg, laboratory data). The Investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the eCRF.
- The investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.
- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.
- Monitoring details describing strategy, including definition of study-critical data items and processes (eg, risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-Based Monitoring), methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques (central, remote, or on-site monitoring) are included in the Monitoring Plan.
- AstraZeneca or designee is responsible for medical oversight throughout the conduct of the study which includes clinical reviews of study data in accordance with the currently approved protocol. Monitoring details describing clinical reviews of study data from a medical perspective are included in more detail in the Monitoring Plan.
- AstraZeneca or designee is responsible for the data management of this study including quality checking of the data.
- AstraZeneca assumes accountability for actions delegated to other individuals (eg, Contract Research Organisations).
- Study monitors will perform ongoing source data verification as per the Monitoring Plan to confirm that data entered into the eCRF by authorised site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of patients are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.
- Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for a minimum of 25 years after study archiving or as required by local regulations, according to the AstraZeneca Global Retention And Disposal (GRAD) Schedule. No records may be destroyed during the retention period without the written approval of AstraZeneca. No records may be transferred to another location or party without written notification to AstraZeneca.

A 8 Source Documents

- Source documents provide evidence for the existence of the patient and substantiate the integrity of the data collected. Source documents are filed at the Investigator's site.
- Data reported on the eCRF or entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The Investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.
- Definition of what constitutes source data can be found in the Central Monitoring Plan. All information in original records and certified copies of original records of clinical findings, observations, or other activities in a clinical study necessary for the reconstruction and evaluation of the study are defined as source documents. Source data are contained in source documents (original records or certified copies).

A 9 Study and Site Start and Closure

The study start date is the date on which the clinical study will be open for recruitment of patients. The first act of recruitment is the first site open and will be the study start date.

The Sponsor designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the Sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The Investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the Sponsor or Investigator may include but are not limited to:

- Failure of the Investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the Sponsor's procedures, or GCP guidelines
- Inadequate recruitment of patients by the Investigator
- Discontinuation of further study treatment development

If the study is prematurely terminated or suspended, the Sponsor shall promptly inform the Investigators, the IECs/IRBs, the regulatory authorities, and any Contract Research Organisation(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The Investigator shall promptly inform the patient and should assure appropriate patient therapy and/or follow-up.

Patients from terminated sites will have the opportunity to be transferred to another site to continue the study.

A 10 Publication Policy

- The results of this study may be published or presented at scientific meetings. If this is foreseen, the Investigator agrees to submit all manuscripts or abstracts to the Sponsor before submission. This allows the Sponsor to protect proprietary information and to provide comments.
- The Sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the Sponsor will generally support publication of multicentre studies only in their entirety and not as individual site data. In this case, a co-ordinating Investigator will be designated by mutual agreement.
- Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

Appendix B Adverse Events: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

B 1 Definition of Adverse Events

An AE is the development of any untoward medical occurrence (other than progression of the malignancy under evaluation) in a patient or clinical study patient administered a study treatment and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable and unintended sign (eg, an abnormal laboratory finding), symptom (eg, nausea, chest pain), or disease temporally associated with the use of a medicinal product, whether or not considered related to the study treatment.

The term AE is used to include both serious and non-serious AEs and can include a deterioration of a pre-existing medical occurrence. An AE may occur at any time, including run-in or washout periods, even if no study treatment has been administered.

B 2 Definitions of Serious Adverse Event

An SAE is an AE occurring during any study phase (ie, run-in, treatment, washout, follow-up), that fulfils one or more of the following criteria:

- Results in death.
- Is immediately life-threatening.
- Requires in-patient hospitalisation or prolongation of existing hospitalisation.
- Results in persistent or significant disability or incapacity.
- Is a congenital abnormality or birth defect.
- Is an important medical event that may jeopardise the patient or may require medical treatment to prevent one of the outcomes listed above.

AEs for **malignant tumours** reported during a study should generally be assessed as **Serious** AEs. If no other seriousness criteria apply, the “Important Medical Event” criterion should be used. In certain situations, however, medical judgment on an individual event basis should be applied to clarify that the malignant tumour event should be assessed and reported as a **Non-Serious** AE. For example, if the tumour is included as medical history and progression occurs during the study, but the progression does not change treatment and/or prognosis of the malignant tumour, the AE may not fulfil the attributes for being assessed as serious, although reporting of the progression of the malignant tumour as an AE is valid and should occur. Also, some types of malignant tumours, which do not spread remotely after a routine treatment that does not require hospitalisation, may be assessed as Non-Serious; examples in adults include Stage 1 basal cell carcinoma and Stage 1A1 cervical cancer removed via cone biopsy.

The above instruction applies only when the malignant tumour event in question is a new malignant tumour (ie, it is *not* the tumour for which entry into the study is a criterion and that is being treated by the study treatment and is not the development of new or progression of existing metastasis to the tumour under study). Malignant tumours that – as part of normal, if rare, progression – undergo transformation (eg, Richter's transformation of B cell chronic lymphocytic leukaemia into diffuse large B cell lymphoma) should not be considered a new malignant tumour.

Life threatening

"Life-threatening" means that the patient was at immediate risk of death from the AE as it occurred or it is suspected that use or continued use of the product would result in the patient's death. "Life-threatening" does not mean that had an AE occurred in a more severe form it might have caused death (eg, hepatitis that resolved without hepatic failure).

Hospitalisation

Outpatient treatment in an emergency room is not in itself a SAE, although the reasons for it may be (eg, bronchospasm, laryngeal oedema). Hospital admissions and/or surgical operations planned before or during a study are not considered AEs if the illness or disease existed before the patient was enrolled in the study, provided that it did not deteriorate in an unexpected way during the study.

Important medical event or medical treatment

Medical and scientific judgment should be exercised in deciding whether a case is serious in situations where important medical events may not be immediately life threatening or result in death, hospitalisation, disability or incapacity but may jeopardise the patient or may require medical treatment to prevent one or more outcomes listed in the definition of serious. These should usually be considered as serious.

Simply stopping the suspect drug does not mean that it is an important medical event; medical judgment must be used.

- Angioedema not severe enough to require intubation but requiring IV hydrocortisone treatment
- Hepatotoxicity caused by paracetamol (acetaminophen) overdose requiring treatment with N-acetylcysteine
- Intensive treatment in an emergency room or at home for allergic bronchospasm
- Blood dyscrasias (eg, neutropenia or anaemia requiring blood transfusion, etc.) or convulsions that do not result in hospitalisation
- Development of drug dependency or drug abuse

Intensity rating scale:

The grading scales found in the revised NCI CTCAE latest version 5 will be utilised for all events with an assigned CTCAE grading. For those events without assigned CTCAE grades, the recommendation in the CTCAE criteria that converts mild, moderate and severe events into CTCAE grades should be used. A copy of the CTCAE can be downloaded from the Cancer Therapy Evaluation Program website (http://ctep.cancer.gov). The applicable version of CTCAE should be described clearly.

It is important to distinguish between serious and severe AEs. Severity is a measure of intensity whereas seriousness is defined by the criteria in Appendix B 2. An AE of severe intensity need not necessarily be considered serious. For example, nausea that persists for several hours may be considered severe nausea, but not a SAE unless it meets the criteria shown in Appendix B 2. On the other hand, a stroke that results in only a limited degree of disability may be considered a mild stroke but would be a SAE when it satisfies the criteria shown in Appendix B 2.

B 3 A Guide to Interpreting the Causality Question

When making an assessment of causality consider the following factors when deciding if there is a “reasonable possibility” that an AE may have been caused by the drug.

- Time Course. Exposure to suspect drug. Has the patient actually received the suspect drug? Did the AE occur in a reasonable temporal relationship to the administration of the suspect drug?
- Consistency with known drug profile. Was the AE consistent with the previous knowledge of the suspect drug (pharmacology and toxicology) or drugs of the same pharmacological class? Or could the AE be anticipated from its pharmacological properties?
- De-challenge experience. Did the AE resolve or improve on stopping or reducing the dose of the suspect drug?
- No alternative cause. The AE cannot be reasonably explained by another aetiology such as the underlying disease, other drugs, other host or environmental factors.
- Re-challenge experience. Did the AE reoccur if the suspected drug was reintroduced after having been stopped? AstraZeneca would not normally recommend or support a re-challenge.
- Laboratory tests. A specific laboratory investigation (if performed) has confirmed the relationship.

In difficult cases, other factors could be considered such as:

- Is this a recognised feature of overdose of the drug?
- Is there a known mechanism?

Causality of "related" is made if following a review of the relevant data, there is evidence for a "reasonable possibility" of a causal relationship for the individual case. The expression "reasonable possibility" of a causal relationship is meant to convey, in general, that there are facts (evidence) or arguments to suggest a causal relationship.

The causality assessment is performed based on the available data including enough information to make an informed judgment. With no available facts or arguments to suggest a causal relationship, the event(s) will be assessed as "not related".

Causal relationship in cases where the disease under study has deteriorated due to lack of effect should be classified as no reasonable possibility.

B 4 Medication Error, Drug Abuse and Drug Misuse

Medication Error

For the purposes of this clinical study a medication error is an unintended failure or mistake in the treatment process for an AstraZeneca study treatment that either causes harm to the patient or has the potential to cause harm to the patient.

A medication error is not lack of efficacy of the drug, but rather a human or process related failure while the drug is in control of the study site staff or patient.

Medication error includes situations where an error:

- occurred.
- was identified and intercepted before the patient received the drug.
- did not occur, but circumstances were recognised that could have led to an error.

Examples of events to be reported in clinical studies as medication errors:

- Drug name confusion.
- Dispensing error eg, medication prepared incorrectly, even if it was not actually given to the patient.
- Drug not administered as indicated, eg, wrong route or wrong site of administration.

- Drug not taken as indicated eg, tablet dissolved in water when it should be taken as a solid tablet.
- Drug not stored as instructed eg, kept in the fridge when it should be at room temperature.
- Wrong patient received the medication (excluding IRT errors).
- Wrong drug administered to patient (excluding IRT errors).

Examples of events that **do not** require reporting as medication errors in clinical studies:

- Errors related to or resulting from IRT – including those which lead to one of the above listed events that would otherwise have been a medication error.
- Patient accidentally missed drug dose(s) eg, forgot to take medication.
- Accidental overdose (will be captured as an overdose).
- Patient failed to return unused medication or empty packaging.
- Errors related to background and rescue medication, or SOC medication in open label studies, even if an AstraZeneca product.

Medication errors are not regarded as AEs but AEs may occur as a consequence of the medication error. Any events of medication error, with or without associated AEs, are to be captured in the eCRF in Rave and forwarded to the DES using the Medication Error Report Form.

Drug Abuse

For the purpose of this study, drug abuse is defined as the persistent or sporadic intentional, non-therapeutic excessive use of study intervention for a perceived reward or desired non-therapeutic effect.

Any events of drug abuse, with or without associated AEs, are to be captured and forwarded to the DES using the Drug Abuse Report Form. This form should be used both if the drug abuse happened in a study participant or if the drug abuse regards a person not enrolled in the study (such as a relative of the study participant).

Examples of drug abuse include but are not limited to:

- The drug is used with the intent of getting a perceived reward (by the study participant or a person not enrolled in the study)
- The drug in the form of a tablet is crushed and injected or snorted with the intent of getting high.

Drug Misuse

Drug misuse is the intentional and inappropriate use (by a study participant) of study intervention for medicinal purposes outside of the authorised product information, or for unauthorised study interventions, outside the intended use as specified in the protocol, and includes deliberate administration of the product by the wrong route.

Events of drug misuse, with or without associated AEs, are to be captured and forwarded to the DES using the Drug Misuse Report Form. This form should be used both if the drug misuse happened in a study participant or if the drug misuse regards a person not enrolled in the study (such as a relative of the study participant).

Examples of drug misuse include but are not limited to:

- The drug is used with the intention to cause an effect in another person
- The drug is sold to other people for recreational purposes
- The drug is used to facilitate assault in another person
- The drug is deliberately administered by the wrong route
- The drug is split in half because it is easier to swallow, when it is stated in the protocol that it must be swallowed whole
- Only half the dose is taken because the study participant feels that they were feeling better when not taking the whole dose
- Someone who is not enrolled in the study intentionally takes the drug.

Appendix C Handling of Human Biological Samples

C 1 Chain of Custody

A full chain of custody is maintained for all samples throughout their lifecycle.

The Investigator at each centre keeps full traceability of collected biological samples from the patients while in storage at the centre until shipment or disposal (where appropriate) and records relevant processing information related to the samples whilst at site.

The sample receiver keeps full traceability of the samples while in storage and during use until used or disposed of or until further shipment and keeps record of receipt of arrival and onward shipment or disposal.

AstraZeneca or delegated representatives will keep oversight of the entire life cycle through internal procedures, monitoring of study sites, auditing or process checks, and contractual requirements of external laboratory providers

Samples retained for further use will be stored in the AstraZeneca-assigned biobanks or other sample archive facilities and will be tracked by the appropriate AstraZeneca Team for the remainder of the sample life cycle.

If required, AstraZeneca will ensure that remaining biological samples are returned to the site according to local regulations or at the end of the retention period, whichever is the sooner.

C 2 Withdrawal of Informed Consent for Donated Biological Samples

If a patient withdraws consent specifically to the subsequent use of donated biological samples, the samples will be disposed of/destroyed/repatriated, and the action documented. If samples are already analysed, AstraZeneca is not obliged to destroy the results of this research. The patient will be presented with the option to opt out of the subsequent use of the donated samples during the withdrawal process. If the patient decides to opt out, then the donated samples will be disposed of. If the patient withdraws consent without opting out for the subsequent use of the donated samples, then the samples will be used as per protocol.

Following withdrawal of consent for biological samples, further study participation should be considered in relation to the withdrawal processes outlined in the informed consent.

The Investigator:

- Ensures patient’s withdrawal of informed consent to the use of donated samples is highlighted immediately to AstraZeneca or delegate.
- Ensures that relevant human biological samples from that patient, if stored at the study site, are immediately identified, disposed of as appropriate, and the action documented.

- Ensures that the patient and AstraZeneca are informed about the sample disposal.

AstraZeneca ensures the organisation(s) holding the samples is/are informed about the withdrawn consent immediately and that samples are disposed of or repatriated as appropriate, and the action documented and study site notified.

C 3 International Airline Transportation Association (IATA) 6.2 Guidance Document

LABELLING AND SHIPMENT OF BIOHAZARD SAMPLES

IATA (https://www.iata.org/whatwedo/cargo/dgr/Pages/download.aspx) classifies infectious substances into 3 categories: Category A, Category B or Exempt

Category A Infectious Substances are infectious substances in a form that, when exposure to it occurs, is capable of causing permanent disability, life-threatening or fatal disease in otherwise healthy humans or animals.
Category A pathogens are, for example, Ebola, Lassa fever virus. Infectious substances meeting these criteria which cause disease in humans or both in humans and animals must be assigned to UN 2814. Infectious substances that cause disease only in animals must be assigned to UN 2900.

Category B Infectious Substances are infectious substances that do not meet the criteria for inclusion in Category A. Category B pathogens are, for example, Hepatitis A, C, D, and E viruses. They are assigned the following UN number and proper shipping name:

- UN 3373 – Biological Substance, Category B
- are to be packed in accordance with UN3373 and IATA 650

Exempt – Substances that do not contain infectious substances or substances which are unlikely to cause disease in humans or animals are not subject to these Regulations unless they meet the criteria for inclusion in another class.

- Clinical study samples will fall into Category B or exempt under IATA regulations.
- Clinical study samples will routinely be packed and transported at ambient temperature in IATA 650 compliant packaging. (https://www.iata.org/whatwedo/cargo/dgr/Documents/DGR-60-EN-PI650.pdf).
- Biological samples transported in dry-ice require additional dangerous goods specification for the dry-ice content.

Appendix D Optional Genomics Initiative Sample

D 1 Use/Analysis of DNA

- AstraZeneca intends to collect and store DNA for genetic research to explore how genetic variations may affect clinical parameters, risk and prognosis of diseases, and the response to medications. This genetic research may lead to better understanding of diseases, better diagnosis of diseases or other improvements in health care and to the discovery of new diagnostics, treatments or medications. Therefore, where local regulations and IRB/IEC allow, a blood sample will be collected for DNA analysis from consenting patients.
- This optional genetic research may consist of the analysis of the structure of the patient's DNA, ie, the entire genome.
- The results of genetic analyses may be reported in a separate study summary.
- The Sponsor will store the DNA samples in a secure storage space with adequate measures to protect confidentiality.
- The samples will be retained while research on T-DXd continues but no longer than 15 years from the end of the study (as defined in the CSP) or other period as per local requirements.

D 2 Genetic Research Plan and Procedures

Selection of genetic research population

- All patients will be asked to participate in this genetic research. Participation is voluntary and if a patient declines to participate there will be no penalty or loss of benefit. The patient will not be excluded from any aspect of the main study.

Inclusion criteria

For inclusion in this genetic research, patients must fulfil all of the inclusion criteria described in the main body of the CSP and provide informed consent for the Genomics Initiative sampling and analyses.

Exclusion criteria

- Exclusion from this genetic research may be for any of the exclusion criteria specified in the main study or any of the following:
  - Previous allogeneic bone marrow transplant.
  - Transfusion of non-leukocyte depleted blood or blood component within 120 days of genetic sample collection.

Withdrawal of consent for genetic research

Patients may withdraw from this genetic research at any time, independent of any decision concerning participation in other aspects of the main study. Voluntary withdrawal will not prejudice further treatment. Procedures for withdrawal are outlined in Section 7.2 of the core CSP.

Collection of samples for genetic research

The blood sample for this genetic research will be obtained from the patient's CCI [redacted] may withdraw due to an AE. If for any reason the sample is CCI [redacted] CCI [redacted]. Only one sample should be collected per patient for genetics during the study.

Coding and storage of DNA samples

- The processes adopted for the coding and storage of samples for genetic analysis are important to maintain patient confidentiality. Samples may be stored for a maximum of 15 years from the end of the study (as defined in the CSP), after which they will be destroyed. DNA is a finite resource that is used up during analyses. Samples will be stored and used until no further analyses are possible or the maximum storage time has been reached.
- An additional second code will be assigned to the sample either before or at the time of DNA extraction replacing the information on the sample tube. Thereafter, the sample will be identifiable only by the second, unique number. This number is used to identify the sample and corresponding data at the AstraZeneca genetics laboratories, or at the designated organisation. No personal details identifying the individual will be available to any person (AstraZeneca employee or designated organisations working with the DNA).
- The link between the patient enrolment/randomisation code and the second number will be maintained and stored in a secure environment, with restricted access at AstraZeneca or designated organisations. The link will be used to identify the relevant DNA samples for analysis, facilitate correlation of genotypic results with clinical data, allow regulatory audit, and permit tracing of samples for destruction in the case of withdrawal of consent.

Ethical and regulatory requirements

The principles for ethical and regulatory requirements for the study, including this genetics research component, are outlined in Appendix A.

Informed consent

The genetic component of this study is optional and the patient may participate in other components of the main study without participating in this genetic component. To participate in the genetic component of the study the patient must sign and date both the consent form for the main study and the addendum for the Genomics Initiative component of the study. Copies

of both signed and dated consent forms must be given to the patient and the original filed at the study centre. The Principal Investigator(s) is responsible for ensuring that consent is given freely and that the patient understands that they may freely withdrawal from the genetic aspect of the study at any time.

Patient data protection

- AstraZeneca will not provide individual genotype results to patients, any insurance company, any employer, their family members, general physician unless required to do so by law.
- Extra precautions are taken to preserve confidentiality and prevent genetic data being linked to the identity of the patient. In exceptional circumstances, however, certain individuals might see both the genetic data and the personal identifiers of a patient. For example, in the case of a medical emergency, an AstraZeneca physician or an Investigator might know a patient's identity and also have access to his/her genetic data. Regulatory authorities may require access to the relevant files, though the patient's medical information and the genetic files would remain physically separate.

Data management

- Any genetic data generated in this study will be stored at a secure system at AstraZeneca and/or designated organisations to analyse the samples.
- AstraZeneca and its designated organisations may share summary results (such as genetic differences from groups of individuals with a disease) from this genetic research with other researchers, such as hospitals, academic organisations or health insurance companies. This can be done by placing the results in scientific databases, where they can be combined with the results of similar studies to learn even more about health and disease. The researchers can only use this information for health-related research purposes. Researchers may see summary results but they will not be able to see individual patient data or any personal identifiers.
- Some or all of the clinical datasets from the main study may be merged with the genetic data in a suitable secure environment separate from the clinical database.

Statistical methods

The number of patients that will agree to participate in the genetic research is unknown. It is therefore not possible to establish whether sufficient data will be collected to allow a formal statistical evaluation or whether only descriptive statistics will be generated. A SAP may be prepared where appropriate.

Appendix E Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy's Law

E 1 Introduction

This Appendix describes the process to be followed in order to identify and appropriately report PHL cases and HL cases. It is not intended to be a comprehensive guide to the management of elevated liver biochemistries.

During the course of the study the Investigator will remain vigilant for increases in liver biochemistry. The Investigator is responsible for determining whether a patient meets potential PHL criteria at any point during the study.

All sources of laboratory data are appropriate for the determination of PHL and HL events; this includes samples taken at scheduled study visits and other visits including central and all local laboratory evaluations even if collected outside of the study visits; for example, PHL criteria could be met by an elevated ALT from a central laboratory **and/or** elevated TBL from a local laboratory.

The Investigator will also review AE data (eg, for AEs that may indicate elevations in liver biochemistry) for possible PHL events.

The Investigator participates, together with AstraZeneca clinical project representatives, in review and assessment of cases meeting PHL criteria to agree whether HL criteria are met. HL criteria are met if there is no alternative explanation for the elevations in liver biochemistry other than drug-induced liver injury caused by the study treatment.

The Investigator is responsible for recording data pertaining to PHL/HL cases and for reporting SAEs and AEs according to the outcome of the review and assessment in line with standard safety reporting processes.

E 2 Definitions

PHL

Aspartate aminotransferase or ALT ≥ 3×ULN **together with** TBL ≥ 2×ULN at any point during the study following the start of study treatment irrespective of an increase in ALP.

HL

AST or ALT ≥ 3×ULN **together with** TBL ≥ 2×ULN, where no other reason, other than the study treatment, can be found to explain the combination of increases, eg, elevated ALP indicating cholestasis, viral hepatitis, another drug.

For PHL and HL the elevation in transaminases must precede or be coincident with (ie, on the same day) the elevation in TBL, but there is no specified timeframe within which the elevations in transaminases and TBL must occur.

E 3 Identification of Potential Hy's Law Cases

In order to identify cases of PHL it is important to perform a comprehensive review of laboratory data for any patient who meets any of the following identification criteria in isolation or in combination:

- ALT ≥ 3×ULN.
- AST ≥ 3×ULN.
- TBL ≥ 2×ULN.

Local laboratories being used

The Investigator will, without delay, review each new laboratory report and if the identification criteria are met will:

- Notify the AstraZeneca representative
- Determine whether the patient meets PHL criteria (see Section E 2 Definitions within this Appendix for definition) by reviewing laboratory reports from all previous visits.
- Promptly enter the laboratory data into the laboratory eCRF.

E 4 Follow-up

E 4.1 Potential Hy's Law Criteria Not Met

If the patient does not meet PHL criteria the Investigator will:

- Inform the AstraZeneca representative that the patient has not met PHL criteria.
- Perform follow-up on subsequent laboratory results according to the guidance provided in the CSP.

E 4.2 Potential Hy's Law Criteria Met

If the patient does meet PHL criteria the Investigator will:

- Determine whether PHL criteria were met at any study visit prior to starting study treatment (See Section E 6).
- Notify the AstraZeneca representative who will then inform the central Study Team.

- Within 1 day of PHL criteria being met, the Investigator will report the case as an SAE of PHL; serious criterion "Important medical event" and causality assessment "yes/related" according to CSP process for SAE reporting.
- For patients that met PHL criteria prior to starting study treatment, the Investigator is not required to submit a PHL SAE unless there is a significant change# in the patient's condition.
- The Study Physician contacts the Investigator, to provide guidance, discuss and agree an approach for the study patients' follow-up (including any further laboratory testing) and the continuous review of data.
- Subsequent to this contact the Investigator will:
  - Monitor the patient until liver biochemistry parameters and appropriate clinical symptoms and signs return to normal or baseline levels, or as long as medically indicated. Completes follow-up SAE Form as required.
  - Investigate the aetiology of the event and perform diagnostic investigations as discussed with the Study Physician.
  - Complete the 3 Liver eCRF modules as information becomes available.

#A "significant" change in the patient's condition refers to a clinically relevant change in any of the individual liver biochemistry parameters (ALT, AST or TBL) in isolation or in combination, or a clinically relevant change in associated symptoms. The determination of whether there has been a significant change will be at the discretion of the Investigator, this may be in consultation with the Study Physician if there is any uncertainty.

E 5 Review and Assessment of Potential Hy's Law Cases

The instructions in this section should be followed for all cases where PHL criteria are met.

As soon as possible after the biochemistry abnormality was initially detected, the Study Physician contacts the Investigator in order to review available data and agree on whether there is an alternative explanation for meeting PHL criteria other than drug-induced liver injury caused by the study treatment, to ensure timely analysis and reporting to health authorities within 15 calendar days from date PHL criteria was met. The AstraZeneca Global Clinical Lead or equivalent and Global Safety Physician will also be involved in this review together with other subject matter experts as appropriate.

According to the outcome of the review and assessment, the Investigator will follow the instructions below.

Where there is an agreed alternative explanation for the ALT or AST and TBL elevations, a determination of whether the alternative explanation is an AE will be made and subsequently whether the AE meets the criteria for a SAE:

- If the alternative explanation is **not** an AE, record the alternative explanation on the appropriate eCRF.
- If the alternative explanation is an AE/SAE: update the previously submitted PHL SAE and AE eCRFs accordingly with the new information (reassessing event term; causality and seriousness criteria) following the AstraZeneca standard processes.

If it is agreed that there is **no** explanation that would explain the ALT or AST and TBL elevations other than the study treatment:

- Send updated SAE (report term "Hy's Law") according to AstraZeneca standard processes.
  - The "Medically Important" serious criterion should be used if no other serious criteria apply.
  - As there is no alternative explanation for the HL case, a causality assessment of "related" should be assigned.

If, there is an unavoidable delay, of over 15 calendar days in obtaining the information necessary to assess whether or not the case meets the criteria for HL, then it is assumed that there is no alternative explanation until such time as an informed decision can be made:

- Provides any further update to the previously submitted SAE of PHL, (report term now "Hy's Law case") ensuring causality assessment is related to study treatment and seriousness criterion is medically important, according to CSP process for SAE reporting.
- Continue follow-up and review according to agreed plan. Once the necessary supplementary information is obtained, repeat the review and assessment to determine whether HL criteria are still met. Update the previously submitted PHL SAE report following CSP process for SAE reporting, according to the outcome of the review and amending the reported term if an alternative explanation for the liver biochemistry elevations is determined.

E 6 Actions Required When Potential Hy's Law Criteria are Met Before and After Starting Study Intervention

This section is applicable to patients with liver metastases who meet PHL criteria on study treatment, having previously met PHL criteria at a study visit prior to starting study treatment.

At the first on-study treatment occurrence of PHL criteria being met the Investigator will determine if there has been a **significant change** in the patient's condition compared with the last visit where PHL criteria were met.

- If there is no significant change no action is required.
- If there is a significant change, notify the AstraZeneca representative, who will inform the central Study Team, then follow the subsequent process described in Section E 4.2.

E 7 Actions Required for Repeat Episodes of Potential Hy's Law

This section is applicable when a patient meets PHL criteria on study treatment and has already met PHL criteria at a previous on study treatment visit.

The requirement to conduct follow-up, review and assessment of a repeat occurrence(s) of PHL is based on the nature of the alternative cause identified for the previous occurrence.

The Investigator should determine the cause for the previous occurrence of PHL criteria being met and answer the following question:

Was the alternative cause for the previous occurrence of PHL criteria being met found to be the disease under study eg, chronic or progressing malignant disease, severe infection or liver disease or did the patient meet PHL criteria prior to starting study treatment and at their first on-study treatment visit as described in Section E 6 of this Appendix?

If **No**: follow the process described in Section E 4.2 for reporting PHL as an SAE.

If **Yes**: Determine if there has been a significant change in the patient's condition compared with when PHL criteria were previously met.

- If there is no significant change no action is required.
- If there is a significant change follow the process described in Section E 4.2 for reporting PHL as an SAE.

E 8 Laboratory Tests

The list below represents the standard, comprehensive list of follow-up tests which are recommended but not mandatory when using a central laboratory. When local laboratories are used, this list may be modified according to clinical judgement. Any test results must be recorded.

Hy's Law lab kit for central laboratories

Additional standard chemistry and coagulation tests	GGT LDH Prothrombin time INR
Viral hepatitis	IgM anti-HAV IgM and IgG anti-HBc
	HbsAg HCV DNA [a] IgM and IgG anti-HCV HCV DNA [a] IgM anti-HEV HEV RNA
Other viral infections	IgM & IgG anti-CMV IgM & IgG anti-HSV IgM & IgG anti-EBV
Alcoholic hepatitis	Carbohydrate-deficient transferrin [b]
Autoimmune hepatitis	Antinuclear antibody Anti-liver/kidney microsomal antibody Anti-smooth muscle antibody
Metabolic diseases	Alpha-1-antitrypsin Ceruloplasmin Iron Ferritin Transferrin [b] Transferrin saturation

[a] HCV RNA; HCV DNA are only tested when IgG anti-HCV is positive or inconclusive.
[b] Carbohydrate-deficient transferrin and transferrin are not available in China. Study teams should amend this list accordingly.

CMV = cytomegalovirus; DNA = deoxyribonucleic acid; EBV = Epstein-Barr virus; GGT = gamma glutamyl transferase; HAV = hepatitis A virus; HBc = hepatitis B core antigen; HbsAg = hepatitis B surface antigen; HBV = hepatitis B virus; HCV = hepatitis C virus; HEV = hepatitis E virus; HSV = herpes simplex virus; IgG = immunoglobulin G; IgM = immunoglobulin M; INR = international normalised ratio; RNA = ribonucleic acid.

E 9 References

Aithal GP, Watkins PB, Andrade RJ, Larrey D, Molokhia M, Takikawa H, et al. Case definition and phenotype standardisation in drug-induced liver injury. Clin Pharmacol Ther 2011;89(6):806-15.

Food and Drug Administration. Guidance for industry: Drug-induced liver injury: premarketing clinical evaluation. July 2009. Available from: URL: https://www.fda.gov/downloads/guidances/UCM174090.pdf. Accessed 08 October 2019.

Appendix F ASCO/CAP Guidelines on HER2 Testing in Breast Cancer

This protocol will determine HER2 status using guidelines from the American Society of Clinical Oncology (Table 19, per ASCO/CAP 2018 guidelines).

Table 19 Summary of Guideline Recommendations

Procedure	Recommendations
Optimal algorithm for HER2 testing	HER2 is positive if IHC 3+. HER2 IHC 2+ is equivocal (the revised definition of IHC 2+ [equivocal] is invasive breast cancer with "Weak to moderate complete membrane staining observed in > 10% of tumour cells) and triggers ISH testing. Diagnostic criteria by ISH is defined by HER2 copy number (signals/cell) and, for dual probe assay, by HER2/CEP17 ratio: For dual probe assay, HER2-positive is defined as: • (Group 1)OR(Group 2 AND concurrent IHC 3+)OR(Group 3 AND concurrent IHC 2+ or 3+)OR(Group 4 AND concurrent IHC 3+) Where Groups 1-4 are defined: • Group 1: ratio ≥ 2.0 and ≥ 4.0 signals/cell • Group 2: ratio ≥ 2.0 and < 4.0 signals/cell • Group 3: ratio < 2.0 and ≥ 6.0 signals/cell • Group 4: ratio < 2.0 and ≥ 4.0 to < 6.0 signals/cell • Group 5: ratio < 2.0 and < 4.0 signals/cell For tumours in groups 2-4 by dual probe assay, definitive diagnosis will be made based on additional work-up as detailed in ASCO/CAP guidelines 2018. For single probe assay, HER2-positive is defined as: • (HER2 copy number ≥ 6.0 signals/cell)OR(HER2 copy number ≥ 4.0 and < 6.0 signals/cell AND concurrent IHC 3+)OR(HER2 copy number ≥ 4.0 and < 6.0 signals/cell AND concurrent dual probe Group 1). • All other tumours will be defined as HER2- as defined by ASCO/CAP 2018. These definitions depend on laboratory documentation of the following: 1 Proof of initial testing validation in which positive and negative HER2 categories are 95% concordant with alternative validated method or same validated method for HER2 2 Ongoing internal QA procedures 3 Participation in external proficiency testing 4 Current accreditation by valid accrediting agency

Table 19 Summary of Guideline Recommendations

Procedure	Recommendations
ISH rejection criteria	Fixation for fewer than 6 hours or longer than 48 hours is not recommended Test is rejected and repeated if: • Controls are not as expected • Observer cannot find and count at least two areas of invasive tumour • > 25% of signals are unscorable due to weak signals • > 10% of signals occur over cytoplasm • Nuclear resolution is poor • Autofluorescence is strong Interpretation done by counting at least 20 cells; a pathologist must confirm that counting involved invasive tumour Sample is subjected to increased counting and/or repeated if equivocal; report must include guideline-detailed elements
IHC rejection criteria	Fixation for fewer than 6 hours or longer than 48 hours is not recommended Test is rejected and repeated or tested by FISH if: • Controls are not as expected • Artifacts involve most of sample • Sample has strong membrane staining of normal breast ducts (internal controls) • Interpretation follows guideline recommendation • Positive HER2 result requires homogeneous, dark circumferential (chicken wire) pattern in > 30% of invasive tumour • Interpreters have method to maintain consistency and competency Sample is subjected to confirmatory FISH testing if equivocal based on initial results Report must include guideline-detailed elements
Optimal tissue handling requirements	• Time from tissue acquisition to fixation should be as short as possible; samples for HER2 testing are fixed in 10% neutral buffered formalin for 6-72 hours. Samples should be sliced at 510 mm intervals after appropriate gross inspection and margins designation and placed in sufficient volume of neutral buffered formalin • Sections should ideally not be used for HER2 testing if cut > 6 weeks earlier; this may vary with primary fixation or storage conditions • Time to fixation and duration of fixation if available should be recorded for each sample • Any exception to this process must be included in report

Table 19 Summary of Guideline Recommendations

Procedure	Recommendations
Optimal internal validation procedure	• Validation of test must be done before test is offered • Initial test validation requires 25-100 samples tested by alternative validated method in the same laboratory or by validated method in another laboratory • Proof of initial testing validation in which positive and negative HER2 categories are 95% concordant with alternative validated method or same validated method for HER2 • Ongoing validation should be done biannually
Optimal internal QA procedures	• Initial test validation • Ongoing quality control and equipment maintenance • Initial and ongoing laboratory personnel training and competency assessment • Use of standardised operating procedures including routine use of control materials • Revalidation of procedure if changed • Ongoing competency assessment and education of pathologists
Optimal external proficiency assessment	• Participation in external proficiency testing program with at least two testing events (mailings)/year • Satisfactory performance requires at least 90% correct responses on graded challenges for either test • Unsatisfactory performance will require laboratory to respond according to accreditation agency program requirements
Optimal laboratory accreditation	• Onsite inspection every other year with annual requirement for self-inspection • Reviews laboratory validation, procedures, QA results and processes, results and reports • Unsatisfactory performance results in suspension of laboratory testing for HER2 for that method

CEP17 = chromosome 17 centromere; FISH = fluorescent *in situ* hybridisation; HER2 = human epidermal growth factor receptor 2; IHC = immunohistochemistry; QA = quality assurance

Excerpt from: Wolff et al 2018

The following tables provide additional information from the ASCO/CAP Guidelines for what needs to be reported for IHC and FISH (Table 20 and Table 21).

Table 20 Reporting Elements for IHC (Local HER2 Test)

Patient identification information
Physician identification
Date of service
Specimen identification (case and block number)
Specimen site and type
Specimen fixative type
Time to fixation (if available)
Duration of fixation (if available)

Table 20 Reporting Elements for IHC (Local HER2 Test)

Patient identification information
Antibody clone/vendor
Method used (test/vendor and if FDA approved)
Image analysis method (if used)
Controls (high protein expression, low-level protein expression, negative protein expression, internal)
Adequacy of sample for evaluation
Results Percentage of invasive tumour cells exhibiting complete membrane staining Uniformity of staining: present/absent Homogeneous, dark circumferential pattern: present/absent
Interpretation Positive (for HER2 protein expression); equivocal (FISH will be done and reported); negative (for HER2 protein expression); not interpretable
Comment If an FDA-approved method is used, it should be stated; if the FDA-approved method has been modified, a statement in the report should be included indicating what modifications were made and that the changes have been validated; if the test is not FDA approved or an FDA-approved test has been modified, a clear statement must be made that the laboratory reporting results takes responsibility for test performance

IHC = immunohistochemistry; FDA = US Food and Drug Administration; FISH = fluorescent *in situ* hybridisation; HER2 = human epidermal growth factor receptor 2.

Table 21 Reporting Elements for FISH (Local HER2 Test)

Patient identification information
Physician identification
Date of service
Specimen identification (case and block number)
Specimen site and type
Specimen fixative type
Time to fixation (if available)
Duration of fixation (if available)
Probe(s) identification
Method used (specifics of test/vendor and if FDA approved)
Image analysis method
Controls (amplified, equivocal, and nonamplified, internal)
Adequacy of sample for evaluation (adequate number of invasive tumour cells present)

Table 21 Reporting Elements for FISH (Local HER2 Test)

Patient identification information
Results Number of invasive tumour cells counted Number of observers Average number of HER2 signals/nucleus or tile Average number of CEP17 chromosome probes/nucleus or tile Ratio of average HER2 signals/CEP17 probe signals Note: Tile is unit used for image system counting
Interpretation Positive (amplified); equivocal; negative (not amplified); not interpretable; if IHC is being done because of problems with assay or results, this should also be indicated
Comment If an FDA-approved method is used, it should be noted; if the FDA-approved method has been modified, a statement in the report should be included indicating what modifications were made and that the changes have been validated; if the test is not FDA-approved or an FDA-approved test has been modified, a clear statement must be made that the laboratory reporting results takes responsibility for test performance

CEP17 = chromosome 17 centromere; FISH = fluorescent *in situ* hybridisation; FDA = US Food and Drug Administration; HER2 = human epidermal growth factor receptor 2; IHC = immunohistochemistry.

Excerpt from: Wolff et al 2018

Appendix G Guidelines for Evaluation of Objective Tumour Response Using RECIST 1.1 Criteria (Response Evaluation Criteria in Solid Tumours)

Introduction

This Appendix details the implementation of RECIST 1.1 guidelines (Eisenhauer et al 2009). Investigator assessments will use the RECIST 1.1 guidelines described in this Appendix.

Imaging modalities and acquisition specifications for RECIST 1.1

A summary of the imaging modalities that can be used for tumour assessment of TLs, NTLs and NLs is provided in Table 22.

Table 22 Summary of Imaging Modalities for Tumour Assessment

Target Lesions	Non-Target Lesions	New Lesions
CT MRI	CT MRI Plain X-ray Chest X-ray	CT MRI Plain X-ray Chest X-ray Bone scan (Scintigraphy) ^{18}F-fluoro-deoxyglucose-PET/CT

CT = computed tomography; PET/CT = positron emission tomography/CT; MRI = magnetic resonance imaging.

Computed Tomography and Magnetic Resonance Imaging

CT with IV contrast is the preferred imaging modality (although MRI with IV contrast is acceptable if CT is contraindicated) to generate reproducible anatomical images for tumour assessments (ie, for measurement of TLs, assessment of NTLs, and identification of NLs). It is essential that the same correct imaging modality, image acquisition parameters (eg, anatomic coverage, imaging sequences, etc), imaging facility, tumour assessor (eg, radiologist), and method of tumour assessment (eg, RECIST 1.1) are used consistently for each patient throughout the study. The use of the same scanner for serial scans is recommended, if possible. It is important to follow the image collection/tumour assessment schedule as closely as possible (refer to the SoA in the respective modules), and this on-study imaging schedule MUST be followed regardless of any delays in dosing or missed imaging visits. If an unscheduled assessment is performed (eg, to investigate clinical signs/symptoms of progression) and the disease has not progressed, every attempt should be made to perform the subsequent scan acquisitions at the next scheduled imaging visit.

Due to its inherent rapid acquisition (seconds), CT is the imaging modality of choice. Body scans should be performed with breath-hold scanning techniques, if possible. Therefore, CT of the chest is recommended over MRI due to significant motion artefacts (eg, heart, major blood

vessels, breathing) associated with MRI. MRI has excellent contrast and spatial and temporal resolutions; however, there are many image acquisition variables involved in MRI, which greatly impact image quality, lesion conspicuity, and measurement. Furthermore, the availability of MRI is variable globally. The modality used at follow-up should be the same as was used at baseline, and the lesions should be measured/assessed on the same pulse sequence. In general, local oncology diagnostic imaging parameters are applied for scan acquisition. It is beyond the scope of this Appendix to prescribe specific MRI pulse sequence parameters for all scanners, body parts, and diseases.

The most critical CT and MRI image acquisition parameters for optimal tumour evaluation are anatomic coverage, contrast administration, slice thickness, and reconstruction interval.

a. Anatomic coverage: Optimal anatomic coverage for most solid tumours is the chest-abdomen (-pelvis). Coverage should encompass all areas of known predilection for metastases in the disease under evaluation and should additionally investigate areas that may be involved based on signs and symptoms of individual patients. Because a lesion later identified in a body part not scanned at baseline would be considered as a NL representing PD, careful consideration should be given to the extent of imaging coverage at baseline and at subsequent follow-up timepoints. This will enable better consistency not only of tumour measurements but also identification of new disease.

Required anatomical regions to be imaged for assessment of tumour burden (TLs and/or NTLs) at baseline and follow-up visits vary according to the study, and these timepoints are specified in the SoA in the respective modules. Examples include the following:

- IV contrast-enhanced CT of chest-abdomen (including the entire liver and both adrenal glands) (-pelvis).
- Non-contrast CT of chest and IV contrast-enhanced abdomen (including the entire liver and both adrenal glands) (-pelvis).
- IV contrast-enhanced MRI of the head and neck.
- IV contrast-enhanced MRI of the brain.

For chest-abdomen (-pelvis) imaging, the following are scanning options in decreasing order of preference, with additional options (2 to 4) for consideration when patients have sensitivity to IV contrast or have compromised renal function:

1. Chest-abdomen (-pelvis) CT with IV CT contrast (most preferred).
2. Chest CT without IV-contrast + abdomen (-pelvis) MRI with IV MRI contrast, if CT IV contrast (iodine based) is medically contraindicated at any time during the study.

3. Chest-abdomen (-pelvis) CT without IV contrast, if both IV CT and MRI contrast are medically contraindicated or the patient has compromised renal function.
4. Chest-abdomen (-pelvis) MRI with IV MRI contrast, if CT cannot be performed at any time during the study.

b. IV contrast administration: Optimal visualisation and measurement of metastases in solid tumours require consistent administration (dose and rate) of IV contrast as well as timing of scanning. An adequate volume of a suitable contrast agent should be given so that the tumour lesions are demonstrated to best effect and a consistent method is used on subsequent examinations for any given patient. Oral contrast is recommended to help visualise and differentiate structures in the abdomen and pelvis.

c. Slice thickness and reconstruction interval: It is recommended that CT or MRI scans be acquired/reconstructed as contiguous (no gap) slices with ≤ 5 mm thickness throughout the entire anatomic region of interest for optimal lesion measurements. Exceptionally, particular institutions may perform medically acceptable scans at slice thicknesses > 5 mm. If this occurs, the minimum size of measurable lesions at baseline should be twice the slice thickness of the baseline scans.

For CT scans, all window settings should be included in the assessment, particularly in the thorax where lung and soft tissue windows should be considered. When measuring lesions, the TL should be measured on the same window setting for repeated examinations throughout the study.

Brain Scan

All patients in this study will receive an IV contrast-enhanced MRI of the brain at screening/baseline. Regularly scheduled follow-up brain scans are mandatory for all patients who were enrolled with baseline stable or active brain metastases. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.

Chest X-ray

Chest X-ray assessment will not be used for the assessment of TLs. Chest X-ray can, however, be used to assess NTLs and to identify the presence of NLs. However, there is preference that a higher resolution modality, such as CT, be used to confirm the presence of NLs.

Plain X-ray[2]

Plain X-ray may be used as a method of assessment for bone NTLs and to identify the presence of new bone lesions.

Isotopic bone scan

Bone lesions identified on an isotopic bone scan at baseline and confirmed by CT, MRI, or X-ray at baseline should be recorded as NTLs and followed by the same method per baseline assessment (CT, MRI, or X-ray).

Isotopic bone scans may be used as a method of assessment to identify the presence of new bone lesions at follow-up visits. NLs may be recorded in case positive hot-spots appear on a bone scan that were not present on a previous bone scan; however, a newly observed equivocal hot-spot on a bone scan that cannot be verified with correlative imaging (CT, MRI, or X-ray) of the same anatomical region shall not be the only trigger for a PD assessment at that time point.

^{18}F-Fluoro-deoxyglucose-PET/CT

^{18}F-fluoro-deoxyglucose positron emission tomography (PET)/CT scans may be used as a method for identifying new extrahepatic lesions (but not intrahepatic lesions) for RECIST 1.1 assessments according to the following algorithm: NLs will be recorded where there is positive ^{18}F-Fluoro-deoxyglucose uptake[3] not present on baseline or prior ^{18}F-fluoro-deoxyglucose-PET scan or in a location corresponding to a NL on a companion CT/MRI collected close in time to the ^{18}F-fluoro-deoxyglucose-PET scan. The PET portion of the PET/CT introduces additional data that may bias an Investigator if it is not routinely or serially performed. Therefore, if there is no baseline or prior ^{18}F-fluoro-deoxyglucose-PET scan available for comparison, and no evidence of NLs on companion CT/MRI scans, then follow-up CT/MRI assessments should continue as per the regular imaging schedule to verify the unequivocal presence of NLs.

At present, low-dose or attenuation correction CT portions of a combined ^{18}F-fluoro-deoxyglucose-PET/CT scan are of limited use in anatomically based efficacy assessments, and it is therefore suggested that they should not substitute for dedicated diagnostic contrast-enhanced CT scans for tumour measurements by RECIST 1.1. In

[2] In this study, a patient with non-measurable, bone-only disease assessed by X-ray at screening must have lytic or mixed lytic bone lesions; ie, bone-only patients with exclusively sclerotic (blastic) bone lesions in the absence of measurable disease are not eligible.

[3] A positive ^{18}F-fluoro-deoxyglucose-PET scan lesion should be reported only when an uptake (eg, standard uptake value) greater than twice that of the surrounding tissue or liver is observed.

exceptional situations, if a site can document that the CT performed, as part of a PET/CT examination, is of identical diagnostic quality (with IV contrast) to a dedicated diagnostic CT scan, then the CT portion of the PET/CT can be used for RECIST 1.1 tumour assessments. Caution that this is not recommended because the PET portion of the CT introduces additional (PET) data that may bias an Investigator if it is not routinely or serially performed.

Ultrasound

Ultrasound examination will not be used for RECIST 1.1 assessment of tumours as it is not a reproducible acquisition method (operator dependent), is subjective in interpretation, and may not provide an accurate assessment of the true tumour size. Tumours identified by ultrasound will need to be assessed by correlative CT or MRI anatomical scan.

Other tumour assessments

Clinical examination

Clinical examination of skin/surface lesions (by visual inspection or manual palpation) will not be used for RECIST 1.1 assessments. Tumours identified by clinical examination will need to be assessed by correlative CT or MRI anatomical scans.

Endoscopy and laparoscopy

Endoscopy and laparoscopy will not be used for tumour assessments as they are not validated in the context of tumour assessment.

Histology and cytology

Histology or tumour markers on tumour biopsy samples will not be used as part of the tumour response assessment as per RECIST 1.1.

Results of cytological examination for the neoplastic origin of any effusion (eg, ascites, pericardial effusion, and pleural effusion) that appears or worsens during the study will not be used as part of the tumour response assessment as per RECIST 1.1.

Furthermore, an overall assessment of CR (all other disease disappears/reverts to normal) would be changed to PR if an effusion remains present radiologically.

Measurability of tumour lesions at baseline

RECIST 1.1 measurable lesions at baseline

- A tumour lesion that can be accurately measured at baseline as ≥ 10 mm in the longest diameter for non-nodal lesions or ≥ 15 mm in short-axis[4] diameter for lymph node lesions with IV contrast-enhanced CT or MRI and that is suitable for accurate repeated measurements. Please see additional RECIST 1.1 guidance below on measurability of intrahepatic hepatocellular carcinoma lesions and porta hepatis lymph nodes.

- Brain metastasis. For this study, patients in Module 6 and Module 7 may include active brain lesions as RECIST 1.1 TLs at baseline. All patients in this study will receive an IV contrast-enhanced MRI of the brain at screening/baseline. Regularly scheduled follow-up brain scans are mandatory for all patients who were enrolled with baseline active brain metastases.

Non-measurable lesions at baseline

- Truly non-measurable lesions include the following:
  - Bone lesions (see exception below for soft tissue component).
  - Leptomeningeal disease.
  - Ascites, pleural effusion, or pericardial effusion.
  - Inflammatory breast disease.
  - Lymphangitic involvement of skin or lung.
- All other lesions, including small lesions (longest diameter < 10 mm or pathological lymph nodes with ≥ 10 mm to < 15 mm short-axis diameter at baseline).[5]
- Previously irradiated lesions.[6]
- Brain metastasis. [All patients in this study will receive an IV contrast-enhanced MRI of the brain at screening/baseline]. Regularly scheduled follow-up brain scans are mandatory for all patients in Modules 0-5 who were enrolled with baseline stable brain metastases.

Special considerations regarding lesion measurability at baseline

- Bone lesions:

[4] The short axis is defined as the longest in-plane axis perpendicular to the long axis.

[5] Lymph nodes with < 10 mm short-axis diameter are considered non-pathological and should not be recorded or followed as NTLs.

[6] Localised post-radiation changes that affect lesion size may occur. Therefore, lesions that have been previously irradiated are typically considered non-measurable and as NTL at baseline and followed up as part of the NTL assessment.

  - Bone scan, PET scan, or plain X-ray are not considered adequate imaging techniques to measure bone lesions; however, these techniques can be used to confirm the presence or disappearance of bone lesions.
  - Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, can be considered measurable if the soft tissue component meets the definition of measurability.
  - Blastic lesions are considered non-measurable.[7]
- Cystic lesions thought to represent cystic metastases can be considered measurable lesions if they meet the criteria for measurability from a radiological point of view, but if non-cystic lesions are present in the same patient, these should be selected over cystic lesions as TLs.

RECIST 1.1 TL selection at baseline

A maximum of 5 measurable lesions, with a maximum of 2 lesions per organ (including lymph nodes collectively considered as a single organ), representative of all lesions involved should be identified as TLs at baseline. TLs should be selected on the basis of their size (longest diameter for non-nodal lesions or short-axis diameter for nodal lesions), but in addition should be those that lend themselves to reproducible repeated measurements. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement, in which circumstance the next largest lesion that can be measured reproducibly should be selected.

For patients enrolled in this study with active brain metastases, at least 1 active brain lesion should be considered for inclusion as a TL at baseline.

Lymph nodes, in any location (local/regional and distant), are collectively considered as a single organ, with a maximum of 2 lymph nodes as TLs. A bilateral organ (eg, adrenal glands), a segmented organ (eg, liver), or a multilobed organ (eg, lung) is each considered as a single organ.

The site and location of each TL should be documented, as well as the longest axis diameter for non-nodal lesions (or short-axis diameter for lymph nodes). All measurements should be recorded in millimetres. At baseline, the sum of the diameters for all TLs will be calculated

[7] In this study, a patient with non-measurable, bone-only disease assessed by X-ray at screening must have lytic or mixed lytic bone lesions; ie, bone-only patients with exclusively sclerotic (blastic) bone lesions in the absence of measurable disease are not eligible.

and reported as the baseline sum of diameters. At follow-up visits, the sum of diameters for all TLs will be calculated and reported as the follow-up sum of diameters.

Special cases for TL assessment at baseline

- For TLs measurable in 2 or 3 dimensions, always report the longest diameter. For pathological lymph nodes measurable in 2 or 3 dimensions, always report the short-axis diameter.
- When lymph nodes are coalesced and no longer separable in a conglomerate mass, the vector of the longest diameter should be used to determine the perpendicular vector for the maximal short-axis diameter of the coalesced mass. Non-nodal lesions that coalesce should similarly be assessed by the longest axis diameter.
- Tumour lesions selected for newly acquired screening biopsy should not be selected as TLs, unless imaging occurred at least approximately 2 weeks after biopsy, allowing time for healing.
- If the CT/MRI slice thickness used is > 5 mm, the minimum size of measurable disease at baseline should be twice the slice thickness of the baseline scan.
- If a lesion has completely disappeared, the diameter should be recorded as 0 mm. If a lesion appears in the same location on a subsequent scan, it will be recorded as a NL.

RECIST 1.1 NTL selection at baseline

All other lesions, including non-measurable lesions and surplus measurable lesions, not recorded as TLs should be identified as NTLs at baseline. Measurements of these lesions are not required, but the presence or absence of each should be noted throughout follow-up.

Evaluation of tumour response and progression

RECIST 1.1 TL assessment at follow-up

This section defines the criteria used to determine objective tumour visit response for RECIST 1.1-defined TLs (see also Table 23). The imaging modality, location, and scan date of each TL identified previously at baseline should be documented at follow-up visits with the long-axis diameter for non-nodal lesions or short-axis diameter for lymph node lesions. All measurements should be rounded to and recorded in whole millimetres. The sum of the diameters for all TLs at each follow-up visit will be compared with the baseline sum of diameters (for response or SD) or to the smallest prior (nadir) sum of diameters (for progression).

Special cases for TL assessment at follow-up:

- If a lesion has completely disappeared, the diameter should be recorded as 0 mm. If a lesion appears in the same location on a subsequent scan, it will be recorded as an NL.

- If a TL splits into 2 or more parts, the sum of the diameters of those parts should be recorded.
- If 2 or more TLs merge, then the sum of the diameters of the combined lesion should be recorded for 1 of the lesions and 0 mm recorded for the other lesion(s). If the merged TLs are non-nodal lesions, record the long-axis diameter of the merged lesion. If pathologic lymph nodes coalesce and are no longer individually separable within a conglomerate mass, the vector of the longest diameter of the coalesced mass should be used to determine the perpendicular vector for the maximal short-axis diameter.
- If a TL is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. If an accurate measure can be given, this should be recorded, even if it is below 5 mm.
- If a TL cannot be measured accurately due to it being too large, provide an estimate of the size of the lesion. The choice of "Too large to measure" in the CRF will trigger an overall visit response of PD.
- When a TL has had an unscheduled, non-protocol intervention, the following apply:
  - Target Lesion Intervention may include radiotherapy, embolization, excisional biopsy, surgery, etc. that is not a part of study treatment and might adversely affect the size of that Target lesion.
  - If an Intervention on a Target Lesion is ticked in the case report form, the diameter of the lesion is still recorded (0 mm if no longer present) and is included in the sum of diameters.
  - If a Target Lesion Intervention is ticked, the Intervention must be reported for all subsequent assessments of that Target lesion.
  - If a Target Lesion has an Intervention, the only Overall Visit Responses allowed to be recorded by the Investigator are NE or PD, with PD if the sum of diameters exceeds a 20% increase and at least a 5mm absolute increase in the visit sum of diameters compared to the previous minimum (nadir) sum of diameters.
  - No visit with a recorded Target Lesion Intervention can be used as the minimum (nadir) sum of diameters.

Table 23 RECIST 1.1 Evaluation of Target Lesions

CR	Disappearance of all TLs since baseline. Any pathological lymph nodes selected as TLs must have a reduction in short-axis diameter to < 10 mm.
PR	At least a 30% decrease in the sum of the diameters of TL, taking as reference the baseline sum of diameters.
SD	Neither sufficient decrease in the sum of diameters to qualify for PR nor sufficient increase to qualify for PD.

PD	At least a 20% increase in the sum of diameters of TLs, taking as reference the smallest previous sum of diameters (nadir). This includes the baseline sum if that is the smallest on study. In addition to the relative increase of 20%, the sum must demonstrate an absolute increase of at least 5 mm from nadir.
NE	Only relevant if any of the TLs at follow-up were not assessed or NE (eg, missing anatomy) or had a lesion intervention at this visit. Note: If the sum of diameters meets the PD criteria, PD overrides NE as a TL response.
Not applicable	Only relevant if no TLs present at baseline.

CR = complete response; NE = not evaluable; PD = progression of disease; PR = partial response; SD = stable disease; TL = target lesion.

RECIST 1.1 NTL assessment at follow-up

All other lesions (or sites of disease) not recorded as TLs should be identified as NTLs at baseline. Measurements are not required for these lesions, but their status should be followed at subsequent visits. At each visit, an overall assessment of the NTL response should be recorded by the Investigator.

To achieve "unequivocal progression" on the basis of NTLs, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in TLs, the overall tumour burden has increased sufficiently to merit unequivocal progression by NTLs. A modest "increase" in the size of 1 or more NTLs is usually not sufficient to qualify for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PD of target disease will therefore be extremely rare (Table 24).

Table 24 RECIST 1.1 Evaluation of Non-Target Lesions

CR	Disappearance of all NTLs since baseline. All lymph nodes must be non-pathological in size (< 10 mm short axis).
Non CR/non PD	Persistence of 1 or more NTLs.
PD	Unequivocal progression of existing NTLs. Unequivocal progression may be due to an important progression in 1 lesion only or in several lesions. In all cases, the progression MUST be clinically significant for the physician to consider changing (or stopping) therapy.
NE	Only relevant when 1 or some of the NTLs were not assessed and, in the Investigator's opinion, they are not able to provide an evaluable overall NTL assessment at this visit. Note: For patients without TLs at baseline, this is relevant if any of the NTLs were not assessed at this visit and the progression criteria have not been met.
Not applicable	Only relevant if no NTLs present at baseline.

CR = complete response; NE = not evaluable; NTL = non-target lesion; PD = progression of disease; TL = target lesion.

RECIST 1.1 NL identification at follow-up

Details, including the imaging modality, the date of scan, and the location of any NLs will also be recorded in the CRF. The presence of 1 or more NLs is assessed as progression. The finding of a NL should be unequivocal, ie, not attributable to differences in scanning technique, change in imaging modality, or findings thought to represent something other than tumour. If a NL is equivocal, for example because of its small size, the treatment and tumour assessments should be continued until the previously (pre-existing) NL has been assessed as unequivocal at a follow-up visit, and then the progression date should be declared using the date of the initial scan when the NL first appeared.

A lesion identified at a follow-up assessment in an anatomical location that was not scanned at baseline is considered a NL and will indicate PD.

RECIST 1.1 evaluation of overall visit response at follow-up

Derivation of overall visit response as a result of the combined assessment of TLs, NTLs, and NLs uses the algorithm shown in Table 25.

Table 25 RECIST 1.1 Overall Visit Response

Target Lesions	Non-Target Lesions	New Lesions	Overall Visit Response
CR	CR	No	CR
CR	NA	No	CR
NA	CR	No	CR
CR	Non CR/Non PD	No	PR
CR	NE	No	PR
PR	Non PD or NE or NA	No	PR
SD	Non PD or NE or NA	No	SD
NA	Non-CR/Non-PD	No	SD (non-CR/non-PD)
NE	Non PD or NE	No	NE
NA	NE	No	NE
NA	NA	No	NED
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

Non-CR/Non-PD for overall response if only NTL (no TLs) are present at baseline.
Note: An overall assessment of CR (all other disease disappears/reverts to normal) would be changed to PR if ascites remains present radiologically.
CR = complete response; NA = not applicable (only relevant if there were no TLs at baseline or NTLs at baseline), NE = not evaluable; NED = no evidence of diseases (only relevant if there were neither target lesions nor non-target lesions at baseline); NTL = non-target lesion; PD = progression of disease; PR = partial response; SD = stable disease; TL = target lesion.

The following overall visit responses are possible depending on the extent of tumour disease at baseline:

- For patients with TLs (at baseline): CR, PR, SD, PD, or NE.
- For patients with NTLs only (at baseline): CR, Non-CR/Non-PD, PD, or NE.
- For patients with no disease at baseline: no evidence of disease (available as an option in the eCRF), PD, or NE.

References

Eisenhauer EA, Therasse P, Bogaerts J, Schwartz LH, Sargent D, Ford R, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). Eur J Cancer 2009;45(2):228-47.

Appendix H Contraception Requirements

Contraception requirements for this study are as follows.

H 1 Female Patient of Child-bearing Potential

Women not of childbearing potential are defined as those who are surgically sterile (ie, bilateral salpingectomy, bilateral oophorectomy, or complete hysterectomy) or who are post-menopausal.

Women will be considered post-menopausal if they have been amenorrhoeic for 12 months without an alternative medical cause. The following age-specific requirements apply:

- Women < 50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of all hormonal replacement therapy, or had radiation-induced menopause with last menses > 1 year ago, or had chemotherapy-induced menopause with last menses > 1 year ago.
- Women ≥ 50 years of age would be considered post-menopausal if they have been amenorrhoeic for 12 months or more following cessation of all hormonal replacement therapy and if they have luteinizing hormone and follicle-stimulating hormone levels in the post-menopausal range for the institution.

Women of childbearing potential who are not totally sexually abstinent (ie, refraining from heterosexual intercourse during the entire period of risk associated with study treatments) and intend to be sexually active with a non-sterilised male partner must use at least 1 highly effective method of contraception (Table 26). They should continue to use it throughout the total duration of the drug treatment and the drug washout period (7 months after the last dose of study treatment). Cessation of birth control after this point should be discussed with a responsible physician.

Non-sterilised male partners of a woman of childbearing potential must use a male condom plus spermicide (condom alone in countries where spermicides are not approved) throughout this period. Cessation of birth control after this point should be discussed with a responsible physician. Periodic abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. Total sexual abstinence is an acceptable method, provided it is the preferred and usual lifestyle choice of the participant. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical trial. Female participants should refrain from breastfeeding throughout this period. Women must not donate, or retrieve for their own use, ova from the time of screening and throughout the study treatment period, and for at least 7 months after the final study drug administration.

H 2 Male Patients with a Female Partner of Child-bearing Potential

Non-sterilised male patients (including males sterilised by a method other than bilateral orchidectomy, eg, vasectomy) who intend to be sexually active with a female partner of child-bearing potential must be using an acceptable method of contraception such as male condom plus spermicide (condom alone in countries where spermicides are not approved) from the time of screening throughout the total duration of the study and the drug washout period to prevent pregnancy in a partner.

For Modules 0, 1, 5, 6 and 7, the drug washout period is 4 months; for Modules 2, 3 and 4, the drug washout period is 6 months.

Periodic abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. Total sexual abstinence is an acceptable method, provided it is the preferred and usual lifestyle choice of the participant. Male patients should refrain from sperm donation or banking throughout this period.

Vasectomised (ie, sterile) males are considered fertile and should still use a male condom plus spermicide as indicated above during the clinical study.

Even if the female partner is pregnant, male patients should still use a condom plus spermicide (where approved), as indicated above during the clinical study, if there is a concern about damaging the developing foetus from drug in ejaculate.

Female partners (of child-bearing potential) of male patients must also use a highly effective method of contraception throughout this period (Table 26).

H 3 Highly Effective Methods of Contraception

Highly effective methods of contraception, defined as one that results in a low failure rate (ie, less than 1% per year) when used consistently and correctly, are described in Table 26. Note that some contraception methods are not considered highly effective (eg, male or female condom with or without spermicide; female cap, diaphragm, or sponge with or without spermicide; non copper containing intrauterine device; progestogen-only oral hormonal contraceptive pills where inhibition of ovulation is not the primary mode of action [excluding Cerazette/desogestrel which is considered highly effective]; and triphasic combined oral contraceptive pills). Female patients of childbearing potential must use a non-hormonal method of contraception; hormonal methods of contraception are only allowed for female partners (of childbearing potential) of male patients.

Table 26 Highly Effective Methods of Contraception (< 1% Failure Rate)

Non-Hormonal Methods (female participants and female partners [of childbearing potential] of male participants)	Hormonal Methods (female partners [of childbearing potential] of male participants ONLY)
• Total sexual abstinence (evaluate in relation to the duration of the clinical study and the preferred and usual lifestyle choice of the patient) • Vasectomised sexual partner (with patient assurance that partner received post-vasectomy confirmation of azoospermia) • Tubal occlusion • Intrauterine device (provided coils are copper-banded)	• Injection: Medroxyprogesterone injection (eg, Depo-Provera®) [a] • Levonorgestrel-releasing intrauterine system (eg, Mirena®) [a] • Progesterone T intrauterine device • Implants: Etonogestrel-releasing implants (eg, Implanon® or Norplant®) • Intravaginal devices: Ethinylestradiol/etonogestrel-releasing intravaginal devices (eg, NuvaRing®) • Combined pill: Normal and low dose combined oral contraceptive pill • Patch: Norelgestromin/ethinylestradiol-releasing transdermal system (eg, Ortho Evra®) • Mini pill: Progesterone based oral contraceptive pill using desogestrel: Cerazette® is currently the only highly effective progesterone-based pill

Note: for hormonal methods to be considered “highly effective”, they must inhibit ovulation.

Appendix I Patient-reported Outcomes

I 1 MDASI

Patient Reported Outcome Questionnaire: MDASI was removed due to copyrights.

I 2 SGRQ-I

Patient Reported Outcome Questionnaire: SGRQ-I was removed due to copyrights.

Patient Reported Outcome Questionnaire: SGRQ-I was removed due to copyrights.

Patient Reported Outcome Questionnaire: SGRQ-I was removed due to copyrights.

Patient Reported Outcome Questionnaire: SGRQ-I was removed due to copyrights.

I 3 CANTAB Test Battery

REACTION TIME (RTI): PSYCHOMOTOR SPEED DURATION: ~3 MIN

RTI provides assays of motor and mental response speeds, as well as measures of movement time, reaction time and response accuracy. In this 5-choice reaction time test the participant will see 5 circles at the top of the screen and a button at the bottom of the screen. For each trial, the participant must hold down the button at the bottom of the screen until one of the five circles above flashes yellow. Once this has occurred, the participant must then release their finger and touch the circle which just flashed, as quickly as possible. Voiceover instructions introduce the task and explain how the task is completed. The subject is informed that 'Both speed and accuracy are important, so go as quickly as you can, whilst trying not to make any mistakes and that 'You will need to use the index finger of your dominant hand only: do not use two hands'. The subject will then be asked to practice holding the down the button at the bottom of the screen and is again informed that they should hold this button down until a yellow spot flashes in one of the circles above. The subject will then be presented with 10 practice trials, 2 from each of the 5 locations at the top of the screen, in a randomised order. After the participant has completed the 10 training problems, the voiceovers instructions introduce the assessed phase of the task during which the participant will be presented with 30 trials, 6 from each of the 5 locations at the top of the screen, in a randomised order to complete.



PAIRED ASSOCIATES LEARNING (PAL): VISUAL EPISODIC MEMORY DURATION: ~ 8 MIN

PAL assesses visual memory and new learning and is a sensitive tool for accurate assessment of episodic memory. Boxes are displayed on the screen and open one by one in a randomised order to reveal patterns hidden inside. The patterns are then displayed in the middle of the screen, one at a time, and the participant must touch the box where the pattern was originally located. As the test progresses so the stages become more difficult as the number of patterns to be remembered increases. Voiceover instructions introduce the task and explain how the task is completed. It is explained to the participant that 'Some of these boxes will open up one by one to reveal some patterns. Your task is to remember where each pattern is located'. The participant will then be presented with a training problem in which 6 boxes appear on screen,



but only 2 patterns are hidden behind the boxes. After the participant has completed the training problem, the voiceover instructions introduce the assessed phase of the task during which the participant will be presented with 4 problems of varying levels of difficulty to complete. During the first problem 6 boxes will be shown on screen but they will only contain 2 patterns, the next trial will contain 4 patterns, the next will contain 6 patterns and during the final trial the number of boxes and patterns will increase to 8. If the participant fails to complete any of the assessed problems, the boxes will open up again to reveal the patterns locations. If the participant fails to complete any assessed trial after 4 attempts, the task will terminate.

SPATIAL WORKING MEMORY (SWM): WORKING MEMORY DURATION: ~ 4 MIN

SWM requires retention and manipulation of visuospatial information. This test has notable executive function demands, and measures strategy use as well as errors. In this task the participant must search for tokens hidden in boxes. The participant must touch a box to open the box to reveal either a yellow token or an empty box. Once the participant has found a yellow token, they must touch the outline of the right-hand side of the screen to 'store' it. The participant must then continue searching through the boxes until all the tokens have been found. Voiceover instructions introduce the task and explain how the task is completed. The episodic memory paired associates learning (PAL) working memory spatial working memory participant is informed that 'The tokens will never be in the same box twice, so do not go back to a box if you have already found a token there'. The participant will then be asked to complete 2 practice trials, each containing 3 boxes/ tokens. If the participant returns to a box where a token has already been found, they are reminded that the tokens will never be in the same box twice, so do not go back to a box if you have already found a token there.



I 4 MDASI-BT

Patient Reported Outcome Questionnaire: MDASI-BT was removed due to copyrights.

Patient Reported Outcome Questionnaire: MDASI-BT was removed due to copyrights.

I 5 NANO

Patient Reported Outcome Questionnaire: NANO was removed due to copyrights.

Patient Reported Outcome Questionnaire: NANO was removed due to copyrights.

Appendix J Concomitant Medications

J 1 Guidance Regarding Potential Interactions with Concomitant Medications

The use of any natural/herbal products or other "folk remedies" should be discouraged, but use of these products, as well as use of all vitamins, nutritional supplements, and all other concomitant medications must be recorded in the eCRF.

Paclitaxel

In a Phase 1 trial using escalating doses of paclitaxel (110– 200 mg/m^2) and cisplatin (50 or 75 mg/m^2) given as sequential infusions, myelosuppression was more profound when paclitaxel was given after cisplatin than with the alternate sequence (ie, paclitaxel before cisplatin). PK data from these patients demonstrated a decrease in paclitaxel clearance of approximately 33% when paclitaxel was administered following cisplatin. The metabolism of paclitaxel is catalysed by CYP P450 isoenzymes CYP2C8 and CYP3A4.

Caution should be exercised when paclitaxel is concomitantly administered with known substrates (eg, midazolam, buspirone, felodipine, lovastatin, eletriptan, sildenafil, simvastatin, and triazolam), inhibitors (eg, atazanavir, clarithromycin, indinavir, itraconazole, ketoconazole, nefazodone, nelfinavir, ritonavir, saquinavir, and telithromycin), and inducers (eg, rifampin and carbamazepine) of CYP3A4. Caution should also be exercised when paclitaxel is concomitantly administered with known substrates (eg, repaglinide and rosiglitazone), inhibitors (eg, gemfibrozil), and inducers (eg, rifampin) of CYP2C8. Potential interactions between paclitaxel, a substrate of CYP3A4, and protease inhibitors (ritonavir, saquinavir, indinavir, and nelfinavir), which are substrates and/or inhibitors of CYP3A4, have not been evaluated in clinical trials. Reports in the literature suggest that plasma levels of doxorubicin (and its active metabolite doxorubicinol) may be increased when paclitaxel and doxorubicin are used in combination.

J 2 Restricted, Prohibited, and Permitted Concomitant Medications/Therapies with T-DXd

Restricted, prohibited, and permitted concomitant medications/therapies are described in Table 27, Table 28, and Table 29. Refer also to the dose modification guidelines for management of study treatment-related toxicities in Section 6.6. Patients must be instructed not to take any medications, including over-the-counter products, without first consulting with the investigator.

Table 27 Restricted medications/therapies

Medication/class of drug/therapy	Usage (including limits for duration permitted and special situations in which it's allowed)
Tobacco products, e-cigarettes and vaping	Use of tobacco products, e-cigarettes and vaping is strongly discouraged but not prohibited. Any prior or current use of these products should be recorded in the eCRF.

eCRF = electronic case report form.

Table 28 Prohibited medications/therapies

Prohibited medication/class of drug/therapy	Usage
Chloroquine or hydroxychloroquine	Concomitant treatment with chloroquine or hydroxychloroquine is not allowed during the study treatment.
Any concurrent chemotherapy, study treatment, biologic, radiotherapy (except palliative radiotherapy to areas other than chest, after consultation with the Study Physician)	Must not be given concomitantly while the patient is on study treatment
Live vaccine during the study and up to 30 days after the last dose of study treatment.	Patients who have received live, attenuated vaccine within 30 days prior to the first dose of T-DXd will be excluded.

Prohibited medication/class of drug/therapy	Usage
Immunosuppressive medications, including corticosteroids, methotrexate, azathioprine, and tumour necrosis factor-alpha blockers.	T-DXd cannot be administered when the patient is taking immunosuppressive medications, including corticosteroids, with the following exceptions: • short-term courses (< 2 weeks); • low to moderate dose; • long-term, alternate-day treatment with short-acting preparations; • maintenance physiologic doses (replacement therapy); • administered topically (skin or eyes), by aerosol, or by intra-articular, bursal, or tendon injection. • treatment with corticosteroids to prevent or treat hypersensitivity reactions to radiographic contrast agents is allowed. A temporary period of steroid treatment will be allowed for different indications after discussion with the Study Physician (eg, chronic obstructive pulmonary disease, radiation, nausea, etc). • patients with bronchopulmonary disorders may use bronchodilators if only administered intermittently. • use of immunosuppressive medications for the management of study treatment-related AEs or in patients with contrast allergies is acceptable.
EGFR TKIs	Should not be given concomitantly. Should be used with caution in the 90 days post last dose of durvalumab. Increased incidences of pneumonitis (with third generation EGFR TKIs) and increased incidence of transaminase increases (with 1st generation EGFR TKIs) has been reported when durvalumab has been given concomitantly.
Hormone replacement therapy and hormonal methods of contraception	Must not be given concomitantly while the patient is on study treatment.

COVID-19 = coronavirus 2019-nCoV; EGFR = epidermal growth factor receptor; T-DXd = trastuzumab deruxtecan; TKI = tyrosine kinase inhibitor.

Table 29 Supportive medications/therapies

Supportive medication/class of drug/therapy	Usage
Prophylactic anti-emetic agents	Trastuzumab deruxtecan is emetogenic, which includes delayed nausea and/or vomiting. Recommended that patients receive prophylactic anti-emetic agents prior to infusion of T-DXd and on subsequent days. A 2 or 3-drug combination regimen (e.g., dexamethasone with either a 5-HT3 receptor antagonist and/or an NK1 receptor antagonist, as well as other medicinal products as indicated) for prevention of chemotherapy-induced nausea and vomiting (CINV) should be considered and administered in accordance with the prescribing information or institutional guidelines.
Best supportive care (including antibiotics, anti-emetics, anti-diarrhoeal medication, G-CSF, nutritional support, correction of metabolic disorders, optimal symptom control, and pain management [including palliative radiotherapy, etc]) except for those medications identified as "prohibited," as listed above	Should be used, when necessary, for all patients except for those medications identified as "prohibited," as listed in Table 28.
Corticosteroids and/or bisphosphonates for the treatment of bone and brain metastases and for the treatment of specific adverse drug reactions (refer to Toxicity Management Guidelines document)	Permitted ≤ 10 mg/day of prednisone or its equivalent
Inactivated viruses, such as those in the influenza vaccine	Permitted
Coumadin anticoagulants	Permitted
Hematopoietic growth factors	Hematopoietic growth factors may be used for prophylaxis or treatment based on the clinical judgment of the Investigator
Required for management of other medical conditions	As required except for those identified as "prohibited," as listed in Table 28

CNS = central nervous system; G-CSF = granulocyte-colony stimulating factor; T-DXd = trastuzumab deruxtecan.

J 3 Restricted, Prohibited, and Permitted Concomitant Medications/Therapies with Tucatinib

Strong CYP3A Inducers or Moderate CYP2C8 Inducers: Avoid concomitant use.

Strong CYP2C8 Inhibitors: Avoid concomitant use; reduce TUKYSA dose if concomitant use cannot be avoided.

CYP3A Substrates: Avoid concomitant use with CYP3A substrates, where minimal concentration changes may lead to serious or life-threatening toxicities.

P-gp Substrates: Consider reducing the dose of P-gp substrates, where minimal concentration changes may lead to serious or life-threatening toxicities.

Tucatinib is a substrate and an inhibitor of P-gp and also inhibits BCRP in vitro. The clinical implications of the interaction between tucatinib and P-gp or BCRP transporters are not currently known. Drugs that are inhibitors of substrates of P-gp and drugs that are substrates of BCRP should be used with caution in combination with tucatinib. For additional details on these medicines, refer to local prescribing information for tucatinib.

Table 30 Selected Strong Inhibitors and Inducer of CYP2C8 and Their Elimination Half-Lives

Drug [a,b]	**Elimination Half-life** [c]
Strong Inhibitors	
Clopidogrel	6 hours
Gemfibrozil	1–2 hours
Montelukast	3–6 hours
Trimethoprim	8-12 hours
Strong Inducer	
Rifampin	3-5 hours

[a] FDA. “Drug Development and Drug Interactions: Table of Substrates, Inhibitors and Inducers” (http://www.fda.gov/Drugs/DevelopmentApprovalProcess/DevelopmentResources/DrugInteractionsLabeling/ucm093664.htm#potency).

[b] EMA. “Guideline on the investigation of drug interactions” http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2012/07/WC500129606.pd

[c] Drug package insert

Note: This table is prepared to provide examples of clinical substrates and not intended to be an exhaustive list.

Table 31 Selected Strong Inhibitors or Inducers of CYP3A4 and Their Elimination Half-Lives

Drug [a, b, c]	Elimination Half-life [d]
Strong Inhibitors	
Macrolide Antibiotics	
Clarithromycin	3–7 hours
Troleandomycin	2 hours
Azole Antifungals	
Itraconazole	16-28 hours (single dose), 34-42 hours (repeat dose)
Ketoconazole (systemic)	2–8 hours
Voriconazole	Dose dependent
Posaconazole	27–35 hours
Other	
Nefazodone	2–4 hours
Diltiazem	3-4 hours
White grapefruit juice	~ 4-5 hours[e]
Strong Inducers	
Barbiturates	Variable
Carbamazepine	25–65 hours (single dose), 12-17 hours (repeat dose)
Phenytoin	7–42 hours
Rifampin	3–4 hours (single dose), 2-3 hours (repeat dose)
St. John's Wort	9–43 hours[f]

[a] FDA. "Drug Development and Drug Interactions: Table of Substrates, Inhibitors and Inducers" (http://www.fda.gov/Drugs/DevelopmentApprovalProcess/DevelopmentResources/DrugInteractionsLabeling/u cm093664.htm#potency).

[b] EMA. "Guideline on the investigation of drug interactions" http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2012/07/WC500129606.pdf Strong CYP3A inhibitors are defined as those drugs that increase the AUC of oral midazolam or other CYP3A substrates ≥ 5-fold. Ritonavir, indinavir, nelfinavir, atazanivir, and saquinavir are also strong CYP3A3 inhibitors, but would not be used in this study as patients with known HIV are excluded.

[c] Drug package insert

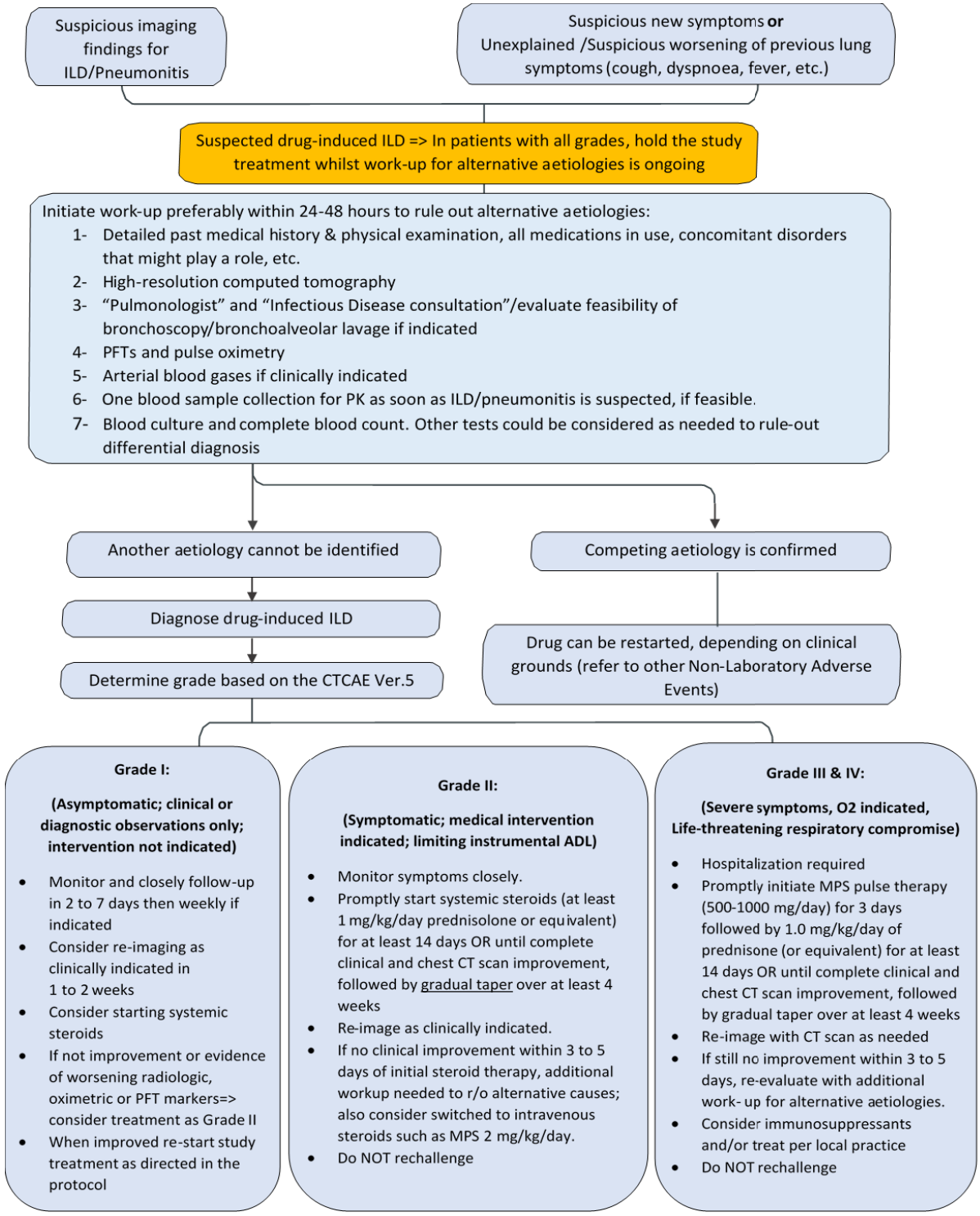
Bailey et al., Br J Clin Pharmacol 1998; 46: 101-110

Kerb et al., Antimicrob Agent & Chemother 1996, 40(9): 2087-2093

AUC = area under the concentration-time curve; HIV = human immunodeficiency virus

Note: This table is prepared to provide examples of clinical substrates and not intended to be an exhaustive list.

Appendix K Guidance for Management of Patients with Drug-Induced ILD/Pneumonitis



ADL = activities of daily living; CTCAE = Common Terminology Criteria for Adverse Events; ILD = interstitial lung disease; MPS = methylprednisolone; PFT = pulmonary function test; PK = pharmacokinetics.

Appendix L Toxicity Management Guidelines

This Appendix contains TMG information relevant for the treatments and treatment combinations used in this study.

TMGs for study treatments

- **Durvalumab:** The most current version of the TMGs entitled "Dosing Modification and Toxicity Management Guidelines for Immune-Mediated, Infusion-Related, and NonImmune Mediated Reactions of Durvalumab Monotherapy or Combination Therapy with Tremelimumab or Tremelimumab Monotherapy" is provided to the investigative site as an Annex document and is maintained within the Site Master File. Each of the toxicities should be treated with maximum supportive care (including withholding the agent suspected of causing the toxicity if required).
- **Pertuzumab:** Toxicities of pertuzumab may be managed according to the local prescribing information.
- **Paclitaxel:** Toxicities of paclitaxel may be managed according to the local prescribing information.
- **Tucatinib:** Toxicities of tucatinib may be managed according to the local prescribing information.
- **T-DXd:** Please refer to Table 32 below.

For treatment combinations, guidance indicating which TMGs to follow depending on the toxicity are provided below:

- **T-DXd monotherapy:** If any toxicity is judged by the treating physician to be related to T-DXd, the T-DXd management guidelines should be followed.

Some known overlapping toxicities, may on theoretical grounds be potentiated in the combination arms and particular attention should be paid to these. Examples for each combination are:

- **Potential overlapping toxicities for T-DXd + durvalumab combinations:**

CCI



CCI

- **Potential overlapping toxicities for T-DXd + pertuzumab combinations:**

CCI

- **Potential overlapping toxicities for T-DXd + paclitaxel combinations:**

CCI

- **Potential overlapping toxicities for T-DXd + durvalumab + paclitaxel combinations:**

CCI

- **Potential overlapping toxicities for T-DXd + tucatinib combinations:**

CCI



If these overlapping toxicities occur, consult the TMGs or local prescribing information for the combination drugs. The Investigator should use clinical judgement when attributing a toxicity to 1 or more of the drugs and consider following the more conservative approach.

If the symptoms promptly resolve with supportive care, consideration should be given to continuing the same dose of the assigned study treatment along with appropriate continuing supportive care. If medically appropriate, dose modifications are permitted for T-DXd. In addition, guidelines for T-DXd dose modifications are provided in Section 6.6.1. In the event of toxicity that cannot be managed by following the TMGs for T-DXd, consider stopping treatment with T-DXd.

All dose modifications should be documented with clear reasoning and documentation of the approach taken. Dose reductions are not permitted without prior agreement with the Study Physician.

All dose modifications (interruption, reduction, and/or discontinuation) should be based on the worst preceding toxicity. Specific criteria for interruption, re-initiation, dose reduction, and/or discontinuation of T-DXd are listed in the table below, which is applicable only to TEAEs that are assessed as related to use of T-DXd by the Investigator. For non-drug related TEAEs, follow standard clinical practice. Appropriate clinical experts should be consulted as deemed necessary.

If AE causality is possibly, probably, or definitely related to T-DXd, the most conservative toxicity guidance should be adopted. Discussion with the Sponsor's Study Physician is recommended in such cases.

Table 32 Toxicity Management Guidelines for T-DXd

CCI

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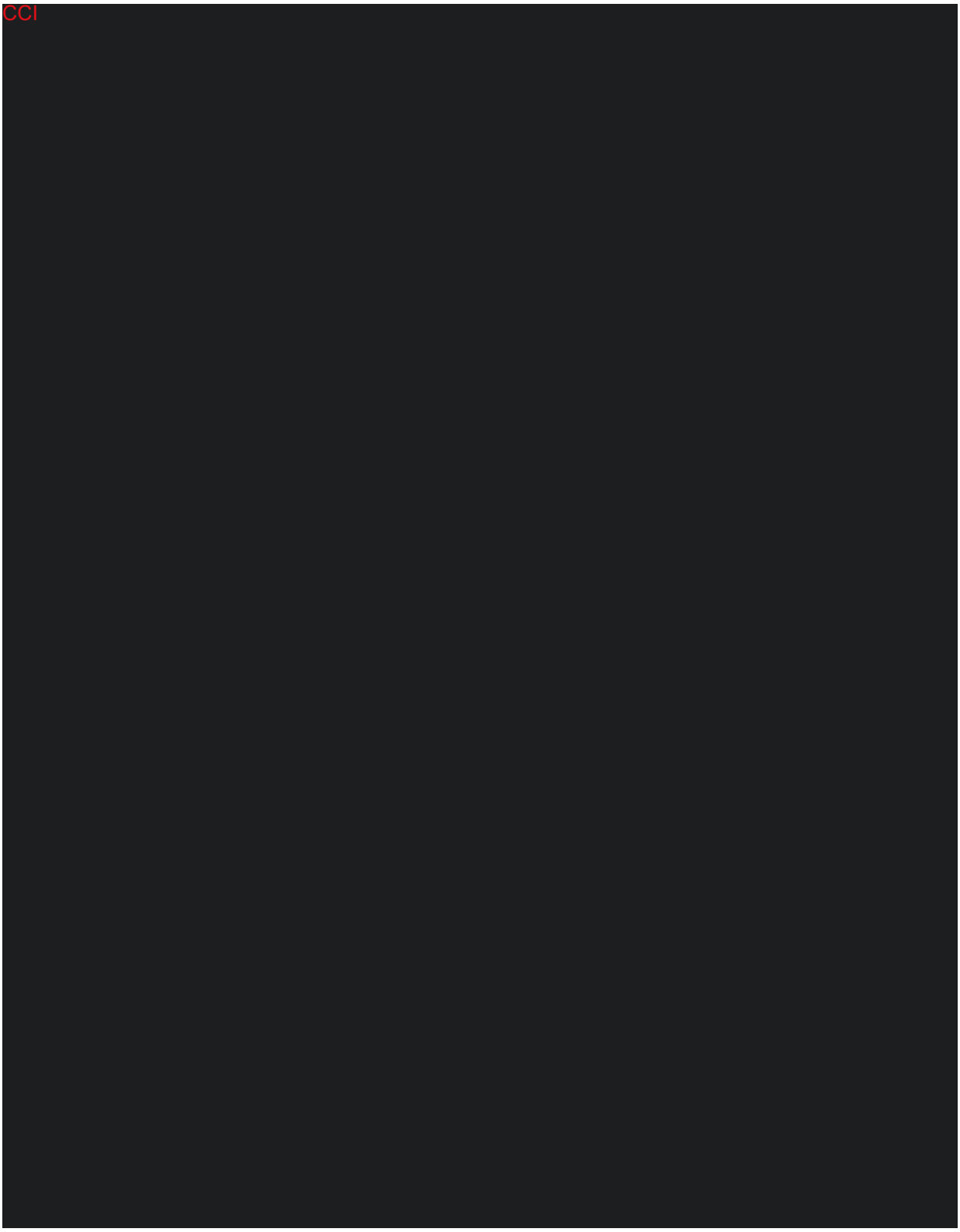
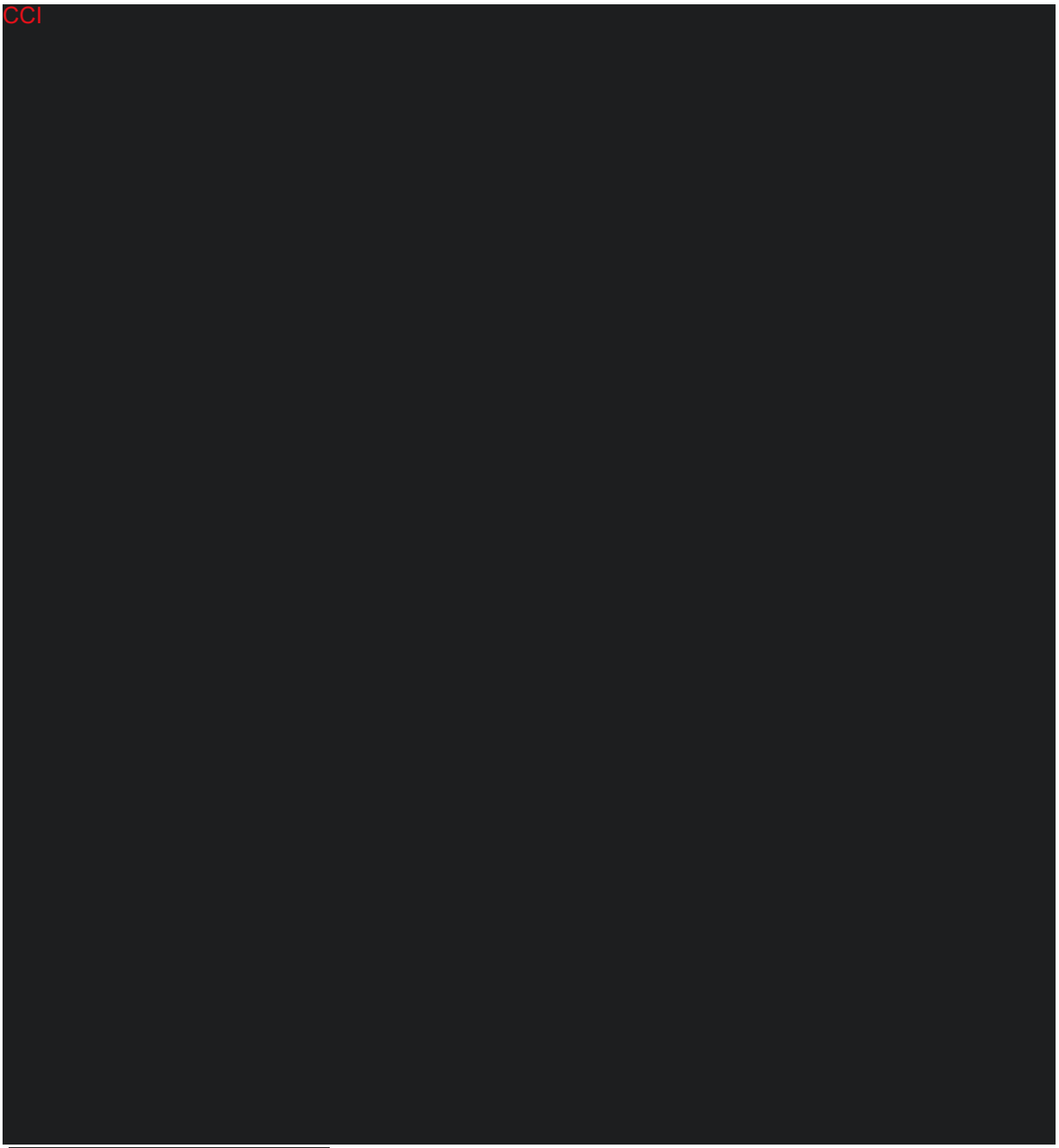


Table 32 Toxicity Management Guidelines for T-DXd

CCI



CCI

Note: All dose modifications should be based on the worst preceding toxicity.
AE = adverse event; ALT = alanine transaminase; AST = aspartate transaminase; CHF = congestive heart failure; CT = computer tomography; CTCAE = Common Terminology Criteria for Adverse Events; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; NSAIDS = non-steroidal anti-inflammatory drugs; PK = pharmacokinetics; SpO_2 = pulse oximetry; TMGs = toxicity management guidelines; ULN = upper limit of normal.

Appendix M Instructions Related to Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2/COVID-19)

M 1 Eligibility, Concomitant Medication, T-DXd Dose Modification and PK Sampling Text Relevant to SARS-CoV-2

Inclusion criteria

The following inclusion criteria has been added (see also Section 5.1):

- Has adequate treatment washout period before randomisation, defined as:
    a. Chloroquine/Hydroxychloroquine: > 14 days

Prior and Concomitant Medications

In addition to Section 6.5, the following text is relevant for participants with SARS-CoV-2:

Concomitant treatment with chloroquine or hydroxychloroquine is not allowed during the study treatment. If treatment with chloroquine or hydroxychloroquine treatment is absolutely required for SARS-CoV-2, study treatment must be interrupted.

Pharmacokinetic sampling

In addition to the PK samples specified in the SoA in Section 1.3, patients receiving T-DXd who receive chloroquine or hydroxychloroquine during the treatment period should have T-DXd PK blood samples taken at the following timepoints:

- On the day that chloroquine/hydroxychloroquine treatment starts, before any chloroquine/hydroxychloroquine is administered.
- Before chloroquine/hydroxychloroquine dosing on Day 3 or 4
- On the last day of chloroquine/hydroxychloroquine treatment
- Pre-dose on the day of restarting T-DXd treatment, if T-DXd is restarted.

Following the above blood PK draws, if T-DXd is restarted, routine T-DXd PK blood sample collection will continue per the SoA in Section 1.3.

SARS-CoV-2/COVID-19 Assessment(s)

Due to the potential impact of coronavirus disease 2019 (COVID-19, due to severe acute respiratory syndrome coronavirus 2 [SARS CoV-2]), on subject safety, the following SARS-Cov-2 infection assessment and dose modification and management plan for subjects with confirmed or suspected COVID-19 while being treated with trastuzumab deruxtecan should be followed. Dose modifications will be based on the worst CTCAE grade. **Use CTCAE version**

5.0 general grading criteria to evaluate COVID-19. All dose modifications (discontinuation, interruptions or reductions) must be recorded on the AE and drug administration eCRFs.

SARS-CoV-2 Infection Assessment(s)

All confirmed or suspected SARS-CoV-2 infection events must be recorded in the eCRF. If a subject presents to the clinic with symptoms suggestive of COVID-19, infection should be confirmed via NAAT (nucleic acid amplification test such as RT-PCR) or rapid antigen testing. SARS-CoV-2 antigen testing can be used to confirm infection, but not to rule it out.

Serum samples will be collected and may be used for SARS-CoV-2 testing from each subject who provides consent. Samples will be collected prior to the study drug infusion, shipped to a central laboratory, and stored there until the testing is required.

If participants consent, the remaining serum samples will also be stored for future analysis.

Sample collection, preparation, handling, storage, and shipping instructions are provided in the Study Laboratory Manual.

Dose modification criteria for suspected or confirmed SARS-CoV-2 infection

All confirmed or suspected SARS-CoV-2 infection events must be recorded in the eCRF. Dose modifications will be based on the worst CTCAE grade. All interruptions or modifications must be recorded on the AE and drug administration eCRFs. **Please use CTCAE v5.0 general grading criteria to evaluate COVID-19.** All dose modifications (discontinuation, interruptions or reductions) must be recorded on the AE and drug administration eCRFs.

Dose modification criteria

If asymptomatic or symptomatic COVID-19 is suspected, interrupt trastuzumab deruxtecan and rule out COVID-19 per local guidance.

- If COVID-19 is ruled out, follow dose modification and management guidelines as outlined in the protocol.
- If COVID-19 is confirmed or diagnosis is suspected after evaluation, follow dose modification as outlined below and manage COVID-19 per local guidance until recovery of COVID-19. COVID-19 recovery is defined as no respiratory signs/symptoms of COVID-19, and completely or nearly resolved chest CT findings which are equivalent to CT Severity Score of 1 (CT Severity Score of 1 = Subtle Ground Glass Opacities and very few findings)*. Then follow below dose modifications:

** Luger, Anna K., et al. "Chest CT of lung injury 1 year after COVID-19 Pneumonia: The CovILD study." Radiology (2022).*

COVID-19 Dose Modification Criteria

COVID-19 Worst Toxicity NCI-CTCAE Version 5.0 Grade (unless otherwise specified)	**Schedule Modification for trastuzumab deruxtecan**
Grade 1	After recovery, resume study treatment(s)at the same dose
Grade 2	After recovery, resume study treatment(s) at the same dose if chest CT findings are completely resolved Reduce by 1 dose level if chest CT findings are nearly resolved (equivalent to CT Severity Score of 1)
Grade 3	After recovery, reduce by 1 dose level if chest CT findings are completely resolved Discontinue study drug if chest CT findings are **not** completely resolved
Grade 4	Discontinue study drug

COVID-19 = coronavirus disease 2019; CT = computed tomography; CT Severity Score of 1 = Subtle Ground Glass Opacities and very few findings

Closely monitor signs/symptoms after resuming T-DXd, initially with a weekly phone call or site visit for 6 weeks.

- In addition to the recommendations outlined in the table above, Investigators may consider dose modifications of the study drug according to the subject's condition and after discussion with the study Medical Monitor or designee.
- If an event is suspected to be drug related ILD/pneumonitis, manage per protocol ILD/pneumonitis management guideline.

M 2 Benefit Risk Considerations for COVID-19

The emergence of the coronavirus 2019-nCoV (COVID-19) presents a potential safety risk for patients. Several risk mitigation factors have been implemented in this study. Notably, the eligibility criteria will exclude patients with COVID-19 infections (see CSP Section 5.2).

Due to the potential overlapping impact of T-DXd and COVID-19 on the lung, the Sponsor has also provided in this Appendix, a dose modification and management plan for patients with confirmed or suspected COVID-19 who are being treated with T-DXd.

With these measures in place, it is considered the anticipated potential benefits for the patients enrolled in this study outweigh the potential risks.

Appendix N RANO Criteria for Assessment of Primary CNS Tumours

Selected sections from RANO-BM criteria are provided below. For detailed guidelines regarding minimum requirements for brain imaging, and assessment of response, see Image Acquisition Guidelines (IAG).

CNS vs non-CNS assessments: Assignment of CNS response and progression is independent of non-CNS response and progression. Non-CNS response and progression will be assessed by RECIST 1.1. Assessment of CNS response and progression will be per the RANO-BM criteria. At each assessment point where a CNS assessment occurs, the MRI should be coincident with a corresponding non-CNS assessment.

Measurable lesions: Up to 5 measurable CNS lesions may be identified as TLs. A sum of longest diameters for all TLs will be recorded at baseline and at each subsequent measurement. Definition of measurable disease:

- Contrast enhancing lesion that can be measured accurately in at least one dimension with a minimum size of 10 mm, and visible in 2 or more axial slices that are ≤ 5 mm apart with 0 mm skip.
- The diameter perpendicular to the longest diameter must be ≥ 5 mm (although this is not recorded).
- Cysts and surgical cavities are not measurable unless there is a nodular component measuring ≥ 10 mm in the longest diameter and ≥ 5 mm in the perpendicular plane.
- Lesions previously treated with local therapy (SRS, WBRT, or surgery) may only be considered measurable if progression has occurred since the time of local treatment. If lesions are present which have not been treated previously with local therapy, these are preferred TLs.
  - All other lesions should be considered non-measurable and listed as NTLs including dural metastases, bony skull metastases, and cystic-only lesions. At each assessment, these should be classified as present, absent, or unequivocal progression.

Lesions < 5 mm: For lesions which are sufficiently small (but still present) to be assigned an exact measure, a default value of 5 mm should be recorded. If a lesion disappears, the value should be recorded as 0 mm.

Coalescence of lesions: Lesions might coalesce during treatment. As lesions coalesce, a plane between them may be maintained that would aid in obtaining maximum longest diameter of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximum longest diameter for the coalesced lesion.

New lesions: The finding of a new CNS lesion should be unequivocal and not due to technical or slice variation. A NL is one that was not present on prior scans. If the MRI is obtained with

slice thickness of 1.5 mm or less, the new lesion should also be visible in axial, coronal, and sagittal reconstructions of 1.5 mm or thinner projections. If a NL is equivocal, for example because of its small size (ie, ≤ 5 mm), continued therapy can be considered, and a follow-up assessment will clarify if it really is new disease. If repeated scans confirm a NL, progression should be declared using the date of the initial scan showing the NL.

Treatment effect from radiation or surgery/pseudo-progression: Patients who have been treated with radiation or surgery, for whom there has been radiographic evidence of enlargement of TLs or NTLs, which do not necessarily represent tumour progression. If radiographic evidence of progression exists, but clinical evidence indicates that the radiological changes are due to treatment effect (and not to progression of cancer), additional evidence is needed to distinguish between true progression and treatment effect, in which a single MRI alone is insufficient. Patients can be continued on protocol therapy pending further investigation as follows:

- The equivocal scan should initially be recorded as “not evaluable”.
- The scan can be repeated at the next protocol-scheduled assessment or sooner, and generally within about 6 weeks. An Investigator can choose a shorter time interval if progressive symptoms or other clinical concerns arise.
- Continued tumour growth might be consistent with radiographic progression, in which case the patient should go off study. If the initial “not evaluable” scan is now considered to be consistent with PD, the date of progression should be recorded as the date of the initial scan where progression was suspected, and the initial scan should now be recorded as PD.
- Stabilization and shrinkage of a lesion can be consistent with treatment effect, in which case the patient can stay in the study.
- For patients with equivocal results even on the next restaging scan, the scan can be repeated again at a subsequent protocol-scheduled assessment or sooner. Equivocal scans must all initially be recorded as “not evaluable”.

Corticosteroid use: In the absence of clinical deterioration related to the tumour, an increase in corticosteroid dose alone should not be used as a sole determinant of progression. Patients with stable imaging results and whose corticosteroid dose has increased for reasons other than clinical deterioration related to the tumour do not qualify as having SD or PD. These patients should be observed closely, and if their corticosteroid dose can be reduced back to baseline, they will be considered as having SD, but if further clinical deterioration related to the tumour becomes apparent, they will be considered as having clinical progression. The definition of clinical deterioration is left to the discretion of the treating physician. For patients who initiate or increase the dose of corticosteroids but who are not considered to have clinical or radiographic progression, scans taken during the use of corticosteroids should be recorded as “not evaluable” regardless of radiographic measurements.

RESPONSE ASSESSMENT OF TARGET AND NON-TARGET LESIONS

An additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review.

Target lesions

Complete response

Disappearance of all CNS target lesions sustained for at least 4 weeks; with no new lesions, no use of corticosteroids, and patient is stable or improved clinically.

Partial response

At least a 30% decrease in the sum longest diameter of CNS target lesions, taking as reference the baseline sum longest diameter sustained for at least 4 weeks; no new lesions; stable to decreased corticosteroid dose; stable or improved clinically.

Progressive disease

At least a 20% increase in the sum longest diameter of CNS target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, at least one lesion must increase by an absolute value of 5 mm or more to be considered progression.

Stable disease

Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum longest diameter while on study.

Non-target lesions

Non-target lesions should be assessed qualitatively at each of the timepoints specified in the protocol.

Complete response

Requires all of the following: disappearance of all enhancing CNS non-target lesions, no new CNS lesions.

Non-complete response or non-PD

Persistence of one or more non-target CNS lesion or lesions.

Progressive disease

Any of the following: unequivocal progression of existing enhancing non-target CNS lesions, new lesion(s) (except while on immunotherapy-based treatment), or unequivocal progression

of existing tumour-related non-enhancing (T2/FLAIR) CNS lesions. In the case of immunotherapy-based treatment, new lesions alone may not constitute PD.

Summary of the Response Criteria for CNS Metastases Proposed by RANO-BM

	Complete Response	Partial Response	Stable Disease	Progressive Disease
Target lesions	None	≥ 30% decrease in sum longest distance relative to baseline	< 30% decrease relative to baseline but < 20% increase in sum longest distance relative to nadir	≥ 20% increase in sum longest distance relative to nadir; at least one lesion must increase by an absolute value of 5mm or more
Non-target lesions	None	Stable or improved	Stable or improved	Unequivocal progressive disease
New lesions	None	None	None	Present
Corticosteroids	None	Stable or decreased	Stable or decreased	Not applicable [a]
Clinical status	Stable or improved	Stable or improved	Stable or improved	Worse
Requirement for response	All	All	All	Any

[a] increase in corticosteroids alone will not be taken into account in determining progression in the absence of persistent clinical deterioration.

CNS and non-CNS Response Assessment

CNS (RANO-BM)	Non-CNS (RECIST 1.1)	Response
Complete response, partial response, or stable disease	Complete response, partial response, or stable disease	Record as CNS and non-CNS complete response, partial response, or stable disease
Complete response, partial response, or stable disease	Progressive disease	Record as CNS complete response, partial response, or stable disease; record as non-CNS progressive disease
Progressive disease	Complete response, partial response, or stable disease	Record as CNS progressive disease; record as non-CNS complete response, partial response, or stable disease

CNS = central nervous system; RANO-BM = Response Assessment in NeuroOncology – Brain Metastases; RECIST 1.1 = Response Evaluation Criteria in Solid Tumours, Version 1.1

Bi-Compartmental Progression-Free Survival

CNS (RANO-BM)	Non-CNS (RECIST 1.1)	Bi-Compartmental PFS	Note
Complete response, partial response, or stable disease	Progressive disease	Record as PFS event	Record as non-CNS progressive disease
Progressive disease	Complete response, partial response, or stable disease	Record as PFS event	Record as CNS progressive disease
Progressive disease	Progressive disease	Record as PFS event	Record as both CNS and non-CNS progressive disease

CNS = central nervous system; PFS = progression-free survival; RANO-BM = Response Assessment in NeuroOncology – Brain Metastases; RECIST 1.1 = Response Evaluation Criteria in Solid Tumours, Version 1.1.

In this trial, if a patient is found to have PD due to CNS progression per RANO-BM, but does not have PD outside the CNS per RECIST 1.1, the patient may undergo local therapy (radiotherapy or surgery) to the progressive CNS lesion(s), and continue on study treatment after local therapy after discussion with and approval from the AstraZeneca Study Physician or delegate. The patient will remain on the same treatment arm assigned initially, and may continue on study until a second PD event occurs (either CNS or non-CNS).

N 1 Abbreviations

Abbreviation or Special Term	Explanation
ADA	anti-drug antibody
ADC	antibody-drug conjugate
ADCC	antibody-dependent cell-mediated cytotoxicity
AE	adverse event
AESI	adverse event of special interest
ALP	alkaline phosphatase
ALT	alanine aminotransferase
ASCO	American Society of Clinical Oncology
AST	aspartate aminotransferase
BCRP	Breast cancer resistance protein
BICR	blinded independent central review
BMFS	brain metastases-free survival
CANTAB	Cambridge Neuropsychological Test Automated Battery
CAP	College of American Pathologists

Abbreviation or Special Term	Explanation
CHF	congestive heart failure
CI	confidence interval
C_{min}	minimum plasma concentration
CNS	central nervous system
CNS-DoR	central nervous system duration of response
CNS-ORR	central nervous system overall response rate
CNS-PFS	central nervous system progression-free survival
COPD	chronic obstructive pulmonary disease
COVID-19	coronavirus 2019-nCoV
CR	complete response
CRF	case report form
CSP	Clinical Study Protocol
CSR	Clinical Study Report
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
CCI [redacted]	
CTFG	Clinical Trial Facilitation Group
CYP	cytochrome P450
DAR	drug to antibody ratio
DCO	data cut-off
DLCO	diffusion capacity of the lungs for carbon monoxide
DLT	dose-limiting toxicity
DNA	deoxyribonucleic acid
DoR	duration of response
DU	current dose is unacceptably toxic
EAS	efficacy analysis set
EC	Ethics Committee, synonymous to Institutional Review Board (IRB) and Independent Ethics Committee (IEC)
ECG	electrocardiogram
ECHO	echocardiogram
ECOG PS	Eastern Cooperative Oncology Group Performance Status
eCRF	electronic case report form
EAS	efficacy analysis set
EDC	electronic data capture
EMA	European Medicines Agency
EOT	end of treatment

Abbreviation or Special Term	Explanation
ePRO	electronic patient-reported outcome
FAS	full analysis set
FDA	Food and Drug Administration
FFPE	formalin-fixed and paraffin-embedded
FISH	fluorescent *in situ* hybridisation
GCP	Good Clinical Practice
G-CSF	granulocyte-colony stimulating factor
HBc	hepatitis B core antigen
HbsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
HER2	human epidermal growth factor receptor 2
HIV	human immunodeficiency virus
HL	Hy's Law
HOSPAD	hospital admission
HR	hazard ratio
HRCT	high resolution computed tomography
HRQoL	health-related quality of life
HRT	Hormone replacement therapy
IATA	International Airline Transportation Association
IB	Investigator's Brochure
ICF	informed consent form
ICH	International Council for Harmonisation
iCRO	imaging Contract Research Organisation
IEC	Independent Ethics Committee
International co-ordinating Investigator	If a study is conducted in several countries the international co-ordinating Investigator is the Investigator co-ordinating the Investigators and/or activities internationally.
IHC	immunohistochemistry
ILD	interstitial lung disease
imAE	immune-mediated adverse event
IND	Investigational New Drug
IPF	idiopathic pulmonary fibrosis
IRB	Institutional Review Board
IRT	Interactive Response Technology
ISH	*in situ* hybridisation

Abbreviation or Special Term	Explanation
ITT	Intent to treat
IV	intravenous
LVEF	left ventricular ejection fraction
mAb	monoclonal antibody
MBC	metastatic breast cancer
MDASI	MD Anderson Symptom Inventory
MDASI-BT	MDASI Brain Tumour
MedDRA	Medical Dictionary for Regulatory Activities
MOA	mechanism of action
mOS	median overall survival
mPFS	median progression-free survival
MRI	magnetic resonance imaging
mRNA	messenger ribonucleic acid
mTPI-2	modified toxicity probability interval-2
MUGA	Multiple gated acquisition scans
NANO	Neurologic Assessment in Neuro-Oncology
NCI	National Cancer Institute
NE	not evaluable
NL	new lesion
NSCLC	non-small cell lung cancer
NTL	non-target lesion
NYHA	New York Heart Association
ORR	objective response rate
OS	overall survival
PAL	paired associates learning
CCI	[redacted]
pCR	pathologic complete response
PD	progression of disease
PD-1	programmed cell death protein 1
PD-L1	programmed death-ligand 1
PEI	Paul-Ehrlich-Institut
PET	positron emission tomography
PFS	progression-free survival
PFS2	progression-free survival after next-line treatment
PFT	pulmonary function test
P-gp	P-glycoprotein

Abbreviation or Special Term	Explanation
PHL	Potential Hy's Law
PK	pharmacokinetic
PR	partial response
PRO	patient reported outcome
QTcF	QT interval corrected by Fridericia's formula
QW	every week
Q/3/9W	every 3/6/9 weeks
RANO	Response Assessment in Neuro-Oncology
RANO-BM	Response Assessment in Neuro-Oncology – Brain Metastases
RECIST 1.1	Response Evaluation Criteria in Solid Tumours, Version 1.1
RNA	ribonucleic acid
RP2D	recommended phase 2 dose
RTI	Reaction time
SAE	serious adverse event
SAF	safety analysis set
SAP	Statistical Analysis Plan
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SCLC	small cell lung cancer
SD	stable disease
SGRQ-I	St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis
SmPC	Summary of Product Characteristics
SoA	schedule of activities
SOC	standard of care
SpO_2	pulse oximetry
SRC	Safety Review Committee
SRS	Stereotactic radiosurgery
SWM	Spatial working memory
TBL	total bilirubin
CCI [redacted]	
T-DM1	trastuzumab emtansine
T-DXd	trastuzumab deruxtecan
TEAE	treatment-emergent adverse event
TKI	tyrosine kinase inhibitor
TL	target lesion
TMG	toxicity management guidelines

Abbreviation or Special Term	Explanation
TNBC	triple-negative breast cancer
ULN	upper limit of normal
US	United States
USPI	United States Prescribing Information
WBRT	whole brain radiation therapy
1L	First line
1-2 L	First or second line
2L	Second line

Appendix V Protocol Amendment History

The Protocol Amendment Summary of Changes Table for the current amendment is located directly before the table of contents.

Amendment 3 (19 September 2023)

Overall Rationale for the Amendment:

The main reasons for CSP amendment 3 are to comply with regulatory requirements (e.g. EU CTR) and to align the CSP as per the latest AstraZeneca-specific protocol templates. Reasons to amend the CSP also include aligning language on dose modification and interruption criteria with the most recent version of the T-DXd IB (v9) and to align with changes in the interim analysis and final analysis plans. Additional administrational changes and clarifications or corrections regarding study assessments have also been implemented. The rationale for each of these changes is provided in the table below.

Section #	Description of Change	Brief Rationale
SUBSTANTIAL CHANGES		
Core CSP Section 1.1 Section 9.4 Section 9.5	Addition of new interim analysis.	To provide clarification of study analyses.
Core CSP Section 1.1 Section 9.4	Updated wording on final analysis.	To clarify end of study timelines.
Core CSP Section 2.3.1	Included wording on hepatic impairment and ILD education in patients with renal impairment.	To align with T-DXd IB and label update regarding ILD risk in patients with moderate renal impairment at baseline.
Core CSP **Appendix G** **Appendix T (Module 6)** **Appendix U (Module 7)** Section 5.1	Updated language in inclusion criterion #12 and Appendix G to reflect active brain metastases as a target lesion in Module 6 and Module 7.	Clarifications related to inclusion of active brain metastasis as RECIST 1.1 TLs.
Core CSP Section 8.3.16.1	Updated Dose Interruption and Modification/TMGs.	To clarify guidelines on dose delays and dose modifications.
Appendix L	Added additional clinical laboratory tests for patients with pulmonary toxicity.	ILD management guidance updated to align with additional risk mitigation measures included in the TMGs.
Appendix O (Module 1) **Appendix R (Module 4)** Section 8.3.16.2	Added wording on holding or discontinuation of durvalumab and removed wording on dose reduction.	To clarify required actions for treatment modifications for specific imAEs

Section #	Description of Change	Brief Rationale
NON-SUBSTANTIAL CHANGES		
Core CSP Title Page Section 1.1 – Synopsis **Appendix O (Module 1)** **Appendix P (Module 2)** **Appendix Q (Module 3)** **Appendix R (Module 4)** **Appendix S (Module 5)** **Appendix T (Module 6)** **Appendix U (Module 7)** Title Page	Inclusion of EU CT number in 'Regulatory Agency Identifier Number(s)'.	Update required to comply with regulatory requirement (e.g. EU CTR).
Core CSP **Appendix O (Module 1)** **Appendix P (Module 2)** **Appendix Q (Module 3)** **Appendix R (Module 4)** **Appendix S (Module 5)** **Appendix T (Module 6)** **Appendix U (Module 7)** Section 1.3	Updated SoA table to include vital sign and SpO_2 EI check at C4 and subsequent cycles.	To ensure any abnormalities before and after each infusion are captured and to provide clarity.
Core CSP **Appendix O (Module 1)** **Appendix P (Module 2)** **Appendix Q (Module 3)** **Appendix R (Module 4)** **Appendix S (Module 5)** **Appendix T (Module 6)** **Appendix U (Module 7)** Section 1.3	Removed CTC collection at progression.	Sample collection method no longer suitable for analysis.
Core CSP **Appendix O (Module 1)** **Appendix P (Module 2)** **Appendix Q (Module 3)** **Appendix R (Module 4)** **Appendix S (Module 5)** **Appendix T (Module 6)** **Appendix U (Module 7)** Section 1.3 Section 8.6.1	Removal of blood sample for plasma exploratory clinical benefit or safety analysis.	Plasma samples not a suitable medium for the anticipated analyses.
Core CSP Section 4.4	Clarified definition of the end of study according to European Union	To comply with regulatory requirement (e.g. EU CTR).

Section #	Description of Change	Brief Rationale
	and Food and Drug Administration requirements.	
	Added wording on alternative drug supply possibilities and potential roll-over or safety extension study.	To discuss alternative product supply options when available, and to allow for the possibility of treatment continuation of participants, at the time of the final DCO.
Core CSP Section 2.3.1 **Appendix J2** Table 29	Added wording on T-DXd being emetogenic.	T-DXd is classified as an emetogenic anticancer agent as per Core Data Sheet prophylaxis advised according to International guidelines.
Core CSP Section 5.1 **Appendix M1**	Updated treatment washout period for chloroquine/Hydroxychloroquine from ≥ 14 days to > 14 days.	To clarify the washout period of chloroquine/hydroxychloroquine.
Core CSP **Appendix O (Module 1)** **Appendix R (Module 4)** Section 6.1	Minor change to labelling and packaging requirements.	Update required to comply with regulatory requirement (e.g. EU CTR).
Core CSP Section 6.2.1	Added information on density of study intervention.	Sites have requested density when performing gravimetric preparations.
	Revised wording on preparation of study intervention.	To provide further clarification on handling and preparation of T-DXd.
Appendix O (Module 1) Section 6.2.2 Section 6.2.3 **Appendix R (Module 4)** Section 6.2.2 Section 6.2.4	Revised wording on preparation and storage of study intervention.	To provide further clarification on handling and preparation of durvalumab.
Core CSP Section 6.5	Added wording on standard endocrine therapy, already present in Section 6.5 of all combination modules.	Clarification that use of standard endocrine therapy is allowed for all modules.
Core CSP Section 6.7	Added further details on study treatment after end of study.	To provide details on SAE reporting and participant follow-up after final DCO.
	Added wording on potential roll-over or safety extension study and alternative drug supply possibilities.	To discuss alternative product supply options when available, and to allow for the possibility of treatment continuation of participants, at the time of the final DCO.

Section #	Description of Change	Brief Rationale
Core CSP Section 8.1.1	Updated text to follow the tumour assessment schedule relative to randomization.	Clarification of reference timepoint for tumour assessments.
Core CSP Section 8.1.5	Removed wording on exact number of patients diagnosed with ILD/pneumonitis to be interviewed.	ILD/pneumonitis interviews needed to reach data saturation not achievable within study timelines.
Core CSP Section 8.1.5.3	Deleted text indicating that at-home pulse oximetry testing will be performed at approximately the same time each day.	At approximately the same time each day is not a requirement.
Core CSP Section 8.2.5.5	Added language for additional clinical laboratory tests to be performed for potential cases of ILD/pneumonitis.	To align with ILD investigation requirements
Core CSP **Appendix O (Module 1)** **Appendix P (Module 2)** **Appendix Q (Module 3)** **Appendix R (Module 4)** **Appendix S (Module 5)** **Appendix T (Module 6)** Section 8.3 **Appendix B4**	Updated title to include drug abuse, drug misuse. Included new subsections. Added Drug Abuse and Drug Misuse definition. Added wording on collection of medication errors.	To define and ensure events of drug misuse and drug abuse during the study are reported. Clarification on how medication errors should be collected.
Core CSP Section 8.4	Indicated more specific timelines for reporting.	To ensure overdoses associated with an SAE are reported within 1 or 5 calendar days.
Core CSP Section 8.5	Revised wording on sample retention timelines.	To be compliant with ICH GCP, and AstraZeneca HBS global standards and policies for sample storage and retention.
Core CSP Section 8.5.2	Inserted wording that baseline samples may also be used for additional method validation activities.	To provide clarity on possible additional uses of patient samples.
Core CSP Section 9.3	Updated Table 17 footnote.	To align with Section 9.4
Core CSP Section 9.4.2.2.6	Updated text to reflect subgroup analyses of PD-L1 expression in Modules 0 and 1 instead of Modules 2 and 4.	To align with rationale for study design noted in Section 4.2.1.
Core CSP Section 9.4.3	Updated definition of TEAE.	To clarify how TEAE are defined within the study.
Appendix A1	Clarification to note that Investigator will be responsible for providing oversight study conduct at site, and	Update required to comply with regulatory requirement (e.g. EU CTR)

Section #	Description of Change	Brief Rationale
	adherence to 21 CFR 312.120, ICH guidelines, IRB/IEC, European Regulation 536/2014 as applicable, and all other applicable local regulations	
	Addition of sub-heading "Regulatory Reporting Requirements for Serious Breaches"	To comply with regulatory requirement (e.g. EU CTR) and global company requirement.
Appendix A4	Updated wording on data protection including personal data breaches.	Additional guidelines informing patients on data protection. Description of personal data breach and guidelines on prevention and addressing personal data breaches.
Appendix A6	Addition of new wording : "Any results both technical and lay summaries for this trial, will be submitted to EU CTIS within a year from global End of Trial Date in all participating countries, due to scientific reasons, as otherwise statistical analysis is not relevant."	To comply with EU CTR requirements.
Appendix A7	• Inclusion of medical oversight details • Clarification on medical oversight process • Updated information about retention timelines of records and documents to [25 years after study archiving or as required by local regulations]	To comply with EU CTR and global company requirements.
Appendix J2 Table 28	Removal of Hydroxychloroquine and chloroquine treatment for COVID-19	Management practices for COVID-19 have changed such that this is no longer required.
Appendix L	Added footnote CCI to table for CCI [redacted] clarification	To add cross-reference to Section-CCI [redacted].
Appendix M	Added clarifications and changes	Updated instructions related to COVID-19, including addition of the CT Severity Score to give further guidance regarding 'nearly resolved' chest CT findings.

Section #	Description of Change	Brief Rationale
Appendix O (Module 1) Mod 1 Table 1	Addition of footnote 'b' to 'TSH (reflex-free T3 or free T4)', and 'Amylase and lipase' rows	To clarify that these labs may be assessed within 3 days before study treatment administration, same as other clinical safety labs
Appendix O (Module 1) Mod 1 Table 2	Addition of footnote d, ADA for durvalumab need not be collected after safety follow-up when subject starts another therapy for cancer.	Durvalumab and pertuzumab ADA is not needed when patient starts another anti-cancer therapy.
Appendix P (Module 2) Mod 2 Table 1	Addition of footnote r, ADA for pertuzumab need not be collected after safety follow-up when participant starts another therapy for cancer.	
Appendix R (Module 4) Mod 4 Table 2	Addition of footnote d, ADA for durvalumab need not be collected after safety follow-up when subject starts another therapy for cancer.	
Appendix O (Module 1) **Appendix R (Module 4)** Section 6.2	Updated wording on preparation, administration, and storage of durvalumab.	Language added to provide clarity on durvalumab handling.
Appendix O (Module 1) **Appendix R (Module 4)** Section 2.2	Updated wording on background of durvalumab.	To provide the latest information on durvalumab approval.
Appendix W	Addition of new appendix.	To detail country-specific requirements to comply with regulatory requirement e.g. EU CTR (consolidated CSP).
General changes		
Throughout	Minor editorial and document formatting revisions.	Minor; therefore, have not been summarised.

Amendment 2 (23 November 2021)

Overall Rationale for the Amendment:

The main reasons for CSP amendment 2 are adding stopping rules for Part 2 combinations modules based on the treatment discontinuation rate and adding an optional pre-screening to permit tumour sample collection in the metastatic setting for biomarker testing prior to the 28-day screening window. Additional administrational changes and clarifications or corrections regarding study assessments have also been implemented. The rationale for each of these changes is provided in the table below.

Section #	Description of Change	Brief Rationale
SUBSTANTIAL CHANGES		
Core CSP Section 1.1 Section 9.5 Section 9.6.1	Updated the interim safety analysis and SRC sections to include Modules 5 and 6. Added details of safety stopping rules for Modules 1 to 6. Noted that no stopping rules will be applied to Modules 0 and 7 and gave rationale. Added new "Stopping Rules" Section 9.5.1.	Safety stopping rules have been added to Modules 1 to 6 to ensure patient safety.
Core CSP Section 1.1 Section 1.2, Figure 1 Study Design Section 4.1.1 **Appendix S (Module 5)** Section 1.1 Section 4.1 Section 6.1, Mod 5 Table 5 Study Treatments in Module 5 Section 6.2.2 **Appendix T (Module 6)** Section 1.1 Section 4.1 Section 4.1.1 Section 6.1, Mod 6 Table 4 Study Treatments in Module 6 Section 6.2.2	Addition of potential central sourcing of tucatinb for Module 5 and Module 6, where feasible.	Update regarding the possibility of central tucatinib supply
Core CSP Section 1.1 Section 1.1, Table 2 Modules 1 to 5 Part 1, Dose Finding Section 1.2, Figure 1 Study Design **Appendix R (Module 4)** Section 1.2 Section 4.1.1	Added text describing that dose escalation data about the combination of T-DXd + durvalumab + paclitaxel from clinical study DESTINY-Breast08 may be used for alternative identification of RP2D in this study in lieu of conducting Module 4 Part 1 in this study.	To allow potential acceptance of RP2D identification for Module 4 from DESTINY-Breast08, which has the same study treatment.
Core CSP Section 1.1 Section 1.2, Figure 1 Study Design Section 4.1.3 Section 8.6.1 Section 8.6.2 Section 9.4	Added note to figure footnotes (Figure 1) and text to Section 8.6.1 and Section 8.6.2 detailing the number of biopsies required for Part 2 along with a description of potential additional enrolment with mandatory paired biopsies if a required number of paired biopsies are not obtained in initial target. **Section 8.6.1 only**: Deleted text referring to paired biopsies becoming mandatory for enrolment onto a module when the total number of remaining slots in Part 2 of the module equals the number of outstanding paired biopsies for the module.	Operational change to study such that all modules will enrol to the prespecified number of patients per cohort before previously optional paired biopsies become mandatory.

Section #	Description of Change	Brief Rationale
	Section 1.1 only: Deleted text noting that any additional patients enrolled for paired biopsies will not be included in the main efficacy analyses but will be included in the safety analyses. **Section 9.4 only**: Added text noting that additional patients may be enrolled for paired biospsies. Data from these additional patients will be included in any subsequent data cut, including the data cut for the CSR regardless of their follow-up time.	Clarifications regarding the ITT population
Core CSP Section 1.3.2, Table 4 Schedule of Activities during the Treatment Period – Module 0 (Part 2 only) Section 8.1.1 **Appendix G** **Appendix N** **Appendix O (Module 1), Appendix P (Module 2), Appendix Q (Module 3), Appendix R (Module 4), Appendix S (Module 5), Appendix T (Module 6), and Appendix U (Module 7)** Section 1.3	Clarified the additional tumour scan following investigator-determined RECIST 1.1 progression, previously required, should now be obtained if feasible; added text that additional scans obtained may be sent for BICR review.	Clarified requirements to confirm radiological progressive disease to if feasible and that the scan may be sent for BICR review.
Core CSP Section 4.1.2.1	Addition of DLT criterion for haematological toxicities. Addition of guidance on the use of G-CSF for haematological toxicities during the DLT evaluation period. Addition of Module 5-specific criteria for DLT-evaluable patient requiring patients to have received 75% or more of nominal tucatinib doses to be considered DLT evaluable.	To provide an updated DLT definition for haematological toxicities, guidance on the use of G-CSF during the DLT evaluation period, and clarity around the module-specific criteria for DLT-evaluable patient.
Core CSP Section 5.1	**Inclusion criterion #4(b)**: Added text detailing the local HER2 result must be from a tumour sample obtained in the metastatic setting.	Clarification that the local HER2-positive results should be obtained from a tumour sample collected in the metastatic setting.
Core CSP Section 5.2	**Exclusion criterion #14**: Deleted text referring to concurrent use of hormonal therapy for non-cancer-related conditions (eg, HRT).	Updated in-line with the recommendation on the non-use of concurrent HRT in female breast cancer patients and to ensure patients do not

Section #	Description of Change	Brief Rationale
		receive HRT while on study treatment.
	Exclusion criterion #16: Change of time frame for prior/concurrent clinical study participation from "12 months" to "30 days".	Updated to allow flexibility for later line patients in Part 1 to have an opportunity to participate in the study.
Core CSP Section 8.3.13 **Appendix S (Module 5)** Section 8.3.13 **Appendix T (Module 6)** Section 8.3.13	Updated tucatinib reference safety information document from USPI to SmPC for the definition of expectedness/listedness.	Update to the reference safety information available for tucatinib.
Core CSP Appendix H 1	Deleted text referring to contraceptive methods for females on HRT and whose menopause status is in doubt.	Updated in-line with the recommendation on the non-use of HRT and hormonal methods of contraception in female breast cancer patients and to ensure patients do not receive HRT or use hormonal methods of contraception while on study treatment.
	Deleted text for women of childbearing potential referring to stable use of chosen method of birth control for a minimum of 3 months before study entry.	Updated to align with the sponsor standards for the T-DXd program.
Core CSP Appendix H 3 Appendix H 3, Table 26 Highly Effective Methods of Contraception (< 1% Failure Rate)	Added text describing that female patients of childbearing potential must use a non-hormonal method of contraception; hormonal methods of contraception are only allowed for female partners of childbearing potential of male patients.	Updated in-line with the recommendation on the non-use of hormonal methods of contraception in female breast cancer patients. The hormonal methods of contraception are only allowed for female partners of male breast cancer patients.
	Updated table header row "Non-Hormonal Methods" to "Non-hormonal Methods (female participants and female partners [of childbearing potential] of male participants)"	Clarification that non-hormonal methods of contraception can be used by both female breast cancer patients and female partners of male breast cancer patients.
	Updated table header row "Hormonal Methods (must inhibit ovulation)" to "Hormonal Methods (female partners [of childbearing potential] of male participants ONLY)".	Clarification that hormonal methods of contraception should be used only by female partners of male breast cancer patients.

Section #	Description of Change	Brief Rationale
	Updated footnote text to specify that for hormonal methods to be considered "highly effective", they must inhibit ovulation.	Updated in-line with health authority feedback and CTFG guidelines on birth control methods.
Core CSP Appendix J 2, Table 28 Prohibited Medications/Therapies Appendix J 2, Table 29 Supportive Medications/Therapies	**Table 28**: Added a row for HRT and hormonal methods of contraception. **Table 29**: Deleted row for hormones for non-cancer-related conditions.	Updated in-line with the recommendation on the non-use of HRT and hormonal methods of contraception in female breast cancer patients and to ensure patients do not receive HRT or use hormonal methods of contraception while on study treatment.
Core CSP Appendix M 1	**Appendix M 1**: Updated to SARS-CoV-2/COVID-19 assessment text to note that participant must not have any signs or symptoms of COVID-19 infection for at least 2 weeks and nearly or completely resolved chest CT findings.	Updated to clarify and ensure that study participants do not have any signs or symptoms of COVID-19 infection for at least 2 weeks and nearly or completely resolved chest CT findings before treatment with study drugs.
NON-SUBSTANTIAL CHANGES		
Core CSP Section 1.1, Table 1 Objectives and Endpoints Section 3, Table 6 Objectives and Endpoints Section 9.4.2.2.1 Section 9.4.2.2.2	Updated the secondary endpoint for ORR and PFS as applicable to all modules in Part 1 as well as in Part 2.	Extending efficacy reports to cover both Part 1 and Part 2 of the study for complete information.
	Updated secondary objective language to specify that anti-tumour activity of T-DXd combinations will be assessed in Part 1 and that T-DXd monotherapy will be assessed in Part 2.	Patients in Part 1 are not necessarily treated at the RP2D.
Core CSP Section 1.1 Section 9.4	**Section 1.1 only**: Added more specificity to the text that an interim analysis will be performed separately in Part 2 of Modules 0 to 6 after the first [redacted] evaluable patients (for Modules 0 to 4) or the first [redacted] evaluable patients (for Modules 5 to 6) all have had the opportunity for 2 months of treatment or have discontinued or withdrawn from treatment. **Section 9.4 only**: Edited module reference for interim analysis in Part 2 from "Modules 0 to 4" to "Modules 0 to 6".	Updated language to specify stopping criteria based on safety signal for Modules 1 to 6 in response to requests from regulatory agencies.
Core CSP Section 1.3 Section 6.3.1 Appendix A3	Added pre-screening text detailing an optional pre-screening ICF.	To permit tumour tissue sample collection for central PD-L1 testing and/or central retrospective HER2 status testing prior to the 28-day screening window

Section #	Description of Change	Brief Rationale
Core CSP Section 1.3, Table 3 Schedule of Activities for Screening/Baseline – All Modules Section 8.3.1	Deleted previous footnote 'g' ("Within 3 days before first dose of study treatment') from the following procedures: SpO_2, full physical examination, ECOG PS, vital signs, weight, clinical safety laboratory assessment, and coagulation.	Correct typographical error indicating screening assessments were to be completed within 3 days before C1D1.
	Deleted previous footnote 'k' ("This sample CCI CCI ") from the biomarker assessment description for newly acquired tissue sample from the metastatic setting. Added text to footnote 'j' to specify that tumour sample collection does not need to be repeated if the patient went through pre-screening and a tumour sample was collected at pre-screening.	Correct typographical error indicating tumour samples for eligibility biomarker assessments could be collected on CCI
	Section 1.3, Table 3 and Section 8.3.1: Added new footnote 'g' describing the requirement to report SAEs directly related to tissue screening procedure during the pre-screening period.	Provide clarification due to inclusion of optional pre-screening in the study.
Core CSP Section 1.3.2, Table 4 Schedule of Activities during the Treatment Period – Module 0 (Part 2 only) Section 8.1.1 Section 8.2.5.2 **Appendix O (Module 1), Appendix P (Module 2), Appendix Q (Module 3), Appendix R (Module 4), Appendix S (Module 5), Appendix T (Module 6), and Appendix U (Module 7)** Section 1.3	Deleted footnote 'b' in Core and Modules 1, 2, 5, 6, and 7 and deletion of footnote 'c' in Modules 3 and 4, "Within 3 days before study treatment administration" from the vital signs and SpO_2 row. **Core Section 8.2.5.2 only**: Deleted text referring to SpO_2 evaluation within 3 days before study treatment administration; added text noting that SpO_2 assessments will follow the same guidance described for vital signs.	Updated vital sign and SpO_2 based on details added to Section 8.2.2
	Specified cycle details in the C4 column cells for the PK and immunogenicity assessments, as required.	To align with the detail given in the table footnotes.
	Edited C4 BI timepoint for blood sample for plasma biomarker assessment (CCI to include "CCI New text CCI CCI in alignment with RECIST assessments)".	Clarification that CCI sample collection after CCI aligns with both CCI and CCI RECIST assessments.
Core CSP Section 2.3.1.1	Updated text with most recent potential risks of T-DXd.	Updated risks to align with known safety profile of T-DXd.
Core CSP Section 5	Added definitions for enrolled and randomised patients. Clarified that patients in Part 1 of Modules 1 to 5 will be Sponsor-assigned to treatment and that patients are ramdomised in Part 2.	Clarification of study population.
Core CSP Section 5.1	**Core inclusion criterion #10**: Added text detailing all parameters must meet the inclusion criteria on the same day and must be the most recent results available.	Clarification to help ensure patients meet all laboratory eligibility criteria at study entry.

Section #	Description of Change	Brief Rationale
Section 8.3.14.2 (refer to row for inclusion criterion #16) Appendix H 1 and Appendix H 2 (refer to row for inclusion criteria #15 and #16) **Appendix P (Module 2), Appendix Q (Module 3), and Appendix R (Module 4)** Section 5.1	**Core inclusion criterion #13 (Table 9, Adequate Treatment Washout Periods)**: Updated text for radiation therapy washout periods to specify "other *anatomical* areas". Added a new row for targeted agents and small molecules. Deleted Part 1 restriction for chemotherapy, immunotherapy, and hormonal therapy. Added hormone replacement therapy.	Updated in response to health authority feedback and to align with CTFG guidelines on contraception.
	Core inclusion criteria #15 and #16, Appendix H 1, Appendix H 2, and Appendix P (Module 2), Appendix Q (Module 3), Appendix R (Module 4) inclusion criterion #16: Added text noting that not engaging in sexual activity is an acceptable practice, provided it is the preferred and usual lifestyle choice of the participant.	Updated in response to health authority feedback and to align with CTFG guidelines of contraception.
	Core inclusion criterion #16, Section 8.3.14.2: Added the module numbers to the text for contraception duration for males after the last dose. **Core inclusion criterion #16, Core Section 8.3.14.2, and Appendix P (Module 2), Appendix Q (Module 3), Appendix R (Module 4) inclusion criterion #16:** Added text noting that preservation of sperm should be considered prior to enrolment in this study.	Updated in response to health authority feedback.
	Core inclusion criterion #17: Added text noting that preservation of ova should be considered prior to enrolment in this study.	Updated in response to health authority feedback.
	Core inclusion criterion #19(a): Deleted "prednisone ≤ 10 mg/day" and added "dexamethasone ≤ 2 mg/day".	Updated language for consistency with corticosteroid treatment for brain metastasis.
	Core inclusion criterion #20(a): Modified requirement for time since stereotactic radiosurgery or gamma knife treatment to be at least 14 days before first dose from prior requirement of at least 7 days prior to first dose	Updated for consistency of radiotherapy washout periods across inclusion criteria (Table 9).
Core CSP Section 5.2	**Exclusion criterion #5**: Added text for serological evidence of viral infection to confirm known HIV infection, or active hepatitis B or C infection. Added text noting that patients with past or resolved hepatitis B virus infection who are anti-HBc positive are eligible only if they are HBsAg negative.	Updated to clarify that serological evidence of viral infection is needed to confirm known HIV or active hepatitis B or C infection, and to allow patients with past or resolved hepatitis B virus infection who are anti-HBc positive and HBsAg negative to participate in the study.

Section #	Description of Change	Brief Rationale
	Exclusion criterion #12: Added text stating that mRNA and replication deficient adenoviral vaccines are not considered attenuated live vaccines.	Updated to clarify mRNA and replication deficient adenoviral vaccines are not considered live vaccines.
	Exclusion criterion #23: Deleted text excluding male or female patients of reproductive potential who are not willing to employ effective birth control from screening to 7 months after the last dose of study treatment.	Removed text that is intended to be covered by inclusion criteria #15 and #16. Exclusion criterion #23 is specific to female patients who are pregnant or breast feeding.
	Exclusion criterion #24: Deleted previous criterion 'Require therapy with warfarin or other coumarin derivatives (non-coumarin anticoagulants are welcome)'.	Removed restriction of warfarin or other coumarin derivatives for consistency with the study treatments administered (Table 29).
Core CSP Section 5.3	Added 'ova' to the specification for considering preserving sperm/ova before enrolment in the study.	Updated in response to health authority feedback.
Core CSP Section 6.3.1	Added 'pre-screening' in the description of investigators keeping a log of patients who enter the study. Added text confirming E-code assignment at pre-screening or screening.	Inclusion of an additional time period for the collection of tumour samples prior to study start, if needed.
Core CSP Section 6.6.1.1	Deleted text from the T-DXd dose modification details "Any dose delayed longer than 7 days should be withheld, and the patient should receive the next scheduled dose".	Removed to allow study treatment to resume after dose delays longer than 7 days.
Core CSP Section 7.1	Edited "Patients may be discontinued from the study…" to "Patients must be discontinued from the study…"	To provide more specific guidance on when a patient should be discontinued from study treatment and to align with edit to local CSP addendum language requested by PEI in Germany.
Core CSP Section 8.1.5.2 Section 8.1.5.3 Section 8.1.5.4	Added text noting that the assessments described (ie, MDASI symptom diary, at home pulse oximetry, and SGRQ-I) correspond to the study exploratory objective and will not be shared with the site during the course of the trial and that these data will not be used for any clinical care or treatment decisions during the trial, and the analyses results may be presented in a separate report (other than the CSR).	Clarification of principles of data sharing.
Core CSP Section 8.1.6.2	Added text noting that MDASI-BT assessments will only be completed by participants if a linguistically validated version is available in their language for the country in which they live.	Updated to reflect specificities of translation availability.

Section #	Description of Change	Brief Rationale
Core CSP Section 8.2.2 **Appendix O (Module 1), Appendix P (Module 2), Appendix Q (Module 3), Appendix R (Module 4),** Section 8.2.2	Updated text regarding the timing of vital sign assessments before and after each study treatment. Added core- and module-specific instructions for vital sign measurements on the first day of study treatment administration and following subsequent infusions of study treatment.	Revised instructions for vital sign and SpO_2 measurements to provide clarity on the timing of assessment before and after infusion of each study treatment.
Core CSP Section 8.2.4 Section 8.3.1 Section 8.3.6 Appendix E2 Appendix E3 Appendix L, Table 32 Toxicity Management Guidelines for T-DXd	Updated total bilirubin threshold for liver biochemistry from "> 2 × ULN" to "≥ 2 × ULN".	Updated in accordance with FDA guidance.
Core CSP Section 9.3, Table 17 Population for Analysis	Added description for the EAS for Part 1 population.	Extending efficacy reports to cover both Part 1 and Part 2 of the study for complete information.
	Addition of footnote 'b' for unscheduled PK assessments related to ILD/pneumonitis and intake of chloroquine or hydroxychloroquine for the PK Analysis set	Clarification about the statistical analysis for alignment with the SAP.
Core CSP Section 9.4.1, Table 18 Summary of Outcome Variables and Analysis Populations	Deleted 'Part 2' from the in-text description of the efficacy analyses performed on the FAS or EAS. Updated Table 18 to remove Part 2 from the efficacy data row; deleted ORR and DoR for the FAS and added ORR and DoR for the EAS.	Extending efficacy reports to cover both Part 1 and Part 2 of the study for complete information.
Core CSP Section 9.4.2.2.6	Deleted PFS2 from module subgroup analysis description. Added text for the summary method and presentation of subgroup analysis.	Number of PFS2 events are expected to be too small to support any formal statistical analyses.
Core CSP Appendix L, Table 32 Toxicity Management Guidelines for T-DXd	Deleted footnote CCI [redacted]	Updated Table 32 CCI [redacted]
Core CSP Appendix M 1 Appendix M 2	**Appendix M1**: Deleted "[nasopharyngeal swab/saliva]" from sample kit description. Deleted "Serum, [nasopharyngeal swab/saliva], and PK" from sample collection description.	Updated due to operational considerations.

Section #	Description of Change	Brief Rationale
	Appendix M2: Deleted text referring to chloroquine and hydroxychloroquine use for COVID-19.	During the study, chloroquine and hydroxychloroquine are considered prohibited medications.
Appendix O (Module 1) Section 2.2.1 Section 2.3.1 Section 4.2.3 **Appendix R (Module 4)** Section 2.2.1 Section 2.3.1	Deleted references to 'urothelial carcinoma' (Modules 1 and 4, Section 2.2.1), 'urothelial cancer' (Modules 1 and 4, Section 2.3.1), and 'bladder cancer' (Module 1, Section 4.2.3) as indications for durvalumab.	Alignment with current USPI for durvalumab.
Appendix O (Module 1) Section 6.6.2 **Appendix R (Module 4)** Section 6.6.2	Edited time between 2 consecutive doses of durvalumab (based on half-life) from not less than 21 days to not less than 19 days.	Revised to align time between 2 consecutive doses of durvalumab with T-DXd.
Appendix O (Module 1) Section 8.3.11.2 **Appendix R (Module 4)** Section 8.3.11.2	Deleted hormone replacement therapy from list of interventions for AESIs for durvalumab.	Updated in-line with the recommendation on the non-use of HRT in female breast cancer patients and to ensure patients do not receive HRT while on study treatment.
Appendix S (Module 5) Section 1.3, Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and Part 2) – Module 5 **Appendix T (Module 6)** Section 1.3, Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only) – Module 6	Edited footnote 'j' for pre-dose PK samples to specify that the sample should be taken immediately or up to 2 hours (previously stated as 6 hours) prior to tucatinib dose on D1 of C1, 2, and 4.	Based on the current understanding of tucatinib PK, a 6-hour pre-dose sampling window would provide sample outside of expected C_{min} range. Therefore, the sampling window was amended to up to 2 hours before the tucatinib dosing.
Appendix S (Module 5) Section 4.1.1	Correction to include dose level (-2) in-text for the description of patients who will be enrolled beyond the first cohort of [redacted] patients.	Typographical error excluded dose level (-2) from the description of dose levels evaluated beyond the first cohort of [redacted] patients.
Appendix T (Module 6) Section 5.2, exclusion criterion #27 **Appendix U (Module 7)** Section 5.2, exclusion criterion #27	Edited systemic corticosteroid use for control of brain metastasis symptoms from > 10 mg prednisone daily dose to > 2 mg dexamethasone daily dose.	Updated language for consistency with corticosteroid treatment for brain metastasis.
Appendix V	Added a separate appendix for prior protocol amendment history for amendments prior to the current version.	Updated in line with the latest late phase oncology CSP sponsor template.
General changes throughout CSP		
Throughout	Minor editorial and document formatting revisions.	Minor; therefore, have not been summarised.

Amendment 1 (15 December 2020)

Overall Rationale for the Amendment:

The primary change in Amendment 1 is to add 3 new modules. Module 6 and Module 7 will include a patient population with active brain metastases. Modules 5 and 6 will include treatment combination of T-DXd with tucatinib.

Additional changes implemented in this protocol amendment relate to updates to the objectives and endpoints, eligibility criteria and clarifications or corrections regarding study assessments and dosing. The rationale for each of these changes is provided in the table below.

Section # and Name	Description of Change	Brief Rationale
Substantial Changes		
Addition of information relating to new Module 5, **Core CSP-** throughout **Module 5** (Appendix S) – T-DXd + tucatinib.	Addition of tucatinib as a combination treatment.	To investigate the treatment of patients with stable or no brain metastases at baseline with T-DXd in combination with tucatinib.
Addition of information relating to new Module 6, **Core CSP-** throughout **Module 6** (Appendix T) – T-DXd + tucatinib in patients with active brain metastases	Addition of tucatinib as a combination treatment.	To investigate the treatment of patients with active brain metastases at baseline with T-DXd in combination with tucatinib.
Addition of information relating to new Module 7 **Module 7** (Appendix U) – T-DXd monotherapy in patients with active brain metastases	Addition of T-DXd monotherapy in patients with active brain metastases at baseline.	To investigate the treatment of patients with active brain metastases at baseline with T-DXd monotherapy.
Addition of tucatinib as a combination drug		
Core CSP **Section 2.3** Benefit/Risk Assessment **Section 4.2.3** Rationale for T-DXd Combinations **Appendix J: Concomitant Medications** **Appendix L:** Toxicity Management Guidelines	Addition of tucatinib as a combination drug (Modules 5 and 6) with additions to the core csp including supportive text for rationale, background, benefit risks and toxicities of tucatinib.	To investigate the treatment of patients with T-DXd in combination with tucatinib.

Section # and Name	Description of Change	Brief Rationale
Additional Clinical Outcome Assessments for patients with stable brain metastases		
Core CSP **Modules 0 to 4 Schedule of Activities During the Treatment Period** **Core CSP** **Section 8.1.6** Additional Clinical Outcomes Assessments **Section 9.4.4.4** Clinical Outcome Assessments **Appendix I** Patient-reported Outcomes	Addition of Questionnaires for CANTAB MDASI-BT NANO	Additional assessments to monitor patients with stable brain metastases at baseline more carefully.
Core CSP **Appendix N** RANO criteria for assessment of primary CNS tumours	Additional appendix. Patients with active brain metastases will be reviewed by RANO criteria.	Additional criteria relevant to inclusion of patients with active brain metastases at baseline (Modules 6 and 7).
Objectives and Endpoints		
Core CSP **Secondary objectives/ endpoints** **Section 3** **Table 6:** Objectives and Endpoints **Section 8.5.1** Pharmacokinetics **Section 9:** Statistical Considerations **Section 9.3** **Table 17** Populations for Analysis	To assess the PK of combination treatments.	Endpoints have been adjusted because of inclusion of new Modules 5,6, and 7.
Exploratory objectives/ endpoints **Section 9.4.4.4** Clinical Outcome Assessments **Section 9.4.4.5** BMFS **Section 9.4.4.6** CNS Efficacy Endpoints (Modules 6 and 7 Part 2 Only) **Section 9.4.4.7** Time to second progression beyond CNS progression	BMFS, added for patients who enrol without CCI (Modules 0-5). Additional CCI endpoints: CCI, CCI (per RECIST 1.1 and per RANO), time to next CCI therapy and time to CCI. Time from the date of randomisation to second progression and time from the date of the first isolated CCI to second progression for patients with CCI CCI at baseline.	Endpoints have been adjusted because of inclusion of new Modules 5, 6 and 7.
Study Population		
Section 5.1 Inclusion criteria	Inclusion criteria # 18 to 20 CNS Inclusion: Additional Inclusion criteria added for CNS.	Definition of active brain metastases added to Modules 6 and 7.

Section # and Name	Description of Change	Brief Rationale
Core CSP **Section 5.1 Inclusion criteria** **Appendix H1**	**Inclusion criteria 14 and 15** Updates made to contraception restrictions for female patients.	Alignment with current project requirements.
Modules 2, 3, and 4 **Section 5.1 Inclusion criteria** **Core CSP Appendix H2**	**Inclusion Criteria 16** Non-sterilised male patients who are sexually active with a female partner of child-bearing potential must use a condom with spermicide from screening to **6 months** after the final dose of study treatment.	Alignment with current project requirements.
Section 5.2 Exclusion Criteria	Exclusion criteria # 10 Additional criteria for CNS exclusions.	Definition of active brain metastases added to Modules 6 and 7.
	Exclusion criteria # 15 (b) Additional exclusion of prior treatment with tucatinib.	No prior treatment with tucatinib allowed in study.
	Exclusion criteria # 24 Exclusion of therapy with warfarin or other coumarin derivatives.	No exposure to warfarin allowed.
	Exclusion criteria # 25 Inability to swallow or absorb pills.	Patients will be potentially randomised to receive tucatinib tablets (Modules 5 and 6).
	Exclusion criteria # 26 Exclusion of recent use of strong CYP3A4 or CYP2C8 inhibitor or strong CYP3A4 or CYP2C8 inducers.	Alignment with guidance on tucatinib use.
	# 27 Inability to undergo contrast MRI of the brain.	To align with requirement for patients to undergo assessment.
Study Intervention		
Section 7.1: Discontinuation of Study Intervention **Section 7.1.3** Continuation on Study Treatment After CNS-Only Progression	Addition that those patients who experience isolated progression in the brain and undergo local therapy are allowed to continue study treatment until second progression.	To align with common procedures for patients with brain metastases.
Appendix J2 Restricted, Prohibited, and Permitted Concomitant Medications/Therapies	Additional text in Table 29 of guidance concerning supportive use of prophylactic anti-emetic agents.	To align with current project level text and to provide clarity.
Study Assessment		
Section 8.2.4 Clinical Safety Laboratory Assessments **Section 4.1.2.1** DLT Definitions **Appendix E** Time Period and Frequency for Collecting AE and SAE Information	Changes in the criteria for potential Hy's law cases for AST or ALT from > 3 × ULN to ≥ 3 × ULN.	Alignment with current project safety requirements.

Section # and Name	Description of Change	Brief Rationale
Synopsis	Updates made to be consistent with main body of CSP.	For consistency with main body of CSP.
Schema	Updated to reflect the additional Modules and additional endpoints.	For consistency with study design.
Module 1 and 4: **Table 2:** Schedule of Activities for Safety Follow-up and Long-term Follow-up Module 1.	Blood sampling frequency reduced for T-DXd and durvalumab ADA analysis in serum.	Alignment with current project requirements.
Core CSP Table 11 Laboratory Safety Variables	Addition of missing urinalysis assessments Bilirubin, Ketones, Colors and appearance, Glucose, pH. Addition of missing Gamma glutamyltransferase assessment	Alignment with current durvalumab project-specific safety requirements.
Core CSP **Modules 0 to 4** Schedule of Activities During the Treatment Period **Section 8.1.1** Imaging and Tumour Assessments	Addition of the requirement to perform a confirmatory scan of PD.	Alignment with current project requirements.
Core CSP Section 9.4.2.2.6	The addition of subgroup analysis by brain metastases status at baseline.	To investigate efficacy in patients with and without brain metastasis.
Non-substantial changes		
Modules 0 to 4 Schedule of Assessments During the Treatment Period	Change in scheduled ECHO/MUGA from 'C4 and subsequent cycles until PD' to 'C5 (and every 4 cycles)'.	Alignment with current project requirements.
Core CSP Schedule of Activities for Screening/Baseline - -All Modules **Modules 0 to 4** Schedule of Activities During the Treatment Period **Core CSP** **Section 8.1.1** Imaging Tumour Assessments	Removal of option for brain imaging to be performed by CT-imaging. The contrast will be gadolinium.	Better imaging quality with contrast MRI.
Modules 0 to 4 Schedule of Activities For Screening/Baseline – All Modules	Troponin assessments Removal of central troponin testing. All assessments will be done locally. For T-DXd monotherapy, troponin will be measured at screening, EOT and as clinically indicated. For combination modules, troponin will be measured at each visit in addition to screening and EOT.	Troponin is not being assessed centrally in this study. To align with current project-safety requirements.
Section 6.2.1 T-DXd Preparation and Administration	Addition of text around the preparation and storage of T-DXd.	Added in response to FDA feedback.

Section # and Name	Description of Change	Brief Rationale
Modules 1 and 4 **Section 6.6.2** Durvalumab	Addition of the instruction not to dose durvalumab less than 21 days after the previous dose of durvalumab, based on the durvalumab half-life.	Alignment with current durvalumab project-specific safety requirements.
Core CSP **Appendix M** Instructions Related to Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2/COVID-19)	Update to dose modification criteria in the event of SARS-CoV-2 infection, and addition of instructions for testing for SARS-CoV-2 in participants presenting with symptoms.	Alignment with T-DXd project safety requirements.
General changes throughout CSP		
Throughout	Minor changes and clarifications to CSP wording.	To align with project standard or to provide clarification.
Throughout	Minor editorial and document formatting revisions.	Minor; therefore, have not been summarised.

Appendix W Country-Specific Requirements

European Union Country Specific Requirements

W 1Germany

Section	Description of Change with Reason
NA	According to AstraZeneca's procedures the Clinical Study Protocol (CSP) version 1.0 from 29 Jun 2020 was signed by the sponsor electronically only. This Addendum to the CSP contains the wet-ink signatures of the sponsor and the National Co-Ordinating Investigator as requested by local regulatory authority.
NA	The dose-finding phase (Part 1) of the combination-treatment modules is not performed in Germany. German sites only take part in the dose-expansion phase (Part 2) of this study.
5.2 Exclusion Criteria	The following change implements a requirement from the responsible local Ethics committee: In accordance with the German drug law (AMG § 40 Abs. 1 S. 3 Nr. 4) patients are not eligible to participate in this trial if they are committed to an institution by virtue of an order issued by the judicial or the administrative authorities.
1.1 Protocol Synopsis 1.2 Schema (footnote) 4.1.1 Modular Protocol Structure Appendix S (Module 5) Appendix T (Module 6)	Treatment modules 5 and 6 with Tucatinib will not be opened in Germany until CMC documentation and labels for Tucatinib are submitted and approved.
1.3 Schedule of Activities 8.2.5 Echocardiograms/Multi-gated Acquisition Scan 8.3.5 Adverse Events Based on Examinations and Tests 8.3.11 Adverse Events of Special Interest 8.3.16.1 Specific Toxicity Management and Dose Modification Information - T-DXd 9.4.3 Safety	LVEF will be measured only by echocardiogram (ECHO). Multiple gated acquisition (MUGA) scan will not be allowed in Germany to avoid unnecessary exposure to radiation. This has already been discussed during previous studies and is in accordance with Federal Office for Radiation Protection (BfS) in Germany

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Clinical Study Protocol	
Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix O

Module 1 – T-DXd + Durvalumab

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

EU CT: 2023-505309-18-00

This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

PROTOCOL AMENDMENT SUMMARY OF CHANGES TABLE

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Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 1 of this study will evaluate the PD-L1 inhibitor durvalumab in combination with T-DXd in patients with HER2-positive advanced/MBC. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2. The RP2D for Part 2 of this module may come from BEGONIA (NCT03742102), a 1L study in metastatic TNBC patients with HER2-low expression (defined as IHC 2+/ISH- and IHC 1+).

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 1 Table 1) and follow-up period (Mod 1 Table 2) for Module 1 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.5	
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
Weight	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
ECOG PS	X [b]				X[b]		X[b]		X [b]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [c]													
Pregnancy test [d]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [b]		X	X	X[b]		X[b]		X [b]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3	

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
ECHO/MUGA (LVEF) [f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [g] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [s]	As clinically indicated										X	8.2.5.4	
TSH (reflex-free T3 or free T4)	X[b]				X[b]		X[b]		X[b]		X	Table 11	
Amylase and lipase	X[b]				X[b]		X[b]		X[b]		X	Table 11	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycles 4, 6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for durvalumab PK testing [j]	X				X				X (Cycles 4, 6, and 8)				8.5.1
Post-dose blood sample for durvalumab PK testing [k]		X				X				X (C4 only)			8.5.1

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for T-DXd ADA assessment in serum[l]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Blood sample for durvalumab ADA assessment in serum[l,r]	X				X				X (C4, then every 4 cycles)		X		8.5.2
Study treatments administered													
T-DXd	CCI or 5.4 mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Durvalumab	CCI mg will be administered intravenously on D1 of each CC week cycle												6.6.2
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following										X	7.1.2, 8.1.1	

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	investigator determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.												
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [m] (if not collected at screening)	X [n]											8.6.1, 8.6.2	
Paired on-treatment biopsy					X[n]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	
Blood sample for CCI [redacted]	CCI [redacted]											8.6.1	

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
									RECIST assessments)				
Blood sample CCI	CCI											8.6.1	
Blood sample for gene CCI CCI												8.6.1	
Blood sample for DNA/CCI CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI CCI [o]												8.6.1	
Blood sample to analyse CCI CCI												8.6.1	

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	CCI [redacted]												
Blood sample for CCI [redacted] (CCI [redacted]												8.7	
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [q]	X											8.1.5.1	
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6, and Month 12 for patients with stable brain metastases at baseline)		X	8.1.6.1	

Mod 1 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 1

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2	
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening laboratory assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.
[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd.
[i] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.

[j] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for durvalumab.
[k] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for durvalumab.
[l] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[m] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the core CSP Section 8.6.1 and 8.6.2).
[n] Refer to the core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[o] At time of suspected CCI an additional sample should be taken.
[p] The sample for genetic research will be obtained at CCI. If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI. Only CCI CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.
[q] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use in case needed, in accordance with device training.
[r] ADA for durvalumab need not to be collected after safety follow-up when subject starts another therapy for cancer.
[s] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in the core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB = Cambridge Neuropsychological Test Automated Battery; CNS = central nervous system; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI; D = day; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; T3 = triiodothyronine; T4 = thyroxine; CCI CCI T-DXd = trastuzumab deruxtecan; TSH; thyroid stimulating hormone.

Mod 1 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 1

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
Vital signs	X				8.2.2	
Weight	X	X	X		8.2.1	
Full physical examination	X				8.2.1	
SpO_2	X				8.2.5.2	
ECOG PS	X	X	X		8.2.5.3	
12-lead ECG	X				8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X	X	X		8.2.4, Table 11	
Troponin [c]	X	X	X		Table 11	
Urinalysis	As clinically indicated				Table 11	
Coagulation tests	As clinically indicated				Table 11	
Pregnancy test	X	As clinically indicated			8.2.4	
Blood sample for T-DXd ADA analysis in serum	X				8.5.2	
Blood sample for durvalumab ADA analysis in serum [d]			X			8.5.2
Concomitant medications	X	X	X		6.5	
AEs/SAEs	X	X	X		8.3	
PFS2 assessment	X	X	X	X	7.1.2.1, 8.1.3	
Health care resource use (HOSPAD)	X				8.8	

Mod 1 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 1

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: To be completed daily up to and including the first survival visit.				8.1.5.2	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit				8.1.5.3	
MDASI-BT			X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)		8.1.6.2	
NANO			X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)		8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.				8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)				8.1.5.5	
Survival status	X	X	X	X	7.1.2.1, 8.1.4	

Mod 1 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 1

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
Subsequent cancer therapy following discontinuation of study treatment	X	X	X	X	8.1.4	
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.				8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.				8.1.1	

[a] If EOT is > 30 days (± 3 days) after last treatment, then the EOT assessments can also function as the 30-day (± 3 days) follow-up visit. Note that as durvalumab is used in this module, the safety follow-up period is 90 days after last dose.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.
[d] ADA for durvalumab need not be collected after safety follow-up when subject starts another therapy for cancer.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

The development of immune checkpoint inhibitors is an important milestone in the field of oncology. Tumour cells evade immunosurveillance and progress through different mechanisms, including activation of immune checkpoint pathways that suppress anti-tumour immune responses. Immune checkpoint inhibitors reinvigorate anti-tumour immune responses by interrupting co-inhibitory signalling pathways and promote immune-mediated elimination of tumour cells. Immune checkpoint inhibitors are now approved for the treatment of a broad range of tumour types, with additional indications expected in the near future (Wei et al 2018).

Several ongoing clinical trials are investigating the use of immune checkpoint inhibitors with PD-1 and PD-L1 inhibitors for triple-negative breast cancer (Schmid et al 2018, Schmid et al 2020). However, the role of PD-L1 expression in HER2-positive breast cancer remains unclear. Retrospective studies have investigated the relationship between PD-L1 status and clinical outcomes, suggesting that PD-L1 expression may be associated with response to neoadjuvant chemotherapy with trastuzumab in patients with HER2-positive breast cancer (Kurozumi et al 2019). Addition of immune checkpoint inhibitors to HER2-targeted therapy has shown promise in HER2-positive MBC. Studies adding pembrolizumab (Loi et al 2019) and atezolizumab (Emens et al 2019a) to SOC treatments have shown improved efficacy in a PD-L1-positive patient population. T-DXd is currently being explored in study DS8201-A-U105 (NCT03523572), a Phase 1b, multicentre, open-label, non-randomised study in combination with nivolumab (an anti-PD-1 antibody) in patients with HER2-expressing advanced breast cancer.

More studies are needed to confirm these observations and to understand the role of PD-L1 in HER2-positive breast cancer.

2.2.1 Durvalumab

Durvalumab is a mAb of the IgG 1 kappa subclass that blocks the interaction of PD-L1 (but not programmed cell death ligand-2) with PD-1 on T cells and CD80 (B7.1) on immune cells. It is being developed by AstraZeneca for use in the treatment of cancer. The proposed MOA for durvalumab is interference in the interaction of PD-L1 with PD-1 and CD80. Blockade of PD-L1/PD-1 and PD-L1/CD80 interactions releases the inhibition of immune responses, including those that may result in tumour elimination. *In vitro* studies demonstrate that durvalumab antagonises the inhibitory effect of PD-L1 on primary human T cells resulting in the restored proliferation of interferon gamma (Stewart et al 2015). *In vivo* studies have shown

that durvalumab inhibits tumour growth in xenograft models via a T-cell dependent mechanism (Stewart et al 2015). Based on these data, durvalumab is expected to stimulate the patient's anti-tumour immune response by binding to PD-L1 and shifting the balance toward an anti-tumour response. Durvalumab has been engineered to reduce antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity.

Durvalumab is approved in some countries as monotherapy for unresectable Stage III NSCLC (following chemoradiation therapy), and in combination with chemotherapy for extensive-stage small cell lung cancer and for locally advanced or metastatic biliary tract cancer (Imfinzi 2022). Durvalumab is also approved when administered in combination with tremelimumab for unresectable hepatocellular carcinoma, and also in combination with tremelimumab plus chemotherapy for participants with metastatic NSCLC without EGFR or ALK mutations.

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Durvalumab

Durvalumab is a PD-L1 inhibitor, which has shown generally good tolerability (Antonia et al 2017) and has been approved for the treatment of NSCLC and SCLC (Antonia et al 2017, Paz-Ares et al 2019). Studies examining durvalumab in TNBC have shown improved outcomes with a favourable profile (Dalenc et al 2019, Loibl et al 2019). Pre-clinical data show that T-DXd in combination with a check point inhibitor increases tumour growth inhibition compared with T-DXd or check point inhibitor monotherapy in a syngeneic mouse model (T-DXd- IB). The anti-tumour efficacy of cancer immunotherapy, through enhancement of T-cell mediated cytotoxicity, has been shown to be improved by topoisomerase I inhibitors (Antonia et al 2017), therefore, the combination of durvalumab therapy and T-DXd should be investigated.

2.3.2 Potential Risks of T-DXd + Durvalumab

For information on all identified and potential risks with durvalumab, please refer to the current version of the durvalumab IB.

At this time, there are limited safety data regarding the combination of durvalumab and T-DXd. T-DXd and durvalumab share common overlapping toxicities, such as the risks of ILD/pneumonitis and diarrhoea/colitis, which may theoretically be potentiated in the combination arm. Although there are no data on T-DXd and durvalumab in combination, the ongoing combination study of T-DXd and CCI is fully enrolled, has been CCI (unpublished data). This platform study design aims to minimise potential risks of this novel treatment combination. Careful monitoring of pulmonary function to evaluate for ILD/pneumonitis will be carried out throughout the study. Early intervention for immune-mediated diarrhoea/colitis

will be applied. TMGs are provided in Section 8.3.16 in this module, and in the core CSP Appendix L, with further guidance for Investigators available in the T-DXd and durvalumab IBs.

In addition, monoclonal antibodies are not metabolised through classic hepatic enzyme pathways; therefore, no PK interaction is anticipated.

2.3.3 Overall Benefit/Risk Conclusion

The strong efficacy data obtained with T-DXd monotherapy in the heavily pretreated patients with HER2-positive tumours in DESTINY-Breast01 (Modi et al 2020) warrants exploration of T-DXd combinations. Combinations with the immune checkpoint inhibitors are particularly intriguing, as they may provide durable clinical benefit without adding the side-effects seen with combinations of multiple cytotoxic drugs (Chai et al 2019). In breast cancer, recent studies combining chemotherapy and immunotherapy in TNBC have demonstrated added efficacy without significantly impairing the tolerability (Schmid et al 2018, Schmid et al 2020). Therefore, an investigation of T-DXd and durvalumab in HER2-positive advanced/MBC is warranted.

There is potential when combining T-DXd and durvalumab for overlapping toxicities, such as ILD/pneumonitis and diarrhoea/colitis. However, mitigations for early detection and management of ILD/pneumonitis have been included in this protocol. Patients with prior pulmonary co-morbidities will be excluded from the study. Patients will also be undergoing pulmonary function tests to reveal occult pulmonary diseases, Investigators will be trained to promptly identify early signs of ILD/pneumonitis and commence appropriate management. TMGs have been amended to focus on prompt treatment of ILD/pneumonitis. Similarly, TMGs have been applied for the treatment of diarrhoea/colitis from both T-DXd and durvalumab perspective.

Taken together, the sponsor considers that the anticipated clinical benefit with adding durvalumab to T-DXd will outweigh potential risks. The safety precautions built into this protocol will maximise the likelihood of a favourable benefit/risk profile.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC.

For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients will be treated with T-DXd + durvalumab.

4.1.1 Module 1 Part 1

The combination of T-DXd (5.4 mg/kg Q3W) and durvalumab (1120 mg Q3W) is being studied in TNBC patients in the BEGONIA clinical trial (NCT03742102). If the RP2D is established after review by the SRC in the BEGONIA study, Module 1, Part 1 will not be performed in this study.

In Part 1, if performed in this study, approximately [redacted] to [redacted] patients with locally assessed HER2positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled in each dose level to determine the RP2D for the T-DXd + durvalumab combination. Dose escalation/de-escalation will be determined by using a mTPI-2 algorithm (refer to the core CSP Table 7). In Part 1 in this module, [redacted] patients will be enrolled at dose level (1) (Mod 1 Table 3). Dose level (1) will be defined as the RP2D if no more than one DLT is observed from the first [redacted] patients. The dose will be de-escalated to level (-1) if there is evidence indicating intolerable toxicity with dose level (1). Beyond the first batch of [redacted] patients, patients will be enrolled in batches of [redacted] at dose level (1) or (-1) following the mTPI-2 algorithm. Dose level decisions and determination of the RP2D will be made by the SRC (core CSP Section 9.6). Additional details on the dose-finding period and combination dose-finding approach are provided in the core CSP Section 4.1.1.

Mod 1 Table 3 Module 1 Part 1, Dose Finding

mTPI-2 escalation/de-escalation	
Dose Level (1)	**Dose Level (-1)**
T-DXd dose level (1): 5.4 mg/kg IV infusion Q3W Durvalumab [redacted] mg IV Q[redacted]W	T-DXd dose level (-1): [redacted] mg/kg IV infusion Q3W Durvalumab [redacted] mg IV Q[redacted]W

IV = intravenous; mTPI-2 = modified toxicity probability index-2; Q3W = every 3 weeks; T-DXd = trastuzumab deruxtecan.

4.1.2 Module 1 Part 2

In Part 2, approximately 40 to 75 patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/MBC (ie, 1L) will be enrolled to determine evidence of anti-tumour activity. Patients will be treated with T-DXd + durvalumab at the RP2D from Part 1 (Mod 1 Table 4).

Mod 1 Table 4 Module 1 Part 2, Dose Expansion

Study Treatment	Dose
T-DXd [a]	5.4 mg/kg or CCI mg/kg IV infusion Q3W
Durvalumab	CCI mg IV Q[redacted]W

[a] The final dose will be based on the RP2D from BEGONIA, or as determined from Part 1 of this module.
IV = intravenous; Q3W = every 3 weeks; RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

4.2.3 Rationale for T-DXd + Durvalumab in HER2-positive Advanced/MBC

Tumour mutational burden, the presence of tumour infiltrating lymphocytes, and PD-L1 expression have been shown to be prognostic factors and predictors of response to immune checkpoint inhibitors in breast cancer (Emens et al 2019b, Luen et al 2017, Schmid et al 2018, Schmid et al 2020). These factors are generally more prevalent among triple negative- and HER2-positive breast cancers than in hormone receptor-positive tumours. While the role of PD-L1 expression is still under investigation in HER2-positive breast cancer, clinical studies have already shown improved efficacy in HER2-positive advanced breast cancer when immune checkpoint inhibitors have been added to SOC (Emens et al 2019b, Loi et al 2019). Additionally, preclinical data show that T-DXd in combination with an immune checkpoint inhibitor enhances tumour growth inhibition compared with either agent alone in a syngeneic mouse model (T-DXd IB). In addition, administration of T-DXd resulted in upregulation of PD-L1 and major histocompatibility complex Class I on tumour cells *in vivo* (Iwata et al 2017). Thus, HER2-positive breast cancer is an important setting in which to explore the efficacy of cancer immunotherapy (Holgado et al 2018).

Immune checkpoint inhibitors, such as durvalumab, are ideal partners for combination therapy, with likely added durability of benefit without significant added toxicities (Domchek et al 2019). Durvalumab is a PD-L1 inhibitor, which is generally well tolerated (Antonia et al 2017) and has been approved for the treatment of NSCLC and SCLC (Antonia et al 2017, Paz-Ares et al 2019); early phase studies have been completed examining durvalumab in TNBC (Dalenc et al 2019, Loibl et al 2019). The anti-tumour efficacy of cancer immunotherapy, through enhancement of T-cell mediated cytotoxicity, has been shown to be improved by topoisomerase I inhibitors (McKenzie et al 2018), therefore the combination of durvalumab therapy with T-DXd should be investigated.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP Section 4.3.1.

4.3.2 Durvalumab Dose Rationale

Based on available PK and safety data for weight-based doses of 10 mg/kg Q2W, 20 mg/kg Q4W, and simulated PK data for fixed doses of 750 mg Q2W and 1500 mg Q4W, the fixed-dosing regimen was predicted to provide comparable exposure to the body weight based dosing regimens. The proposed dosing schedule of CCI mg Q[CCI]W is aligned with the standard fixed dosing of 1500 mg durvalumab Q4W, which is supported by efficacy, safety, and tolerability data across multiple studies in multiple tumour types. To conform to the T-DXd schedule in this module, the Sponsor proposes to use a standard durvalumab combination therapy dose and ratio at a Q[CCI]W dosing schedule.

The safety of a Q3W dosing schedule of durvalumab in combination with chemotherapy has been explored in the ongoing Study NCT02658214, where tremelimumab is administered at 75 mg in combination with 1120 mg durvalumab Q3W, followed by 1120 mg durvalumab Q3W. The combination was considered tolerable and manageable. The 1120 mg dose of durvalumab is the Q3W equivalent of the standard 1500 mg Q4W dose.

Durvalumab has a flat exposure-response (efficacy and safety) relationship. NCT02537418 is an ongoing Phase 1 study of durvalumab ± tremelimumab in combination with multiple standard platinum-based chemotherapy regimens in patients with incurable advanced or metastatic cancer. The dose regimens in the study included a fixed dose cohort of durvalumab 1125 mg + tremelimumab 56 mg Q3W concurrent with platinum-based doublet chemotherapy, and subsequently a cohort of durvalumab 1500 mg + tremelimumab 75 mg Q3W concurrent with chemotherapy. In general, all regimens were tolerable and manageable at all dose levels. In terms of the safety data reported, there was no evidence of a dose response, including in the 1500 mg + 75 mg dose cohort (Juergens et al 2017).

In addition, PK modelling has been carried out to predict the effect of switching from a Q4W regimen to a Q3W regimen for durvalumab at a 1500 mg fixed dose and suggests that a Q3W schedule does not impose a significant increased safety risk based on expected durvalumab exposures.

A dose of CCI mg rather than CCI mg is being used for practical reasons (ie, due to the vial sizes available).

For additional details on the preclinical and clinical data that informed durvalumab dose selection, refer to the current durvalumab IB. For information on dose modifications for durvalumab, see Section 6.6 in this module.

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with durvalumab.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.6 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for durvalumab. The study treatments in this module are shown in Mod 1 Table 5.

Mod 1 Table 5 Study Treatments in Module 1

Intervention Name	T-DXd	Durvalumab
Type	Refer to the core CSP Table 10	Monoclonal antibody
Dosage Presentation		Vial
Unit Dose Strength(s)		Concentrate for solution for infusion, CCI mg/mL (CCI mg/vial)
Dosage Level(s)		CCI mg
Route of Administration		IV
Use		Experimental
Sourcing		Provided centrally by the Sponsor
Packaging and Labelling		Study Treatment will be provided in multiple vials in carton. Each of the vials and carton will be labelled as required per country requirements [a]

[a] Label text prepared for durvalumab (MEDI4736) will show the product name as "MEDI4736" or "durvalumab (MEDI4736)" depending upon the agreed product name used in the approved study master label document. All naming conventions are correct during this transitional period.

CSP = clinical study protocol ; IV = intravenous; T-DXd = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration and monitoring.

T-DXd will be administered first. Durvalumab (see Section 6.2.2 in this module) will be administered approximately one hour (maximum 2 hours) after the end of the infusion of T-DXd. If there are no clinically significant infusion reactions within the first cycle, and at the discretion of the Investigator, then for all other cycles, durvalumab can be given no less than 15 minutes after the infusion of T-DXd has finished.

6.2.2 Durvalumab Preparation and Administration

Durvalumab will be supplied by AstraZeneca as a CCI mg vial concentrate for solution for infusion. The solution contains CCI mg/mL durvalumab, CCI mM histidine/ histidine hydrochloride, CCI mM trehalose dihydrate, and CCI % (w/v) CCI ; it has a pH of CCI and density of CCI g/mL. The label-claim volume is CCI mL.

Durvalumab is a sterile, clear to opalescent, colourless to slightly yellow solution, free from visible particles. Each vial selected for dose preparation should be inspected.

Preparation of Durvalumab Doses for Administration with an IV Bag

The dose of durvalumab for administration must be prepared by the Investigator or site's designated study intervention manager using aseptic technique and following local practices and site requirements. The total time from needle puncture of the durvalumab vial to the start of administration must not exceed 24 hours. Of this time, not more than 4 hours may be at room temperature, with the remaining time at 2°C to 8°C (36°F to 46°F), otherwise a new dose must be prepared from new vials.

Administration of Durvalumab

A dose of CCI mg will be administered using an IV bag containing 0.9% (w/v) saline or 5% (w/v) dextrose, with a final durvalumab concentration ranging from 1-15 mg/mL, and delivered through an IV administration set with a 0.2 or 0.22 µm filter. Add CCI mL (ie, CCI mg) of durvalumab to the IV bag. The IV bag size should be selected such that the final concentration is within 1-15 mg/mL. The bag should be mixed by gently inverting to ensure homogeneity of the dose in the bag.

The use of elastometric pumps and use of a pneumatic tube for transport of either a vial or a final prepared product have not been studied; please contact AstraZeneca before using.

Standard infusion time for durvalumab is 1 hour (± 10 minutes), however if there are interruptions, the total allowed time should not exceed 8 hours at room temperature.

Do not co-administer other drugs through the same infusion line.

The IV line will be flushed according to local practices to ensure the full dose is administered. Infusion time does not include the final flush time.

If either preparation time or infusion time exceeds the time limits, a new dose must be prepared from new vials. Durvalumab does not contain preservatives; any unused portion of the vial must be discarded immediately after use.

6.2.3 Storage

For T-DXd storage information refer to the core CSP Section 6.2.2.

Durvalumab vials are stored at 2°C to 8°C (36°F to 46°F) and must not be frozen. The investigator, or an approved representative (eg, pharmacist), will ensure that all study intervention is stored in a secured area, in refrigerated temperatures (2°C to 8°C; 36°F to 46°F) and in accordance with applicable regulatory requirements. A temperature log will be used to record the temperature of the storage area. Temperature excursions outside the permissible range listed in the clinical supply packaging are to be reported to the monitor upon detection. A calibrated temperature monitoring device will be used to record the temperature conditions in the drug storage facility. Storage conditions stated in the IB may be superseded by the label storage.

Study intervention must be kept in original packaging until time of preparation to prevent prolonged light exposure.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after patients have completed 8 or more cycles of T-DXd for patients with HR-positive tumours, as per local practice.

Restricted, prohibited, and permitted concomitant medications are described in the core CSP Appendix J. Refer also to the core CSP Section 6.5 for further details.

6.5.1 Rescue Medication

As a result of imAEs that could potentially be experienced by patients on durvalumab, steroids and other immunosuppressant rescue medication has to be made available to this patient population. The 2 products that fall into the category of immunosuppressants are infliximab

(eg, for colitis) and mycophenolate (eg, for hepatitis). AstraZeneca supply chain will be responsible for sourcing these 2 rescue medications to the sites if local regulations prevent the use of infliximab and mycophenolate in this indication. These rescue medications must be receipted, controlled, and dispensed by the pharmacist and stored according to the labelled storage conditions, with temperature excursions reported accordingly.

If these rescue medications are centrally sourced: If required for use as a result of an imAE, then the IRT will provide to the pharmacists the kit identification number to be allocated to the patient at the time.

6.6 Dose Modification

6.6.1 T-DXd

The combination of T-DXd and durvalumab may increase the risk or severity of potential overlapping toxicities. Furthermore, there is potential for new, unanticipated toxicities. The Investigator should use clinical judgement when attributing a toxicity to one or more of the agents. For dose modifications that apply to toxicities that are attributable to T-DXd refer to the core CSP Section 6.6.1.

6.6.2 Durvalumab

Dose delays are permitted for durvalumab (see Section 8.3.16.2 in this module). However, dose reduction is not permitted. Dose delays may be entered into the eCRF as unscheduled visits.

Subsequent time between 2 consecutive doses cannot be less than 19 days, based on the half-life of durvalumab.

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 in this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1.

7.1.1 Procedures for Discontinuation of Study Treatment

Refer to the core CSP Section 7.1.1.

7.1.2 Follow-up of Patients Post Discontinuation of Study Treatment

The follow-up period for durvalumab is 90-days after last dose of durvalumab. The follow-up period for T-DXd is 40 days (+ 7days) after last dose of T-DXd.

Additional assessments to be performed at the time of the 90-day safety follow-up are detailed in the SoA in Section 1.3.

Refer to the core CSP Section 7.1.2 and to the SoA in Section 1.3 for more information regarding follow up of patients to PFS2 and for follow-up for survival.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for module specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1 for all subsections.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.2.2 Vital Signs

Vital signs will be performed at timepoints as specified in the SoA. Temperature, pulse rate, respiratory rate, and blood pressure will be assessed.

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the patient in a quiet setting without distractions (eg, television, cell phones).

Vital signs will be evaluated and recorded in eCRF. Standard infusion time for T-DXd is approximately 90 minutes for first infusion and 30 minutes thereafter if the first infusion was well tolerated; however, if there are interruptions during infusion, the total allowed infusion time should not exceed 3 hours at room temperature.

If the infusion takes longer than 90 minutes, then blood pressure and pulse measurements should follow the principles as described below or be taken more frequently if clinically indicated. An hour observation period is recommended after the first infusion of T-DXd and the combination partner treatments that are administered intravenously.

First Infusion of Study Treatment

On the first day of study treatment administration, patients' vital signs will be monitored and collected/recorded in the eCRF before and after infusion of each study treatment as follows:

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 30 minutes ± 10 minutes after end of the T-DXd infusion (during the 1-hour observation period after the infusion). Additionally, approximately 30 minutes ± 10 minutes after end of the durvalumab infusion (during the 1-hour observation period after the durvalumab infusion).

Subsequent Infusions of Study Treatment

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 7 minutes ± 5 minutes after end of the T-DXd infusion (during the 15-minute observation period after the infusion). Additionally, approximately 15 minutes ± 5 minutes after end of the durvalumab infusion (during the 30-minute observation period after the durvalumab infusion).

Vital signs during initial and subsequent T-DXd and durvalumab infusions should be monitored, and any clinically significant findings at other times to those described above should be reported via an unscheduled visit form. Any changes in vital signs should be recorded as an AE (if applicable). Situations in which vital signs results should be reported as AEs are described in core CSP Section 8.3.5.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2.

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

8.3.11.1 T-DXd

Refer to the core CSP Section 8.3.11 for AESIs related to T-DXd.

8.3.11.2 Durvalumab

AESIs for durvalumab include, but are not limited to, events with a potential inflammatory or immune-mediated mechanism and which may require more frequent monitoring and/or interventions such as steroids, and/or immunosuppressants. These AESIs are being closely monitored in clinical studies with durvalumab monotherapy and combination therapy. An imAE is defined as an AE that is associated with drug exposure and is consistent with an immune-mediated MOA and where there is no clear alternate aetiology. Serologic, immunologic, and histologic (biopsy) data, as appropriate, should be used to support an imAE

diagnosis. Appropriate efforts should be made to rule out neoplastic, infectious, metabolic, toxin, or other aetiologic causes of the imAE.

If the Investigator has any questions regarding an event being an imAE, the Investigator should promptly contact the Study Physician.

AESIs observed with anti-PD-L1/PD-1 agents such as durvalumab include the following:

- Diarrhoea/colitis and intestinal perforation
- Pneumonitis/ILD
- Endocrinopathies (ie, events of hypophysitis/hypopituitarism, adrenal insufficiency, hyper and hypothyroidism, and Type I diabetes mellitus)
- Hepatitis/transaminase increases
- Nephritis/blood creatinine increases
- Pancreatitis/serum lipase and amylase increases
- Rash/dermatitis
- Myocarditis
- Myositis/polymyositis
- Neuropathy/neuromuscular toxicity (eg, Guillain-Barré syndrome and myasthenia gravis)
- Other inflammatory responses that are rare/less frequent with a potential immune-mediated aetiology include, but are not limited to, pericarditis, sarcoidosis, uveitis, and other events involving the eye, skin, haematological, and rheumatological events, vasculitis, non-infectious meningitis and non-infectious encephalitis. It is possible that events with an inflammatory or immune-mediated mechanism could occur in nearly all organs.

In addition, infusion-related reactions and hypersensitivity/anaphylactic reactions with a different underlying pharmacological aetiology are also considered AESIs.

Further information on these risks (eg, presenting symptoms) can be found in the current durvalumab IB. More specific guidelines for their evaluation and treatment are described in detail in Section 8.3.16 in this module. These guidelines have been prepared by the Sponsor to assist the Investigator in the exercise of his/her clinical judgment in treating these types of toxicities. These guidelines apply to AEs considered causally related to the study treatment by the reporting Investigator.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12. All SAEs that occur in patients still receiving study treatment within 90 days following the last dose of durvalumab after the final DCO must be reported as detailed in the core CSP Section 8.3.13. If durvalumab is discontinued and the patient continues to receive T-DXd, SAEs within 40 days (+ 7 days) following the last dose of T-DXd after the final DCO must be reported as detailed in the core CSP Section 8.3.13.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd
- The current IB for durvalumab

8.3.14 Pregnancy

Refer to the core CSP Section 8.3.14.

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in the core CSP Appendix K, and the T-DXd TMGs in the core CSP Appendix L.

8.3.16.2 Toxicity Management and Dose Modification Information – Durvalumab

Comprehensive TMGs have been developed to assist Investigators with the recognition and management of toxicities associated with the use of the immune-checkpoint inhibitor durvalumab (PD-L1 inhibitor). Additionally, these guidelines are applicable when durvalumab is used alone or in combination and is administered concurrently or sequentially with other anti-cancer drugs (ie, antineoplastic chemotherapy, targeted agents), as part of a protocol specific treatment regimen. The TMGs provide information for the management of immune-mediated reactions, infusion-related reactions, and non-immune-mediated reactions that may be observed with checkpoint inhibitor monotherapy or combination checkpoint inhibitor regimens, with specific instructions for dose modifications (including discontinuations) and treatment interventions. Investigators are advised, however, to use local practice guidelines and consult local references for the management of toxicities observed with other cancer treatment.

The most current version of the TMGs entitled "Dosing Modification and Toxicity Management Guidelines for Immune-Mediated, Infusion-Related, and Non-Immune Mediated Reactions of Durvalumab Monotherapy or Combination Therapy with Tremelimumab or Tremelimumab Monotherapy" is provided to the investigative site as an Annex document and is maintained within the Site Master File.

Dose modifications with regards to hold or discontinuation of durvalumab are required actions for the management of potential immune-mediated events or non-immune-mediated adverse reactions, as described in the Dosing Modification and Toxicity Management Guideline; permanent discontinuation of study treatment for these adverse events is considered a study-specific requirement.

Patients should be thoroughly evaluated and appropriate efforts should be made to rule out neoplastic, infectious, metabolic, toxic, or other etiologic causes of the imAE. Serologic, immunologic, and histologic (biopsy) data, as appropriate, should be used to support an imAE diagnosis. In the absence of a clear alternative aetiology, events should be considered potentially immune-related.

Following the first dose of study treatment, subsequent administration of durvalumab can be modified based on toxicities observed as described in the Dosing Modification and Toxicity Management Guidelines. These guidelines have been prepared by the Sponsor to assist the Investigator in the exercise of his/her clinical judgment in treating these types of toxicities. These guidelines apply to AEs considered causally related to durvalumab by the reporting Investigator.

In addition, there are certain circumstances in which durvalumab should be permanently discontinued (see the core CSP Section 7.1).

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of durvalumab in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd and durvalumab, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total anti-HER2 antibody, MAAA-1181a and durvalumab in serum at timepoints as specified in the SoA in this module. Additional PK blood samples will be collected for patients who receive

treatment with chloroquine/hydroxychloroquine during the treatment period of the study. Refer to the core CSP Module M for further details.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd and durvalumab in serum will be collected as specified in the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9.

9.1 Statistical Hypotheses

Refer to the core CSP Section 9.1.

9.2 Sample Size Determination

Refer to the core CSP Section 9.2.

9.3 Populations for Analyses

Refer to the core CSP Section 9.3.

9.4 Statistical Analyses

Refer to the core CSP Section 9.4.

9.5 Interim Analyses

For this module, an interim analysis will be conducted in Part 2 as outlined in the core CSP Section 9.5. Enrolment will not be halted while waiting for the first 20 evaluable patients to reach the 2-month landmark. No more than 40 patients will be enrolled before the interim

analysis is completed. An SRC will review the safety data from this study, as well as that from HUDSON (NCT03334617), BEGONIA (NCT03742102), and DS8201-A-U105 (NCT03523572). The totality of data, and pre-specified criteria, will be considered in the decision to enrol more patients.

9.6 Data Monitoring Committee

Refer to the core CSP Section 9.6.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol

Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix P

Module 2 – T-DXd + Pertuzumab

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

EU CT: 2023-505309-18-00

This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

PROTOCOL AMENDMENT SUMMARY OF CHANGES TABLE

DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 2 of this study will evaluate the anti-HER2 humanised mAb pertuzumab in combination with T-DXd in patients with HER2-positive advanced/MBC. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2.

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 2 Table 1) and follow-up period (Mod 2 Table 2) for Module 2 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.5	
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [b]				X [b]		X [b]		X [b]		X	8.2.1	
Weight	X [b]				X [b]		X [b]		X [b]		X	8.2.1	
ECOG PS	X [b]				X [b]		X [b]		X [b]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [c]													
Pregnancy test [d]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [b]		X	X	X [b]		X [b]		X [b]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3	

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
ECHO/MUGA (LVEF) [f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [g] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [s]	As clinically indicated										X	8.2.5.4	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycles 4, 6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for pertuzumab PK testing [j]	X				X				X (C4 only)				8.5.1
Post-dose blood sample for pertuzumab PK testing [k]		X				X				X (C4 only)			8.5.1

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for T-DXd ADA assessment in serum [l]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Blood sample for pertuzumab ADA assessment in serum [l,r]	X				X				X (C4, then every 4 cycles)		X		8.5.2
Study treatments administered													
T-DXd	5.4 or [CCI] mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Pertuzumab	1x loading dose of [CCI] mg, followed by [CCI] mg administered by intravenous infusion on D1 of each 3-week cycle												6.2.2
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until										X	7.1.2, 8.1.1	

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.												
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [m] (if not collected at screening)	X [n]											8.6.1, 8.6.2	
Paired on-treatment biopsy					X [n]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for CCI	CCI [redacted]											8.6.1	
Blood sample to CCI												8.6.1	
Blood sample for CCI												8.7	
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [q]	X											8.1.5.1	
MDASI symptom diary	To be completed C1D1 before dosing and daily thereafter											8.1.5.2	

Mod 2 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 2

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
At-home pulse oximetry	To be completed C1D1 before dosing and daily thereafter (resting) To be completed C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6 and Month 12 for patients with stable brain metastases at baseline).		X	8.1.6.1	
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2	
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3	
SGRQ-I	To be completed after diagnosis with ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.

[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd.
[i] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for T-DXd and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[j] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 C1, 2 and 4 for pertuzumab.
[k] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for pertuzumab.
[l] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[m] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the core CSP Section 8.6.1 and 8.6.2).
[n] Refer to the core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[o] At time of suspected CCI , an additional sample should be taken.
[p] The sample for genetic research will be obtained at CCI . If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI . Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.
[q] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use, in accordance with device training.
[r] ADA for pertuzumab need not be collected after safety follow-up when participant starts another therapy for cancer.
[s] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in the core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB = Cambridge Neuropsychological Test Automated Battery; CNS = central nervous system; COVID-19 = coronavirus 2019-nCoV CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI ; D = day; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI ; T-DXd = trastuzumab deruxtecan.

Mod 2 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 2

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Vital signs	X		8.2.2	
Weight	X		8.2.1	
Full physical examination	X		8.2.1	
SpO_2	X		8.2.5.2	
ECOG PS	X		8.2.5.3	
12-lead ECG	X		8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4, Table 11	
Troponin [c]	X		Table 11	
Urinalysis	As clinically indicated		Table 11	
Coagulation tests	As clinically indicated		Table 11	
Pregnancy test	X	As clinically indicated	8.2.4	
Blood sample for T-DXd ADA analysis in serum	X		8.5.2	
Blood sample for pertuzumab ADA analysis in serum [d]		X		8.5.2
Concomitant medications	X		6.5	
AEs/SAEs	X		8.3	
PFS2 assessment	X	X	7.1.2.1, 8.1.3,	
Health care resource use (HOSPAD)	X		8.8	

Mod 2 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 2

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: • To be completed daily up to and including the first survival visit.		8.1.5.2	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit • To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3	
MDASI-BT	X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)		8.1.6.2	
NANO	X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)		8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5	
Survival status	X	X	7.1.2.1, 8.1.4	
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4	

Mod 2 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 2

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1	

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+ 7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.
[d] ADA for pertuzumab need not to be collected after safety follow-up when participant starts another therapy for cancer.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Pertuzumab is a recombinant humanised mAb that targets the extracellular dimerisation domain (Subdomain II) of HER2 and, thereby, blocks ligand-dependent heterodimerisation of HER2 with other HER family members, including epidermal growth factor receptor, HER3, and HER4. As a result, pertuzumab inhibits ligand-initiated intracellular signaling through two major signal pathways, mitogen-activated protein kinase, and phosphoinositide 3-kinase. Inhibition of these signaling pathways can result in cell growth arrest and apoptosis, respectively. In addition, pertuzumab mediates ADCC. Pertuzumab is approved in combination with trastuzumab and docetaxel for treatment of patients with HER2-positive MBC who have not received prior anti-HER2 therapy or chemotherapy for metastatic disease. It is also approved for the use in combination with trastuzumab and chemotherapy as neoadjuvant treatment of patients with HER2-positive, locally advanced, inflammatory, or early-stage breast cancer and adjuvant treatment of patients with HER2-positive early breast cancer at high risk of recurrence (Perjeta SmPC 2020).

Pertuzumab combined with trastuzumab is the SOC in neoadjuvant, adjuvant and MBC. In combination with trastuzumab and chemotherapy, pertuzumab demonstrated an mPFS of 18.5 months and mOS of 56.5 months in the first-line metastatic setting (Swain et al 2015), and a pathological complete response rate of 46% in the neoadjuvant setting (Gianni et al 2012), establishing the combination of taxane, trastuzumab and pertuzumab as the SOC in both indications. In addition, pertuzumab is approved in the adjuvant setting in patients with high-risk features (von Minckwitz et al 2017).

Combination of T-DM1 and pertuzumab has shown synergistic cytotoxic activity in cell culture and enhanced anti-tumour activity in xenograft models (Phillips et al 2014). In the first-line Phase 3 trial in HER2-positive MBC, the efficacy of the combination of T-DM1 and pertuzumab was shown to be non-inferior against a combination of taxane and trastuzumab. A better safety profile was observed in the T-DM1 + pertuzumab arm (Perez et al 2017).
CCI [redacted] (data on file). Thus, exploration of the T-DXd + pertuzumab combination in a Phase 2 setting is warranted.

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Pertuzumab

Pertuzumab combined with trastuzumab and chemotherapy forms the backbone of SOC in neoadjuvant, adjuvant and metastatic HER2-positive breast cancer. Recent 8-year follow-up results of the CLEOPATRA trial have shown that the OS rate was 37% for patients receiving pertuzumab (Swain et al 2020), demonstrating how pertuzumab has changed the natural history of HER2-positive breast cancer.

Dual targeting of HER2 with the combination of the ADC, T-DM1, and pertuzumab in cell culture and mouse xenograft models has resulted in enhanced anti-tumour activity (Phillips et al 2014). Additionally, T-DM1 binding to cell surface HER2 and internalisation of HER2-bound T-DM1 from the cell surface into the cytoplasm was increased in the presence of pertuzumab in preclinical gastric cancer models (Yamashita-Kashima et al 2012). In clinical studies; in first-line metastatic HER2-positive breast cancer patients, T-DM1 with or without pertuzumab showed non-inferior PFS, and better tolerability than trastuzumab plus a taxane (Perez et al 2017). T-DM1 and pertuzumab also demonstrated promising results in a neoadjuvant Phase 2 study, where the pCR rate with T-DM1 and pertuzumab demonstrated an improvement over SOC (DeMichele et al 2016). However, when T-DM1 and pertuzumab were further evaluated in a Phase 3 neoadjuvant study, the pCR rate (44%) did not meet the pCR rate of SOC (56%), although it was well tolerated and resulted in fewer AEs (Hurvitz et al 2018).

CCI [redacted] (data on file). While in clinical trials the combination of T-DM1 and pertuzumab has not resulted in superiority of the ADC combination compared to SOC, the safety profile of T-DM1 + pertuzumab appeared to be improved compared to SOC. Taken together and in view of the strong efficacy profile of T-DXd in late-line HER2-positive MBC, these data suggest T--DXd + pertuzumab combination could be efficacious and could provide a de-escalation strategy from standard chemotherapy and its side effects (Okines 2017).

2.3.2 Potential Risks of T-DXd + Pertuzumab

There is potential for overlapping toxicities when combining T-DXd and pertuzumab. Cardiotoxicity is a known side-effect for most HER2-targeted agents (Florido et al 2017). Interestingly however, the studies conducted with T-DXd to date have not identified an increased risk of cardiotoxicity (T-DXd IB). In addition, the adjuvant study investigating the addition of pertuzumab to trastuzumab and chemotherapy did not report increased rates of cardiotoxicity (von Minckwitz et al 2017). Cardiac monitoring with serial ECHO/MUGAs, ECGs and troponin tests has been added to this CSP throughout the treatment period and follow-up to diagnose any cardiac events early.

A second potential overlapping toxicity is diarrhoea. The addition of pertuzumab to trastuzumab and chemotherapy has increased the risk of diarrhoea. However, in clinical trials, diarrhoea has been generally manageable with common antidiarrheals and has not affected patients' ability to receive treatment (Bines et al 2018). Guidance for the management of diarrhoea is provided in the overlapping TMGs (core CSP Appendix L).

2.3.3 Overall Benefit/Risk Conclusion

Given the promising T-DXd monotherapy efficacy seen in the late-line HER2-positive patients and the synergistic preclinical data on T-DXd + pertuzumab combination, it is anticipated that the combination will provide additional efficacy benefit. Interventions have been placed in this CSP to prevent cardiotoxicity and diarrhoea. Cardiac monitoring will be done at regular intervals throughout the study and diarrhoea will be managed with anti-diarrhoeal medications. With these preventive mitigations, the benefits in this study are estimated to outweigh the risks.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients will be treated with T-DXd + pertuzumab.

4.1.1 Module 2 Part 1

In Part 1, approximately [redacted] to [redacted] patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled in each dose level to determine the RP2D. Dose-escalation/de-escalation will be determined by using a mTPI-2 algorithm (refer to the core CSP Table 7). In Part 1 in this module, [redacted] patients will be enrolled at dose level (1) (Mod 2 Table 3). Dose level (1) will be defined as the RP2D if no more than one DLT is observed from the first [redacted] patients. The dose will be de-escalated to level (-1) if there is evidence indicating intolerable toxicity with dose level (1). Beyond the first batch of [redacted] patients will be enrolled in batches of [redacted] at dose level (1) or (-1) following the mTPI-2 algorithm. Dose level decisions and determination of the RP2D will be made by a study specific SRC. Additional details on the dose-finding period and combination dose-finding approach are provided in the core CSP Section 4.1.1.

Mod 2 Table 3 Module 2 Part 1, Dose Finding

mTPI-2 escalation/de-escalation	
Dose Level (1)	**Dose Level (-1)**
T-DXd dose level (1): 5.4 mg/kg IV infusion Q3W Pertuzumab: 1x loading dose CCI mg, CCI mg IV infusion Q3W	T-DXd dose level (-1): CCI mg/kg IV infusion Q3W Pertuzumab: 1x loading dose CCI mg, CCI mg IV infusion Q3W

IV = intravenous; mTPI-2 = modified toxicity probability index-2; Q3W = every 3 weeks; T-DXd = trastuzumab deruxtecan.

4.1.2 Module 2 Part 2

In Part 2, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/MBC (ie, 1L) will be enrolled to determine evidence of anti-tumour activity. Approximately 40 to 50 patients will be randomised and treated with the RP2D determined from Part 1 (Mod 2 Table 4).

Mod 2 Table 4 Module 2 Part 2, Dose Expansion

Investigational Product	Dose
T-DXd	RP2D from Part 1
Pertuzumab	1x loading dose CCI mg, CCI mg IV infusion Q3W

IV= intravenous; Q3W = every 3 weeks; RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan.

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

4.2.3 Rationale for T-DXd + Pertuzumab in HER2-positive Advanced/MBC

CCI (data on file). While in clinical trials the combination of T-DM1 and pertuzumab did not result in superiority of the ADC combination compared to SOC, the safety profile of the T-DM1 + pertuzumab combination appeared to be improved compared to SOC (Perez et al 2017). Taken together and in view of the strong efficacy profile of T--DXd in late-line HER2-positive MBC these data suggest a T-DXd + pertuzumab combination could be efficacious and could provide a de-escalation strategy from standard chemotherapy and its side effects (Okines 2017). Given the difference in drugs conjugated to the HER2 mAb between T-DM1 and T-DXd, synergy with combination

partners may be very different. Based on preclinical and clinical data with pertuzumab combinations, an exploration of T--DXd and pertuzumab in combination is warranted. It is hypothesised the efficacy T-DXd monotherapy has already demonstrated in heavily pretreated patients will be further improved in early-line patients with the addition of pertuzumab.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP Section 4.3.1.

4.3.2 Pertuzumab Dose Rationale

Pertuzumab will be administered at the approved dose. For further information refer to the local prescribing information.

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

Specific eligibility criteria for Module 2 are listed below.

5.1 Inclusion Criteria

Refer to core module for the other inclusion criteria. The below inclusion criterion is applicable to this module only.

16 Non-sterilised male patients who are sexually active with a female partner of child-bearing potential must use a condom with spermicide from screening to **6 months** after the final dose of study treatment. Not engaging in sexual activity for the duration of the study and drug washout period is an acceptable practice provided it is the preferred and usual lifestyle choice of the participant; however, periodic or occasional abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. It is strongly recommended for the female partners of a male patient to also use at least one highly effective method of contraception throughout this period, as described in the Core CSP Table 26. In addition, male patients should refrain from fathering a child, or freezing or donating sperm during the study and **for 6 months** after the last dose of study treatment. Preservation of sperm should be considered prior to enrolment in this study.

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with pertuzumab.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.6 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for pertuzumab. The study treatments in this module are shown in Mod 2 Table 5.

Mod 2 Table 5 Study Treatments in Module 2

Intervention Name	T-DXd	Pertuzumab
Type	Refer to the core CSP Table 10	Monoclonal Antibody
Dosage Presentation		As sourced locally [a]
Unit Dose Strength(s)		Variable
Dosage Level(s)		1x loading dose of CCI mg, then CCI mg
Route of Administration		IV
Use		Experimental
Sourcing		Sourced locally by site [a]
Packaging and Labelling		Sourced locally by site [a]

[a] Under certain circumstances when local sourcing is not feasible, pertuzumab may be supplied centrally through AstraZeneca.
CSP = clinical study protocol; IV = intravenous; T-DXd = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration and monitoring. T-DXd will be administered as described in the core CSP Section 6.2.1.

T-DXd will be administered first, followed by pertuzumab (see Section 6.2.2), with approximately an hour between infusions.

6.2.2 Pertuzumab Preparation and Administration

Pertuzumab will be prepared and administered as outlined in the prescribing information or treatment guidance in general use by the investigating site. Under certain circumstances when local sourcing by the study site is not feasible, AstraZeneca will centrally supply the drug, which will be labelled with local language translated text in accordance with regulatory guidelines.

The first infusion of pertuzumab will be administered over 60 minutes (± 10 minutes). Patients will be monitored for any AEs during and for at least 60 minutes following the first pertuzumab infusion. If the first infusion of pertuzumab on Cycle 1 Day 1 is tolerated, subsequent infusions may be administered over 30 minutes (± 10 minutes).

Patients will be monitored for any AEs during and for at least 30 minutes following each pertuzumab infusion in subsequent cycles.

6.2.3 Storage

For T-DXd storage information refer to the core CSP Section 6.2.2. For pertuzumab storage information refer to the prescribing information.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after patients with HR-positive tumours have completed 8 or more cycles of T-DXd, as per local practice.

Refer to the core CSP Appendix J for details on T-DXd safety-specific restrictions, including restricted, prohibited, and permitted concomitant medications. Refer to the local prescribing information for pertuzumab with regard to warnings, precautions, contraindications, and prohibited medications during treatment.

6.5.1 Rescue Medication

Not applicable to this module.

6.6 Dose Modification

6.6.1 T-DXd

For dose modifications that apply to toxicities attributable to T-DXd refer to the core CSP Section 6.6.1.

6.6.2 Pertuzumab

Dose modification for pertuzumab is not permitted.

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 in this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1, and to the pertuzumab local prescribing information for definitions of any AE that meets criteria for discontinuation.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1 for all subsections.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.2.2 Vital Signs

Vital signs will be performed at timepoints as specified in the SoA. Temperature, pulse rate, respiratory rate, and blood pressure will be assessed.

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the patient in a quiet setting without distractions (eg, television, cell phones).

Vital signs will be evaluated and recorded in eCRF. Standard infusion time for T-DXd is approximately 90 minutes for first infusion and 30 minutes thereafter if the first infusion was well tolerated; however, if there are interruptions during infusion, the total allowed infusion time should not exceed 3 hours at room temperature.

If the infusion takes longer than 90 minutes, then blood pressure and pulse measurements should follow the principles as described below or be taken more frequently if clinically indicated. An hour observation period is recommended after the first infusion of T-DXd and the combination partner treatments that are administered intravenously.

First Infusion of Study Treatment

On the first day of study treatment administration, patients' vital signs will be monitored and collected/recorded in the eCRF before and after infusion of each study treatment as follows:

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 30 minutes ± 10 minutes after end of the T-DXd infusion (during the 1-hour observation period after the infusion). Additionally, approximately 30 minutes ± 10 minutes after end of the pertuzumab infusion (during the 1-hour observation period after the pertuzumab infusion).

Subsequent Infusions of Study Treatment

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 7 minutes ± 5 minutes after end of the T-DXd infusion (during the 15-minute observation period after the infusion). Additionally, approximately 15 minutes ± 5 minutes after end of the pertuzumab infusion (during the 30-minute observation period after the pertuzumab infusion).

Vital signs during initial and subsequent T-DXd and combination partner infusions should be monitored, and any clinically significant findings at other times to those described above should be reported via an unscheduled visit form. Any changes in vital signs should be recorded as an AE (if applicable). Situations in which vital signs results should be reported as AEs are described in core CSP Section 8.3.5.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2.

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

No AESIs will be considered for pertuzumab. Refer to the core CSP Section 8.3.11 for details on AESIs related to T-DXd.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd

- The label of the current effective version of the SmPC Perjeta, Roche Products Limited.

8.3.14 Pregnancy

Any pregnancies occurring while a patient is receiving study treatment or up to 7 months after the last dose of study treatment must be reported to the Sponsor within 24 hours of learning of the pregnancy. Patients in this module should follow the pertuzumab local prescribing information relating to contraception, the time limits for such precautions, and any additional restrictions. Refer to the core CSP Section 8.3.14 for further details.

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in the core CSP Appendix K, and the T-DXd TMGs in the core CSP Appendix L.

8.3.16.2 Toxicity Management and Dose Modification Information – Pertuzumab

Toxicities of pertuzumab may be managed according to the local prescribing information.

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of pertuzumab in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd and pertuzumab, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information, including details on overdose with T-DXd. Refer to the local prescribing information for treatment in case of an overdose related to pertuzumab.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total anti-HER2 antibody, MAAA-1181a, and pertuzumab in serum at timepoints as specified in the SoA in this module. Additional PK blood samples will be collected for patients who receive treatment with chloroquine/hydroxychloroquine during the treatment period of the study. Refer to the core CSP Section Appendix M.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd and pertuzumab in serum will be collected as specified in the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9 for all subsections.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol

Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix Q

Module 3 – T-DXd + Paclitaxel

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

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This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

PROTOCOL AMENDMENT SUMMARY OF CHANGES TABLE

DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 3 of this study will evaluate paclitaxel chemotherapy in combination with T-DXd in patients with HER2-positive advanced/MBC. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2.

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 3 Table 1) and follow-up period (Mod 3 Table 2) for Module 3 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs [b]	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.5	
SpO_2 [b]	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [c]				X [c]		X [c]		X [c]		X	8.2.1	
Weight	X [c]				X [c]		X [c]		X [c]		X	8.2.1	
ECOG PS	X [c]				X [c]		X [c]		X [c]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [b,d]													
Pregnancy test [e]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [c]		X	X	X [c]		X [c]		X [c]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
12-lead ECG [f]	X [c]				X		X		X		X	8.2.3	
ECHO/MUGA (LVEF) [g]									X		X C5 (and every 4 cycles [± 7 days])	8.2.5.1	
Troponin [h] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [s]	As clinically indicated										X	8.2.5.4	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [i]	X				X				X (C4, C6, and C8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [j]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for paclitaxel PK testing [k]	X				X				X (C4 only)				

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Post-dose blood sample for paclitaxel PK testing [l]		X				X				X (C4 only)			
Blood sample for T-DXd ADA assessment in serum[m]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Study treatments administered													
T-DXd	[redacted] or [redacted] mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Paclitaxel	[redacted] mg/m^2 or [redacted] mg/m^2 will be administered intravenously QW in 3-week cycles												6.2.2
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following investigator-determined RECIST 1.1 progression, an additional										X	7.1.2, 8.1.1	

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.												
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [n] (if not collected at screening)	X [o]											8.6.1, 8.6.2	
Paired on-treatment biopsy					X [o]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (at progression only)	8.6.2	

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI t												8.61	
Blood sample for CCI												8.6.1	

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for CCI CCI [p]	CCI [redacted]											8.6.1	
Blood sample to CCI												8.6.1	
Blood sample for CCI												8.7	
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [r]	X											8.1.5.1	
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	

Mod 3 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) – Module 3

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6, and Month 12 for patients with stable brain metastases at baseline)		X	8.1.6.1	
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2	
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Laboratory, SpO_2, and vital sign assessments on D8 and D15 of cycle visits when patients receive Paclitaxel will need to be recorded in RAVE.
[c] Within 3 days before study treatment administration.
[d] If screening assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[e] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at end of treatment.

[f] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[g] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[h] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[i] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd,
[j] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for T-DXd and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[k] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 C1, 2 and 4 for paclitaxel.
[l] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for paclitaxel.
[m] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[n] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the core CSP Section 8.6.1 and 8.6.2).
[o] Refer to the core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[p] At time of suspected CCI , an additional sample should be taken.
[q] The sample for genetic research will be obtained at CCI . If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI . Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.
[r] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use, in accordance with device training.
[s] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in the core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB = Cambridge Neuropsychological Test Automated Battery; CNS = central nervous system; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI ; D = day; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI ; T-DXd = trastuzumab deruxtecan.

Mod 3 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 3

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Vital signs	X		8.2.2	
Weight	X		8.2.1	
Full physical examination	X		8.2.1	
SpO_2	X		8.2.5.2	
ECOG PS	X		8.2.5.3	
12-lead ECG	X		8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4	
Troponin [c]	X		Table 11	
Urinalysis	As clinically indicated		Table 11	
Coagulation tests	As clinically indicated		Table 11	
Pregnancy test	X	As clinically indicated	8.2.4	
Blood sample for T-DXd- ADA analysis in serum	X		8.5.2	
Concomitant medications	X		6.5	
AEs/SAEs	X		8.3	
PFS2 assessment	X	X	7.1.2.1, 8.1.3	
Health care resource use (HOSPAD)	X		8.8	
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: • To be completed daily up to and including the first survival visit.		8.1.5.2	

Mod 3 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 3

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit • To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3	
MDASI-BT		X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.2	
NANO		X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5	
Survival status	X	X	7.1.2.1, 8.1.4	
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4	

Mod 3 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 3

Procedures	Safety F/U (40 days [+7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1	

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+ 7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Similar to other breast cancer subtypes, taxanes are the SOC and backbone of HER2-positive breast cancer treatment in the neoadjuvant (Gianni et al 2012), adjuvant (Slamon et al 2011) and metastatic settings (Swain et al 2015). Taxanes enhance the polymerisation of tubulin, stabilizing microtubules and blocking cells in the G2/M phase of the cell cycle, inhibiting the formation of the mitotic apparatus (Horwitz 1994). Thus, the mechanisms of cytotoxicity of taxanes are different from those of DXd, which is the chemotherapy component of the T-DXd ADC and inhibits topoisomerase I.

2.2.1 Paclitaxel

Paclitaxel is an anti-microtubule agent that promotes the assembly of microtubules from tubulin dimers and stabilises microtubules by preventing depolymerisation. This stability results in the inhibition of the normal dynamic reorganisation of the microtubule network that is essential for vital interphase and mitotic cellular functions. In addition, paclitaxel induces abnormal arrays or “bundles” of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis (Paclitaxel USPI 2018).

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Paclitaxel

Taxanes, including paclitaxel and docetaxel, have been a cornerstone of breast cancer treatment over the past three decades (Gradishar 2012). Not surprisingly, they are also part of the SOC in HER2-positive breast cancer in the neoadjuvant, adjuvant and metastatic settings. Docetaxel in combination with trastuzumab and pertuzumab was approved for the neoadjuvant treatment of HER2-positive breast cancer based on the pCR rate of 45.8% (Gianni et al 2012). Docetaxel was also part of the chemotherapy backbone in the adjuvant HER2-positive breast cancer trial testing the addition of pertuzumab to trastuzumab. The study showed a 3-year invasive disease-free survival of 92.0% (von Minckwitz et al 2017). In addition, docetaxel was used in the first-line metastatic HER2-positive breast cancer trial combined with trastuzumab and pertuzumab. The study showed an mPFS of 18.5 months (Baselga et al 2012).

Paclitaxel inhibits the microtubule dynamics, resulting in cell cycle arrest in G2/M phase, leading to cell death (Gradishar 2012). The payload in T-DXd, is an exatecan derivative, which is a potent topoisomerase I inhibitor (Nakada et al 2016). It is hypothesised that the two different MOAs will complement each other resulting in increased efficacy signal. CCI

CCI [redacted] (data on file). Taken together these data suggest T-DXd- + paclitaxel has the potential to offer substantial benefit in early-line MBC.

2.3.2 Potential Risks of T-DXd + Paclitaxel

The MOA is similar in for both docetaxel and paclitaxel, but the toxicity profiles are slightly different. In an adjuvant chemotherapy trial comparing paclitaxel to docetaxel in node-positive early breast cancer, a similar 3-year disease-free survival was seen, but more favourable toxicity profile, particularly for haematological toxicities, was seen with paclitaxel (Saloustros et al 2014). Thus, paclitaxel will be evaluated in this study.

In terms of the risks of paclitaxel, haematological side-effects are frequently seen. The frequency of all grade neutropenia is 90%, anaemia 78% and thrombocytopenia 20%. Also, alopecia is commonly observed with paclitaxel (87% all grades). In addition, neuropathy is frequently seen with taxanes (paclitaxel 60%) and is sometimes severe (paclitaxel 3%). Gastrointestinal toxicities include nausea, vomiting and diarrhoea with the all-grade frequencies of 52%, 52%, and 38%, respectively (Paclitaxel USPI 2018).

The potential overlapping toxicities with paclitaxel and T-DXd include haematological and gastrointestinal toxicities (mainly nausea and vomiting), as well as ILD/pneumonitis. Dose-escalation/de-escalation will be applied to find the optimal dose for the combination therapy to mitigate the risk of severe cytopenias. In addition, the use of G-CSF will be allowed as per local practice. Gastrointestinal toxicities will be managed as per standard practice. ILD/pneumonitis risk with paclitaxel is low and the toxicity management guidelines outlined for ILD/pneumonitis by T-DXd will be applied.

2.3.3 Overall Benefit/Risk Conclusion

To date, T-DXd has shown the most promising efficacy results in the heavily pretreated population compared to any other HER2-targeted therapy. CCI [redacted] CCI [redacted] (data on file). Given that taxanes are the cornerstone in the treatment regimens of all subtypes of breast cancer, including HER2-positive MBC, investigation of the T-DXd + paclitaxel combination is warranted.

While the combination of T-DXd + paclitaxel may demonstrate an increased frequency of haematological and gastrointestinal toxicities, in particular, it is believed these can be managed through appropriate dose selection and TMGs. In the breast cancer SOC, no single drug has been able to reach sufficient efficacy to provide higher rates of cure than combination therapies. Therefore, T-DXd combinations with chemotherapy partners should be explored. The obvious chemotherapy combination partner is paclitaxel, which is already used

world-wide. With a rigorous monitoring of the toxicities and early intervention to manage them, the benefits are anticipated to outweigh the risks.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients will be treated with -T-DXd- + paclitaxel.

4.1.1 Module 3 Part 1

In Part 1, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled. The combination of T-DXd with paclitaxel may be associated with acute haematological toxicity, therefore patients will be enrolled in batches of [CCI] until the RP2D can be defined or the maximum cohort size ([CCI] patients per dose level) is reached. T-DXd will be combined with paclitaxel at dose level (1) (Mod 3 Table 3). The dose can be escalated to dose level (2) or de-escalated to dose level (1) following the mTPI-2 algorithm (see the core CSP Table 7). Dose level decisions and determination of the RP2D will be made by a study specific SRC. Additional details on the dose-finding period and combination dose-finding approach are provided in the core CSP Section 4.1.1.

Mod 3 Table 3 Module 3 Part 1, Dose Finding

mTPI-2 escalation/de-escalation		
Dose Level (1)	**Dose Level (2)**	**Dose Level (-1)**
T-DXd: [CCI] mg/kg IV infusion Q3W Paclitaxel: [CCI] mg/m^2 IV QW	T-DXd: [CCI] mg/kg IV infusion Q3W Paclitaxel: [CCI] mg/m^2 IV QW	T-DXd: [CCI] mg/kg IV infusion Q3W Paclitaxel: [CCI] mg/m^2 IV QW

IV = intravenous; mTPI-2 = modified toxicity probability index-2; QW = every week; Q3W = every 3 weeks; T-DXd- = trastuzumab deruxtecan.

4.1.2 Module 3 Part 2

In Part 2, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/MBC (ie, 1L) will be enrolled to

determine evidence of anti-tumour activity. Approximately 40 to 50 patients will be randomised and treated with the RP2D determined from Part 1 (Mod 3 Table 4).

Mod 3 Table 4 Module 3 Part 2, Dose Expansion

Study Treatment	Dose
T-DXd	RP2D from Part 1
Paclitaxel	RP2D from Part 1

RP2D = recommended phase 2 dose; T-DXd- = trastuzumab deruxtecan.

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

4.2.3 Rationale for T-DXd + Paclitaxel in HER2-positive Advanced/MBC

The combination of docetaxel (a taxane) and irinotecan (a topoisomerase I inhibitor) has been evaluated in advanced oesophageal cancer. In this setting, the data were promising, with efficacy demonstrated in chemotherapy-naive and previously exposed patients. The combination was well tolerated, with the primary toxicities being either haematological or gastrointestinal (Burtness et al 2009). Docetaxel and irinotecan have also been combined in a Phase 3 randomised study in metastatic pancreatic cancer, where little efficacy benefit was reported, and the combination resulted in diarrhoea as the most common Grade 3/4 AE (Burtness et al 2016).

Preclinical studies have shown an additive effect when combining paclitaxel with trastuzumab and synergy when combining docetaxel with trastuzumab (Pegram 2001). CCI [redacted] (data on file). While the clinical efficacy of paclitaxel is similar to that of docetaxel in HER2-positive breast cancer, paclitaxel is the taxane of choice for this study due to its more favourable toxicity profile (Saloustros et al 2014). Another ADC, T-DM1, albeit having a different chemotherapy payload than T-DXd, has been evaluated in combination with paclitaxel. In a Phase 2 neoadjuvant study, a combination of paclitaxel, lapatinib, and T-DM1 was more efficacious than a combination of trastuzumab, pertuzumab and paclitaxel, especially in oestrogen-receptor positive patients. This novel combination was well tolerated, suggesting that taxanes and ADCs can be combined safely (Patel et al 2019).

Based on the potential synergy between taxanes and topoisomerase I inhibitors, an exploration of T-DXd and paclitaxel in combination is warranted. It is hypothesised that the efficacy T-DXd monotherapy has already demonstrated in heavily pretreated patients may be further enhanced with the addition of paclitaxel.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP Section 4.3.1.

4.3.2 Paclitaxel Dose Schedule Rationale

In the dose-finding phase of this module (Part 1), paclitaxel will be administered at a dose of [CCI] mg/m^2 (dose level 1, Mod 3 Table 3), with an option to escalate to [CCI] mg/m^2 (dose level 2), QW as opposed to the United States Prescribing Information labelled paclitaxel dose of 175 mg/m^2 Q3W. In a meta-analysis of randomised, controlled studies in advanced breast cancer (5 studies, 1471 patients), which compared weekly administration of paclitaxel with Q3W administration, the analysis concluded that there was an improvement in OS with weekly paclitaxel (pooled HR, 0.78; 95% CI: 0.67, 0.89; p=0.001). Additionally, the incidence of neutropenia, neutropenic fever, peripheral neuropathy, and other SAEs was significantly lower with weekly paclitaxel compared with paclitaxel Q3W (Mauri et al 2010). Based on this meta-analysis and to be aligned to standard practice among breast cancer clinicians, the weekly regimen is proposed for this study.

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

Specific eligibility criteria for Module 3 are listed below.

5.1 Inclusion Criteria

Refer to core module for the other inclusion criteria. The below inclusion criterion is applicable to this module only.

16 Non-sterilised male patients who are sexually active with a female partner of child-bearing potential must use a condom with spermicide from screening to **6 months** after the final dose of study treatment. Not engaging in sexual activity for the duration of

the study and drug washout period is an acceptable practice provided it is the preferred and usual lifestyle choice of the participant; however, periodic or occasional abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. It is strongly recommended for the female partners of a male patient to also use at least one highly effective method of contraception throughout this period, as described in the Core CSP Table 26. In addition, male patients should refrain from fathering a child, or freezing, or donating sperm during the study and **for 6 months** after the last dose of study treatment. Preservation of sperm should be considered prior to enrolment in this study.

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with paclitaxel.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.6 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for paclitaxel. The study treatments in this module are shown in Mod 3 Table 5.

Mod 3 Table 5 Study Treatments in Module 3

Intervention Name	T-DXd	Paclitaxel
Type	Refer to the core CSP Table 10	Chemotherapy
Dosage Presentation		As sourced locally [a]
Unit Dose Strength(s)		Variable
Dosage Level(s)		[redacted] or [redacted] mg/m^2
Route of Administration		IV
Use		Experimental
Sourcing		Sourced locally by site [a]
Packaging and Labelling		Sourced locally by site [a]

[a] Under certain circumstances when local sourcing is not feasible, paclitaxel may be supplied centrally through AstraZeneca.
CSP = clinical study protocol; IV = intravenous; T-DXd = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration, and monitoring.

T-DXd will be administered first followed by paclitaxel, with approximately one hour between infusions.

6.2.2 Paclitaxel Preparation and Administration

Paclitaxel will be prepared and administered as outlined in the prescribing information or treatment guidance in general use by the investigating site. Under certain circumstances when local sourcing by the study site is not feasible, AstraZeneca will centrally supply the drug, which will be labelled with local language translated text in accordance with regulatory guidelines.

All patients should be premedicated prior to paclitaxel administration in order to prevent severe hypersensitivity reactions. Such premedication may consist of dexamethasone 20 mg PO administered approximately 12 and 6 hours before paclitaxel, diphenhydramine (or its equivalent) 50 mg IV 30 to 60 minutes prior to paclitaxel, and cimetidine (300 mg) or ranitidine (50 mg) IV 30 to 60 minutes before paclitaxel.

6.2.3 Storage

For T-DXd storage information refer to the core CSP Section 6.2.2. For paclitaxel storage information refer to the prescribing information.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after 8 cycles of T-DXd or discontinuation of paclitaxel (whichever occurs later) for patients with HR-positive tumours, as per local practice.

Refer to the core CSP Section 6.5 and the core CSP Appendix J for details on T-DXd safety-specific restrictions; restricted, prohibited, and permitted therapies. Refer to the local prescribing information for paclitaxel with regard to warnings, precautions, contraindications, and prohibited medications during treatment and to the core CSP Appendix J1.

6.5.1 Rescue Medication

Not applicable for this module.

6.6 Dose Modification

6.6.1 T-DXd

For dose modifications that apply to toxicities that are attributable to T-DXd refer to the core CSP Section 6.6.1.

6.6.2 Paclitaxel

Dose modification for paclitaxel should be done according to the local prescribing information.

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 of this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1, and to the paclitaxel prescribing information for definitions of any AE that meets criteria for discontinuation.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 of this module. Details for module-specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1 for all subsections.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.2.2 Vital Signs

Vital signs will be performed at timepoints as specified in the SoA. Temperature, pulse rate, respiratory rate, and blood pressure will be assessed.

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the patient in a quiet setting without distractions (eg, television, cell phones).

Vital signs will be evaluated and recorded in eCRF. Standard infusion time for T-DXd is approximately 90 minutes for first infusion and 30 minutes thereafter if the first infusion was well tolerated; however, if there are interruptions during infusion, the total allowed infusion time should not exceed 3 hours at room temperature.

If the infusion takes longer than 90 minutes, then blood pressure and pulse measurements should follow the principles as described below or be taken more frequently if clinically indicated. An hour observation period is recommended after the first infusion of T-DXd and the combination partner treatments that are administered intravenously.

First Infusion of Study Treatment

On the first day of study treatment administration, patients' vital signs will be monitored and collected/recorded in the eCRF before and after infusion of each study treatment as follows:

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 30 minutes ± 10 minutes after end of the T-DXd infusion (during the 1-hour observation period after the infusion). Additionally, approximately 30 minutes ± 10 minutes after end of the paclitaxel infusion (during the 1-hour observation period after the paclitaxel infusion).

Subsequent Infusions of Study Treatment

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 7 minutes ± 5 minutes after end of the T-DXd infusion (during the 15-minute observation period after the infusion). Additionally, approximately 15 minutes ± 5 minutes after end of the paclitaxel infusion (during the 30-minute observation period after the paclitaxel infusion).

Vital signs during initial and subsequent T-DXd and combination partner infusions should be monitored, and any clinically significant findings at other times to those described above should be reported via an unscheduled visit form. Any changes in vital signs should be recorded as an AE (if applicable). Situations in which vital signs results should be reported as AEs are described in core CSP Section 8.3.5.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2.

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

No AESIs will be considered for paclitaxel. Refer to the core CSP Section 8.3.11 for AESIs related to T-DXd.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd
- The label of the current effective version of the SmPC paclitaxel, Hospira UK Limited.

8.3.14 Pregnancy

Patients in this module should follow the rules for T-DXd (refer to the core CSP Section 8.3.14).

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in the core CSP Appendix K, and the TMGs in the core CSP Appendix L.

8.3.16.2 Toxicity Management and Dose Modification Information – Paclitaxel

Investigators should follow the local prescribing information and standard clinical practice regarding management of paclitaxel-related toxicities. Anti-emetic therapy prior to and after paclitaxel treatment should be given as needed on a prophylactic and treatment basis in compliance with institutional standards; based on the currently available clinical safety data, it is recommended that patients receive anti-emetic agents prior to infusion of T-DXd and on subsequent days. Granulocyte colony stimulating factors in compliance with the standards of the centre should be given as needed for secondary prophylaxis or treatment of neutropenia.

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of paclitaxel in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd and paclitaxel, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information, including details on overdose with T-

DXd. Refer to the local prescribing information for treatment in case of an overdose related to paclitaxel.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total HER2-antibody, and MAAA-1181a in serum, and paclitaxel in plasma as per the SoA in this module. Refer to the core CSP Section 8.5.1 for further details. Additional PK blood samples will be collected for patients who receive treatment with chloroquine/hydroxychloroquine during the treatment period of the study. Refer to the core CSP Appendix M for further details.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd in serum will be collected as per the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9 for all subsections.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol

Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix R

Module 4 – T-DXd + Durvalumab + Paclitaxel

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

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This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

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DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 4 of this study will evaluate paclitaxel chemotherapy in combination with T-DXd and durvalumab in patients with HER2-positive advanced/MBC. For further details on the study synopsis, refer to the core CSP Section 1.1.

The initiation of this module is ungated on the completion of Part 1 of Module 3 (T-DXd + paclitaxel).

1.2 Schema

The study schema is provided in the core CSP Section 1.2. The RP2D for Part 2 of this module may come from DESTINY-Breast08 (NCT04556773), a Phase 1b study in metastatic breast cancer patients with HER2-low expression (defined as IHC 2+/ISH- and IHC 1+).

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 4 Table 1) and follow-up period (Mod 4 Table 2) for Module 4 are shown below. Details of assessments and procedures are described in Section 8 in the core CSP unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs [b]	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.5	
SpO_2 [b]	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [c]				X [c]		X [c]		X [c]		X	8.2.1	
Weight	X [c]				X [c]		X [c]		X		X	8.2.1	
ECOG PS	X [c]				X [c]		X [c]		X [c]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [b, d]													
Pregnancy test [e]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [c]		X	X	X [c]		X [c]		X [c]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
12-lead ECG [f]	X [c]				X		X		X		X	8.2.3	
ECHO/MUGA (LVEF) [g]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [h] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [v]	As clinically indicated										X	8.2.5.4	
TSH (reflex-free T3 or free T4)	X				X		X		X		X	Table 11	
Amylase and lipase	X				X		X		X		X	Table 11	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [i]	X				X				X (Cycles 4,6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [j]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for durvalumab PK testing [k]	X				X				X (Cycles 4,6, and 8)				8.5.1

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Post-dose blood sample for durvalumab PK testing [l]		X				X				X (C4 only)			8.5.1
Pre-dose blood sample for paclitaxel PK testing [m]	X				X				X (C4 only)				8.5.1
Post-dose blood sample for paclitaxel PK testing [n]		X				X				X (C4 only)			8.5.1
Blood sample for T-DXd- ADA assessment in serum [o]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Blood sample for durvalumab ADA assessment in serum [j,u]	X				X				X (C4, then every 4 cycles)		X		8.5.2
Study treatments administered													
T-DXd	[redacted] or [redacted] mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Durvalumab	[redacted] mg will be administered intravenously on D1 of each [redacted] week cycle												6.2.2
Paclitaxel	[redacted] mg/m^2 or [redacted] mg/m^2 will be administered intravenously QW in 3-week cycles												6.2.3
Tumour assessments													

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.										X	7.1.2, 8.1.1	
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either	X [q]											8.6.1, 8.6.2	

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
a newly acquired biopsy or the most recent archival tumour [p] (if not collected at screening)													
Paired on-treatment biopsy					X [q]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI												8.6.1	

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
CCI													
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for serum CCI [r]												8.6.1	
Blood sample to CCI												8.6.1	
Blood sample for CCI												8.7	
Other assessments and procedures													

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [t]	X											8.1.5.1	
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6, and Month 12 for patients with stable brain metastases at baseline)		X	8.1.6.1	
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2	

Mod 4 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) -Module 4

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 1 day		D1 ± 1 day		D1 ± 1 day				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Laboratory, SpO_2, and vital sign assessments on D8 and D15 of cycle visits when patients receive paclitaxel will need to be recorded in RAVE.
[c] Within 3 days before study treatment administration.
[d] If screening assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[e] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at end of treatment.
[f] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[g] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[h] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[i] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd.
[j] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[k] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for durvalumab.
[l] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for durvalumab.
[m] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 C1, 2 and 4 for paclitaxel.

[n] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4 for paclitaxel.
[o] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[p] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the Core CSP, Section 8.6.1 and 8.6.2).
[q] Refer to the Core CSP, Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[r] At time of suspected CCI , an additional sample should be taken.
[s] The sample for genetic research will be obtained at CCI . If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI . Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.
[t] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use, in accordance with device training.
[u] ADA for durvalumab need not to be collected after safety FU when subject starts another therapy for cancer.
[v] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in Section 8 of the core CSP.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB = Cambridge Neuropsychological Test Automated Battery; CNS = central nervous system; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI ; D = day; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; T3 = triiodothyronine; T4 = thyroxine; CCI CCI T-DXd = trastuzumab deruxtecan; TSH; thyroid stimulating hormone.

Mod 4 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 4

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
Vital signs	X				8.2.2	
Weight	X	X	X		8.2.1	
Full physical examination	X				8.2.1	
SpO_2	X				8.2.5.2	
ECOG PS	X	X	X		8.2.5.3	
12-lead ECG	X				8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X	X	X		8.2.4, Table 11	
Troponin [c]	X	X	X		Table 11	
Urinalysis	As clinically indicated				Table 11	
Coagulation tests	As clinically indicated				Table 11	
Pregnancy test	X	As clinically indicated			8.2.4	
Blood sample for T-DXd- ADA analysis in serum	X				8.5.2	
Blood sample for durvalumab ADA analysis in serum[d]			X			8.5.2
Concomitant medications	X	X	X		6.5	
AEs/SAEs	X	X	X		8.3	
PFS2 assessment	X	X	X	X	7.1.2.1, 8.1.3	

Mod 4 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 4

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
Health care resource use (HOSPAD)	X				8.8	
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: To be completed daily up to and including the first survival visit.				8.1.5.2	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit				8.1.5.3	
MDASI-BT	X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)				8.1.6.2	
NANO	X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)				8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.				8.1.5.4	

Mod 4 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 4

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)				8.1.5.5	
Survival status	X	X	X	X	7.1.2.1, 8.1.4	
Subsequent cancer therapy following discontinuation of study treatment	X	X	X	X	8.1.4	
Tumour assessment by RECIST 1.1 (CT or MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.				8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.				8.1.1	

Mod 4 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 4

Procedure	Follow-up period Time since last dose of study intervention (months)				Section in core CSP	Section in this module
	1 (30 days) [a]	2	3	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]		
Window	± 3 days	± 1 week	± 1 week			

[a] If EOT is > 30 days (± 3 days) after last treatment, then the EOT assessments can also function as the 30-day (± 3 days) follow-up visit. Note that as durvalumab is used in this module, the safety follow-up period is 90 days after last dose.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.
[d] ADA for durvalumab need not be collected after safety follow-up when subject starts another therapy for cancer.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd- = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Combinations of chemotherapies and immune checkpoint inhibitors have recently become the SOC in many tumour types including breast cancer. In first-line metastatic TNBC, the addition of atezolizumab to nab-paclitaxel improved the mPFS from 5.0 months to 7.5 months in patients with PD-L1-positive tumours (Schmid et al 2018). In the neoadjuvant setting, the addition of pembrolizumab improved the rates of pathological CR from 51.2% to 64.8%, regardless of the PD-L1 status (Schmid et al 2020). In HER2-positive breast cancer, the KATE2 study has investigated the combination of ADC and immune checkpoint inhibitors. This study, conducted in heavily pretreated patients, showed an improvement of mPFS in patients with PD-L1-positive tumours when atezolizumab was added to T-DM1 (8.5 months combination vs 4.1 months T-DM1 alone) (Emens et al 2019). A pivotal study (NCT03199885) in the first-line HER2-positive MBC combining taxane, pertuzumab, trastuzumab and atezolizumab is ongoing.

2.2.1 Durvalumab

Durvalumab is a mAb of the IgG 1 kappa subclass that blocks the interaction of PD-L1 (but not programmed cell death ligand-2) with PD-1 on T cells and CD80 (B7.1) on immune cells. It is being developed by AstraZeneca for use in the treatment of cancer. The proposed MOA for durvalumab is interference in the interaction of PD-L1 with PD-1 and CD80. Blockade of PD-L1/PD-1 and PD-L1/CD80 interactions releases the inhibition of immune responses, including those that may result in tumour elimination. In vitro studies demonstrate that durvalumab antagonises the inhibitory effect of PD-L1 on primary human T cells resulting in the restored proliferation of interferon gamma (Stewart et al 2015). In vivo studies have shown that durvalumab inhibits tumour growth in xenograft models via a T-cell dependent mechanism (Stewart et al 2015). Based on these data, durvalumab is expected to stimulate the patient's anti-tumour immune response by binding to PD-L1 and shifting the balance toward an anti-tumour response. Durvalumab has been engineered to reduce antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity.

Durvalumab is approved in some countries as monotherapy for unresectable Stage III NSCLC (following chemoradiation therapy), and in combination with chemotherapy for extensive-stage small cell lung cancer and for locally advanced or metastatic biliary tract cancer (Imfinzi 2022). Durvalumab is also approved when administered in combination with tremelimumab for unresectable hepatocellular carcinoma, and also in combination with

tremelimumab plus chemotherapy for participants with metastatic NSCLC without EGFR or ALK mutations.

2.2.2 Paclitaxel

Paclitaxel is an anti-microtubule agent that promotes the assembly of microtubules from tubulin dimers and stabilises microtubules by preventing depolymerisation. This stability results in the inhibition of the normal dynamic reorganisation of the microtubule network that is essential for vital interphase and mitotic cellular functions. In addition, paclitaxel induces abnormal arrays or "bundles" of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis.

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Durvalumab + Paclitaxel

Durvalumab is a PD-L1 inhibitor, which has shown generally good tolerability (Antonia et al 2017) and has been approved for the treatment of NSCLC and SCLC (Antonia et al 2017, Paz-Ares et al 2019). Studies examining durvalumab in TNBC have shown improved outcomes with an excellent safety profile (Dalenc et al 2019, Loibl et al 2019). Combinations of chemotherapies and immune checkpoint inhibitors have recently become the SOC in TNBC (Schmid et al 2018, Schmid et al 2020). An ongoing study is examining the combination of taxane, pertuzumab, trastuzumab and atezolizumab in the first-line metastatic HER2-positive breast cancer (NCT03199885). Given that paclitaxel in combination with other anti-cancer agents is SOC in all breast cancer subtypes, including HER2-positive cancers, and based on the promising results from the KATE2 study combining immune checkpoint inhibitors and an ADC (Emens et al 2019), the exploration of the triplet combination T-DXd + durvalumab + paclitaxel is warranted.

2.3.2 Potential Risks of T-DXd + Durvalumab + Paclitaxel

The potential overlapping toxicities with paclitaxel and T-DXd include haematological and gastrointestinal toxicities (mainly nausea and vomiting), as well as ILD/pneumonitis. Dose--escalation/de-escalation will be applied to find the optimal dose for the combination therapy to mitigate the risk of severe cytopenias. In addition, the use of G-CSF will be allowed as per local practice. Gastrointestinal toxicities will be managed as per standard practice. Whilst the ILD/pneumonitis risk with paclitaxel is low, the addition of durvalumab to T-DXd- + paclitaxel may increase the risk of ILD/pneumonitis. Updated TMGs outlined for ILD/pneumonitis (Appendix L) aim to facilitate early detection and treatment of ILD/pneumonitis and mitigate the risk of adverse clinical outcomes.

An ongoing combination study of T-DXd and the CCI [redacted] CCI [redacted] is fully enrolled; the combination CCI [redacted] CCI [redacted] (unpublished data).

As the majority of the potential overlapping toxicities are expected to manifest between paclitaxel and T-DXd in this platform study, the dose-finding results from the T-DXd + paclitaxel module will be used to determine the starting doses for these two drugs in the triplet. Durvalumab will be added subsequently, and at least 6 patients will be studied before the expansion phase. As such the study design aims to minimise potential risks of this novel treatment combination. Careful monitoring of pulmonary function to evaluate for ILD/pneumonitis will be carried out throughout the study. Early intervention for immune-mediated diarrhoea/colitis will be applied. The use of G-CSF is allowed for the haematological toxicities.

2.3.3 Overall Benefit/Risk Conclusion

T-DXd has shown the most promising efficacy results in the heavily pretreated population compared to any other HER2-targeted therapy. Given that taxanes are a cornerstone in the treatment regimens of all subtypes and stages of breast cancer, including HER2-positive early and MBC, investigation of the T-DXd + paclitaxel combination is warranted. The addition of durvalumab is justified based on the promising results in the late-line setting of HER2-positive breast cancer, where immune checkpoint inhibitors have improved the mPFS in patients with PD-L1-positive tumours (Emens et al 2019). It is also expected that the cytotoxic effects of both paclitaxel and T-DXd will induce cancer cell death and release neoantigens, increasing the targets for T cells activated by immune checkpoint inhibitors. Preclinical studies have also shown PD-L1 upregulation post T-DXd exposure on cell lines (Iwata et al 2017).

The combination of T-DXd + paclitaxel may potentially demonstrate an increased frequency of haematological and gastrointestinal toxicities. Therefore the study has been designed, with this in mind, with the aim of optimising the combination doses whilst minimising toxicity. The addition of durvalumab to the T-DXd + paclitaxel combination may further increase the risk of ILD/pneumonitis. Notably, one of the factors associated with ILD in the late-line HER2-positive MBC data was greater than 10 lines of prior treatment, and ILD may be less frequent in earlier line populations. Nonetheless, careful monitoring of symptoms and lung imaging will be employed to identify patients at risk promptly and facilitate early intervention. To achieve maximal efficacy moving to earlier lines of treatment, investigation of combinations that combine multiple MOAs, such as T-DXd + durvalumab + paclitaxel, has the potential to provide substantial benefit with a favourable safety profile.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients will be treated with -T-DXd- + durvalumab + paclitaxel. This module will be ungated by RP2D determination in Module 3 Part 1.

4.1.1 Module 4 Part 1

The combination of T-DXd (CCI mg/kg Q3W), durvalumab (CCI mg QCCIW), and paclitaxel (CCI mg/m^2 QW) is being studied in HER2-low patients in the DESTINY-Breast08 clinical trial (NCT04556773). If the RP2D is established after review by the SRC in the DESTINY-Breast08 study, Module 4 Part 1 may not be performed in this study.

In Part 1, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled. The combination of T-DXd with paclitaxel may be associated with acute haematological toxicity, therefore patients will be enrolled in batches of CCI until the RP2D can be defined or the maximum cohort size (CCI patients per dose level) is reached. Durvalumab will be combined with T-DXd + paclitaxel (Mod 4 Table 3), the starting dose level will be determined by the SRC taking into account data from Part 1 of Module 3 (T-DXd + paclitaxel), and data from Part 1 of Module 1 (T-DXd- + durvalumab) or BEGONIA. The dose can be escalated to dose level (2) or de-escalated to dose level (-1) following the mTPI-2 algorithm (see Table 8 in the core CSP). Dose level decisions and determination of the RP2D will be made by a study-specific SRC. Additional details on the dose-finding period and combination dose-finding approach are provided in Section 4.1.1 in the core CSP.

Mod 4 Table 3 Part 1 – Module 4, Dose Finding

mTPI-2 escalation/de-escalation		
Dose Level (1)	Dose Level (2)	Dose Level (-1)
T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QCCIW Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QCCIW Paclitaxel: CCI mg/m^2 IV QW	T-DXd: CCI mg/kg IV infusion Q3W Durvalumab: CCI mg IV QCCIW Paclitaxel: CCI mg/m^2 IV QW

Note: Starting dose level for durvalumab and paclitaxel will be determined by the SRC and will take into account data from BEGONIA/Module 1 Part 1 (refer to Module 1 Section 4.1.1) and Module 3 Part 1 (refer to Module 3 Section 4.1.1).
IV = intravenous; mTPI-2 = modified toxicity probability index-2; QW = every week; Q3W = every 3 weeks; T DXd = trastuzumab deruxtecan.

4.1.2 Module 4 Part 2

In Part 2, patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/MBC (ie, 1L) will be enrolled to determine evidence of anti-tumour activity. Approximately 40 to 50 patients will be enrolled and treated when the RP2D from Module 3 is available (Mod 4 Table 4)

Mod 4 Table 4 Part 2 – Module 4, Dose Expansion

Study Treatment	Dose
T-DXd	RP2D from Part 1
Durvalumab	CCI mg IV Q CCI W
Paclitaxel	RP2D from Part 1

Q CCI W = every CCI weeks; RP2D = recommended phase 2 dose; T-DXd- = trastuzumab deruxtecan.

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

4.2.3 Rationale for T-DXd + Durvalumab + Paclitaxel in HER2-positive Advanced/MBC

The addition of immune checkpoint inhibitors to SOC combination treatments has improved the outcomes in many tumour types, including NSCLC (Antonia et al 2017, Antonia et al 2018), SCLC (Paz-Ares et al 2019) as well as TNBC (Schmid et al 2018). T-DXd monotherapy efficacy in late-line HER2-positive patients leading to FDA approval (Modi et al 2020) warrants exploration of combination therapies. Taxanes are the SOC in early HER2-positive breast cancer (Gianni et al 2012) and in the first -line metastatic setting. Immune checkpoint inhibitors have shown promising efficacy in late-line metastatic HER2-positive breast cancer (Emens et al 2019) and, similar to the case in TNBC, it is quite likely that the addition of immune checkpoint inhibitors will further increase the efficacy of SOC in HER2-positive breast cancers. HER2-positive breast cancers are rich in TILs, and this may reflect their immunogenicity (Luen et al 2017). CCI [redacted] (data on file). Exploration of the triplet combination of T-DXd + durvalumab + paclitaxel in HER2-positive MBC is therefore justified.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP Section 4.3.1.

4.3.2 Durvalumab Dose Rationale

Based on available PK and safety data for weight-based doses of 10 mg/kg Q2W, 20 mg/kg Q4W, and simulated PK data for fixed doses of 750 mg Q2W and 1500 mg Q4W, the fixed-dosing regimen was predicted to provide comparable exposure to the body weight based dosing regimens. The proposed dosing schedule is aligned with the standard fixed dosing of 1500 mg durvalumab, which is supported by efficacy, safety, and tolerability data across multiple studies in multiple tumour types. To conform to the chemotherapy schedule in this module, the Sponsor proposes to use a standard durvalumab combination therapy dose and ratio at a Q[CCI]W dosing schedule.

The safety of a Q3W dosing schedule in combination with chemotherapy has been explored in the ongoing Study NCT02658214, where tremelimumab is administered at 75 mg in combination with 1120 mg durvalumab Q3W, followed by 1120 mg durvalumab Q3W. The combination has been declared tolerable and manageable. The 1120 mg dose of durvalumab is the Q3W equivalent of the standard 1500 mg Q4W dose.

Durvalumab has a flat exposure-response (efficacy and safety) relationship. NCT02537418 is an ongoing Phase 1 study of durvalumab ± tremelimumab in combination with multiple standard platinum-based chemotherapy regimens in patients with incurable advanced or metastatic cancer. The dose regimens in the study included a fixed dose cohort of durvalumab 1125 mg + tremelimumab 56 mg Q3W concurrent with platinum-based doublet chemotherapy, and subsequently a cohort of durvalumab 1500 mg + tremelimumab 75 mg Q3W concurrent with chemotherapy. In general, all regimens were tolerable and manageable at all dose levels. In terms of the safety data reported, there was no evidence of a dose response, including in the 1500 mg + 75 mg dose cohort (Juergens et al 2017).

In addition, PK modelling has been carried out to predict the effect of switching from a Q4W regimen to a Q3W regimen for durvalumab at a 1500 mg fixed dose and suggests that a Q3W schedule does not impose a significant increased safety risk based on expected durvalumab exposures.

A dose of [CCI] mg rather than [CCI] mg is being used for practical reasons (ie, due to the vial sizes available).

For additional details on the preclinical and clinical data that informed durvalumab dose selection, refer to the current durvalumab IB. For information on dose modifications for durvalumab, see Section 6.6 in this module.

4.3.3 Paclitaxel Schedule Rationale

In the dose-finding phase of this module (Part 1), paclitaxel will be administered at a dose of [redacted] mg/m^2 (dose level 1, Mod 4 Table 3), with an option to escalate to [redacted] mg/m^2 (dose level 2), QW as opposed to the United States Prescribing Information labelled paclitaxel dose of 175 mg/m^2 Q3W. In a meta-analysis of randomised, controlled studies in advanced breast cancer (5 studies, 1471 patients), which compared weekly administration of paclitaxel with Q3W administration, the analysis concluded that there was an improvement in OS with weekly paclitaxel (pooled HR, 0.78; 95% CI: 0.67, 0.89; p=0.001). Additionally, the incidence of neutropenia, neutropenic fever, peripheral neuropathy, and other SAEs was significantly lower with weekly paclitaxel compared with paclitaxel Q3W (Mauri et al 2010). Based on this meta-analysis and to be aligned to standard practice among breast cancer clinicians, the weekly regimen is proposed for this study.

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to this section and subsections in the core CSP Section 5.

Specific eligibility criteria for Module 4 are listed below.

5.1 Inclusion Criteria

Refer to core module for the other inclusion criteria. The below inclusion criterion is applicable to this module only.

16 Non-sterilised male patients who are sexually active with a female partner of child-bearing potential must use a condom with spermicide from screening to **6 months** after the final dose of study treatment. Not engaging in sexual activity for the duration of the study and drug washout period is an acceptable practice provided it is the preferred and usual lifestyle choice of the participant; however, periodic or occasional abstinence, the rhythm method, and the withdrawal method are not acceptable methods of contraception. It is strongly recommended for the female partners of a male patient to also use at least one highly effective method of contraception throughout this period, as described in the Core CSP Table 26. In addition, male patients should refrain from fathering a child, or freezing or donating sperm during the study and **for 6 months** after

the last dose of study treatment. Preservation of sperm should be considered prior to enrolment in this study.

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with durvalumab and paclitaxel.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.1 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for durvalumab and paclitaxel. The study treatments in this module are shown in Mod 4 Table 5.

Mod 4 Table 5 Study Treatments in Module 4

Intervention Name	T-DXd	Durvalumab	Paclitaxel
Type	Refer to Table 10 in the core CSP	Monoclonal antibody	Chemotherapy
Dosage Presentation		Vial	As sourced locally [a]
Unit Dose Strength(s)		Concentrate for solution for infusion, 50 mg/mL (500 mg/vial)	Variable
Dosage Level(s)		CCI mg	CCI or CCI mg/m^2
Route of Administration		IV	IV
Use		Experimental	Experimental
Sourcing		Provided centrally by the Sponsor	Sourced locally by site [b]
Packaging and Labelling		Study treatment will be provided in multiple vials in carton. Each of the vials and carton will be labelled as required per country requirements [a]	Sourced locally by site [b]

[a] Label text prepared for durvalumab (MEDI4736) will show the product name as "MEDI4736" or "durvalumab (MEDI4736)" depending upon the agreed product name used in the approved study master label document. All naming conventions are correct during this transitional period.
[b] Under certain circumstances when local sourcing is not feasible, paclitaxel may be supplied centrally through AstraZeneca
CSP = clinical study protocol; IV = intravenous; T-DXd = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration and monitoring.

T-DXd will be administered first, followed by durvalumab and then paclitaxel, with approximately one hour between infusions. Durvalumab will be administered approximately one hour (maximum 2 hours) after the end of the infusion of T-DXd. If there are no clinically significant infusion reactions within the first cycle, and at the discretion of the Investigator, then for all other cycles, durvalumab can be given no less than 15 minutes after the infusion of T-DXd has finished.

6.2.2 Durvalumab Preparation and Administration

Durvalumab will be supplied by AstraZeneca as a CCI mg vial concentrate for solution for infusion. The solution contains CCI mg/mL durvalumab, CCI mM histidine/ histidine hydrochloride, CCI mM trehalose dihydrate, and CCI % (w/v) CCI ; it has a pH of CCI and density of CCI g/mL. The label-claim volume is CCI mL.

Durvalumab is a sterile, clear to opalescent, colourless to slightly yellow solution, free from visible particles. Each vial selected for dose preparation should be inspected.

Preparation of Durvalumab Doses for Administration with an IV Bag

The dose of durvalumab for administration must be prepared by the Investigator or site's designated study intervention manager using aseptic technique and following local practices and site requirements. The total time from needle puncture of the durvalumab vial to the start of administration must not exceed 24 hours. Of this time, not more than 4 hours may be at room temperature, with the remaining time at 2°C to 8°C (36°F to 46°F), otherwise a new dose must be prepared from new vials.

Administration of Durvalumab

A dose of CCI mg will be administered using an IV bag containing 0.9% (w/v) saline or 5% (w/v) dextrose, with a final durvalumab concentration ranging from 1-15 mg/mL, and delivered through an IV administration set with a 0.2 or 0.22 µm filter. Add CCI mL (ie, CCI mg) of durvalumab to the IV bag. The IV bag size should be selected such that the final concentration is within 1-15 mg/mL. The bag should be mixed by gently inverting to ensure homogeneity of the dose in the bag.

The use of elastometric pumps and use of a pneumatic tube for transport of either a vial or a final prepared product have not been studied; please contact AstraZeneca before using.

Standard infusion time for durvalumab is 1 hour (± 10 minutes), however if there are interruptions, the total allowed time should not exceed 8 hours at room temperature.

Do not co-administer other drugs through the same infusion line.

The IV line will be flushed according to local practices to ensure the full dose is administered. Infusion time does not include the final flush time.

If either preparation time or infusion time exceeds the time limits, a new dose must be prepared from new vials. Durvalumab does not contain preservatives; any unused portion of the vial must be discarded immediately after use.

6.2.3 Paclitaxel Preparation and Administration

Paclitaxel will be prepared and administered as outlined in the prescribing information or treatment guidance in general use by the investigating site. Under certain circumstances when local sourcing by the study site is not feasible, AstraZeneca will centrally supply the drug, which will be labelled with local language translated text in accordance with regulatory guidelines.

All patients should be premedicated prior to paclitaxel administration in order to prevent severe hypersensitivity reactions. Such premedication may consist of dexamethasone 20 mg PO administered approximately 12 and 6 hours before paclitaxel, diphenhydramine (or its equivalent) 50 mg IV 30 to 60 minutes prior to paclitaxel, and cimetidine (300 mg) or ranitidine (50 mg) IV 30 to 60 minutes before paclitaxel.

6.2.4 Storage

For T-DXd storage information refer to Section 6.2.2 in the core CSP. For paclitaxel storage information refer to the prescribing information.

Durvalumab vials are stored at 2°C to 8°C (36°F to 46°F) and must not be frozen. The investigator, or an approved representative (eg, pharmacist), will ensure that all study intervention is stored in a secured area, in refrigerated temperatures (2°C to 8°C; 36°F to 46°F) and in accordance with applicable regulatory requirements. A temperature log will be used to record the temperature of the storage area. Temperature excursions outside the permissible range listed in the clinical supply packaging are to be reported to the monitor upon detection. A calibrated temperature monitoring device will be used to record the temperature conditions in the drug storage facility. Storage conditions stated in the IB may be superseded by the label storage.

Study intervention must be kept in original packaging until time of preparation to prevent prolonged light exposure.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after 8 cycles of T-DXd or discontinuation of paclitaxel (whichever occurs later) for patients with HR-positive tumours, as per local practice.

Refer to the core CSP Section 6.5 and Appendix J for details on T-DXd safety-specific restrictions; restricted, prohibited, and permitted therapies. Refer to the local prescribing information for paclitaxel with regard to warnings, precautions, contraindications, and prohibited medications during treatment and to the core CSP Appendix J1.

6.5.1 Rescue Medication

As a result of imAEs that could potentially be experienced by patients on durvalumab, steroids and other immunosuppressant rescue medication has to be made available to this patient population. The 2 products that fall into the category of immunosuppressants are infliximab (eg, for colitis) and mycophenolate (eg, for hepatitis). AstraZeneca supply chain will be responsible for sourcing these 2 rescue medications to the sites if local regulations prevent the use of infliximab and mycophenolate in this indication. These rescue medications must be receipted, controlled, and dispensed by the pharmacist and stored according to the labelled storage conditions, with temperature excursions reported accordingly.

If these rescue medications are centrally sourced: If required for use as a result of an imAE, then the IRT will provide to the pharmacists the kit identification number to be allocated to the patient at the time.

6.6 Dose Modification

6.6.1 T-DXd

The combination of T-DXd, durvalumab and/or chemotherapy may increase the risk or severity of potential overlapping toxicities. Furthermore, there is potential for new, unanticipated toxicities. The Investigator should use clinical judgement when attributing a toxicity to one or more of the agents. For dose modifications that apply to toxicities that are attributable to T-DXd refer to the core CSP Section 6.6.1.1.

6.6.2 Durvalumab

Dose delays are permitted for durvalumab (see Section 8.3.16.2 in this module). However, dose reduction is not permitted. Dose delays may be entered into the eCRF as unscheduled visits.

Subsequent time between 2 consecutive doses cannot be less than 19 days, based on the half-life of durvalumab.

6.6.3 Paclitaxel

Dose modification for paclitaxel should be done according to the local prescribing information.

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 in this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1, and to the paclitaxel local prescribing information for definitions of any AE that meets criteria for discontinuation.

7.1.1 Procedures for Discontinuation of Study Treatment

Refer to the core CSP Section 7.1.1.

7.1.2 Follow-up of Patients Post Discontinuation of Study Treatment

The follow-up period for durvalumab is 90-days after last dose of durvalumab. The follow-up period for T-DXd is 40-days (+7days) after last dose of T-DXd.

Additional assessments to be performed at the time of the 90-day safety follow-up are detailed in the SoA in Section 1.3.

Refer to the core CSP Section 7.1.2 and to the SoA in Section 1.3 for more information follow up of patients to PFS2 and for follow-up for survival.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for module specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1 for all subsections.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.2.2 Vital Signs

Vital signs will be performed at timepoints as specified in the SoA. Temperature, pulse rate, respiratory rate, and blood pressure will be assessed.

Blood pressure and pulse measurements will be assessed with a completely automated device. Manual techniques will be used only if an automated device is not available.

Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the patient in a quiet setting without distractions (eg, television, cell phones).

Vital signs will be evaluated and recorded in eCRF. Standard infusion time for T-DXd is approximately 90 minutes for first infusion and 30 minutes thereafter if the first infusion was well tolerated; however, if there are interruptions during infusion, the total allowed infusion time should not exceed 3 hours at room temperature.

If the infusion takes longer than 90 minutes, then blood pressure and pulse measurements should follow the principles as described below or be taken more frequently if clinically indicated. An hour observation period is recommended after the first infusion of T-DXd and the combination partner treatments that are administered intravenously.

First Infusion of Study Treatment

On the first day of study treatment administration, patients' vital signs will be monitored and collected/recorded in the eCRF before and after infusion of each study treatment as follows:

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 30 minutes ± 10 minutes after end of the T-DXd infusion (during the 1-hour observation period after the infusion). Additionally, approximately 30 minutes ± 10 minutes after end of infusion with durvalumab (during the 1-hour observation period after the durvalumab infusion). Another vital sign

collection should occur approximately 30 minutes ± 10 minutes after end of infusion with paclitaxel (during the 1-hour observation period at the end of all infusions).

Subsequent Infusions of Study Treatment

Prior to starting the T-DXd infusion (measured once from approximately 30 minutes before up to 0 minutes [ie, the beginning of the infusion]) and approximately 7 minutes ± 5 minutes after end of the T-DXd infusion (during the 15-minute observation period after the infusion). Additionally, approximately 15 minutes ± 5 minutes after end of infusion with durvalumab (during the 30-minute observation period after the durvalumab infusion). Another vital sign collection should occur approximately 15 minutes ± 5 minutes after end of infusion with paclitaxel (during the 30-minute observation period at the end of all infusions).

Vital signs during initial and subsequent T-DXd and combination partner infusions should be monitored, and any clinically significant findings at other times to those described above should be reported via an unscheduled visit form. Any changes in vital signs should be recorded as an AE (if applicable). Situations in which vital signs results should be reported as AEs are described in core CSP Section 8.3.5.

For any AEs of infusion reactions, the vital signs values should be entered into the eCRF.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3 for all subsections.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2.

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

8.3.11.1 T-DXd

Refer to the core CSP Section 8.3.11 for AESIs related to T-DXd.

8.3.11.2 Durvalumab

AESIs for durvalumab include, but are not limited to, events with a potential inflammatory or immune-mediated mechanism and which may require more frequent monitoring and/or interventions such as steroids, and/or immunosuppressants. These AESIs are being closely monitored in clinical studies with durvalumab monotherapy and combination therapy. An imAE is defined as an AE that is associated with drug exposure and is consistent with an immune-mediated MOA and where there is no clear alternate aetiology. Serologic, immunologic, and histologic (biopsy) data, as appropriate, should be used to support an imAE diagnosis. Appropriate efforts should be made to rule out neoplastic, infectious, metabolic, toxin, or other etiologic causes of the imAE.

If the Investigator has any questions regarding an event being an imAE, the Investigator should promptly contact the Study Physician.

AESIs observed with anti-PD-L1/PD-1 agents such as durvalumab include the following:

- Diarrhoea/colitis and intestinal perforation
- Pneumonitis/ILD
- Endocrinopathies (ie, events of hypophysitis/hypopituitarism, adrenal insufficiency, hyper and hypothyroidism, and Type I diabetes mellitus)
- Hepatitis/transaminase increases
- Nephritis/blood creatinine increases
- Pancreatitis/serum lipase and amylase increases

- Rash/dermatitis
- Myocarditis
- Myositis/polymyositis
- Neuropathy/neuromuscular toxicity (eg, Guillain-Barré syndrome and myasthenia gravis)
- Other inflammatory responses that are rare/less frequent with a potential immune-mediated aetiology include, but are not limited to, pericarditis, sarcoidosis, uveitis, and other events involving the eye, skin, haematological, and rheumatological events, vasculitis, non-infectious meningitis and non-infectious encephalitis. It is possible that events with an inflammatory or immune-mediated mechanism could occur in nearly all organs.

In addition, infusion-related reactions and hypersensitivity/anaphylactic reactions with a different underlying pharmacological aetiology are also considered AESIs.

Further information on these risks (eg, presenting symptoms) can be found in the current durvalumab IB. More specific guidelines for their evaluation and treatment are described in detail in Section 8.3.16 in this module. These guidelines have been prepared by the Sponsor to assist the Investigator in the exercise of his/her clinical judgment in treating these types of toxicities. These guidelines apply to AEs considered causally related to the study treatment by the reporting Investigator.

8.3.11.3 Paclitaxel

No AESIs will be considered for paclitaxel.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12. All SAEs that occur in patients still receiving study treatment within 90 days following the last dose of durvalumab after the final DCO must be reported as detailed in Section 8.3.13 of the core CSP. If durvalumab is discontinued and the patient continues to receive T-DXd and/or paclitaxel, SAEs within 40 days (+ 7 days) following the last dose of T-DXd or paclitaxel after the final DCO must be reported as detailed in Section 8.3.13 in the core CSP.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd
- The current IB for durvalumab

- The label of the current effective version of the SmPC paclitaxel, Hospira UK Limited.

8.3.14 Pregnancy

Patients in this module should follow the rules for T-DXd (refer to the core CSP Section 8.3.14).

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in Appendix K of the core CSP, and the T-DXd TMGs in Appendix L of the core CSP.

8.3.16.2 Toxicity Management and Dose Modification Information – Durvalumab

Comprehensive TMGs have been developed to assist Investigators with the recognition and management of toxicities associated with the use of the immune-checkpoint inhibitor durvalumab (PD-L1 inhibitor). Additionally, these guidelines are applicable when durvalumab is used alone or in combination and is administered concurrently or sequentially with other anti-cancer drugs (ie, antineoplastic chemotherapy, targeted agents), as part of a CSP-specific treatment regimen. The TMGs provide information for the management of immune-mediated reactions, infusion-related reactions, and non-immune mediated reactions that may be observed with checkpoint inhibitor monotherapy or combination checkpoint inhibitor regimens, with specific instructions for dose modifications (including discontinuations) and treatment interventions. Investigators are advised, however, to use local practice guidelines and consult local references for the management of toxicities observed with other cancer treatment.

The most current version of the TMGs entitled “Dosing Modification and Toxicity Management Guidelines for Immune-Mediated, Infusion-Related, and Non-Immune Mediated Reactions Durvalumab Monotherapy or Combination Therapy with Tremelimumab or Tremelimumab Monotherapy” is provided to the investigative site as an Annex document and is maintained within the Site Master File.

Dose modifications with regards to hold or discontinuation of durvalumab are required actions for the management of potential immune-mediated events or non-immune-mediated adverse reactions, as described in the Dosing Modification and Toxicity Management Guideline; permanent discontinuation of study treatment for these adverse events is considered a study-specific requirement.

Patients should be thoroughly evaluated, and appropriate efforts should be made to rule out neoplastic, infectious, metabolic, toxic, or other etiologic causes of the imAE. Serologic, immunologic, and histologic (biopsy) data, as appropriate, should be used to support an imAE diagnosis. In the absence of a clear alternative aetiology, events should be considered potentially immune-related.

Following the first dose of study treatment, subsequent administration of durvalumab can be modified based on toxicities observed as described in the Dosing Modification and Toxicity Management Guidelines. These guidelines have been prepared by the Sponsor to assist the Investigator in the exercise of his/her clinical judgment in treating these types of toxicities. These guidelines apply to AEs considered causally related to durvalumab by the reporting Investigator.

In addition, there are certain circumstances in which durvalumab should be permanently discontinued (see Section 7.1 of the core CSP).

8.3.16.3 Toxicity Management and Dose Modification Information – Paclitaxel

Investigators should follow the local prescribing information and standard clinical practice regarding management of paclitaxel-related toxicities. Anti-emetic therapy prior to and after paclitaxel treatment should be given as needed on a prophylactic and treatment basis in compliance with institutional standards; based on the currently available clinical safety data, it is recommended that patients receive anti-emetic agents prior to infusion of T-DXd and on subsequent days. Granulocyte colony stimulating factors in compliance with the standards of the centre should be given as needed for secondary prophylaxis or treatment of neutropenia.

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of durvalumab and paclitaxel in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd, durvalumab and paclitaxel, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information, including details on overdose with T-DXd. Refer to the local prescribing information for treatment in case of an overdose related to paclitaxel.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total anti-HER2 antibody, MAAA-1181a and durvalumab concentrations in serum and paclitaxel in plasma at timepoints as specified in the SoA in this module. Additional PK blood samples will

be collected for patients who receive treatment with chloroquine/hydroxychloroquine during the treatment period of the study. Refer to the core CSP Appendix M for further details.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd and durvalumab in serum will be collected as specified in the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9 for all subsections.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol	
Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix S

Module 5 – T-DXd + Tucatinib

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

EU CT: 2023-505309-18-00

This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

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Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 5 of this study will evaluate T-DXd in combination with tucatinib in patients with HER2-positive advanced/MBC in markets where tucatinib is commercially available or central supply of tucatinib is operationally feasible. This module will enrol patients with either no brain metastases or stable metastases. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2.

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 5 Table 1) and follow-up period (Mod 5 Table 2) for Module 5 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.2	
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
Weight	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
ECOG PS	X [b]				X[b]		X[b]		X [b]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [c]													
Pregnancy test [d]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [b]		X	X	X[b]		X[b]		X [b]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
ECHO/MUGA (LVEF)[f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [g] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [r]	As clinically indicated										X	8.2.5.4	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycle 4, 6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for tucatinib PK testing [j]	X				X				X (C4 only)				8.5.1
Post-dose blood sample for tucatinib PK testing[k]					X 2 hours post dose				X (C4 only) 2 hours post dose				8.5.1
Blood sample for T-DXd ADA assessment in serum[l]	X				X				X (C4, then every 4 cycles)		X	8.5.2	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study treatments administered													
T-DXd	[CCI] or [CCI] mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Tucatinib	[CCI] mg or [CCI] mg bid orally												6.2.2
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients who are enrolled with baseline stable brain metastases. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans). Patients without brain metastases require a brain MRI at EOT. Additional brain scans for subsequent tumour assessments are not required, unless clinically indicated.										X	7.1.2, 8.1.1	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [m] (if not collected at screening)	X^n											8.6.1, 8.6.2	
Paired on-treatment biopsy					X^n							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	
Blood sample for CCI [redacted]	CCI [redacted]											8.6.1	
Blood sample for CCI [redacted]												8.6.1	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
CCI													
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI CCI												8.6.1	
Blood sample for CCI [o]												8.6.1	
Blood sample to CCI												8.6.1	
Blood sample for CCI												8.7	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [q]	X											8.1.5.1	
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6, and Month 12 for patients with stable brain metastases at baseline).		X	8.1.6.1	
MDASI-BT	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.2	
NANO	X		X (to be performed at every visit for patients with stable brain metastases at baseline)								X	8.1.6.3	

Mod 5 Table 1 Schedule of Activities During the Treatment Period (Part 1 and 2) - Module 5

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening laboratory assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.
[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd.
[i] Post-dose PK samples for T-DXd to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[j] Pre-dose PK samples to be taken immediately or up to 2 hours prior to tucatinib dose on D1 of C1, 2, and 4.
[k] Post-dose PK samples for tucatinib to be taken 2 hours after tucatinib dose on D1 of C2 and C4.
[l] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[m] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the core CSP Sections 8.6.1 and 8.6.2).
[n] Refer to the core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[o] At time of suspected CCI [redacted], an additional sample should be taken.

[p] The sample for genetic research will be obtained at CCI If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI . Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.

[q] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use in case needed, in accordance with device training.

[r] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in the core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; bid = twice daily; C = cycle; CANTAB = Cambridge Neuropsychological Test Automated Battery; COVID-19 = coronavirus 2019-nCoV; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI ; DNA = deoxyribonucleic acid; D = day; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI ; T-DXd = trastuzumab deruxtecan.

Mod 5 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 5

Procedure	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Vital signs	X		8.2.2	
Weight	X		8.2.1	
Full physical examination	X		8.2.1	
SpO_2	X		8.2.5.2	
ECOG PS	X		8.2.5.3	
12-lead ECG	X		8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4, Table 11	
Troponin [c]	X		Table 11	
Urinalysis	As clinically indicated		Table 11	
Coagulation tests	As clinically indicated		Table 11	
Pregnancy test	X	As clinically indicated	8.2.4	
Blood sample for T-DXd ADA analysis in serum	X		8.5.2	
Concomitant medications	X		6.5	
AEs/SAEs	X		8.3	
PFS2 assessment	X	X	7.1.2.1, 8.1.3	
Health care resource use (HOSPAD)	X		8.8	

Mod 5 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 5

Procedure	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: To be completed daily up to and including the first survival visit.		8.1.5.2	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3	
MDASI-BT		X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.2	
NANO		X (to be performed at month 3 and month 6 for patients with stable brain metastases at baseline)	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5	
Survival status	X	X	7.1.2.1, 8.1.4	

Mod 5 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 5

Procedure	Safety F/U (40 days [+ 7 days] after last dose) [a]	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4	
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1	

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+ 7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT = MD Anderson Symptom Inventory for brain tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Trastuzumab was the first approved HER2-targeting mAb, which is indicated for both advanced (Slamon et al 2001) and early HER2-positive breast cancer (Piccart-Gebhart et al 2005). The addition of trastuzumab to standard chemotherapy improved mPFS from 4.6 to 7.4 months, and OS from 20.3 months to 25.1 months (Slamon et al 2001).

Combinations of HER2-targeted therapies have become the SOC in neoadjuvant, adjuvant and first-line metastatic disease (Baselga et al 2012, Gianni et al 2012, Tucatinib USPI 2020, von Minckwitz et al 2017) in HER2-positive breast cancer. While the combinations of 2 antibodies, such as trastuzumab and pertuzumab, that bind different extracellular domains of the HER2-receptor resulting in synergistic anti-tumour effect (Scheuer et al 2009), are currently the SOC, there is increasing interest in combining antibodies or ADCs with TKIs of HER2. Preclinical data show enhancement of the payload internalization when combining T-DXd with neratinib in HER2-amplified lung cancer (Li et al 2020). ADCs have previously been combined with TKIs, for example, tucatinib in combination with T-DM1 showed acceptable toxicity and preliminary anti-tumour activity (Borges et al 2018).

2.2.1 Tucatinib

Tucatinib is an oral, potent, reversible HER2-specific TKI. In vitro, tucatinib inhibits phosphorylation of HER2 and HER3, resulting in inhibition of downstream MAPK and AKT signalling and cell proliferation, and has shown anti-tumour activity in HER2 expressing tumour cells. In vivo, tucatinib inhibited the growth of HER2 expressing tumours. The combination of tucatinib and trastuzumab showed increased anti-tumour activity *in vitro* and *in vivo* compared to either drug alone.

Tucatinib combined with trastuzumab and capecitabine showed a median PFS of 7.8 months compared to 5.6 months with placebo combined with trastuzumab and capecitabine in patients including those with brain metastases (Murthy et al 2020). Based on these results tucatinib was approved in heavily pre-treated patients with HER2-positive metastatic breast cancer.

In the United States, tucatinib is approved in combination with trastuzumab and capecitabine for treatment of adult patients with advanced unresectable or metastatic HER2-positive breast cancer, including patients with brain metastases, who have received one or more prior anti-HER2-based regimens in the metastatic setting (tucatinib [TUKYSA ®, Seagen] Tucatinib USPI 2020). As of October 2020, tucatinib is also approved in Australia, Canada, Singapore, and Switzerland.

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Tucatinib

Tucatinib is an oral, potent, reversible HER2-specific TKI that is being developed for HER2-positive breast cancer. Tucatinib selectively inhibits HER2, a feature that differentiates it from the other small molecule HER2-targeted TKIs, all of which are dual inhibitors of both EGFR and HER2. This may potentially lead to improved toxicity profile compared with non-selective HER2-targeting TKIs (Moulder et al 2017).

In the HER2CLIMB study, patients with HER2-positive metastatic breast cancer previously treated with trastuzumab, pertuzumab, and trastuzumab emtansine, who had or did not have brain metastases, were randomly assigned to receive either tucatinib or placebo, in combination with trastuzumab and capecitabine. The mPFS was 7.8 months and 5.6 months, respectively. Overall survival at 2 years was 44.9% in the tucatinib-combination group and 26.6% in the placebo-combination group (hazard ratio for death, 0.66), and the mOS was 21.9 months and 17.4 months, respectively. Among the patients with brain metastases, PFS at 1 year was 24.9% in the tucatinib-combination group and 0% in the placebo-combination group, and the mPFS was 7.6 months and 5.4 months, respectively (Murthy et al 2020).

T-DXd monotherapy has also shown preliminary efficacy in patients with stable brain metastases. In the U201 (DESTINY-Breast01) study, a subset (n=24) of patients with stable brain metastases was found to have a mPFS of 18.1 months, while the ITT population had mPFS of 16.4 months. Overall, only 4 out of 48 patients in the study had disease progression in the CNS (Jerusalem et al 2020).

Given the encouraging efficacy that both tucatinib and T-DXd have demonstrated in heavily pre-treated patients with HER2-positive MBC with or without brain metastases, an exploration of T-DXd + tucatinib combination therapy is justified.

2.3.2 Potential Risks of T-DXd + Tucatinib

Tucatinib is a potent and selective oral HER2-inhibitor. Due to its selectivity, it has been postulated that tucatinib may have a better safety and tolerability profile compared to the other HER-2 targeted TKIs (Pheneger et al 2009). A Phase 1 study of tucatinib in advanced solid tumours suggested a lower incidence and severity of diarrhoea and rash than that typically associated with current dual HER2/EGFR inhibitors (Moulder et al 2017). In the HER2CLIMB study combining tucatinib, trastuzumab and capecitabine, the most common AEs included diarrhoea, palmar–plantar erythrodysesthaesia syndrome, nausea, fatigue, and vomiting. Diarrhoea and elevated aminotransferase levels of grade 3 or higher were more common in the tucatinib-combination group than in the placebo-combination group (Murthy et al 2020). These elevations were typically transient and reversible.

Potential overlapping toxicities between T-DXd and tucatinib are gastrointestinal toxicities (diarrhoea, nausea, and vomiting), hepatic toxicities (ALT and AST increased), decreased appetite, rash, and fatigue. Careful monitoring of the patient symptoms will be applied along with the management of diarrhoea with anti-diarrhoea medications and nausea/vomiting with anti-emetics. Increases in levels of AST and/or ALT should be monitored at each visit. Toxicity management guidelines for T-DXd are provided in the core CSP Section 8.3.16 and Appendix L, with further guidance for Investigators available in the current T-DXd IB. Tucatinib safety details are described in the latest effective local prescribing information for tucatinib. See the core CSP Appendix L for management of overlapping toxicities. The Investigator should use clinical judgement when attributing toxicity to one or more of the agents.

2.3.3 Overall Benefit/Risk Conclusion

Both T-DXd and tucatinib are emerging therapies and have recently been approved for the treatment of HER2-positive MBC (Sections 2.2 and 2.2.1). T-DXd has demonstrated an unparalleled mPFS of 16.4 months in late-line HER2-positive patient population (Modi et al 2020). Tucatinib has shown efficacy particularly in patients with brain metastases with mPFS of 7.6 months versus 5.4 months in patient who did not receive tucatinib (Murthy et al 2020). While T-DXd and tucatinib both target HER2, the mechanisms of actions for these two drugs are different. T-DXd is an ADC with a cytotoxic payload, whereas tucatinib is a TKI, binding to the intracellular part of HER2-receptor. Preclinical data show that the irreversible HER2 TKIs, such as neratinib, can enhance the internalisation of the cytotoxic payload when combined with ADCs (Li et al 2020). CCI
(data on file).

Typical for TKIs, the most common AEs for tucatinib are diarrhoea, nausea, and vomiting. These AEs are also shared with T-DXd. Careful continuous monitoring of gastrointestinal symptoms, in particular, is required. Early intervention with anti-diarrhoea and anti-emetic medications is recommended. In addition, careful monitoring of liver function is required to avoid hepatotoxicity.

Given the encouraging efficacy that both drugs have demonstrated individually in HER2-positive MBC in patients with brain metastases in particular, as well as the measures in place to manage the potential overlapping toxicities of T-DXd and tucatinib, the benefit-risk ratio of this combination is anticipated to be favourable.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients with no or stable brain metastases will be treated with T-DXd + tucatinib. This module of the study will be conducted in countries where tucatinib is commercially available or central supply of tucatinib is operationally feasible.

4.1.1 Module 5 Part 1

In Part 1, approximately CCI patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC in second-line or later (≥ 2L) will be enrolled in each dose level to determine the RP2D for the T-DXd + tucatinib combination. Dose-escalation/de-escalation will be determined by using a mTPI-2 algorithm (refer to the core CSP Section 4.1.2 Table 7). In Part 1 in this module, CCI patients will be enrolled at dose level (1) (Mod 5 Table 3). Dose level (1) will be defined as the RP2D if no more than one DLT is observed from the first CCI patients. If the full doses are not tolerated, then a dose de-escalation will be performed by reducing the tucatinib dose to CCI mg PO bid first (Dose level -1) and then T-DXd dose to CCI mg/kg (Dose level - 2). Beyond the first cohort of CCI patients, patients will be enrolled in cohorts of CCI at dose level (1), (-1), or (-2) following the mTPI-2 algorithm. Dose level decisions and determination of the RP2D will be made by the SRC (core CSP Section 9.6). Additional details on the dose-finding period and combination dose-finding approach are provided in the core CSP Section 4.1.1.

Mod 5 Table 3 Module 5 Part 1, Dose Finding

mTPI-2 Escalation/De-escalation		
Dose Level (1)	**Dose Level (-1)**	**Dose Level (-2)**
T-DXd dose level: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid	T-DXd dose level: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid	T-DXd dose level: CCI mg/kg IV infusion Q3W Tucatinib: CCI mg PO bid

bid = twice daily; IV = intravenous; mTPI-2 = modified toxicity probability index-2; PO = orally; Q3W = every 3 weeks; T-DXd = trastuzumab deruxtecan

4.1.2 Module 5 Part 2

In Part 2, approximately 35 patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer who have not received prior treatment for advanced/MBC (ie, 1L) will be randomised to determine evidence of anti-tumour activity of the combination. Patients will be treated with T-DXd + tucatinib at the RP2D from Part 1 (Mod 5 Table 4).

Mod 5 Table 4 Module 5 Part 2, Dose Expansion

Study Treatment	Dose
T-DXd	RP2D from Part 1
Tucatinib	RP2D from Part 1

RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP 4.2.2.

4.2.3 Rationale for T-DXd + Tucatinib in HER2-positive Advanced/MBC

Combinations of HER2-targeted therapies are the SOC in HER2-positive MBC and have led to improved clinical outcomes compared to single agent HER2-targeted therapy, when combined with chemotherapy (Baselga et al 2012). Combinations that have been studied include antibody-antibody (Baselga et al 2012), antibody-TKI (Johnston et al 2020), ADC-antibody (Perez et al 2017) and ADC-TKI (Borges et al 2018). T-DXd and tucatinib are recent additions to the increasing number of HER2-targeted therapies.

T-DXd has demonstrated a mPFS of 16.4 months (N=184) in the late-line HER2-positive patient population (Modi et al 2020). In addition, a subset (n=24) of patients with stable brain metastases was found to have a mPFS of 18.1 months, which is longer than that seen in the general population, though this result must be interpreted with caution given the small sample size (Jerusalem et al 2020). Tucatinib has shown promising efficacy in HER2-positive MBC with mPFS of 7.6 months versus 5.4 months in patients who did not receive tucatinib (Murthy et al 2020).

In patients with active and stable brain metastases, the median CNS-PFS was 9.9 months in the tucatinib arm versus 4.2 months in the control arm (Lin et al 2020). Both agents have been approved for the treatment of late-line HER2-positive MBC. Tucatinib has previously been combined with another ADC, T-DM1, resulting in a mPFS of 8.2 months in a heavily pre-treated patient population. The combination had an acceptable toxicity profile (Borges et al 2018). In addition to the clinical rationale, preclinical data support testing the combination of T-DXd and tucatinib in clinical trials. Studies have shown that the addition of TKI to ADC can enhance the internalization of the cytotoxic payload (Li et al 2020). Targeting multiple steps in the HER-2 signalling pathway may result in an increased anti-tumour activity.

Since both T-DXd and tucatinib have shown efficacy in HER2-positive MBC and have the potential to synergize with each other, it is important to determine the preliminary efficacy and safety profile for this combination.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP 4.3.1.

4.3.2 Tucatinib Dose Rationale

The starting dose in Part 1 will be CCI mg PO BID for tucatinib. This is the labelled dose for the drug. If the full doses are not tolerated, then a dose de-escalation will be performed by reducing the tucatinib dose to CCI mg PO bid first and then T-DXd dose to CCI mg/kg. The justification for starting with full doses is that the potential overlapping toxicities for this combination are all considered to be manageable, and several measures, including recommendation for anti-diarrhoea supportive medication, have been incorporated into this study protocol to mitigate any potential or identified risks associated with these agents. In addition, a study testing a combination of T-DM1 and tucatinib found that the maximum tolerated dose for tucatinib was 300 mg PO bid (Borges et al 2018).

4.4 End of Study Definition

Refer to the core CSP 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with tucatinib.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.6 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for tucatinib. The study treatments in this module are shown in Mod 5 Table 5.

Mod 5 Table 5 Study Treatments in Module 5

Intervention Name	T-DXd	Tucatinib
Type	Refer to the core CSP Table 10	Tyrosine kinase inhibitor
Dosage Presentation		Tablet
Unit Dose Strength(s)		50 mg and 150 mg
Dosage Level(s)		CCI mg or CCI mg
Route of Administration		Oral
Use		Experimental
Sourcing		Sourced locally by country or centrally [a]
Packaging and Labelling		Sourced locally by country or centrally [a]

[a] Under circumstances when local sourcing is not feasible, tucatinib may be supplied by AstraZeneca or an AstraZeneca-approved supplier

CSP = clinical study protocol; T-DXd- = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration and monitoring.

T-DXd will be administered via infusion. Tucatinib can be administered either before or after T-DXd but it is important that the 12-hour window for twice-daily dosing is taken into account. Tucatinib should be administered at hospital/clinic during visits for T-DXd infusion to allow for PK testing.

6.2.2 Tucatinib Administration

Tucatinib will be administered as outlined in the prescribing information or treatment guidance in general use by the investigating site. Tucatinib may be supplied locally by country or centrally by AstraZeneca or an AstraZeneca-approved supplier and will be labelled with local language translated text in accordance with regulatory guidelines.

Tucatinib will be dispensed to patients either during scheduled visits to the clinic or by delivery to the patient's home.

Dosing and handling instructions and emergency contact details will be provided to patients in writing. Once the patient or his/her caregiver is deemed appropriate for at-home or remote

location administration, they must receive appropriate training. All necessary supplies and instructions for administration of tucatinib will be provided.

Tucatinib is supplied as 50 mg and 150 mg tablets.

6.2.3 Storage

For T-DXd storage information refer to the core CSP Section 6.2.2. For tucatinib storage information refer to the prescribing information.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after patients have completed 8 or more cycles of T-DXd for patients with HER2-positive tumours, as per local practice.

Restricted, prohibited, and permitted concomitant medications are described in the core CSP Appendix J. Refer also to the core CSP Section 6.5 for further details. Refer to the local prescribing information for tucatinib with regard to warnings, precautions, contraindications, and prohibited medications during treatment and to the core CSP Appendix J3.

6.5.1 Rescue Medication

Not applicable to this module.

6.6 Dose Modification

6.6.1 T-DXd

The combination of T-DXd and tucatinib may increase the risk or severity of potential overlapping toxicities. Furthermore, there is potential for new, unanticipated toxicities. The Investigator should use clinical judgement when attributing a toxicity to one or more of the agents. For dose modifications that apply to toxicities that are attributable to T-DXd refer to the core CSP Section 6.6.1.

6.6.2 Tucatinib

Mod 5 Table 6 provides recommended dose levels when dose modifications of tucatinib for a patient are necessary.

Mod 5 Table 6 Dose Reduction Levels of Tucatinib

Dose Level 1	Dose Level -1	Dose Level -2	Dose Level -3
300 mg bid	250 mg bid	200 mg bid	150 mg bid

If CCI mg is the RP2D, then the initial dose reduction level would be CCI mg. Dose reductions CCI CCI mg may be made if considered clinically appropriate by the investigator. Dose reductions below 150 mg are not permitted.
bid = twice daily; RP2D = recommended phase 2 dose

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 in this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for module-specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2.

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

No AESIs will be considered for tucatinib. Refer to the core CSP Section 8.3.11 for details on AESIs related to T-DXd.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd.
- The label of the current effective version of the SmPC for tucatinib, Seagen, Inc.

8.3.14 Pregnancy

Refer to the core CSP Section 8.3.14.

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in the core CSP Appendix K, and the T-DXd TMGs in the core CSP Appendix L.

8.3.16.2 Toxicity Management and Dose Modification Information – Tucatinib

Toxicities of tucatinib may be managed according to the local prescribing information.

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of tucatinib in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd and tucatinib, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information. Refer to the local prescribing information for treatment in case of an overdose related to tucatinib.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total anti-HER2 antibody, and MAAA-1181a in serum, and tucatinib and its metabolite in plasma at timepoints as specified in the SoA in this module. Additional PK blood samples will be collected for patients who receive treatment with chloroquine/hydroxychloroquine during the treatment period of the study. Refer to the core CSP Module M for further details.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd in serum will be collected as specified in the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol	
Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix T

Module 6 – T-DXd + Tucatinib in patients with active brain metastases

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

Regulatory Agency Identifier Number(s)

IND: 151,117

EudraCT: 2019-004531-22

EU CT: 2023-505309-18-00

This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

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DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 6 of this study will evaluate T-DXd in combination with tucatinib in patients with HER2-positive advanced/MBC, in the first-or second-line setting with active brain metastases, in markets where tucatinib is commercially available or central supply of tucatinib is operationally feasible. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2.

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 6 Table 1) and follow-up period (Mod 6 Table 2) for Module 6 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.2	
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
Weight	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
ECOG PS	X [b]				X[b]		X[b]		X [b]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [c]													
Pregnancy test [d]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [b]		X	X	X[b]		X[b]		X [b]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3	

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
ECHO/MUGA (LVEF) [f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [g] (local assessment)		X				X		X		X	X	Table 11, 8.2.5.1	
Ophthalmologic assessments [r]	As clinically indicated										X	8.2.5.4	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycle 4,6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1	
Pre-dose blood sample for tucatinib PK testing [j]	X				X				X (C4 only)				8.5.1
Post-dose blood sample for tucatinib PK testing [k]					X 2 hours post-dose				X (C4 only) 2 hours post-dose				8.5.1

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for T-DXd ADA assessment in serum [1]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Study treatments administered													
T-DXd	CCI or CCI mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Tucatinib	CCI mg or CCI mg bid orally												6.2.2
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 until RECIST 1.1 disease progression. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.											8.1.1	
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local										X	7.1.2, 8.1.1	

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	therapy for CNS-progression should maintain their assessment schedule (including scans).												
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour[m] (if not collected at screening)	X [n]											8.6.1, 8.6.2	
Paired on-treatment biopsy					X [n]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	
Blood sample for CCI [redacted]	CCI [redacted]											8.6.1	

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
	CCI												
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI												8.6.1	
Blood sample for CCI CCI												8.6.1	
Blood sample for CCI [o]												8.6.1	
Blood sample to CCI												8.6.1	

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
					during the day)								
Blood sample for CCI [redacted]	CCI [redacted]											8.7	
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if patient does not have own compatible device) and training [q]	X											8.1.5.1	
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6, and Month 12)		X	8.1.6.1	
MDASI-BT	X				X		X		X		X	8.1.6.2	

Mod 6 Table 1 Schedule of Activities During the Treatment Period (Part 2 only)- Module 6

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
NANO	X				X		X		X		X	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening laboratory assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.
[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.
[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.
[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, day 1 of each cycle 2-3 hours post infusion, and EOT, and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.
[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8 for T-DXd.
[i] Post-dose PK samples for T-DXd to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.
[j] Pre-dose PK samples to be taken immediately or up to 2 hours prior to tucatinib dose on D1 of C1, 2, and 4.
[k] Post-dose PK samples for tucatinib to be taken 2 hours after tucatinib dose on D1 of C2 and C4.
[l] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.
[m] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the Core CSP Sections 8.6.1 and 8.6.2).
[n] Refer to the Core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.
[o] At time of suspected CCI [redacted], an additional sample should be taken.

[p] The sample for genetic research will be obtained at CCI. If, for any reason, the sample is not drawn at CCI it may be taken at CCI CCI. Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.

[q] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use in case needed, in accordance with device training.

[r] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; bid -twice daily; BICR = blinded independent central review; C = cycle; CANTAB, Cambridge Neuropsychological Test Automated Battery; COVID-19 = coronavirus 2019-nCoV; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI; D = day; DNA = deoxyribonucleic acid; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT, MDASI Brain Tumor; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI; T-DXd = trastuzumab deruxtecan.

Mod 6 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 6

Procedure	Safety F/U (40 days [a] [+ 7 days]) after last dose	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Vital signs	X		8.2.2	
Weight	X		8.2.1	
Full physical examination	X		8.2.1	
SpO_2	X		8.2.5.2	
ECOG PS	X		8.2.5.3	
12-lead ECG	X		8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4, Table 11	
Troponin [c]	X		Table 11	
Urinalysis	As clinically indicated		Table 11	
Coagulation tests	As clinically indicated		Table 11	
Pregnancy test	X	As clinically indicated	8.2.4	
Blood sample for T-DXd ADA analysis in serum	X		8.5.2	
Concomitant medications	X		6.5	

Mod 6 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 6

Procedure	Safety F/U (40 days [a] [+ 7 days]) after last dose	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
AEs/SAEs	X		8.3	
PFS2 assessment	X	X	7.1.2.1, 8.1.3	
Health care resource use (HOSPAD)	X		8.8	
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: To be completed daily up to and including the first survival visit.		8.1.5.2	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3	
MDASI-BT		X (3 months & 6 months only)	8.1.6.2	
NANO		X (3 months & 6 months only)	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5	
Survival status	X	X	7.1.2.1, 8.1.4	

Mod 6 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up – Module 6

Procedure	Safety F/U (40 days [a] [+ 7 days]) after last dose	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4	
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1	

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.
[c] During the follow-up period, troponin will be measured locally. If an additional troponin sample is taken, then an ECG also needs to be performed.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT=MDASI Brain Tumor; MRI = magnetic resonance imaging; NANO = Neurologic Assessment in Neuro-Oncology; PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumors; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Refer to Module 5 Section 2.2.

2.3 Benefit/Risk Assessment

2.3.1 Potential Benefits of T-DXd + Tucatinib

Refer to Module 5 Section 2.3.1.

2.3.2 Potential Risks of T-DXd + Tucatinib

Refer to Module 5 Section 2.3.2.

2.3.3 Overall Benefit/Risk Conclusion

Refer to Module 5 Section 2.3.3.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients with active brain metastases will be treated with T-DXd + tucatinib. This module of the study will be conducted in countries where tucatinib is commercially available or central supply of tucatinib is operationally feasible.

There is no Part 1 (dose-finding) in Module 6. Data from Module 5 Part 1 will be used to assess dose finding (refer to Module 5 Section 4.1.1).

4.1.1 Module 6 Part 2

In Part 2, approximately 35 patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer in the first or second-line treatment setting with active brain metastases will be randomised in countries where tucatinib is commercially available or

central supply of tucatinib is operationally feasible to determine evidence of anti-tumour activity. Patients will be treated with T-DXd + tucatinib at the RP2D from Module 5 Part 1 (Mod 6 Table 3).

Mod 6 Table 3 Module 6 Part 2, Dose Expansion

Study Treatment	Dose
T-DXd	RP2D from Module 5 Part 1
Tucatinib	RP2D from Module 5 Part 1

RP2D = recommended phase 2 dose; T-DXd = trastuzumab deruxtecan

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

4.2.3 Rationale for T-DXd + Tucatinib in HER2-positive Advanced/MBC

Combinations of HER2-targeted therapies are the SOC in HER2-positive MBC and have led to improved clinical outcomes compared to single agent HER2-targeted therapy, when combined with chemotherapy (Baselga et al 2012). Combinations that have been studied include antibody-antibody (Baselga et al 2012), antibody-TKI (Johnston et al 2020), ADC-antibody (Perez et al 2017) and ADC-TKI (Borges et al 2018). T-DXd and tucatinib are recent additions to the increasing number of HER2-targeted therapies.

T-DXd has demonstrated a mPFS of 16.4 months (n=184) in the late-line HER2-positive patient population (Modi et al 2020). In addition, a subset (n=24) of patients with stable brain metastases were found to have a mPFS of 18.1 months, which is longer than that seen in the general population, though this result must be interpreted with caution given the small sample size (Jerusalem et al 2020). Tucatinib has shown promising efficacy in HER2-positive MBC with mPFS of 7.6 months versus 5.4 months in patients who did not receive tucatinib (Murthy et al 2020).

In patients with active and stable brain metastases, the median CNS-PFS was 9.9 months in the tucatinib arm versus 4.2 months in the control arm (Lin et al 2020). Both agents have been approved for the treatment of late-line HER2-positive MBC. Tucatinib has previously been combined with another ADC, T-DM1, resulting in a mPFS of 8.2 months in a heavily pre-treated patient population. The combination had an acceptable toxicity profile (Borges et al 2018) CCI [redacted] (data on file). Studies have shown that the addition of

TKI to ADC can enhance the internalization of the cytotoxic payload (Li et al 2020). Targeting multiple sites in the HER-2 signalling pathway may result in an increased anti-tumour activity.

Patients with brain metastases have limited treatment options, representing an unmet medical need. Since both T-DXd and tucatinib have shown efficacy in CNS disease and the two drugs have the potential for synergistic activity, it is important to determine the preliminary efficacy and safety profile for this combination.

4.2.4 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

4.3.1 T-DXd Dose Rationale

Refer to the core CSP Section 4.3.1.

4.3.2 Tucatinib Dose Rationale

Refer to Module 5 Section 4.3.2.

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

Specific eligibility criteria for Module 6 are listed below.

5.1 Inclusion Criteria

Refer to core module for the other inclusion criteria. The below inclusion criteria are applicable to this module only.

8 **Part 2**, Zero or one prior lines of therapy for advanced/MBC allowed. (Additional requirement: at least 12 months since last adjuvant or neoadjuvant HER2-targeted therapy, or chemotherapy required). Prior taxane, trastuzumab and pertuzumab exposure is allowed in the adjuvant or neoadjuvant setting.

12 **Part 2,** At least one lesion, not previously irradiated, that qualifies as a RECIST 1.1 TL at baseline (see Appendix G). Up to 2 active brain lesions not previously irradiated may be selected as RECIST 1.1 TLs at baseline.

CNS Inclusion

18 Patients must have either:

(a) Untreated brain metastasis not needing local therapy. For patients with untreated CNS lesions > 2.0 cm on screening contrast MRI, discussion with and approval from the Study Physician or delegate is required prior to enrolment.

(b) Brain metastasis previously treated with local therapy and have progressed since prior local therapy, provided that there is no clinical indication for immediate re-treatment for local therapy according to the investigator's opinion.

5.2 Exclusion Criteria

CNS Exclusion

27 Ongoing use of systemic corticosteroids for control of symptoms of brain metastasis at a total daily dose of > 2 mg dexamethasone (or equivalent).

28 Any brain lesion thought to require immediate local therapy, including (but not limited to) a lesion in an anatomic site where increase in size or possible treatment-related edema may pose risk to patients (eg, brain stem lesions). Patients who undergo local treatment for such lesions identified screening contrast brain MRI may still be eligible for the study based on criteria described under CNS inclusion criteria 20 (core CSP Section 5.2).

6 STUDY INTERVENTION

Study treatment in this module refers to T-DXd in combination with tucatinib.

6.1 Study Intervention(s) Administered

Refer to the core CSP Section 6.6 for dose modification details for T-DXd, and to Section 6.6 in this module for dose modification details for tucatinib. The study treatments in this module are shown in Mod 6 Table 4.

Mod 6 Table 4 Study Treatments in Module 6

Intervention Name	T-DXd	Tucatinib
Type	Refer to core CSP Table 10	Tyrosine kinase inhibitor
Dosage Presentation		Tablet
Unit Dose Strength(s)		50 mg and 150 mg
Dosage Level(s)		CCI mg or CCI mg
Route of Administration		Oral
Use		Experimental
Sourcing		Sourced locally by country or centrally [a]
Packaging and Labelling		Sourced locally by country or centrally [a]

[a] Under circumstances when local sourcing is not feasible, tucatinib may be supplied by AstraZeneca or an AstraZeneca-approved supplier.
CSP = clinical study protocol; T-DXd = trastuzumab deruxtecan.

6.2 Preparation/Handling/Storage/Accountability of Interventions

Refer to the core CSP Section 6.2 for general considerations for this section.

6.2.1 T-DXd Preparation and Administration

Refer to the core CSP Section 6.2.1 for details on T-DXd preparation, administration and monitoring.

T-DXd will be administered via infusion. Tucatinib can be administered either before or after T-DXd but it is important that the 12-hour window for twice-daily dosing is taken into account. Tucatinib should be administered at hospital/clinic during visits for T-DXd infusion to allow for PK testing.

6.2.2 Tucatinib Administration

Tucatinib will be administered as outlined in the prescribing information or treatment guidance in general use by the investigating site. Tucatinib may be supplied locally by country or centrally by AstraZeneca or an AstraZeneca-approved supplier and will be labelled with local language translated text in accordance with regulatory guidelines.

Tucatinib will be dispensed to patients either during scheduled visits to the clinic or by delivery to the patient's home.

Dosing and handling instructions and emergency contact details will be provided to patients in writing. Once the patient or his/her caregiver is deemed appropriate for at-home or remote

location administration, they must receive appropriate training. All necessary supplies and instructions for administration of tucatinib will be provided.

Tucatinib is supplied as 50 mg and 150 mg tablets.

6.2.3 Storage

For T-DXd storage information refer to the core CSP Section 6.2.2. For tucatinib storage information refer to the prescribing information.

6.3 Measures to Minimise Bias: Randomisation and Blinding

Refer to the core CSP Section 6.3. This is an open-label study. The actual treatment given to patients in Part 1 and Part 2 will be determined by the assignment scheme in the IRT. As with all modules in Part 2, allocation of patients to Module 6 and 7 is randomised and will be determined by the randomisation scheme in the IRT.

6.4 Study Intervention Compliance

Refer to the core CSP Section 6.4.

6.5 Concomitant Therapy

Standard endocrine therapy (aromatase inhibitor or tamoxifen) administered concurrently with the study treatment is allowed after patients have completed 8 or more cycles of T-DXd for patients with HER2-positive tumours, as per local practice.

Restricted, prohibited, and permitted concomitant medications are described in the core CSP Appendix J. Refer also to the core CSP Section 6.5 for further details. Refer to the local prescribing information for tucatinib with regard to warnings, precautions, contraindications, and prohibited medications during treatment and to the core CSP Appendix J3.

6.5.1 Rescue Medication

Not applicable to this module.

6.6 Dose Modification

6.6.1 T-DXd

The combination of T-DXd and tucatinib may increase the risk or severity of potential overlapping toxicities. Furthermore, there is potential for new, unanticipated toxicities. The Investigator should use clinical judgement when attributing a toxicity to one or more of the agents. For dose modifications that apply to toxicities that are attributable to T-DXd refer to the core CSP Section 6.6.1.

6.6.2 Tucatinib

Mod 6 Table 5 provides recommended dose levels when dose modifications of tucatinib for a patient are necessary.

Mod 6 Table 5 Dose Reduction Levels of Tucatinib

Dose Level 1	Dose Level -1	Dose Level -2	Dose Level -3
300 mg bid	250 mg bid	200 mg bid	150 mg bid

If CCI mg is the RP2D, then the initial dose reduction level would be CCI mg. Dose reductions CCI CCI may be made if considered clinically appropriate by the investigator. Dose reductions below 150 mg are not permitted.
bid = twice daily; RP2D = recommended phase 2 dose

6.7 Intervention after the End of the Study

Refer to the core CSP Section 6.7.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

See the SoA in Section 1.3 in this module for data to be collected at the time of discontinuation of study treatment and follow-up and for any further evaluations that need to be completed.

7.1 Discontinuation of Study Intervention

Refer to the core CSP Section 7.1.

7.2 Patient Withdrawal from the Study

Refer to the core CSP Section 7.2.

7.3 Lost to Follow-up

Refer to the core CSP Section 7.3.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for module-specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Refer to the core CSP Section 8.3.1.

8.3.2 Follow-up of AEs and SAEs

Refer to the core CSP Section 8.3.2

8.3.3 Causality Collection

Refer to the core CSP Section 8.3.3.

8.3.4 Adverse Events Based on Signs and Symptoms

Refer to the core CSP Section 8.3.4.

8.3.5 Adverse Events Based on Examinations and Tests

Refer to the core CSP Section 8.3.5.

8.3.6 Hy's Law

Refer to the core CSP Section 8.3.6.

8.3.7 Disease Progression

Refer to the core CSP Section 8.3.7.

8.3.8 Disease Under Study

Refer to the core CSP Section 8.3.8.

8.3.9 New Cancers

Refer to the core CSP Section 8.3.9.

8.3.10 Deaths

Refer to the core CSP Section 8.3.10.

8.3.11 Adverse Events of Special Interest

No AESIs will be considered for tucatinib. Refer to the core CSP Section 8.3.11 for details on AESIs related to T-DXd.

8.3.12 Safety Data to be Collected Following the Final Data Cut-off of the Study

Refer to the core CSP Section 8.3.12.

8.3.13 Reporting of Serious Adverse Events

Refer to the core CSP Section 8.3.13. The reference documents for definition of expectedness/listedness for the study treatments used in this module are:

- The current IB for T-DXd.
- The label of the current effective version of the SmPC for tucatinib, Seagen, Inc.

8.3.14 Pregnancy

Refer to the core CSP Section 8.3.14.

8.3.15 Medication Error, Drug Abuse, and Drug Misuse

Refer to the core CSP Section 8.3.15.

8.3.16 Management of Study Treatment-related Toxicities

8.3.16.1 Toxicity Management and Dose Modification Information – T-DXd

Refer to the core CSP Section 8.3.16, the ILD/pneumonitis management summary flow chart in Appendix K of the core CSP, and the T-DXd TMGs in the core CSP Appendix L.

8.3.16.2 Toxicity Management and Dose Modification Information – Tucatinib

Toxicities of tucatinib may be managed according to the local prescribing information.

8.4 Overdose

Use of 20% more than the intended w/v dose of T-DXd, and doses of tucatinib in excess of those specified in the CSP are considered an overdose. There is currently no specific treatment for overdose of T-DXd and tucatinib, and possible symptoms of overdose are not established. Refer to the core CSP Section 8.4 for more information. Refer to the local prescribing information for treatment in case of an overdose related to tucatinib.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.5.1 Pharmacokinetics

Whole blood samples will be collected for measurement of concentrations of T-DXd, total anti-HER2 antibody, and MAAA-1181a in serum, and tucatinib and its metabolite in plasma

at timepoints as specified in the SoA in this module. Additional PK blood samples will be collected for patients diagnosed with COVID-19. Refer to the core CSP Appendix M.

8.5.2 Immunogenicity Assessments

Blood samples for determination of ADA for T-DXd in serum will be collected as specified in the SoA in this module. Refer to the core CSP Section 8.5.2 for further details.

8.5.3 Pharmacodynamics

Pharmacodynamic parameters are not evaluated in this study.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

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Clinical Study Protocol	
Study Intervention	Trastuzumab deruxtecan (T-DXd, AZD4552, DS-8201a)
Study Code	D967JC00001
Version	5.0
Date	03 December 2024
EudraCT / EU CT Number	2019-004531-22 / 2023-505309-18-00

Appendix U

Module 7 – T-DXd monotherapy in patients with active brain metastases

A Phase 1b/2 Multicentre, Open-label, Modular, Dose-finding and Dose-expansion Study to Explore the Safety, Tolerability, and Anti-tumour Activity of Trastuzumab Deruxtecan (T-DXd) in Combination with other Anti-cancer Agents in Patients with HER2-positive Metastatic Breast Cancer (DESTINY-Breast07)

Sponsor Name: AstraZeneca AB

Legal Registered Address: AstraZeneca AB, 151 85 Södertälje, Sweden

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This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The CSP is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Policy on Bioethics and in compliance with prevailing laws and regulations.

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DOCUMENT HISTORY	
Document	**Date**
Amendment 4 (Version 5.0)	03 December 2024
Amendment 3 (Version 4.0)	19 September 2023
Amendment 2 (Version 3.0)	23 November 2021
Amendment 1 (Version 2.0)	15 December 2020
Original Protocol (Version 1.0)	30 July 2020

Refer to core CSP for summary of changes table for all modules.

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1 PROTOCOL SUMMARY

1.1 Synopsis

Module 7 of this study will evaluate T-DXd monotherapy in patients with HER2-positive advanced/MBC, in the first-or second-line setting with active brain metastases. For further details on the study synopsis, refer to the core CSP Section 1.1.

1.2 Schema

The study schema is provided in the core CSP Section 1.2.

1.3 Schedule of Activities

Refer to the core CSP Section 1.3 for details of screening assessments. The tables for procedures to be conducted in the treatment period (Mod 7 Table 1) and follow-up period (Mod 7 Table 2) for Module 7 are shown below. Details of assessments and procedures are described in the core CSP Section 8 unless otherwise specified. Any treatment-specific procedures are described in Section 8 of this module.

Whenever vital signs and blood draws are scheduled for the same nominal time, the vital signs assessments should occur first. Whenever ECGs, vital signs, and blood draws are scheduled for the same nominal time, ECG assessments should occur first, then vital signs assessments, and then blood draws; the timing of the first 2 assessments should be such that it allows the blood draw (eg, PK blood sample) to occur at the timepoints indicated in the SoA.

When patients do not receive study treatment, patients should undergo a physical examination and vital signs assessment with additional assessments at the discretion of the Investigator. All samples are collected before infusion unless otherwise indicated.

Every effort should be made to minimise the time between treatment assignment and dosing. Dosing should occur no more than 3 days after treatment assignment. If it is anticipated that dosing cannot occur within 3 days, a discussion with the AstraZeneca Study Physician is required. Patients who fail to meet the eligibility criteria will be termed “screen failures”.

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Study procedures and assessments													
Vital signs	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
HRCT of the chest	Only if ILD/pneumonitis is suspected											8.2.5.2	
SpO_2	X	X	X	X	X	X	X	X	X	X	X	8.2.5.2	
Full physical examination	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
Weight	X [b]				X[b]		X[b]		X [b]		X	8.2.1	
ECOG PS	X [b]				X[b]		X[b]		X [b]		X	8.2.5.3	
Concomitant medications	X (at every visit and may be conducted by phone if not tied to a visit)											6.5	
AEs/SAEs	X (at every visit and may be conducted by phone if not tied to a visit)											8.3	
Laboratory assessments [c]													
Pregnancy test [d]	X				X		X		X		X	8.2.4	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X [b]		X	X	X[b]		X[b]		X [b]		X	8.2.4, Table 11	
Urinalysis	As clinically indicated											Table 11	
Coagulation	As clinically indicated											Table 11	
12-lead ECG [e]	X [b]				X		X		X		X	8.2.3	

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
ECHO/MUGA (LVEF) [f]									X C5 (and every 4 cycles [± 7 days])		X	8.2.5.1	
Troponin [g] (local assessment)	As clinically indicated										X	Table 11, 8.2.5.1	
Ophthalmologic assessments [p]	As clinically indicated										X	8.2.5.4	
Pharmacokinetic and Immunogenicity assessments (for patients diagnosed with COVID-19, please see Appendix M for further instructions on potential additional PK sample collections)													
Pre-dose blood sample for T-DXd PK testing [h]	X				X				X (Cycles 4, 6, and 8)			8.5.1	
Post-dose blood sample for T-DXd PK testing [i]		X				X				X (C4 only)		8.5.1	
Blood sample for T-DXd ADA assessment in serum [j]	X				X				X (C4, then every 4 cycles)		X	8.5.2	
Study treatments administered													
T-DXd	5.4 mg/kg will be administered via intravenous infusion on D1 of each 3-week cycle											6.2.1	
Tumour assessments													
Tumour assessment by RECIST 1.1 of chest,	Tumour evaluation scans will be performed at screening (as baseline) with follow-ups Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at											8.1.1	

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
abdomen, pelvis (CT or MRI) and brain (MRI)	Week 57 until RECIST 1.1 disease progression. The on-study imaging schedule MUST be followed regardless of any delays in dosing. If clinical progression/deterioration is suspected, an unscheduled RECIST assessment should be performed. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Additional scans to be completed per standard practice post-progression.												
Brain contrast MRI	Regularly scheduled follow-up brain scans (Q6W ± 1 week from the date of treatment assignment for 48 weeks, and then Q9W ± 1 week, starting at Week 57 thereafter until RECIST 1.1 disease progression) are mandatory for all patients. Following investigator-determined RECIST 1.1 progression, an additional tumour scan will be performed no earlier than 4 weeks and no later than the next regularly scheduled imaging assessment if it is feasible. This additional scan may be sent for BICR review. Patients who receive local therapy for CNS-progression should maintain their assessment schedule (including scans).										X	7.1.2, 8.1.1	
Whole body bone scan	As clinically indicated											8.1.1	
Biomarker assessments													
Tissue sample from the metastatic setting: either a newly acquired biopsy or the most recent archival tumour [k] (if not collected at screening)	X [l]											8.6.1, 8.6.2	

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Paired on-treatment biopsy					X[1]							8.6.1, 8.6.2	
Tumour biopsy on disease progression											X (At progression only)	8.6.2	
Blood sample for CCI [redacted]	CCI [redacted]											8.6.1	
Blood sample for CCI [redacted]												8.6.1	
Blood sample for CCI [redacted]												8.6.1	

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days		D1 ± 2 days				
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
Blood sample for CCI	CCI											8.6.1	
Blood sample for CCI CCI												8.6.1	
Blood sample for CCI [m]												8.6.1	
Blood sample to analyse CCI												8.6.1	
Blood sample for CCI												8.7	
Other assessments and procedures													
Heath care resource use (HOSPAD)	At each scheduled visit, the site should review clinical notes for any non-study related hospital admissions and visits that have occurred.											8.8	
ePRO and pulse oximeter device allocation (ePRO device allocated only if	X											8.1.5.1	

Mod 7 Table 1 Schedule of Activities During the Treatment Period - Module 7 (Part 2 only)

Procedures	C1				C2		C3		C4 and subsequent cycles until PD D1 ± 2 days		EOT or E/D [a]	Section in core CSP	Section in this module
	D1		D8	D15	D1 ± 2 days		D1 ± 2 days						
	BI	EI	(± 1 day)	(± 1 day)	Visits are driven by the treatment dosing schedule: Q3W, unless dosing needs to be held for toxicity reasons								
					BI	EI	BI	EI	BI	EI			
patient does not have own compatible device) and training [0]													
MDASI symptom diary	To be completed on C1D1 before dosing and daily thereafter											8.1.5.2	
At-home pulse oximetry	To be completed on C1D1 before dosing and daily thereafter (resting) To be completed on C1D1 before dosing and Q7D thereafter (walking)											8.1.5.3	
CANTAB	X								X (to be performed at Month 3, Month 6 and Month 12)		X	8.1.6.1	
MDASI-BT	X				X		X		X		X	8.1.6.2	
NANO	X				X		X		X		X	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter											8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)											8.1.5.5	

[a] EOT is the date the Investigator decides to discontinue a patient from study treatment; the visit should occur within 7 days of the decision.
[b] Within 3 days before study treatment administration.
[c] If screening laboratory assessments have been performed within 3 days prior to starting study treatment, they do not have to be repeated at C1D1 if the patient's condition has not changed.
[d] Perform repeat pregnancy tests (urine or serum test per institutional guideline) 72 hours before infusion of each cycle and at EOT.
[e] ECGs will be performed in triplicate only if an abnormality is noted. ECGs will be taken in close succession, while in a supine/semi-recumbent position.

[f] ECHO or MUGA scan assessments will be performed BI on D1 of C5 and then every 4 cycles (± 7 days) (eg, C5, 9, 13…) until EOT. The same method must be used for the patient throughout the study.

[g] Collect blood samples for troponin (preferably high-sensitivity troponin-T) at screening, EOT and if at any time a participant reports signs or symptoms suggesting congestive heart failure, myocardial infarction, or other causes of myocyte necrosis. If an additional troponin sample is taken, then an ECG also needs to be performed.

[h] Pre-dose PK samples to be taken up to 6 hours prior to study treatment administration on D1 of C1, 2, 4, 6 and 8.

[i] End of infusion PK samples to be taken within 15 minutes after the end of infusion on D1 of C1, 2 and 4, and an additional PK sample (for T-DXd only) to be taken on C1D1 at 5 hours (± 2 hours) post-infusion.

[j] Within 8 hours before infusion on D1 of C1, 2 and 4, and then every 4 cycles thereafter until EOT.

[k] For patients who have consented to on-treatment (paired) biopsies this sample could represent the first biopsy of the pair (refer to the Core CSP, Section 8.6.1 and 8.6.2).

[l] Refer to the Core CSP Section 8.6.1 for the minimum number of evaluable paired biopsy samples required for specific modules.

[m] At time of suspected CCI , an additional sample should be taken.

[n] The sample for genetic research will be obtained at CCI . If, for any reason, the sample is not drawn at CCI , it may be taken at CCI CCI . Only CCI should be collected per patient for genetics during the study. Although DNA is stable, early sample collection is preferred to avoid introducing bias through excluding patients who may withdraw due to an AE.

[o] An ePRO device should be charged and ready for the patient prior to their arrival at the site for C1D1, ideally at least the day prior, to ensure it is functioning properly and ready for use in case needed, in accordance with device training.

[p] Ophthalmologic assessments including visual acuity testing, slit lamp examination and fundoscopy will be performed at screening, EOT, and as clinically indicated.

Note: All assessments on treatment days are to be performed prior to study treatment administration, unless otherwise indicated. Data collection following study analysis until the end of the study is described in core CSP Section 8.

ADA = anti-drug antibody; AE = adverse event; BI = before infusion; BICR = blinded independent central review; C = cycle; CANTAB, Cambridge Neuropsychological Test Automated Battery; COVID-19 = coronavirus 2019-nCoV; CSP = clinical study protocol; CT = computed tomography; CTCAE = Common Terminology Criteria for Adverse Events; CCI ; D = day; DNA = deoxyribonucleic acid; E/D = Early Study Treatment/Discontinuation; ECG = electrocardiogram; ECHO = echocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EI = end of infusion; EOT = end of treatment; ePRO = electronic patient-reported outcome; HOSPAD = hospital admission; HRCT = high-resolution computed tomography; ILD = interstitial lung disease; IV = intravenous; LVEF = left ventricular ejection fraction; MDASI = MD Anderson Symptom Inventory; MDASI-BT, MDASI Brain Tumor ; MRI = magnetic resonance imaging; MUGA = multiple gated acquisition scans; NANO = Neurologic Assessment in Neuro-Oncology; CCI ; PD = progression of disease; PK = pharmacokinetic; Q7D = every 7 days; Q3W = every 3 weeks; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAE = serious adverse event; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; SpO_2 = pulse oximetry; CCI ; T-DXd = trastuzumab deruxtecan.

Mod 7 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 7

Procedure	Safety F/U (40 days [a] +[7 days] after last dose)	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Vital signs	X		8.2.2	
Weight	X		8.2.1	
Full physical examination	X		8.2.1	
SpO_2	X		8.2.5.2	
ECOG PS	X		8.2.5.3	
12-lead ECG	X		8.2.3	
Clinical safety laboratory assessments (clinical chemistry, haematology)	X		8.2.4, Table 11	
Urinalysis	As clinically indicated		Table 11	
Coagulation tests	As clinically indicated		Table 11	
Pregnancy test	X	As clinically indicated	8.2.4	
Blood sample for T-DXd ADA analysis in serum	X		8.5.2	
Concomitant medications	X		6.5	
AEs/SAEs	X		8.3	
PFS2 assessment	X	X	7.1.2.1, 8.1.3	
Health care resource use (HOSPAD)	X		8.8	

Mod 7 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 7

Procedure	Safety F/U (40 days [a] +[7 days] after last dose)	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
MDASI symptom diary	For those without ILD/pneumonitis diagnosis: • To be completed daily up to and including the safety F/U For those with an ILD/pneumonitis diagnosis: To be completed daily up to and including the first survival visit.		8.1.5.1	
At-home pulse oximetry	For those without ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the safety F/U • To be completed Q7D relative to C1D1 (walking) up to and including the safety F/U For those with ILD/pneumonitis diagnosis: • To be completed daily (resting) up to and including the first survival visit To be completed Q7D relative to C1D1 (walking) up to and including the first survival visit		8.1.5.3	
MDASI-BT		X (months 3 & 6 only)	8.1.6.2	
NANO		X (months 3 & 6 only)	8.1.6.3	
SGRQ-I	To be completed after diagnosis of ILD/pneumonitis and Q7D thereafter, up to and including the first survival visit.		8.1.5.4	
ILD/pneumonitis qualitative patient interview	To be completed once after diagnosis of ILD/pneumonitis (in selected countries)		8.1.5.5	
Survival status	X	X	7.1.2.1, 8.1.4	

Mod 7 Table 2 Schedule of Activities for Safety Follow-up and Long-term Follow-up - Module 7

Procedure	Safety F/U (40 days [a] +[7 days] after last dose)	Long-term/Survival F/U (every 3 months after the post-treatment F/U ± 14 days) [b]	Section in core CSP	Section in this module
Subsequent cancer therapy following discontinuation of study treatment	X	X	8.1.4	
Tumour assessment by RECIST 1.1 of chest, abdomen, pelvis (CT or MRI) and brain (MRI)	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should continue to have RECIST 1.1 scans performed Q6W ± 1 week for the first 48 weeks from treatment assignment, and then Q9W ± 1 week (starting at Week 57) thereafter until RECIST 1.1 disease progression.		8.1.1	
Whole body bone scan	Patients who permanently discontinue study treatment for reasons other than objective RECIST 1.1 disease progression, withdrawal of consent, closure of study or death (regardless of whether subsequent anticancer therapy was started) should have bone scans as clinically indicated.		8.1.1	

[a] If EOT is > 40 days (+ 7 days) after last treatment, then the EOT assessments can also function as the 40-day (+ 7 days) follow-up visit.
[b] Long-term/Survival F/U visits will be performed until death, withdrawal of consent, or study closure, whichever occurs first.

ADA = anti-drug antibody; AEs = adverse events; C = cycle; CSP = clinical study protocol; CT = computer tomography; D = day; ECG = electrocardiogram; ECOG PS = Eastern Cooperative Oncology Group Performance Status; EOT = end of treatment; F/U = follow-up; HOSPAD = hospital admission; ILD = interstitial lung disease; MDASI = MD Anderson Symptom Inventory; MDASI-BT=MDASI Brain Tumor; MRI = magnetic resonance imaging; NANO, Neurologic Assessment in Neuro-Oncology; PBMC PFS2 = progression-free survival after next-line treatment; Q7D = every 7 days; Q6W = every 6 weeks; Q9W = every 9 weeks; RECIST = Response Evaluation Criteria in Solid Tumours; SAEs = serious adverse events; SpO_2 = pulse oximetry; SGRQ-I = St George's Respiratory Questionnaire for patients with Idiopathic Pulmonary Fibrosis; T-DXd = trastuzumab deruxtecan.

2 INTRODUCTION

2.1 Study Rationale

Refer to the core CSP Section 2.1.

2.2 Background

Refer to the core CSP Section 2.2.

2.3 Benefit/Risk Assessment

Refer to the core CSP Section 2.3 for a benefit/risk assessment of T-DXd in HER2-positive MBC.

2.3.1 Potential Benefits of T-DXd

Refer to the core CSP Section 2.3.4 for details on the potential benefits of T-DXd.

T-DXd monotherapy has shown preliminary efficacy in patients with stable brain metastases. In the U201 (DESTINY-Breast01) study, a subset (n=24) of patients with stable brain metastases was found to have a mPFS of 18.1 months, while the ITT population had mPFS of 16.4 months. Overall, only 4 out of 48 patients in the study had disease progression in the CNS (Jerusalem et al 2020).

Given the encouraging efficacy that T-DXd has demonstrated in heavily pre-treated patients with HER2-positive MBC with or without brain metastases, an exploration of T-DXd is justified.

2.3.2 Potential Risks of T-DXd

Refer to the core CSP Section 2.3.1 for details on the potential risks of T-DXd.

2.3.3 Overall Benefit/Risk Conclusion

T-DXd is an emerging therapy and has recently been approved for the treatment of HER2-positive MBC. T-DXd has demonstrated an unparalleled mPFS of 16.4 months in late line HER2-positive patient population (Modi et al 2020). The subpopulation with stable brain metastases demonstrated a mPFS of 18.1 months, indicating the benefits of T-DXd are also observed in patients with brain metastases (Jerusalem et al 2020).

Given the encouraging efficacy that this drug has demonstrated in HER2-positive MBC in patients with brain metastases, the benefit-risk ratio of this therapy is anticipated to be favourable.

3 OBJECTIVES AND ENDPOINTS

Refer to the core CSP Section 3.

4 STUDY DESIGN

4.1 Overall Design

DESTINY-Breast07 is an open-label, multicentre, Phase 1b/2 study to evaluate the safety, tolerability, and preliminary anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive (IHC 3+ or IHC 2+/ISH+) advanced/MBC. For more details on the overall study design refer to the core CSP Section 4.1. In this module, patients with active brain metastases will be enrolled and treated with T-DXd monotherapy.

There is no Part 1 (dose-finding) in Module 7.

4.1.1 Module 7 Part 2

In Part 2, approximately 35 patients with locally assessed HER2-positive (IHC 3+ or IHC 2+/ISH+) breast cancer in the first-line or second-line setting for advanced/MBC with active brain metastases will be randomised to determine evidence of anti-tumour activity. Patients will be treated with T-DXd monotherapy at the dose of 5.4 mg/kg IV infusion Q3W.

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Study Design

Refer to the core CSP Section 4.2.1.

4.2.2 Rationale for T-DXd in HER2-positive Advanced/MBC

Refer to the core CSP Section 4.2.2.

T-DXd has demonstrated a mPFS of 16.4 months in the late-line HER2-positive ITT patient population (Modi et al 2020). In addition, a subset (n=24) of patients with stable brain metastases were found to have a mPFS of 18.1 months, which is longer than that seen in the general population, though this result must be interpreted with caution given the small sample size (Jerusalem et al 2020).

Given the high prevalence of brain metastases in patients with cancer, their systematic exclusion from clinical trials may result in the assessment of an investigational drug's efficacy or safety in a trial population that is not fully representative of the patient population that will be prescribed the drug in clinical practice. Designing clinical trials that include patients with brain metastases and including this information in the labelling promotes the safe and effective use of these products across a broader patient population likely to use the drug in clinical practice.

Patients with brain metastases have limited treatment options, representing an unmet medical need. Since T-DXd has shown efficacy in CNS disease, it is important to determine the preliminary efficacy and safety profile for this therapy in active brain metastases.

4.2.3 Rationale for Study Endpoints

Refer to the core CSP Section 4.2.4.

4.3 Justification for Dose

Refer to the core CSP Section 4.3.1

4.4 End of Study Definition

Refer to the core CSP Section 4.4.

5 STUDY POPULATION

Refer to the core CSP Section 5.

Specific eligibility criteria for Module 7 are listed below.

5.1 Inclusion Criteria

Refer to core module for the other inclusion criteria. The below inclusion criteria are applicable to this module only.

8 **Part 2**, Zero or one prior lines of therapy for advanced/MBC allowed (additional requirement: at least 12 months since last adjuvant or neoadjuvant HER2-targeted therapy, or chemotherapy required). Prior taxane, trastuzumab and pertuzumab exposure is allowed in the adjuvant or neoadjuvant setting.

12 **Part 2**, At least one lesion, not previously irradiated, that qualifies as a RECIST 1.1 TL at baseline (see Appendix G). Up to 2 active brain lesions not previously irradiated may be selected as RECIST 1.1 TLs at baseline.

CNS Inclusion

18 Patients must have either:

(a) Untreated brain metastasis not needing local therapy. For patients with untreated CNS lesions > 2.0 cm on screening contrast MRI, discussion with and approval from the Study physician or delegate is required prior to enrolment.

(b) Brain metastasis previously treated with local therapy and have progressed since prior local therapy, provided that there is no clinical indication for immediate re-treatment for local therapy according to the investigator's opinion.

5.2 Exclusion Criteria

CNS Exclusion

27 Ongoing use of systemic corticosteroids for control of symptoms of brain metastasis at a total daily dose of > 2 mg dexamethasone (or equivalent).

28 Any brain lesion thought to require immediate local therapy, including (but not limited to) a lesion in an anatomic site where increase in size or possible treatment-related edema may pose risk to patients (eg, brain stem lesions). Patients who undergo local treatment for such lesions identified screening contrast brain MRI may still be eligible for the study based on criteria described under CNS inclusion criteria 20 (see core CSP Section 5.2).

6 STUDY INTERVENTION

Refer to the core CSP Section 6.

7 DISCONTINUATION OF STUDY INTERVENTION AND PATIENT DISCONTINUATION/WITHDRAWAL

Refer to the core CSP Section 7.

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are provided in the SoA in Section 1.3 in this module. Details for module specific assessments and procedures are detailed in the subsections below. Details for all other assessments and procedures are provided in the core CSP Section 8.

8.1 Efficacy Assessments

Refer to the core CSP Section 8.1.

8.2 Safety Assessments

Refer to the core CSP Section 8.2 for all subsections.

8.3 Adverse Events and Serious Adverse Events

Refer to the core CSP Section 8.3.

8.4 Overdose

Refer to the core CSP Section 8.4.

8.5 Human Biological Samples

Refer to the core CSP Section 8.5.

8.6 Human Biological Sample Biomarkers

Refer to the core CSP Section 8.6.

8.7 Optional Genomics Initiative Sample

Refer to the core CSP Section 8.7.

8.8 Medical Resource Utilisation and Health Economics

Refer to the core CSP Section 8.8.

9 STATISTICAL CONSIDERATIONS

Refer to the core CSP Section 9.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Refer to the core CSP Section 10.

11 REFERENCES

Jerusalem et al 2020
Jerusalem G, Park YH, Yamashita T, Hurvitz SA, Chen S, Cathcart J, et al. 1380 CNS metastases in HER2-positive metastatic breast cancer treated with trastuzumab deruxtecan: DESTINY-Breast01 subgroup analyses. Ann Oncol. 2020;31 Suppl 2:S63-S64.

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Modi S, Saura C, Yamashita T, Park YH, Kim SB, Tamura K, et al. Trastuzumab deruxtecan in previously treated HER2-positive breast cancer. N Engl J Med. 2020;382(7):610-21.

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